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PMIC50x0 Power Management IC Specification, Rev 1.0

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PMIC50x0 POWER MANAGEMENT IC SPECIFICATION, Rev. 1
Definition of PMIC5000, PMIC5010 Voltage Regulator
Device for Memory Module Applications

(From JEDEC Board Ballot JCB-20-04, formulated under the cognizance of the JC-40.1 Subcommittee on Digital Logic Families and Applications.)

1 Scope

This standard defines the specifications of interface parameters, signaling protocols, and features for PMIC device as used for memory module applications. The designation PMIC5000, PMIC5010 refers to the device specified by this document.

The purpose is to provide a standard for the PMIC5000, PMIC5010 device for uniformity, multiplicity of sources, elimination of confusion, ease of device specification, and ease of use.

Unless otherwise noted in the document, any illegal operation is not allowed and device operation is not guaranteed.

NOTE The designation PMIC5000, PMIC5010 refers to a portion of the part number designation of a series of commercial logic devices common in the industry. This number is normally preceded by a series of manufacturer specific characters to make up a complete part designation.

2 Device Standard

2.1 Description

The PMIC5000 & PMIC5010 is designed for typical DDR5 RDIMM, DDR5 LRDIMM as well as various types of DDR5 NVDIMM application. The PMIC features four step down switching regulators and three LDO regulators.

The PMIC is designed to support approximately 15 Watts of power. The PMIC is powered from VIN_Bulk input for switching regulators and VIN_Mgmt input for the rest of the PMIC. The PMIC supports selectable interface (I²C or I3C Basic) to fit various application environment. The PMIC device is intended to operate up to 12.5 MHz on a 1.0 V I3C Basic bus or up to 1 MHz on a 1.0 V to 3.3 V I²C bus.

2.2 Common Features summary

Table 1 — PMIC Device Type Summary

Device Type	SWA	SWB	SWC	SWD	Unit
PMIC5000 - Current Capability per Phase	5	5	5	5	A
PMIC5010 - Current Capability per Phase	3	3	3	3	A

- VIN_Bulk input supply range: 4.25 V to 15.0 V
- VIN_Mgmt input supply range: 3.0 V to 3.6 V
- Four step down switching regulators: SWA, SWB, SWC & SWD
- Programmable dual phase and single phase regulator for SWA and SWB
- 3 LDO regulators: VBias, VOUT_1.8V, VOUT_1.0V
- Automatic switchover from VIN_Mgmt input supply to VIN_Bulk input supply
- Error injection capability
- Persistent Error log registers
- Secure mode and programmable of operation
- Independently programmable output voltages, power up and power down sequence for switch regulators
- Input and output power good status reporting mechanism
- VIN_Bulk input supply protection feature: Input over voltage
- Output switch regulators protection feature: Output over voltage, output under voltage, output current limiter
- Output current and power measurement, output current threshold mechanism
- Temperature measurement, temperature warning threshold, critical temperature shutdown
- Multi Time Programmable Non-Volatile Memory
- Programmable and DIMM specific registers for customization
- General Status Interrupt Function
- Flexible Open Drain IO (I²C) and Push Pull (I3C Basic) IO Support

3 Input Supply & Output Regulator Electrical Characteristics

3.1 Input Supply Electrical Characteristics

Table 2 — Input Supply DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Bulk Input Supply Voltage DC Voltage	VIN_Bulk	4.25	12	15	V	1,2
Bulk Input Supply Maximum AC Voltage	VIN_Bulk_AC	-	-	18	V	
Bulk Input Supply Maximum Voltage Start up Overshoot	VIN_Bulk_OS_Startup	-	-	33	V* μ s	3
Bulk Input Supply Voltage Ramp Up Rate	VIN_Bulk_Ramp_Up	0.1	-	3.0	V/ms	4
Bulk Input Supply Voltage Ramp Down Rate	VIN_Bulk_Ramp_Down	0.5	-	1.0	V/ms	5
Management Input Supply Voltage	VIN_Mgmt	3.0	3.3	3.6	V	6
Management Input Supply Ramp Up and Down Rate	VIN_Mgmt_Ramp	0.15		3	V/ms	
Minimum Management Input Supply Current	I _{VIN_Mgmt}	110	-	-	mA	7
Bulk Input Supply Current	I _{VIN_Bulk}		-	2.5	A	8

1. During first power on, the input voltage supply must reach minimum value based on default from register Table 121, "Register 0x1A" [7:5] + 1.0 V for PMIC to detect valid input supply.
2. The PMIC efficiency is optimized for nominal input supply of 12 V or lower. The PMIC efficiency above 13.8 V is degraded and thermal impact must be considered. The lifetime operation above 13.8 V is limited to TBD hours.
3. The area under the curve above VIN_Bulk = 15V. VIN_Bulk_AC spec must also be satisfied.
4. The ramp up rate between 300 mV and 8.0 V.
5. The ramp down rate between 8.0 V and 300 mV.
6. During first power on, the input voltage supply must reach minimum value of 2.8 V for PMIC to detect valid input supply.
7. This is a platform spec. The minimum input current delivered by the platform through the DIMM gold finger to deliver the maximum load on LDO outputs (1.8V LDO output + 1.0V LDO output = 25 mA + 20 mA) plus the current required by the PMIC for its own usage.
8. This is a platform spec. The maximum input current delivered by the platform through the DIMM gold finger.

3.2 Switch Regulator Output Electrical Characteristic

Table 3 — SWA, SWB¹ - Single Phase Regulator; PMIC5000; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Output Voltage	Vout		1.1		V	2
Maximum Continuous DC Current Load	I _{tdc}	0	-	5	A	3
Maximum Peak Instantaneous Current	I _{peakmax}	-	-	6	A	4
Maximum Load Transient	dI/dt	-	-	5	A/Ωs	
Regulator Output DC + AC Voltage Tolerance	Reg_DC_AC_Tol	-2.5%		2.5%		5
Regulator Feedback Set Point Accuracy	FB_Set_Point	-0.75%		0.75%		6

1. Only applicable if Table 171, "Register 0x4F" [0] = '0'.
2. Typical voltage configured in the register Table 128, "Register 0x21," [7:1] for SWA and Table 130, "Register 0x23," [7:1] for SWB.
3. Measured over long period of time. Typically 1 second.
4. Measured over short period of time. Typically $\geq 20 \Omega$ s but less than 50 Ω s.
5. The percentage applies to typical voltage configured in the register. Applies across entire PMIC operating temperature range. The PMIC bulk input supply voltage VIN_Bulk can vary from minimum to maximum value specified in Table 2, "Input Supply DC + AC Specification". The regulator output current load can vary maximum dI/dt value. The output ripple is inclusive in this parameter and not to exceed TBD mV.
6. The percentage applies to typical voltage configured in the register and at a given temperature within operating temperature range. The PMIC bulk input supply voltage VIN_Bulk is fixed at nominal voltage of 12.0 V. The regulator output current load I_{tdc} = 0 A.

Table 4 — SWA + SWB¹ - Dual Phase Regulator; PMIC5000; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Output Voltage	Vout		1.1		V	2
Maximum Continuous DC Current Load	I _{tdc}	0	-	10	A	3
Maximum Peak Instantaneous Current	I _{peakmax}	-	-	12	A	4
Maximum Load Transient	dI/dt	-	-	10	A/μs	
Regulator Output DC + AC Voltage Tolerance	Reg_DC_AC_Tol	-2.5%		2.5%		5
Regulator Feedback Set Point Accuracy	FB_Set_Point	-0.75%		0.75%		6

1. Only applicable if Table 171, "Register 0x4F" [0], = '1'.
2. Typical voltage configured in the register Table 128, "Register 0x21," [7:1].
3. Measured over long period of time. Typically 1 second.
4. Measured over short period of time. Typically $\geq 20 \mu$ s but less than 50 μ s.
5. The percentage applies to typical voltage configured in the register. Applies across entire PMIC operating temperature range. The PMIC bulk input supply voltage VIN_Bulk can vary from minimum to maximum value specified in Table 2, "Input Supply DC + AC Specification". The regulator output current load can vary maximum dI/dt value. The output ripple is inclusive in this parameter and not to exceed TBD mV.
6. The percentage applies to typical voltage configured in the register and at a given temperature within operating temperature range. The PMIC bulk input supply voltage VIN_Bulk is fixed at nominal voltage of 12.0V. The regulator output current load I_{tdc} = 0 A.

Table 5 — SWA, SWB¹ - Single Phase Regulator; PMIC5010; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Output Voltage	Vout		1.1		V	2

Table 5 — SWA, SWB¹ - Single Phase Regulator; PMIC5010; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Maximum Continuous DC Current Load	I_{tdc}	0	-	3	A	3
Maximum Peak Instantaneous Current	$I_{peakmax}$	-	-	3.5	A	4
Maximum Load Transient	dI/dt	-	-	5	A/ μ s	
Regulator Output DC + AC Voltage Tolerance	Reg_DC_AC_Tol	-2.5%		2.5%		5
Regulator Feedback Set Point Accuracy	FB_Set_Point	-0.75%		0.75%		6

1. Only applicable if Table 171, "Register 0x4F" [0] = '0'.
2. Typical voltage configured in the register Table 128, "Register 0x21," [7:1] for SWA and Table 130, "Register 0x23," [7:1] for SWB.
3. Measured over long period of time. Typically 1 second.
4. Measured over short period of time. Typically $\geq 20 \mu$ s but less than 50μ s.
5. The percentage applies to typical voltage configured in the register. Applies across entire PMIC operating temperature range. The PMIC bulk input supply voltage VIN_Bulk can vary from minimum to maximum value specified in Table 2, "Input Supply DC + AC Specification". The regulator output current load can vary maximum dI/dt value. The output ripple is inclusive in this parameter and not to exceed TBD mV.
6. The percentage applies to typical voltage configured in the register and at a given temperature within operating temperature range. The PMIC bulk input supply voltage VIN_Bulk is fixed at nominal voltage of 12.0 V. The regulator output current load $I_{tdc} = 0$ A.

Table 6 — SWA + SWB¹ - Dual Phase Regulator; PMIC5010; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Output Voltage	Vout		1.1		V	2
Maximum Continuous DC Current Load	I_{tdc}	0	-	6	A	3
Maximum Peak Instantaneous Current	$I_{peakmax}$	-	-	7	A	4
Maximum Load Transient	dI/dt	-	-	10	A/ μ s	
Regulator Output DC + AC Voltage Tolerance	Reg_DC_AC_Tol	-2.5%		2.5%		5
Regulator Feedback Set Point Accuracy	FB_Set_Point	-0.75%		0.75%		6

1. Only applicable if Table 171, "Register 0x4F" [0] = '1'.
2. Typical voltage configured in the register Table 128, "Register 0x21," [7:1].
3. Measured over long period of time. Typically 1 second.
4. Measured over short period of time. Typically $\geq 20 \mu$ s but less than 50μ s.
5. The percentage applies to typical voltage configured in the register. Applies across entire PMIC operating temperature range. The PMIC bulk input supply voltage VIN_Bulk can vary from minimum to maximum value specified in Table 2, "Input Supply DC + AC Specification". The regulator output current load can vary maximum dI/dt value. The output ripple is inclusive in this parameter and not to exceed TBD mV.
6. The percentage applies to typical voltage configured in the register and at a given temperature within operating temperature range. The PMIC bulk input supply voltage VIN_Bulk is fixed at nominal voltage of 12.0 V. The regulator output current load $I_{tdc} = 0$ A.

Table 7 — SWC¹ - Single Phase Regulator; PMIC5000; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Output Voltage	V _{out}		1.1		V	2
Maximum Continuous DC Current Load	I _{tdc}	0	-	5	A	3
Maximum Peak Instantaneous Current	I _{peakmax}	-	-	6	A	4
Maximum Load Transient	dI/dt	-	-	5	A/μs	
Regulator Output DC + AC Voltage Tolerance	Reg_DC_AC_Tol	-2.5%		2.5%		5
Regulator Feedback Set Point Accuracy	FB_Set_Point	-0.75%		0.75%		6

1. There is no note. This is intentional.
2. Typical voltage configured in the register Table 132, "Register 0x25," [7:1].
3. Measured over long period of time. Typically 1 second.
4. Measured over short period of time. Typically $\geq 20 \mu\text{s}$ but less than $50 \mu\text{s}$.
5. The percentage applies to typical voltage configured in the register. Applies across entire PMIC operating temperature range. The PMIC bulk input supply voltage VIN_Bulk can vary from minimum to maximum value specified in Table 2, "Input Supply DC + AC Specification". The regulator output current load can vary maximum dI/dt value. The output ripple is inclusive in this parameter and not to exceed TBD mV.
6. The percentage applies to typical voltage configured in the register and at a given temperature within operating temperature range. The PMIC bulk input supply voltage VIN_Bulk is fixed at nominal voltage of 12.0 V. The regulator output current load I_{tdc} = 0 A.

Table 8 — SWC¹ - Single Phase Regulator; PMIC5010; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Output Voltage	V _{out}		1.1		V	2
Maximum Continuous DC Current Load	I _{tdc}	0	-	3	A	3
Maximum Peak Instantaneous Current	I _{peakmax}	-	-	3.5	A	4
Maximum Load Transient	dI/dt	-	-	5	A/μs	
Regulator Output DC + AC Voltage Tolerance	Reg_DC_AC_Tol	-2.5%		2.5%		5
Regulator Feedback Set Point Accuracy	FB_Set_Point	-0.75%		0.75%		6

1. There is no note. This is intentional.
2. Typical voltage configured in the register Table 132, "Register 0x25," [7:1].
3. Measured over long period of time. Typically 1 second.
4. Measured over short period of time. Typically $\geq 20 \mu\text{s}$ but less than $50 \mu\text{s}$.
5. The percentage applies to typical voltage configured in the register. Applies across entire PMIC operating temperature range. The PMIC bulk input supply voltage VIN_Bulk can vary from minimum to maximum value specified in Table 2, "Input Supply DC + AC Specification". The regulator output current load can vary maximum dI/dt value. The output ripple is inclusive in this parameter and not to exceed TBD mV.
6. The percentage applies to typical voltage configured in the register and at a given temperature within operating temperature range. The PMIC bulk input supply voltage VIN_Bulk is fixed at nominal voltage of 12.0 V. The regulator output current load I_{tdc} = 0 A.

Table 9 — SWD¹ - Single Phase Regulator; PMIC5000; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Output Voltage	V _{out}		1.8		V	2
Maximum Continuous DC Current Load	I _{tdc}	0	-	5	A	3
Maximum Peak Instantaneous Current	I _{peakmax}	-	-	6	A	4
Maximum Load Transient	dI/dt	-	-	5	A/μs	
Regulator Output DC + AC Voltage Tolerance	Reg_DC_AC_Tol	-2.5%		2.5%		5
Regulator Feedback Set Point Accuracy	FB_Set_Point	-0.75%		0.75%		6

1. There is no note. This is intentional.
2. Typical voltage configured in the register Table 134, "Register 0x27," [7:1].
3. Measured over long period of time. Typically 1 second.
4. Measured over short period of time. Typically $\geq 20 \mu\text{s}$ but less than $50 \mu\text{s}$.
5. The percentage applies to typical voltage configured in the register. Applies across entire PMIC operating temperature range. The PMIC bulk input supply voltage VIN_Bulk can vary from minimum to maximum value specified in Table 2, "Input Supply DC + AC Specification". The regulator output current load can vary maximum dI/dt value. The output ripple is inclusive in this parameter and not to exceed TBD mV.
6. The percentage applies to typical voltage configured in the register and at a given temperature within operating temperature range. The PMIC bulk input supply voltage VIN_Bulk is fixed at nominal voltage of 12.0 V. The regulator output current load $I_{\text{tdc}} = 0 \text{ A}$.

Table 10 — SWD¹ - Single Phase Regulator; PMIC5010; DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Output Voltage	Vout		1.8		V	2
Maximum Continuous DC Current Load	I_{tdc}	0	-	3	A	3
Maximum Peak Instantaneous Current	I_{peakmax}	-	-	3.5	A	4
Maximum Load Transient	dI/dt	-	-	5	A/ μs	
Regulator Output DC + AC Voltage Tolerance	Reg_DC_AC_Tol	-2.5%		2.5%		5
Regulator Feedback Set Point Accuracy	FB_Set_Point	-0.75%		0.75%		6

1. There is no note. This is intentional.
2. Typical voltage configured in the register Table 134, "Register 0x27," [7:1].
3. Measured over long period of time. Typically 1 second.
4. Measured over short period of time. Typically $\geq 20 \mu\text{s}$ but less than $50 \mu\text{s}$.
5. The percentage applies to typical voltage configured in the register. Applies across entire PMIC operating temperature range. The PMIC bulk input supply voltage VIN_Bulk can vary from minimum to maximum value specified in Table 2, "Input Supply DC + AC Specification". The regulator output current load can vary maximum dI/dt value. The output ripple is inclusive in this parameter and not to exceed TBD mV.
6. The percentage applies to typical voltage configured in the register and at a given temperature within operating temperature range. The PMIC bulk input supply voltage VIN_Bulk is fixed at nominal voltage of 12.0 V. The regulator output current load $I_{\text{tdc}} = 0 \text{ A}$.

3.3 Switch Regulator Efficiency

Table 11 — Efficiency Characteristics; PMIC5000

Switch Node Output	Efficiency (% of Max I_{tdc} Load)			Unit	Notes
	25%	50%	100%		
SWA or SWB (Single Phase Regulator Mode)	≥ 87	≥ 92	≥ 89	%	1,2,3,4,5,6,7,8,9,10
SWA + SWB (Dual Phase Regulator Mode)	≥ 87	≥ 92	≥ 89	%	
SWC	≥ 87	≥ 92	≥ 89	%	
SWD	≥ 87	≥ 92	≥ 89	%	

1. VIN_Bulk = 12 V; VIN_Mgmt = 3.3 V
2. The maximum I_{tdc} as specified in appropriate Tables above.
3. When the efficiency of any given output regulator is being measured, all other switching output regulators are disabled.
4. No external load on VOUT_1.8V, VOUT_1.0V LDO is applied.
5. I²C/I³C Basic bus is pulled High and held static. PWR_GOOD and GSI_n signals are pulled High and held static.
6. The efficiency includes the buck regulator loss, the PCB loss ($\leq 2.5 \text{ m}\Omega$) and see Section 13 for inductor specification assumption for DCR and ACR.
7. Efficiency calculation equation: $(V_{\text{OUT}} * I_{\text{OUT}}) / [(V_{\text{IN_Bulk}} * I_{\text{VIN_Bulk}}) + (V_{\text{IN_Mgmt}} * I_{\text{VIN_Mgmt}})]$; where V_{OUT} , I_{OUT} , $V_{\text{IN_Mgmt}}$, $V_{\text{IN_Bulk}}$, $I_{\text{VIN_Bulk}}$ parameters are actual measured values.
8. Applies at maximum ambient temperature of 65 °C (PMIC Junction temperature of 105 °C). The inductor characteristics noted above applies inductor temperature of 105 °C.

9. The output buck regulator switching frequency can be set to anywhere within 500 KHz to 1000 KHz. For all efficiency qualification testing, the device under test (DUT) must also comply with all PMIC's electrical characteristics (DC+AC) specifications.
10. For input supply rails, probing is done at the input high frequency filter cap (0.1uF) to PMIC pin. For output rail, probing is done at the output cap location at the inductor load side.

Table 12 — Efficiency Characteristics; PMIC5010

Switch Node Output	Efficiency (% of Max I_{tdc} Load)			Unit	Notes
	25%	50%	100%		
SWA or SWB (Single Phase Regulator Mode)	≥ 85	≥ 90	≥ 87	%	1,2,3,4,5,6,7,8,9,10
SWA + SWB (Dual Phase Regulator Mode)	≥ 85	≥ 90	≥ 87	%	
SWC	≥ 85	≥ 90	≥ 87	%	
SWD	≥ 85	≥ 90	≥ 87	%	

- VIN_Bulk = 12 V; VIN_Mgmt = 3.3 V
- The maximum I_{tdc} as specified in appropriate Tables above.
- When the efficiency of any given output regulator is being measured, all other switching output regulators are disabled.
- No external load on VOUT_1.8V, VOUT_1.0V LDO is applied.
- I²C/I³C Basic bus is pulled High and held static. PWR_GOOD and GSI_n signals are pulled High and held static.
- The efficiency includes the buck regulator loss, the PCB loss ($\leq 2.5 \text{ m}\Omega$) and see Section 13 for inductor specification. assumption for DCR and ACR.
- Efficiency calculation equation: $(V_{OUT} * I_{OUT}) / [(V_{IN_Bulk} * I_{VIN_Bulk}) + (V_{IN_Mgmt} * I_{VIN_Mgmt})]$; where V_{OUT} , I_{OUT} , V_{IN_Mgmt} , I_{VIN_Mgmt} , V_{IN_Bulk} , I_{VIN_Bulk} parameters are actual measured values.
- Applies at maximum ambient temperature of 65 °C (PMIC Junction temperature of 105 °C). The inductor characteristics noted above applies inductor temperature of 105 °C.
- The output buck regulator switching frequency can be set to anywhere within 500 KHz to 1000 KHz. For all efficiency qualification testing, the device under test (DUT) must also comply with all PMIC's electrical characteristics (DC+AC) specifications.
- For input supply rails, probing is done at the input high frequency filter cap (0.1uF) to PMIC pin. For output rail, probing is done at the output cap location at the inductor load side.

3.4 LDO Output Regulator Characteristics

Table 13 — LDO Output Regulator DC + AC Specification

Parameter	Symbol	Min	Typ	Max	Unit	Notes
1.8 V LDO Output Voltage	VOUT_1.8V		1.8		V	1
1.8 V LDO Output - Maximum Output Current	I _{tdc} VOUT_1.8V	-	-	25	mA	2
1.0 V LDO Output Voltage	VOUT_1.0V		1.0		V	3
1.0 V LDO Output - Maximum Output Current	I _{tdc} VOUT_1.0V	-	-	20	mA	4

1. Typical voltage is configured in register Table 138, "Register 0x2B," [7:6]. The min and max values are guaranteed to be within ± 100 mV of programmed value.
2. The maximum output current represents the external load and excludes PMIC's own internal current consumption. The specified maximum output current is only applicable after PMIC's 1.8V LDO Power Good status is good (i.e., t1.8V_Ready timing parameter is satisfied). Prior to PMIC's 1.8V LDO Power Good status (i.e. while PMIC is still ramping up the 1.8 V LDO, the maximum output current load shall be limited to maximum of 10 mA.
3. Typical voltage is configured in register Table 138, "Register 0x2B," [2:1]. The min and max values are guaranteed to be within ± 50 mV of programmed value.
4. The maximum output current represents the external load and excludes PMIC's own internal current consumption. The specified maximum output current is only applicable after PMIC's 1.0V LDO Power Good status is good (i.e., t1.0V_Ready timing parameter is satisfied). Prior to PMIC's 1.0V LDO Power Good status (i.e., while PMIC is still ramping up the 1.0 V LDO, the maximum output current load shall be limited to maximum of 5 mA.

3.5 PMIC AC Timing Parameters

Table 14 — PMIC AC Timing Parameters

Parameter	Symbol	Min	Typ	Max	Unit	Notes
Input Supply to GSI_n assertion	tInput_PWR_GOOD_GSI_Assertion			10	μ s	
Input over voltage condition to GSI_n assertion	tInput_OV_GSI_Assertion	-	-	10	μ s	
Input over voltage condition to automatic PMIC VR Disable	tInput_OV_VR_Disable	-	-	20	μ s	
Output Voltage Tolerance to GSI_n assertion	tOutput_PWR_GOOD_GSI_Assertion	-	-	10	μ s	
Output over voltage condition to automatic PMIC VR Disable	tOutput_OV_VR_Disable	-	-	20	μ s	
Output under voltage lockout condition to automatic PMIC VR Disable	tOutput_UV_VR_Disable	-	-	20	μ s	
Output current limiter Warning to GSI_n assertion	tOutput_Current_Limiter	-	-	10	μ s	
High Temperature Warning to GSI_n assertion	tHigh_Temp_Warning	-	-	10	μ s	
Critical Temperature condition to automatic PMIC shut down	tShut_Down_Temp	-	-	10	μ s	
VIN_Mgmt input supply stable to VR Enable Command	tVIN_Mgmt_to_Enable	6.5	-	-	ms	
VIN_Bulk input supply stable to VR Enable Command	tVIN_Bulk_to_Enable	6.5	-	-	ms	
VIN_Mgmt input supply stable to VOUT_1.8V output stable	t1.8V_Ready	-	-	3.5	ms	
VOUT_1.8V output supply stable to VOUT_1.0V output stable	t1.0V_Ready	-	-	1.0	ms	1
VOUT_1.8V output supply to PMIC Management Ready	tManagement_Ready	-	-	3	ms	

Table 14 — PMIC AC Timing Parameters

Parameter	Symbol	Min	Typ	Max	Unit	Notes
VR Enable Command to PMIC output regulator ready	tPMIC_PWR_Good_Out	Figure 15			ms	
VR Disable Command to PMIC Output Regulators Off	tPMIC_Output_Off	Figure 16			ms	
PWR_GOOD Input Low Pulse Width	tPWR_GOOD_Low_Pulse_Width	2	0	0	μs	
PWR_GOOD Input Low Pulse Width Input Filter	tPWR_GOOD_Low_Pulse_Width_Filter	-	-	0.35	μs	
Output Voltage Adjustment in Programmable Mode	$\Delta v/\Delta t$	-	1	-	mV/μs	²

1. This time is added to t_1.8V_Ready parameter to get total time from VIN_Mgmt input supply.
2. See footnote 4 for registers Table 128, “Register 0x21,” [7:1], Table 130, “Register 0x23,” [7:1], Table 132, “Register 0x25,” [7:1], Table 134, “Register 0x27,” [7:1]. The accuracy is $\pm 10\%$

4 I²C, I³C Basic & Interface DC and AC Electrical Characteristics

Table 15 — I²C, I³C and Interface DC Electrical Specification

Parameter	Symbol	Min	Max	Unit	Notes
Input Low Voltage (PWR_GOOD, SDA, SCL)	V _{IL}	-0.3	0.3	V	
Input High Voltage (SDA, SCL)	V _{IH}	0.7	3.6	V	
Input High Voltage (PWR_GOOD)	V _{IH}	1.26	3.6	V	
Output Low Voltage (SDA, GSI_n)	V _{OL}	-	0.3	V	1
Output High Voltage (SDA)	V _{OH}	0.75	-	V	
Output Low Current (SDA, GSI_n, PWR_GOOD)	I _{OL}	-	3	mA	
Output High Current (SDA)	I _{OH}	-3	-	mA	
Output Low Voltage (PWR_GOOD)	V _{OL_PG}	-	0.3	V	2
Output Slew Rate (SDA)	Slew_Rate	0.1	1	V/ns	3

1. The pullup resistor for GSI_n signal may vary and is typically 1K Ohm.
2. PWR_GOOD output is Open Drain output. There is an external pullup resistor to 3.3 V motherboard for standard DDR5 RDIMM/LRDIMM. For other DIMM environment, the external pullup resistor may be pulled to either 1.8 V or 2.5 V or 3.3 V on either on DIMM or on the motherboard.
3. Output slew rate is guaranteed by design and/or characterization. The output slew rate reference load is shown in Figure 4 and Figure 5 shows the timing measurement points. For slew rate measurement, the V_{OH} level shown in Figure 5 is a function of R_{on} value; V_{OH} = {1.0/(R_{on} + 50)} * 50.

Table 16 — Input Capacitance Spec

Parameter	Symbol	Min	Max	Unit	Notes
Input Capacitance (PWR_GOOD, SCL, SDA)	C _{IN}	-	5	pF	

Table 17 — Input Spike Filter Spec

Parameter	Symbol	Test Condition	Min	Max	Unit	Notes
Pulse width of spikes which must be suppressed by the input filter in I ² C mode	t _{sp}	Single glitch, f ≤ 100 KHz	-	-	ns	
		Single glitch, f > 100 KHz	0	50	ns	1

1. T_A = 25 °C; f = 400 KHz. Verified by design and characterization only.

Table 18 — Output Ron

Parameter	Symbol	Min	Max	Unit	Notes
SDA Output Pullup and Pulldown Driver Impedance	R _{ON}	25	100	Ω	1
GSI_n, PWR_GOOD Output Pulldown Driver Impedance		40	100	Ω	1

1. Pulldown Ron = V_{out}/I_{out}. Pullup Ron = (V_{OUT_1.0V} - V_{out})/I_{out}.

Table 19 — I²C and I3C Interface AC Characteristics

Parameter	Symbol	I ² C Mode - Open Drain		I3C Basic Mode Push-Pull ¹		Unit	Notes
		Min	Max	Min	Max		
Clock Frequency	f _{SCL}	0.01	1	0.01	12.5	MHz	
Clock High Pulse Width Time	t _{High}	260		35		ns	
Clock Low Pulse Width Time	t _{Low}	500		35		ns	
Detect Clock Input Low Time	t _{TIMEOUT}	10	50	10	50	ms	
Rise Time	t _R	-	120	-	5	ns	2,3
Fall time	t _F	-	120	0	5	ns	2,3
Data in Setup Time	t _{SU:DAT}	50	-	8	-	ns	2,4
Data in Hold Time	t _{HD:DI}	0	-	3	-	ns	2,4
Start Condition Setup Time	t _{SU:STA}	260	-	12	-	ns	2
Start Condition Hold Time	t _{HD:STA}	260	-	30	-	ns	2
Stop Condition Setup Time	t _{SU:STO}	260	-	12	-	ns	2
Time between Stop Condition and next Start Condition	t _{BUF}	500	-	500	-	ns	2,5
SDA Data Out Hold Time	t _{HD:DAT}	0.5	350	N/A	N/A	ns	6
SCL Falling Clock In to Valid SDA Data Out Time	t _{DOUT}	N/A	N/A	0.5	30	ns	7,8
SCL Rising Clock In to Slave SDA Output Off	t _{DOFFS}	N/A	N/A	0.5	30	ns	7,8,9
SCL Rising Clock In to Master SDA Output Off	t _{DOFFM}	N/A	N/A	0.5	30	ns	7,8,10
SCL Rising Clock In to Master Driving Data Signal Low	t _{CL_r_DAT_f}	N/A	N/A	40	-	ns	11
Bus Available Time (no edges seen on SCL and SDA)	t _{AVAIL}	N/A	N/A	1	-	μs	
Time to issue IBI after an event is detected when Bus is available	t _{IBI_ISSUE}	N/A	N/A	-	15	μs	
Time from Clear Register Status to any I3C Basic operation with Start condition to avoid false IBI generation; PEC disabled	t _{CLR_I3C_CMD_Delay}	N/A	N/A	2.5	-	μs	
Time from Clear Register Status to any I3C Basic operation with Start condition to avoid false IBI generation; PEC enabled		N/A	N/A	15	-	μs	
DEVCTRL CCC Followed by DEVCTRL CCC or Register Read/Write Command Delay	t _{DEVCTRLCCC_DELAY_PEC_DIS}	3	-	3	-	μs	12,13,14
Register Write Command Followed by Register Read Command Delay in PEC Enabled Mode	t _{WR_RD_DELAY_PEC_EN}	N/A	N/A	8	-	μs	15,16,17
SETHID CCC or SETAASA CCC followed by any other CCC or Read/Write Command Delay	t _{I2C_CCC_Update_Delay}	2.5	-	-	-	μs	
RSTDAA CCC or ENEC CCC or DISEC CCC to any other CCC or Read/Write Command Delay	t _{I3C_CCC_Update_Delay}	-	-	2.5	-	μs	
Any CCC followed by RSTDAA CCC delay	t _{CCC_Delay}	N/A	N/A	2.5	-	μs	

1. I²C mode with Open Drain operation follows timing values as shown in I²C Mode - Open Drain column.
2. See Figure 1 for PMIC's input timing definition.
3. See Figure 6 for voltage threshold definition for rise and fall times.
4. The input setup time is referenced from SDA VIL or VIH threshold as shown in Figure 1 to SCL VIH threshold as shown in Figure 1. The input hold time is referenced from SCL VIL threshold as shown in Figure 1 to SDA VIL or VIH threshold as shown in Figure 1.
5. If PEC is enabled, $t_{WR_RD_DELAY_PEC_EN}$ timing parameter applies.
6. The PMIC device guarantees $t_{HD:DAT}$ value in I²C mode of operation. See Figure 3 for PMIC's output timing definitions as well as SCL clock input threshold level and SDA data output threshold levels.
7. The PMIC device must be configured in I3C Basic mode to guarantee t_{DOUT} or t_{DOFFS} or t_{DOFFM} value. See Figure 2 for PMIC's output timing definition as well as SCL clock input threshold level and SDA data output threshold levels.
8. This timing parameter is guaranteed into output timing reference load as shown in Figure 4.
9. The PMIC device must be configured in I3C Basic mode to guarantee t_{DOFFS} value. See Figure 18.
10. The PMIC device must be configured in I3C Basic mode to guarantee t_{DOFFM} value. See Figure 19.
11. See Figure 21.
12. From STOP condition of DEVCTRL CCC to START condition for Register Read or Register Write Command Data Packet delay.
13. The PMIC sends NACK if Host does not satisfy $t_{DEVCTRLCCC_DELAY_PEC_DIS}$ timing parameter.
14. This timing parameter restriction is only applicable when PEC function is disabled in PMIC. If PEC is enabled, this timing parameter does not apply.
15. From STOP condition for Register Write Command Data Packet to START condition for Register Read Command Data Packet delay.
16. This timing parameter restriction is only applicable when PEC function is enabled in PMIC. If PEC is disabled, this timing parameter does not apply.
17. The PMIC sends NACK if Host does not satisfy $t_{WR_RD_DELAY_PEC_EN}$ timing parameter.

The PMIC device follow the I²C or I3C Basic bus timing requirements. Figure 1, Figure 2 and Figure 3 show the timing diagram for Data bus Input and Data Output parameters.

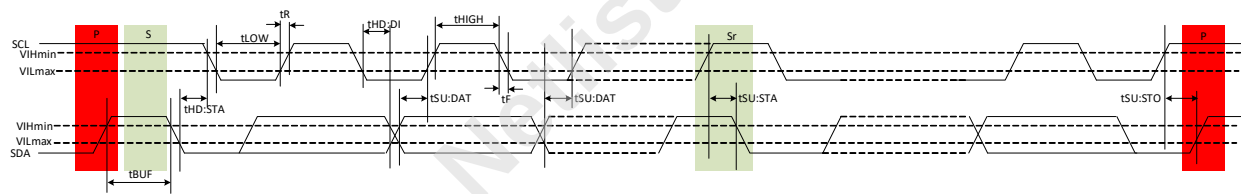


Figure 1 — I²C or I3C Basic Bus AC Input Timing Parameter Definition

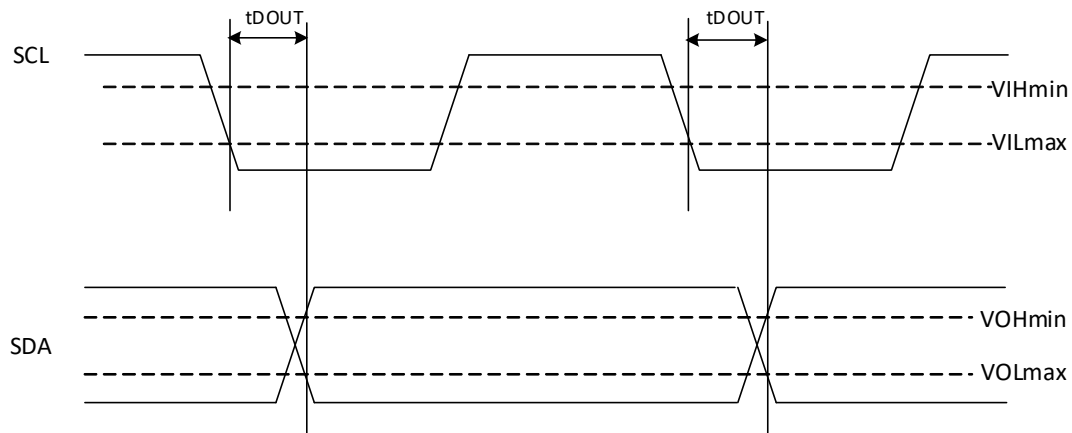


Figure 2 — I3C Basic Bus AC Data Output Timing Parameter Definition

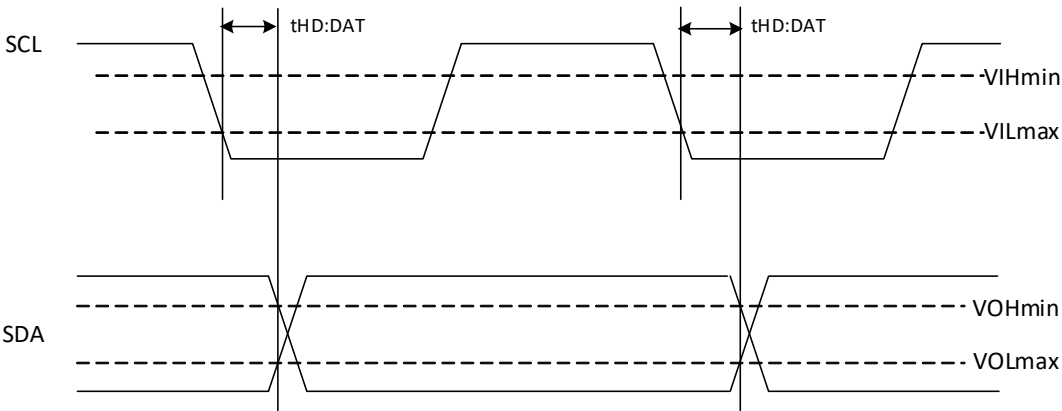


Figure 3 — I²C Bus AC Data Output Timing Parameter Definition

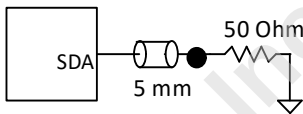


Figure 4 — Output Slew Rate and Output Timing Reference Load

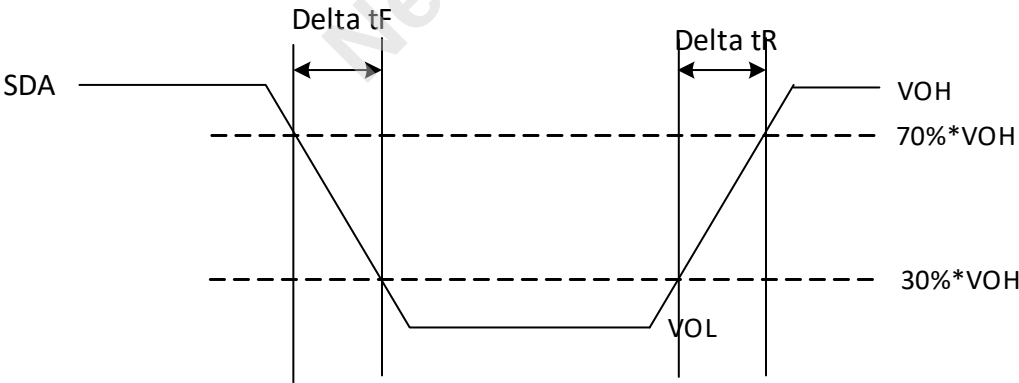


Figure 5 — Output Slew Rate Measurement Points

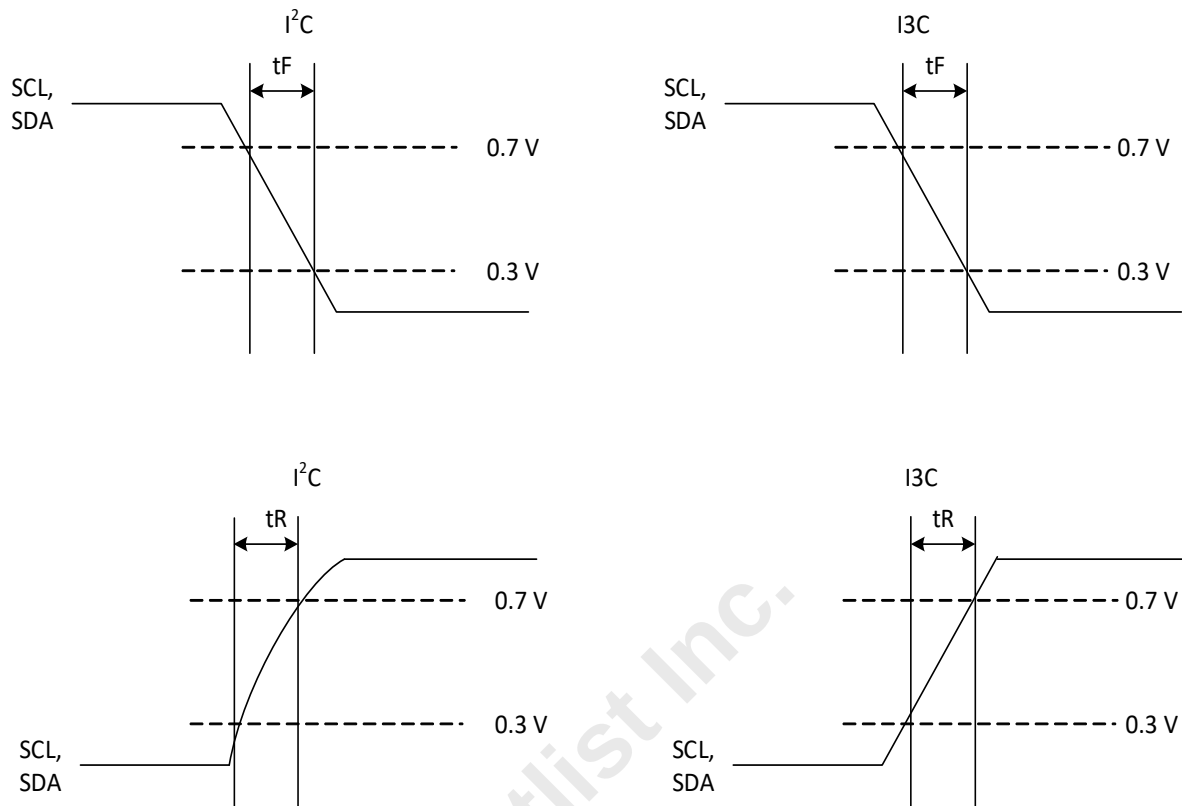


Figure 6 — Rise and Fall Timing Parameter Definition

Table 20 — AC Measurement Conditions¹

Symbol	Parameter	Min	Max	Units
C_L	Load capacitance		40	pF
	Input rise & Fall times - Open Drain	-	TBD	ns
	Input rise and fall times - Push Pull	-	TBD	ns
	Input signal swing levels	0.2 to 0.8		V
	Input and Output timing reference levels	0.3 to 0.7		V

1. This AC measurement condition (Table 20 and Figure 7) is only for the test purpose in lab.

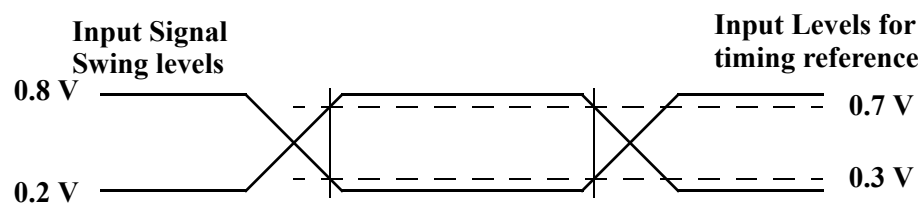


Figure 7 — AC Measurement Waveform

5 Thermal Characteristics

Table 21 — Thermal Characteristics

Pin	Symbol	Maximum Rating	Unit	Notes
Thermal resistance junction to ambient	Θ_{JA}	TBD	$^{\circ}\text{C}/\text{W}$	1
Thermal resistance junction to case	Θ_{JC}	TBD	$^{\circ}\text{C}/\text{W}$	2
Junction operating temperature	T_J	-10 to 125	$^{\circ}\text{C}$	
Ambient operating temperature	T_A	-10 to 65	$^{\circ}\text{C}$	
Storage temperature	T_{STG}	-55 to 150	$^{\circ}\text{C}$	
Lead temperature (soldering, 10s)	T_{LEAD}	300	$^{\circ}\text{C}$	

1. The maximum power dissipation is $P_{D(MAX)} = (T_{JMAX} - T_A) / \Theta_{JA}$. Exceeding the maximum allowable power dissipation results in excessive die temperature and the device will enter thermal shutdown.
2. This thermal rating was calculated on JEDEC JESD51 standard 4 layer board with dimensions 3" x 4.5" in still air conditions. Actual thermal resistance is affected by PCB size, solder joint quality, layer count, copper thickness, air flow, altitude and other unlisted variables.

6 Absolute Maximum Rating

The absolute maximum ratings are stress ratings only. Stresses greater than those listed in Table 22, “Absolute Maximum Rating” and Table 23, “ESD Requirement” and Table 24, “EOS Requirement” may cause permanent damage to the device. Functional operation of the PMIC at absolute maximum ratings is not implied. Exposure to absolute maximum rating condition for extended periods may affect long term reliability.

Table 22 — Absolute Maximum Rating

Pin	Maximum Rating		Unit
	DC	AC	
VIN_BULK	-0.3 to 16.2	TBD (Duration $\leq$ 25 ns)	V
VIN_Mgmt, VBIAS	-0.3 to 6.0	-	V
VOUT_1.8V, VOUT_1.0V	-0.3 to 2.2	-	V
SWA, SWB, SWC, SWD	-0.3 to 16.2	-4.5 to 20 (Duration < 25 ns)	V
SWA_BOOT, SWB_BOOT, SWC_BOOT, SWD_BOOT (to GND)	-0.3 to 21.0	-0.3 to 24 (Duration < 25 ns)	V
SWA_BOOT, SWB_BOOT, SWC_BOOT, SWD_BOOT (to SWx)	-0.3 to 6.0	TBD	V
SWAB_FB_P, SWB_FB_P, SWC_FB_P, SWD_FB_P (to AGND)	-0.3 to 2.2	-	V
SWAB_FB_N, SWC_FB_N, SWD_FB_N/PID	-0.3 to 2.2	-	V
PWR_GOOD, GSI_n	-0.3 to 5.0	-	V
SCL, SDA	-0.3 to 5.0	TBD	V
AGND, PGND	-0.3 to 0.3	-	V

Table 23 — ESD Requirement

Test Model	Pin	Maximum Rating	Unit
HBM	All	± 2000	V
CDM	All	± 500	V

Table 24 — EOS Requirement

Pin	Maximum Rating	Unit	Notes
VIN_BULK	<u>37</u>	V	1,2,3,4
VIN_Mgmt	10	V	

1. The test is performed on DDR5 DIMM module without any input capacitor on VIN_BULK and VIN_Mgmt
2. The input source needs to follow the waveform and condition as shown in Figure 8 and Table 25.
3. Probing is performed at the VIN_BULK and VIN_Mgmt pin of PMIC.
4. Each net test is performed individually.

Table 25 — Input Source Condition

Item	Value	Notes
T (rise from 30% to 90% of peak)	<u>0.72 μs (+ 30%)</u>	
T1 (rise time)	<u>1.2 μs (+ 30%)</u>	T1 = 1.67*T
T2 (duration time to half value)	<u>50 μs (+ 20%)</u>	
Output Impedance	2 Ω	
V_UNDERSHOOT Voltage	30 % Max	

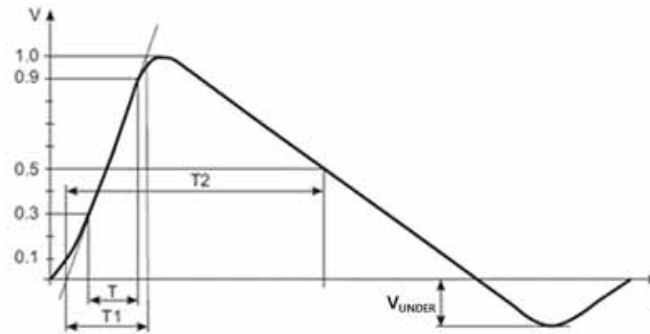


Figure 8 — Impulse Waveform for EOS Test (IEC 61000-4-5)

Netlist Inc.

7 Example Schematic

Figure 9 shows an example schematic when PMIC is configured in dual phase regulator mode. Table 26, “PMIC Schematic Values” shows all component details shown in the schematics. Note that capacitors C25, C26, C27, C29, C30, C31 represents the lump sum of distributed capacitance across the entire DIMM.

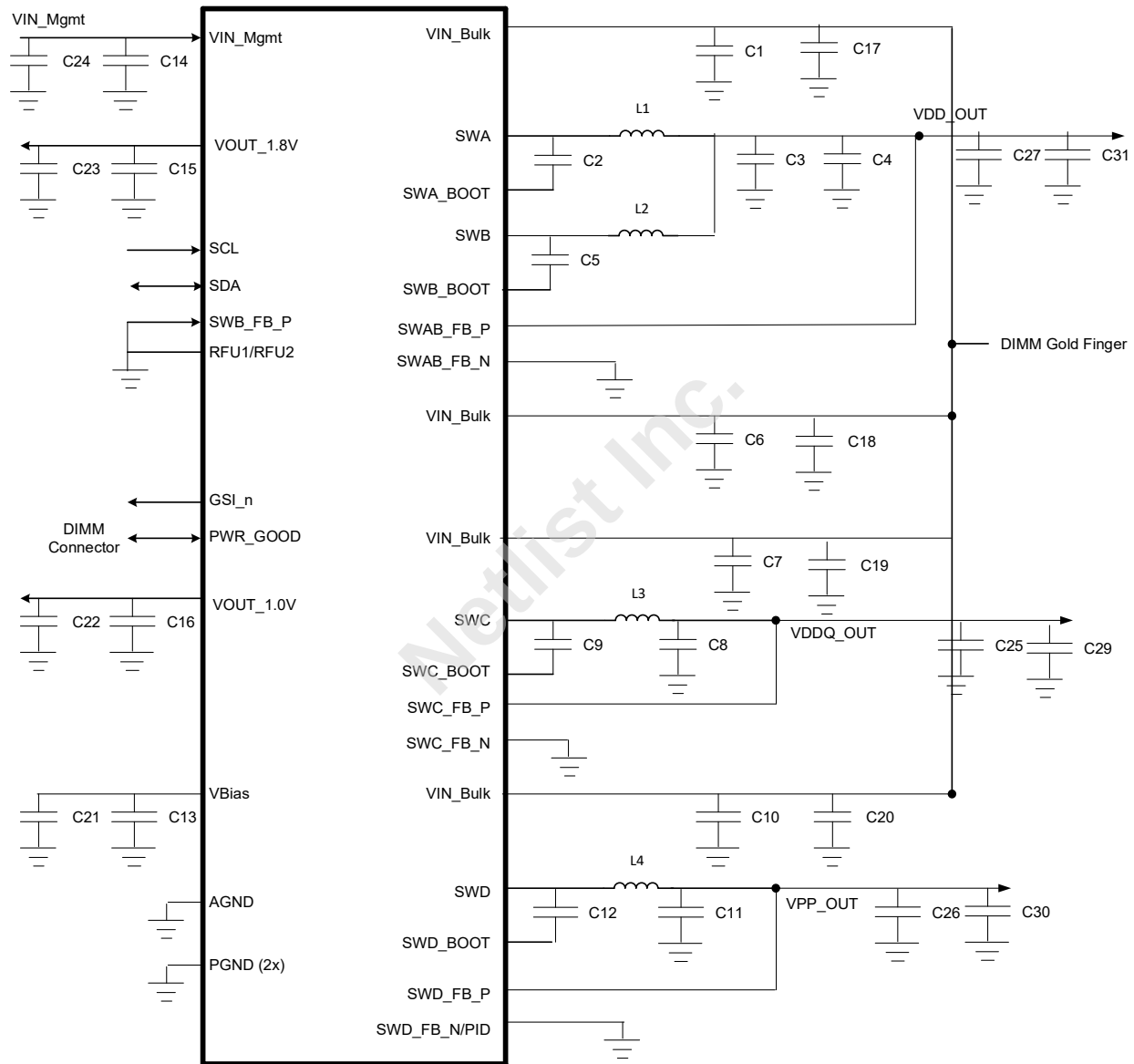


Figure 9 — Dual Phase Regulator Example Schematic

Figure 10 shows an example schematic when PMIC is configured in single phase regulator mode. Table 26, “PMIC Schematic Values” shows all component details shown in the schematics. Note that capacitors C25, C26, C27, C28, C29, C30, C31 and C32 represents the lump sum of distributed capacitance across the entire DIMM.

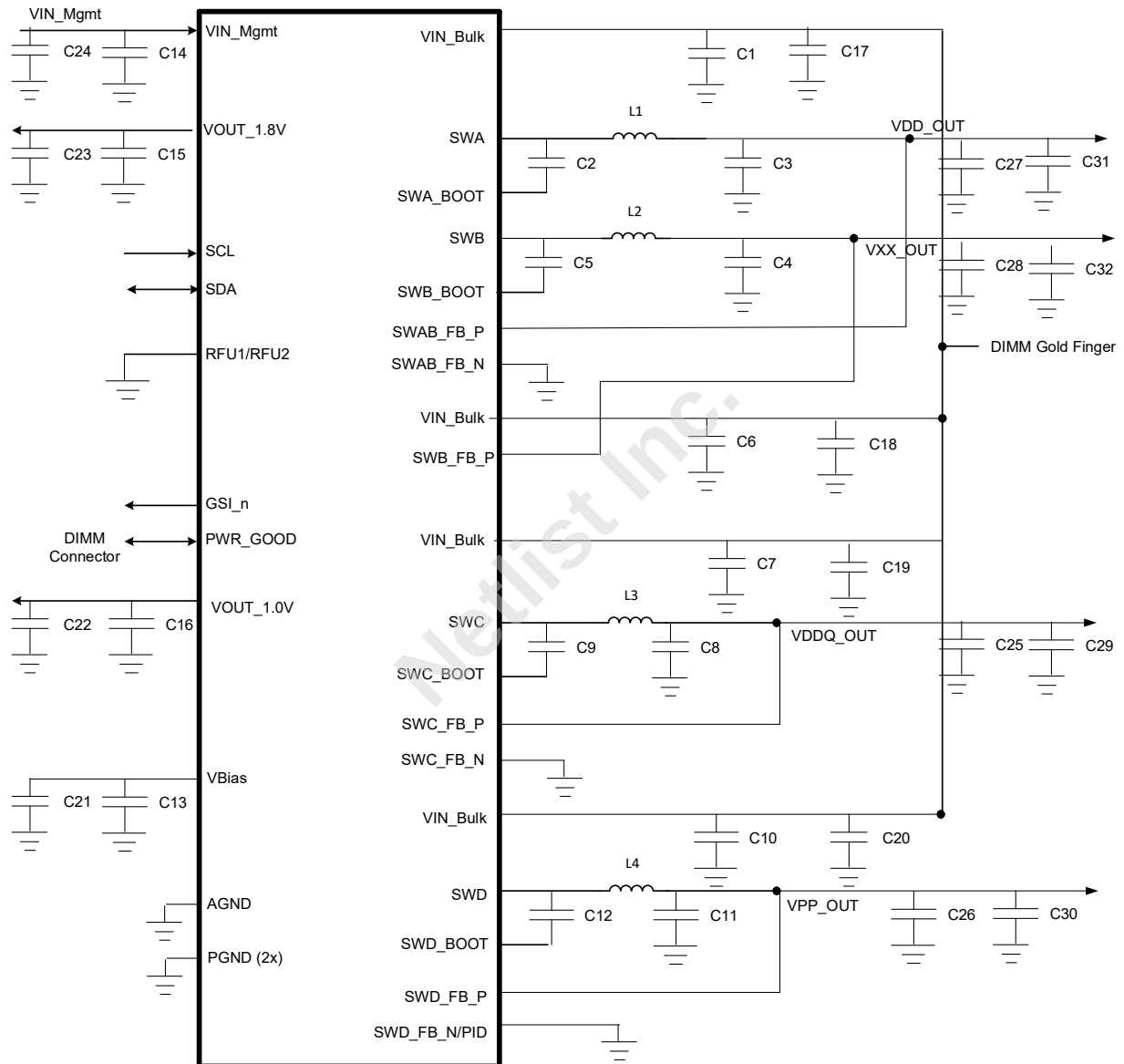


Figure 10 — Single Phase Regulator Example Schematic

Table 26 — PMIC Schematic Values

Component	Dual Phase Regulator Mode		Single Phase Regulator Mode		Unit	Comment
	Value	Physical Size/Rating	Value	Physical Size/Rating		
L1	0.47 - 0.68	4.0 x 4.0 x 2.0	0.47 - 0.68	4.0 x 4.0 x 2.0	μH	
L2	0.47 - 0.68	4.0 x 4.0 x 2.0	0.47 - 0.68	4.0 x 4.0 x 2.0	μH	
L3	0.47 - 0.68	4.0 x 4.0 x 2.0	0.47 - 0.68	4.0 x 4.0 x 2.0	μH	
L4	0.68 - 1.2	4.0 x 4.0 x 2.0	0.68 - 1.2	4.0 x 4.0 x 2.0	μH	
C1	2x; 22	25V	2x; 22	25V	μF	
C2	0.1	16V; 0201	0.1	16V; 0201	μF	
C3	3x; 47	6.3V; 0805	3x; 47	6.3V; 0805	μF	
C4	3x; 47	6.3V; 0805	3x; 47	6.3V; 0805	μF	
C5	0.1	16V; 0201	0.1	16V; 0201	μF	
C6	2x; 22	25V	2x; 22	25V	μF	
C7	2x; 22	25V	2x; 22	25V	μF	
C8	3x; 47	6.3V; 0805	3x; 47	6.3V; 0805	μF	
C9	0.1	16V; 0201	0.1	16V; 0201	μF	
C10	2x; 22	25V	2x; 22	25V	μF	
C11	3x; 47	6.3V; 0805	3x; 47	6.3V; 0805	μF	
C12	0.1	16V; 0201	0.1	16V; 0201	μF	
C13	4.7	6.3V; 0402	4.7	6.3V; 0402	μF	
C14	4.7	6.3V; 0402	4.7	6.3V; 0402	μF	
C15	4.7	6.3V; 0402	4.7	6.3V; 0402	μF	
C16	4.7	6.3V; 0402	4.7	6.3V; 0402	μF	
C17	0.1	25V, X6S; 0201	0.1	25V, X6S; 0201	μF	
C18	0.1	25V, X6S; 0201	0.1	25V, X6S; 0201	μF	
C19	0.1	25V, X6S; 0201	0.1	25V, X6S; 0201	μF	
C20	0.1	25V, X6S; 0201	0.1	25V, X6S; 0201	μF	
C21	0.1	6.3V; 0201	0.1	6.3V; 0201	μF	
C22	0.1	6.3V; 0201	0.1	6.3V; 0201	μF	
C23	0.1	6.3V; 0201	0.1	6.3V; 0201	μF	
C24	0.1	6.3V; 0201	0.1	6.3V; 0201	μF	
C25	175	6.3V	175	6.3V	μF	1
C26	175	6.3V	175	6.3V	μF	
C27	350	6.3V	175	6.3V	μF	
C28	N/A	N/A	175	6.3V	μF	
C29	175	6.3V	175	6.3V	μF	1
C30	175	6.3V	175	6.3V	μF	
C31	350	6.3V	175	6.3V	μF	
C32	N/A		175	6.3V	μF	

1. These capacitor values represents the distributed capacitance for the entire DIMM assuming max I_{tdc} and $I_{peakmax}$ defined in Electrical characteristics for each SWx. For a given DIMM design, the distributed capacitance for each SWx varies and is a function of I_{tdc} and $I_{peakmax}$ current requirement for that DIMM design. See actual DIMM design for total distributed capacitance.

8 Functional Operation

8.1 PMIC Input Voltage Supplies & Ramp Condition

The DDR5 PMIC has two input supply from the platform: VIN_Bulk and VIN_Mgmt.

The VIN_Bulk supply is used by the PMIC for all output regulators except for the VOUT_1.8V & VOUT_1.0V LDO outputs regulators when not in switchover mode. The VIN_Bulk input supply may also be used to generate the internal bias voltage. Note that the VOUT_1.8V LDO output is separate and independent from SWD output, which is for the DRAM VPP rail. The VOUT_1.0V LDO output is separate and independent from SWA, SWB or SWC.

The PMIC internally generates on its own bias voltage (VBias). At first power on, the VIN_Bulk input supply shall reach a minimum threshold voltage value per register Table 121, “Register 0x1A” [7:5] plus 1.0V offset before it can be detected as a valid input supply to the PMIC. The PMIC filters any non-monotonic noise after this threshold. After power on, with valid VIN_Mgmt input supply while PMIC is operating in programmable mode of operation, if VIN_Bulk input supply is removed and re-applied, it must reach the same threshold voltage value per Table 121, “Register 0x1A” [7:5] plus 1.0V offset.

The VIN_Mgmt supply is used to read out its internal non-volatile memory content and to supply VOUT_1.8V & VOUT_1.0V to other devices such as SPD, TS and RCD on the DIMM. At first power on, the VIN_Mgmt supply shall reach a minimum of 2.8V before it can be detected as a valid input supply to the PMIC. At power on, the PMIC drives PWR_GOOD output signal low only when VIN_Mgmt input supply reaches minimum of 2.8V. VIN_Bulk input supply does not trigger PMIC to drive PWR_GOOD output signal low at power on. The VIN_Mgmt supply is strictly a voltage input.

Figure 11 and Figure 12 show PMIC power up sequence when power is first applied. The platform can power up VIN_Bulk and VIN_Mgmt supply in any sequence. Figure 11 shows VIN_Mgmt supply ramps up first prior to VIN_Bulk supply. Figure 12 shows VIN_Bulk supply ramps up first prior to VIN_Mgmt supply. The PMIC does not mandate any specific timing relationship between VIN_Bulk and VIN_Mgmt supply.

The PMIC updates register Table 103, “Register 0x08” [7] when VIN_Bulk input supply status is valid.

8.2 Power Up Initialization Sequence

During power on, the host shall:

1. Ramp up VIN_Mgmt supply; Ramp up VIN_Bulk supply; (No timing relationship between two supplies)
2. Hold VIN_Mgmt supply stable for a minimum of tVIN_Mgmt_to_Enable time
3. Hold VIN_Bulk supply stable for a minimum of tVIN_Bulk_to_Enable time
4. Query the status of the PMIC status register to determine if it is safe to enable VR.
5. If it is safe to enable, send VR Enable command by setting register Table 145, “Register 0x32” [7] = ‘1’ or by issuing DEVCTRL CCC.

Once the VIN_Mgmt supply is valid and stable, the PMIC shall drive VOUT_1.8V & VOUT_1.0V supply within t1.8V_Ready & t1.0V_Ready time. The PMIC shall enable I²C/I³C Basic bus interface function within tManagement_Ready. The host shall not attempt to access the PMIC’s memory registers until tManagement_Ready timing requirement is satisfied. Further the host shall not attempt to issue VR Enable command until tVIN_Mgmt_to_Enable and tVIN_Bulk_to_Enable timing requirement is satisfied. In Figure 11, the PMIC allows access to its memory registers for indefinite period of time as long as VIN_Mgmt input supply is valid and PMIC does not require VIN_Bulk input supply.

The host may read PMIC's own internal memory content prior to ramping VIN_Bulk supply.

The host, prior to issuing VR Enable command, must keep VIN_Mgmt input supply valid as long as VOUT_1.8V & VOUT_1.0V LDO output are required. If VIN_Mgmt input supply is removed or drops below 2.8 V, the PMIC does not guarantee any operation including VOUT_1.8V & VOUT_1.0V LDO output as well as access to its I²C/I³C Basic interface regardless of VIN_Bulk input supply status.

After host issues VR Enable command to the PMIC, the PMIC offers the input supply switchover function. The PMIC has an automatic internal input supply switchover function from VIN_Mgmt input supply to VIN_Bulk input supply. The PMIC triggers the switchover to VIN_Bulk input supply when VIN_Mgmt input supply drops below the threshold set in register Table 142, “Register 0x2F” [7]. The internal input supply switchover is for PMIC’s VOUT_1.8V & VOUT_1.0V LDO output. The PMIC’s I²C/I³C Basic interfaces (SCL/SDA) are kept alive when PMIC switches over to VIN_Bulk input supply. Figure 13 shows automatic internal switchover function when VIN_Mgmt input supply drops below the threshold while maintaining its LDO outputs as well as I²C/I³C Basic interfaces. Under the switchover conditions VIN_Mgmt back-feed voltage shall be less than 0.2 V. While PMIC is in switchover mode to VIN_Bulk, the VIN_Mgmt input supply can re-power backup at any time and PMIC switches back to VIN_Mgmt input supply for its LDO outputs and I²C/I³C interface continues to operate as normal.

The PMIC shall power up its output switch regulators when it registers VR Enable command.

After VR Enable command is registered, the PMIC shall complete the following steps within tPMIC_PWR_GOOD_OUT:

1. Check VIN_Bulk, VIN_Mgmt and VBias Power Good status is valid.
2. Power up itself - PMIC executes Power On Sequence Config 0 to Power On Sequence Config 3 registers and configures PMIC internal registers as programmed in DIMM vendor memory space registers.
3. Power up all enabled output switch regulators and ready for normal operation
4. Update status registers Table 103, “Register 0x08” [5:2] and floats PWR_GOOD signal.

If PMIC PWR_GOOD signal High is not pulled High within tPMIC_PWR_GOOD_OUT time, the host can access the PMIC status registers for detailed information after tPMIC_PWR_GOOD_OUT time. The PMIC may NACK for any host request on I²C or I³C bus after VR Enable command until tPMIC_PWR_GOOD_OUT time expires.

Note that in Figure 11 & Figure 12 the specific sequence of ramping the output regulators (VOUT_A, VOUT_B, VOUT_C and VOUT_D) is for example purpose only. The specific ramp up sequence is configurable through the registers.

Once PMIC’s output regulators are running, the PMIC allows VIN_Bulk input supply to vary. Figure 14 shows that VIN_Bulk can go as low as VIN_Bulk Min value (4.25 V) and PMIC output regulators will continue to operate as normal. The VIN_Mgmt input supply can drop and may re-power back up at any time.

8.3 Power Up Sequence

Figure 11 shows the power up initialization sequence.

After VIN_Bulk supply is valid and stable, the PMIC powers up its output regulators when the system host sends VR Enable command. The PMIC generates its bias voltage (VBias) on its own using VIN_Bulk input supply. Further it executes Power On Sequence Config0 to Config3 as configured in register Table 157, “Register 0x40” to Table 160, “Register 0x43,” to enable its output regulators in the sequence as specified. The PMIC also follows the settings as specified in register 0x44 to register 0x6F for each of the enabled output. The PMIC ensures PWR_GOOD signal is floated within maximum of tPMIC_PWR_GOOD_OUT after registering VR Enable command.

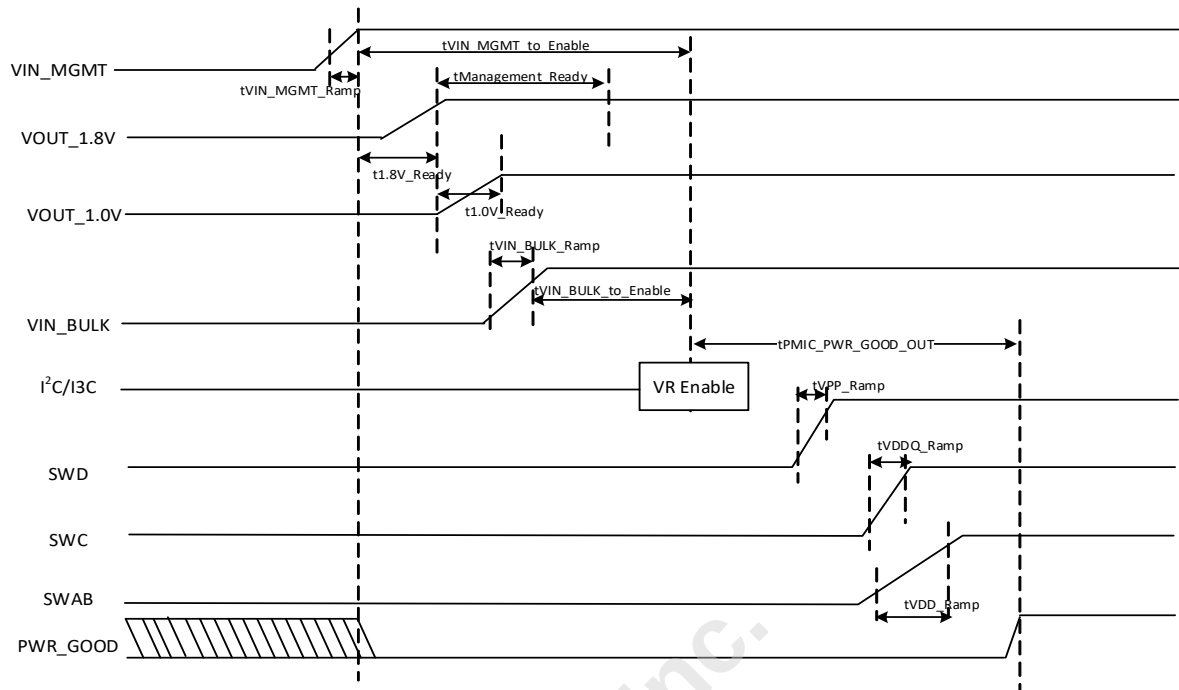


Figure 11 — PMIC Power Up Sequence; VIN_Mgmt followed by VIN_Bulk

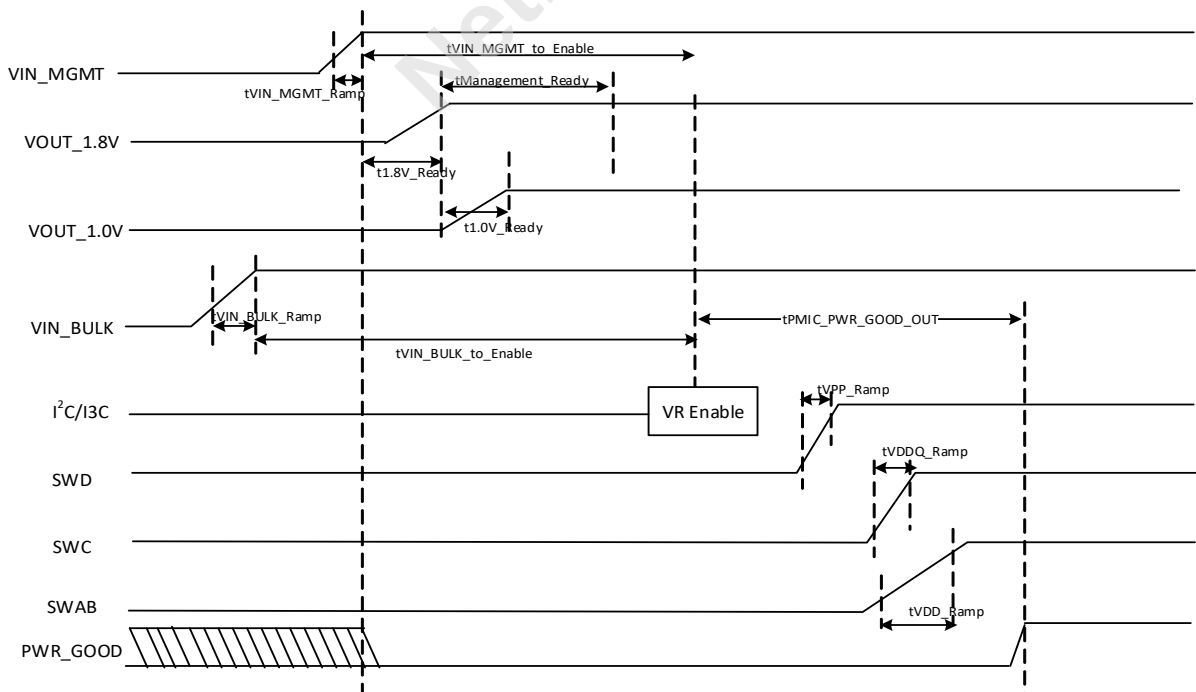


Figure 12 — PMIC Power Up Sequence; VIN_Bulk followed by VIN_Mgmt

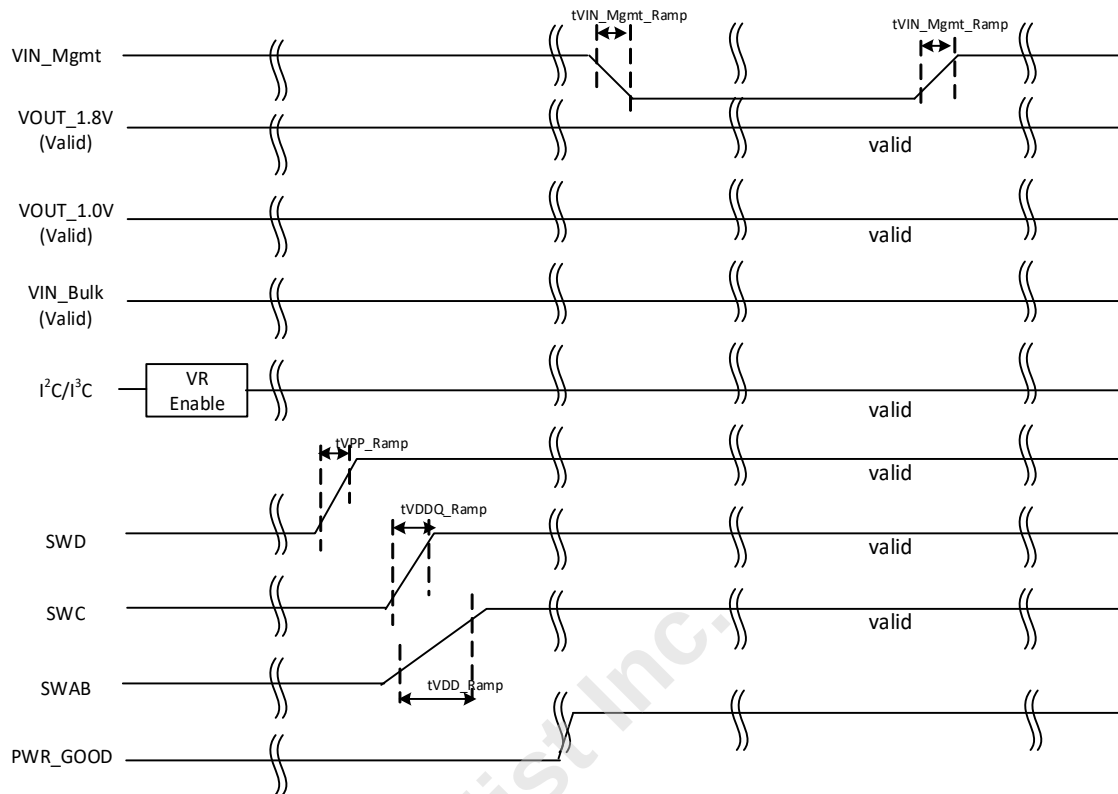


Figure 13 — VIN_Mgmt Input Supply to VIN_Bulk Input Supply Switchover Function

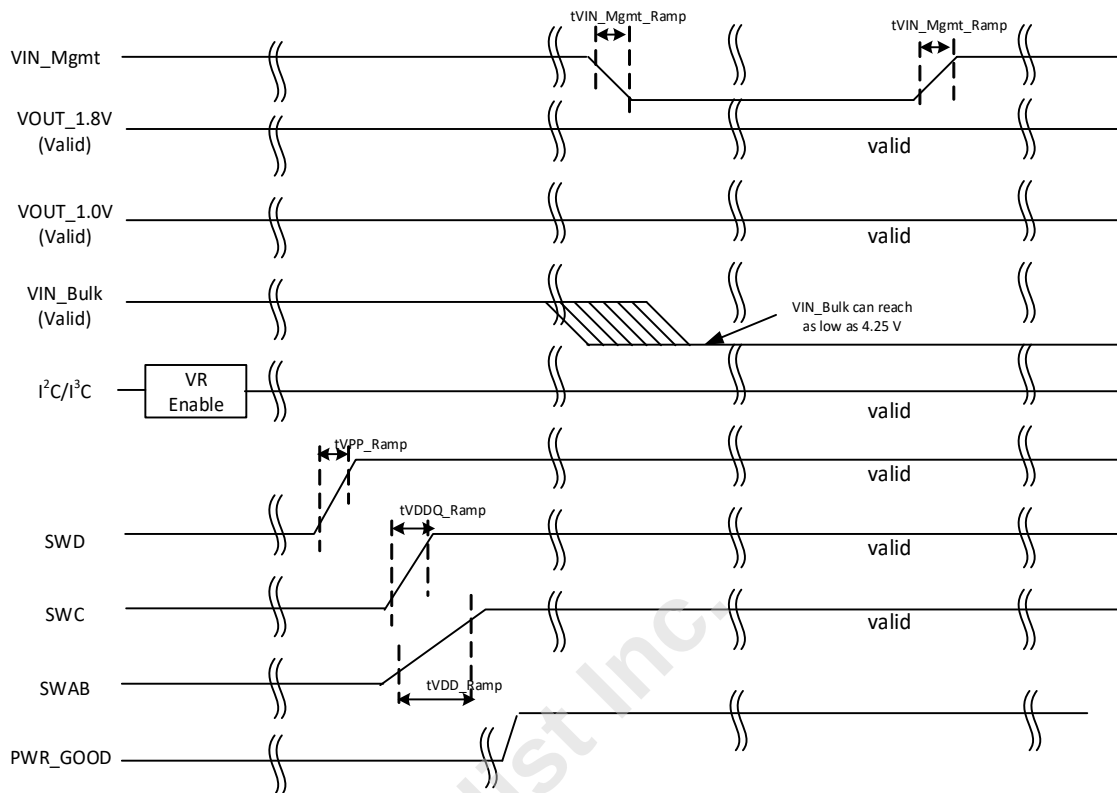


Figure 14 — VIN_Bulk Transition

The timing waveform example in Figure 14 assumes VIN_Bulk threshold is set to 4.25V in Table 121, “Register 0x1A” [7:5] and hence PWR_GOOD signal remains at valid High level.

8.4 Enabling PMIC Output Switch Voltage Regulators

After first power on, the PMIC automatically enables VOUT_1.8V and VOUT_1.0V LDO output regulators and primary I²C/I³C Basic port based on valid VIN_Mgmt input.

The Figure 15 below shows the timing relationship once the PMIC receives VR Enable command and when it floats PWR_GOOD output signal; timing parameter tPMIC_PWR_GOOD_OUT applies. This timing parameter is a sum of maximum soft start time and configured delay for each power on sequence config registers that are executed plus additional 5 ms timing margin error. The waveform shows each buck regulator output soft start time and delay time once the soft start time expires for each power on sequence config0 to power on sequence config3 registers. Note that if more than one regulators are enabled in a power on sequence config register and if those regulators have different soft start time programmed, then the larger value of that soft start time is used as a reference for delay timer to start. Each regulator will still follow different soft start time to turn on the buck regulator.

The specific example in Figure 15 uses only three power on sequence config0 to config2 registers and only one buck regulator is enabled in power on sequence config 0 register and power on sequence config 1 register. The power on sequence config 2 register enables dual phase regulator for SWA and SWB.

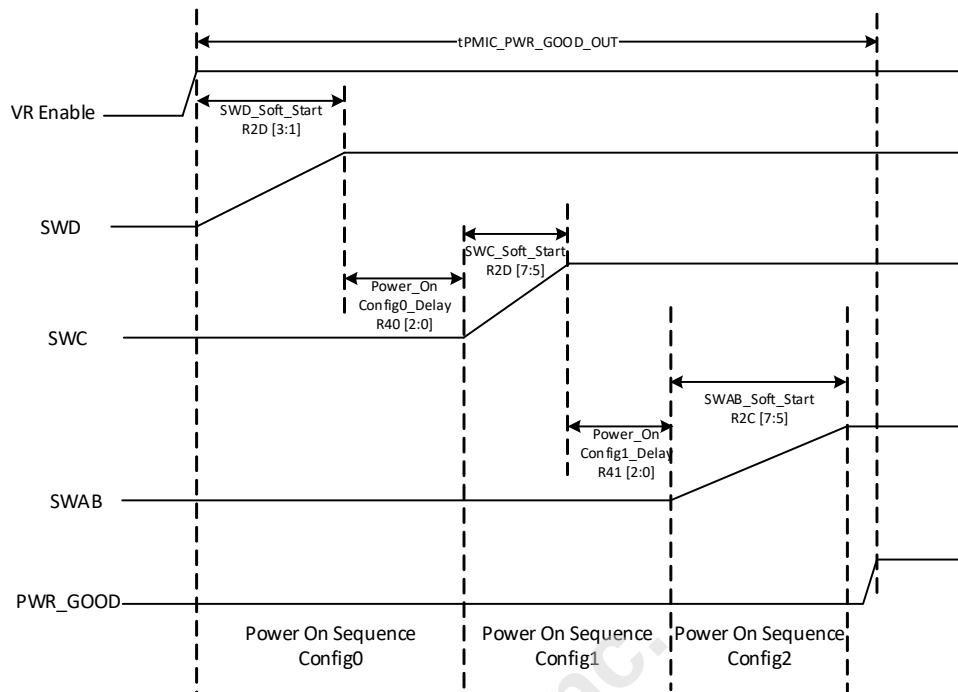


Figure 15 — PMIC Power On Timing

8.5 Secure Mode & Programmable Mode of Operation

Prior to issuing VR Enable command, the host must configure the register Table 142, “Register 0x2F” [2] appropriately as desired. The PMIC offers two modes after VR Enable command is registered.

1. **Programmable Mode** - In this mode, independent of when host issues VR Enable command, the PMIC allows modification to any register in the host region as desired by the host and PMIC will respond appropriately. After VR Enable command is issued to the PMIC and it enters in Programmable mode, the PMIC cannot switch to Secure mode by simply setting the Table 142, “Register 0x2F” [2] = ‘0’. However, after VR Enable command is issued to the PMIC and it enters in Programmable mode, the PMIC does allow to issue VR Disable command without power cycling and then set Table 142, “Register 0x2F” [2] = ‘0’ and issue VR Enable command again and PMIC enters in Secure mode of operation.
2. **Secure Mode** - In this mode, only after host issues VR Enable command, the PMIC does not allow modification of registers Table 116, “Register 0x15” to Table 142, “Register 0x2F”, Table 145, “Register 0x32” [7,5:0] in the host region as well as Table 157, “Register 0x40” to Register 0x6F in the DIMM vendor region. These registers are write protected marked with RED color cells in “Register” column in Table 98, “Host Region - Register Map” & Table 156, “DIMM Vendor Region - Register Map”. The PMIC simply ignores the host request. Throughout this entire specification, when it refers to as PMIC allows access to the registers, it refers to write operation to the registers that are not write protected in secure mode or the PMIC is in programmable mode. There is no restriction for the read operation in secure mode or programmable mode. The host must power cycle the PMIC to make any modification. The PMIC power cycle is defined as complete simultaneous removal of VIN_Bulk and VIN_Mgmt input supply to the PMIC and this definition is applicable to the entire specification. The PMIC does allows modification to any other remaining registers that are not marked in RED color cells in “Register” column in Table 98, “Host Region - Register Map”. The Secure Mode is only applicable once VR Enable command is registered. This is important to because by default Table 142, “Register 0x2F” [2] = ‘0’ when PMIC is first powered up with VIN_Mgmt. Prior to VR Enable command, PMIC allows modification to any registers in the host region.

8.6 Power Down Output Regulators

Regardless of how PMIC's output regulators are turned on, the PMIC's output regulators are powered down as described below depending on mode of operation.

In programmable mode of operation, the PMIC allows host to power down any or all output regulators in programmable mode. The host can disable PMIC's any or all output regulators by any of the three methods as follows:

1. The VR Disable command (Table 145, "Register 0x32" [7] = '0'). The PMIC executes power off sequence config0 (Table 174, "Register 0x58") to power off sequence config3 (Table 177, "Register 0x5B,") to preserve the appropriate voltage relationship as configured by the DIMM vendors. The PMIC keeps the PWR_GOOD signal floating (i.e., it remains High) because this is an intentional command from the host and not a fault condition. Note that host can re-enable the PMIC's output regulator by issuing VR Enable command. The PMIC executes power on sequence config 0 to config 3 registers and keeps the PWR_GOOD signal floating (i.e., it remains High).
2. Configuring one or more bits in Table 142, "Register 0x2F" [6:3] to '0' in any specific sequence that is desired by the host. The PMIC does not execute power off sequence config0 (Table 174, "Register 0x58") to power off sequence config3 (Table 177, "Register 0x5B,") on its own. The PMIC keeps the PWR_GOOD signal floating (i.e it remains High) because this is intentional command from the host and not a fault condition. Note that host can re-enable any of disabled output regulators by configuring one or more bits in Table 142, "Register 0x2F" [6:3] to '1' in any specific sequence that is desired by the host. The PMIC keeps the PWR_GOOD signal floating (i.e it remains High).
3. If Table 145, "Register 0x32" [5] = '1', driving PWR_GOOD input low. The PMIC executes power off sequence config0 (Table 174, "Register 0x58") to power off sequence config3 (Table 177, "Register 0x5B,") to preserve the appropriate voltage relationship as configured by the DIMM vendors.

In Secure mode of operation, the host can disable PMIC's all enabled output regulators by any of the two methods as follows:

1. Power cycle the PMIC.
2. If Table 145, "Register 0x32" [5] = '1', driving PWR_GOOD input low. The PMIC executes power off sequence config0 (Table 174, "Register 0x58") to power off sequence config3 (Table 177, "Register 0x5B,") to preserve the appropriate voltage relationship as configured by the DIMM vendors.

Regardless of the either mode of the operation, the PMIC, on its own, can generate VR Disable command at any time due to one or more events listed in Table 30 under column "Trigger VR Disable". The PMIC executes power off sequence config0 (Table 174, "Register 0x58") to power off sequence config3 (Table 177, "Register 0x5B,") to preserve the appropriate voltage relationship as configured by the DIMM vendors. The PMIC may also reset its Vbias LDO regulator.

The Figure 16 below shows the timing relationship once the PMIC registers VR Disable command internally due to fault. The waveform shows each buck regulator output soft stop time and delay time once the soft stop time expires from each power off sequence config0 to power off sequence config3 registers. Note that if more than one regulators are disabled in a power off sequence config register and if those regulators have different soft stop time programmed, then the larger value of that soft stop time is used as a reference for delay timer to start. Each regulator will still follow different soft stop time to turn off the buck regulator.

The specific example in Figure 16 uses only three power off sequence config0 to config2 registers and only one buck regulator is disabled in power off sequence config 1 register and power off sequence config 2 register. The power off sequence config 0 register disables dual phase regulator for SWA and SWB.

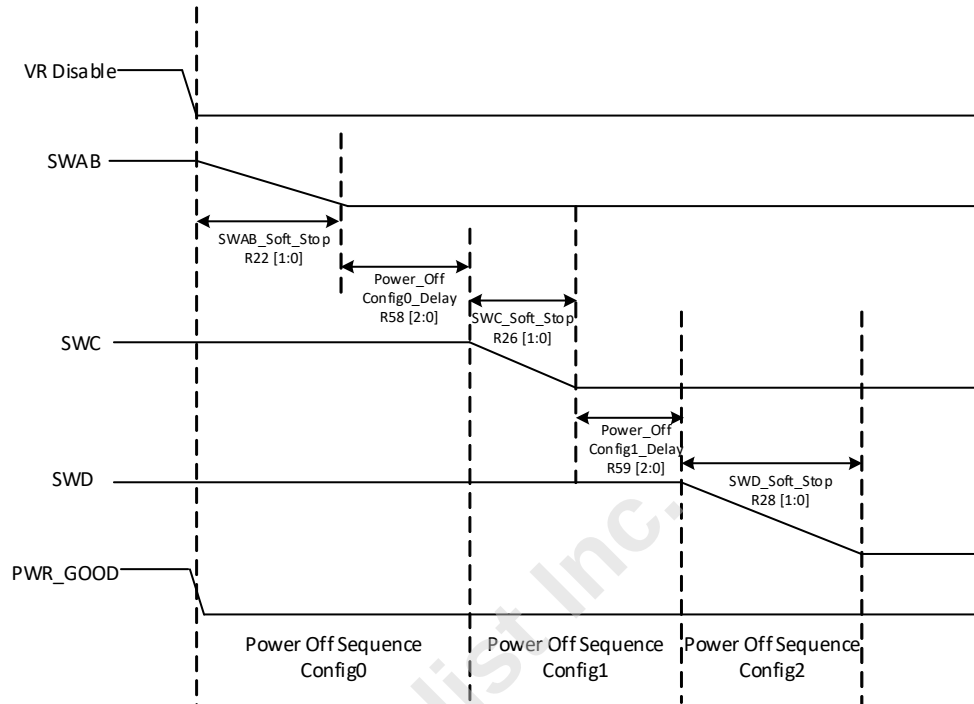


Figure 16 — PMIC Power Off Timing Due to Internal Fault Condition

8.7 Power Down Output Regulators During Power On Sequence

During power on as described in Section 8.2 it is possible that PMIC can trigger VR Disable command on its own as described in Table 30, “Events Interrupt Summary” when one or more regulators are already turned on even while other remaining output regulators are not yet turned on because PMIC has not completed the power on sequence config registers. For these type of cases, the PMIC will not execute the remaining power on sequence config registers and will immediately jump to executing the power off sequence config0 to power off sequence config3 registers. The PMIC will update the status registers and error log registers appropriately as normal because it generated VR Disable command on its own. The PWR_GOOD output signal would remain low.

8.8 Power Good (PWR_GOOD) Signal

The PWR_GOOD output signal type can be configured as either output only or input and output through register Table 145, “Register 0x32” [5]. By default, PWR_GOOD is an input/output signal. The PWR_GOOD signal can only be configured once, at power on, before issuing VR Enable command. The PWR_GOOD signal configuration applies to both secure mode or programmable mode of operation.

Section 8.9, Section 8.10 and Table 27 and Table 28 describes PMIC behavior.

Table 27 — PMIC Operation; PWR_GOOD Type: Input & Output

External PWR_GOOD Input	PMIC's Internal PWR_GOOD	PWR_GOOD Signal	Notes
High	High	High (Power Good) Normal Operation	
High	Low	Low (Power Not Good)	1,2
Low	High	Low; Power Good, Executes VR Disable Command	3
Low	Low	Low; Power Not Good; Internally generated VR Disable Command	

1. The PMIC indicates its own internal status and it may shut down on its own by following Power Off Config0 to Power Off Config3 sequence.
2. This is a transient state and the net result of the PWR_GOOD signal is Low.
3. The PMIC shuts down based on external command by following Power Off Config0 to Power Off Config3 sequence.

Table 28 — PMIC Operation; PWR_GOOD Type: Output Only

External PWR_GOOD Input	PMIC's Internal PWR_GOOD	PWR_GOOD Signal	Notes
X (High or Low)	High	High (Power Good) Normal Operation	
X (High or Low)	Low	Low (Power Not Good)	1

1. The PMIC indicates its own internal status and it may shut down on its own by following Power Off Config0 to Power Off Config3 sequence.

8.9 PWR_GOOD as Output Only Signal

When Table 145, “Register 0x32” [5] = ‘0’, the PWR_GOOD signal type is output only; the input of PWR_GOOD signal is ignored.

The PMIC PWR_GOOD pin indicates status of VIN_Bulk input supply and all output regulators (VOUT_A, VOUT_B, VOUT_C, VOUT_D, VOUT_1.8V, VOUT_1.0V, VBias). The PMIC floats PWR_GOOD pin (i.e., it is High) when VIN_Bulk input supply is valid and all enabled output regulator's (VOUT_A, VOUT_B, VOUT_C, VOUT_D, VOUT_1.8V, VOUT_1.0V, VBias) tolerances are maintained as configured in the appropriate register space. Note that PWR_GOOD pin is not effected based on VIN_Mgmt input supply.

At first power up, when input supply VIN_Mgmt is ramped up and stable, the PMIC keeps PWR_GOOD pin asserted to low however PMIC updates corresponding status register. Once PMIC receives VR Enable command from the host, the PMIC enables all appropriate output regulators and updates corresponding status registers and enters into operating state called as “Regulation”. At this point, PMIC floats PWR_GOOD pin and the external board pullup resistor pulls the pin high as there may be other PMIC on different DIMM may be driving the PWR_GOOD pin low. Once the PWR_GOOD pin is pulled high (i.e no other PMIC is driving the PWR_GOOD pin low), the PMIC remains in “Regulation” state.

Once the PWR_GOOD pin is high, if PMIC detects any condition either on VIN_Bulk input supply or any of the output regulators (VOUT_A, VOUT_B, VOUT_C, VOUT_D, VOUT_1.8V, VOUT_1.0V, VBias) that causes the PMIC to update its status registers to indicate the power status is not good, then PMIC asserts PWR_GOOD pin low and keeps it asserted until the host explicitly takes a specific action corresponding to it. The PMIC does not automatically let the PWR_GOOD pin float (i.e., get High) even if the condition that triggered the PMIC to assert the PWR_GOOD pin no longer exists. In other words, the PMIC's PWR_GOOD pin is latched and once latched, it must be explicitly addressed by the host.

If PMIC is operating in Secure mode of operation:

- VR Disable command (Table 145, “Register 0x32” [7] = ‘0’) or individual buck regulator disable settings (Table 142, “Register 0x2F” [6:3]) has no effect on PMIC output regulators.
- If VIN_Bulk input supply drops below the minimum threshold configured in register but VIN_Mgmt input supply is valid, the PMIC triggers VR Disable command and shuts off all output regulators. The PMIC allows read access to all its configuration registers. The PMIC allows write access to only non-locked configuration registers because PMIC is in secure mode of operation.
- If both VIN_Bulk and VIN_Mgmt input supply is not valid, the PMIC will completely shut off as normal.

If PMIC is operating in Programmable mode of operation:

- VR Disable command (Table 145, “Register 0x32” [7] = ‘0’) or individual buck regulator disable causes PMIC to disable the regulators appropriately.
- If VIN_Bulk input supply drops below the minimum threshold configured in register but VIN_Mgmt input supply is valid, the PMIC triggers VR Disable command shuts off all output regulators. The PMIC will allow read and write access to all its configuration registers because PMIC is in programmable mode of operation.
- If both VIN_Bulk and VIN_Mgmt input supply is not valid, the PMIC will completely shut off as normal.

Regardless of Secure mode or Programmable mode of operation, the PMIC can always drive PWR_GOOD output low at any time to indicate its status.

8.10 PWR_GOOD as Input & Output Signal

When Table 145, “Register 0x32” [5] = ‘1’, the PWR_GOOD signal type is both input and output and is only applicable after host issues VR Enable command.

At first power up, as shown in Section 8.1, the PWR_GOOD output signal is held low by the PMIC. The host, prior to issuing VR Enable command, can configure the register Table 145, “Register 0x32” [5] = ‘1’. When host issues VR Enable command, the PMIC turns on its output regulators and updates corresponding status registers and enters into operating state called “Regulation”. At this point, the PMIC floats PWR_GOOD pin and waits for external board pullup resistor to pull the pin high as there may be other PMIC on different DIMM may be driving the PWR_GOOD pin low. Once the PWR_GOOD pin is pulled high (i.e no other PMIC is driving the PWR_GOOD pin low), the PMIC automatically enters in to operating state called “Bulk Control Link Monitor”.

The PMIC PWR_GOOD pin indicates status of VIN_Bulk input supply and all output regulators (VOUT_A, VOUT_B, VOUT_C, VOUT_D, VOUT_1.8V, VOUT_1.0V, VBias). The PMIC floats PWR_GOOD pin (i.e it is High) when VIN_Bulk input supply is valid and all enabled output regulator’s (VOUT_A, VOUT_B, VOUT_C, VOUT_D, VOUT_1.8V, VOUT_1.0V, VBias) tolerances are maintained as configured in the appropriate register space. Note that PWR_GOOD pin is not effected based on VIN_Mgmt input supply.

Once the PWR_GOOD pin is high, if PMIC detects any condition either on VIN_Bulk input supply or any of the output regulators (VOUT_A, VOUT_B, VOUT_C, VOUT_D, VOUT_1.8V, VOUT_1.0V, VBias) that causes the PMIC to update it status registers to indicate the power status is not good, then PMIC asserts PWR_GOOD pin low and keeps it asserted until the host explicitly takes a specific action corresponding to it. The PMIC does not automatically let the PWR_GOOD pin float (i.e get High) even if the condition that triggered the PMIC to assert the PWR_GOOD pin no longer exists. In other words, the PMIC’s PWR_GOOD pin is latched and once latched, it must be explicitly addressed by the host.

In this “Bulk Control Link Monitor” operating state, the PMIC behavior is as following:

If PMIC is operating in Secure mode of operation:

- VR Disable command (Table 145, “Register 0x32” [7] = ‘0’) or individual buck regulator disable settings (Table 142, “Register 0x2F” [6:3]) has no effect on PMIC output regulators.
- PMIC allows PWR_GOOD input signal low at any time. The host must keep the PWR_GOOD signal low for minimum tPWR_GOOD_Low_Pulse_Width to issue command to PMIC to execute VR Disable. When PMIC detects PWR_GOOD signal low, the PMIC internally triggers VR Disable command and shuts off all output regulators (PMIC executes power off sequence config0 (Table 174, “Register 0x58”) to power off sequence config3 (Table 177, “Register 0x5B,”)); drives PWR_GOOD signal low and unlocks only register Table 145, “Register 0x32” [7]. The PMIC preserves all register contents including the MTP error log registers and keeps all write protect registers locked except for the Table 145, “Register 0x32” [7]. As long as there is valid VIN_Mgmt input supply or VIN_Bulk input supply in switchover mode, the PMIC allows read access to all its configuration registers. The PMIC allows write access to non-locked configuration registers and register Table 145, “Register 0x32” [7]. If host issues VR Enable command, the PMIC executes Power on sequence config0 to Power on sequence config3 registers, floats PWR_GOOD output signal and re-locks register Table 145, “Register 0x32” [7]. The PMIC does not require power cycle to re-enable the PMIC’s output regulators.
- Independent of PWR_GOOD input signal, if VIN_Bulk input supply drops below the minimum threshold configured in Table 121, “Register 0x1A” [7:5] but VIN_Mgmt input supply is valid, the PMIC triggers VR Disable command and shuts off all output regulators. The PMIC allows read access to all its configuration registers. The PMIC allows write access to non-locked configuration registers as PMIC is in secure mode of operation. The PMIC must go through power cycle to re-enable the PMIC’s output regulators.
- If both VIN_Bulk and VIN_Mgmt input supply is not valid, the PMIC will completely shut off as normal.

If PMIC is in Programmable mode of operation:

- VR Disable command (Table 145, “Register 0x32” [7] = ‘0’) or individual buck regulator disable settings (Table 142, “Register 0x2F” [6:3]) causes PMIC to disable the regulators appropriately.
- PMIC allows PWR_GOOD input signal low at any time. The host must keep the PWR_GOOD signal low for minimum tPWR_GOOD_Low_Pulse_Width to issue command to PMIC to execute VR Disable. When PMIC detects PWR_GOOD signal low, the PMIC internally triggers VR Disable command and shuts off all output regulators (PMIC executes power off sequence config0 (Table 174, “Register 0x58”) to power off sequence config3 (Table 177, “Register 0x5B,”)); drives PWR_GOOD signal low. The PMIC preserves all register contents including the MTP error log registers, As long as there is valid VIN_Mgmt input supply or VIN_Bulk input supply in switchover mode, the PMIC allows read and write access to all its configuration registers. The host can issue VR Enable command again to turn on the PMIC’s output regulator and PMIC will execute Power On Config0 to Config3 registers and floats PWR_GOOD output signal.
- Independent of PWR_GOOD input signal, if VIN_Bulk input supply drops below the minimum threshold configured in Table 121, “Register 0x1A” [7:5] but VIN_Mgmt input supply is valid, the PMIC triggers VR Disable command and shuts off all output regulators. The PMIC will allow read and write access to all its configuration registers.
- If both VIN_Bulk and VIN_Mgmt input supply is not valid, the PMIC will completely shut off as normal.

Regardless of PWR_GOOD input in Secure mode or Programmable mode of operation, the PMIC can always drive PWR_GOOD output low at any time to indicate its status as described in Section 8.9.

8.11 GSI_n Signal

TBD

8.12 State Transition Diagram

Following is a summary of high level description of basic PMIC states.

Start State:

- VIN_Mgmt is invalid and VIN_Bulk = X (valid or invalid)
- All registers are reset to specified default values
- The PMIC does not mandate any specific timing relationship requirement between VIN_Mgmt and VIN_Bulk

Configuration state:

- At initial configuration state, all registers follow the register attributes as defined.
- The I²C/I³C bus interface is alive and running.
- When Table 145, “Register 0x32” [7] = ‘1’, the PMIC exits the Configuration state after tPMIC_PWR_GOOD_OUT; PMIC’s regulator reaches steady state.
- The Table 142, “Register 0x2F” [2] bit can only be changed to programmable mode upon the entry to the Configuration state from the Start state prior to VR Enable command. If the PMIC operation requires additional tweaks to the SW[A..D] outputs in Regulation or Bulk Control Link Monitor state, the Table 142, “Register 0x2F” [2] bit must be set to programmable mode in this state prior to setting Table 145, “Register 0x32” [7] = ‘1’.

Regulation state:

- All registers are readable.
- If Table 142, “Register 0x2F” [2] = ‘1’, the PMIC allows to be configured. Changes to PMIC register values require a tSETTLE settling time.
- If Table 142, “Register 0x2F” [2] = ‘0’, registers Table 116, “Register 0x15” to Table 142, “Register 0x2F”, Table 145, “Register 0x32” in the host region as well as Table 157, “Register 0x40” to Register 0x6F are not allowed to be configurable.
- If Table 142, “Register 0x2F” [2] = ‘0’ in this state it can not be changed.
- All enabled output rails are active.

Bulk Control Link Monitor state:

- All registers are readable.
- If Table 142, “Register 0x2F” [2] = ‘1’, the PMIC allows to be configured. Changes to PMIC register values require a tSETTLE settling time.
- If Table 142, “Register 0x2F” [2] = ‘0’, registers Table 116, “Register 0x15” to Table 142, “Register 0x2F”, Table 145, “Register 0x32” in the host region as well as Table 157, “Register 0x40” to Register 0x6F are not allowed to be configurable.
- If Table 142, “Register 0x2F” [2] = ‘0’ in this state it can not be changed.
- The Bulk Control Link Monitor state is used to monitor the PWR_GOOD pin. When the PWR_GOOD pin goes low the PMIC will transition to the Start or Configuration states. This process will set VR_EN==0 and disable the SW[A..D] output rails. The transition logic is described in transitions 11 and 12 of Table 29.

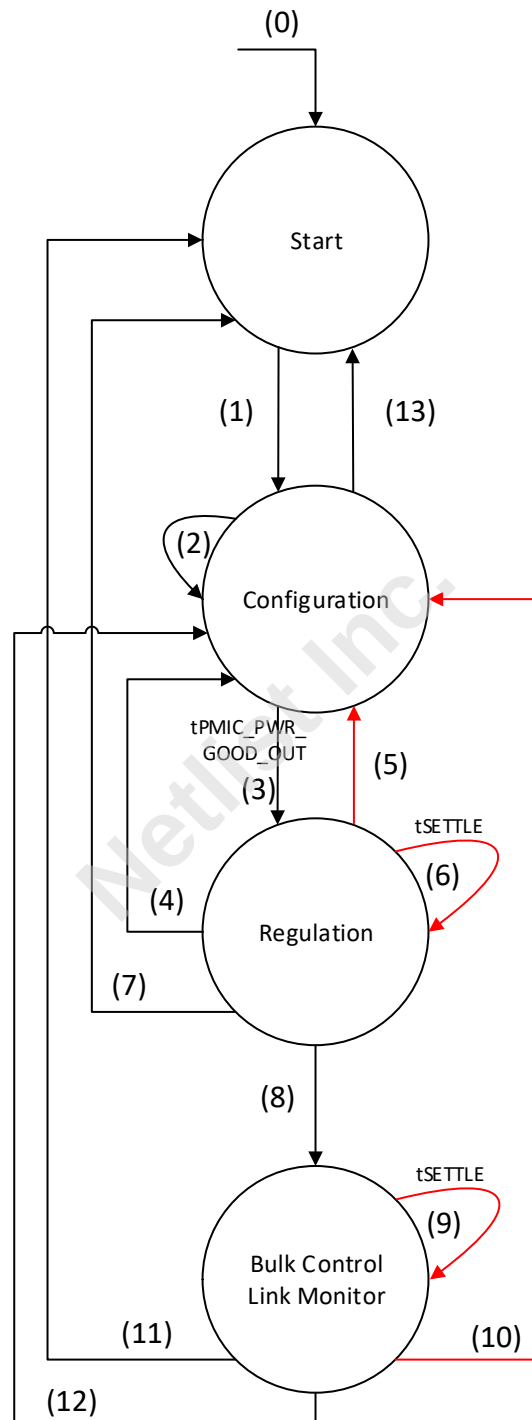


Figure 17 — High Level State Transitions

Table 29 — High Level State Transitions

Transition	Description
(0)	This state indicates the startup condition. VIN_Mgmt and VIN_Bulk are invalid or below thresholds
(1)	The FSM shall transition to the Configuration state when the VIN_MGMT is valid and the VOUT_1.8 LDO is valid. The CSRS are reset to the default state upon exit of the Start state. (VIN_Mgmt $\geq$ VIN_Mgmt_Thresh ¹) && VOUT_1.8V and VOUT_1.0V LDOs are valid
(2)	The FSM shall allow the CSRs to be updated per host commands. (VIN_Mgmt $\geq$ VIN_Mgmt_Thresh ¹) && VOUT_1.8V and VOUT_1.0V LDOs are valid
(3)	The FSM shall exit the Configuration state when the VR_EN bit is set to 1. The PMIC shall enable the configured SW[A..D] rails within the tPMIC_PWR_GOOD_OUT period. The PWR_GOOD output shall be ensured within tPMIC_PWR_GOOD_OUT. VR_EN==1 && tPMIC_PWR_GOOD_OUT satisfied && VOUT_1.0V LDO is valid
(4)	The FSM shall exit the Regulation state and enter the Configuration state when the VIN_MGMT is valid and VIN_Bulk voltage is invalid. VIN_Mgmt $\geq$ VIN_Mgmt_Thresh ¹ && VIN_Bulk < VIN_Bulk_Thresh ²
(5)	The FSM shall exit the Regulation state and return to the Configuration state if SECURE_MODE_n==1 and the VR_EN bit is cleared. The act of setting VR_EN==0 will turn off the SW[A..D] rail outputs. SECURE_MODE_n==1 && VR_EN==0
(6)	While in the Regulation state the PMIC can continue to be configured if SECURE_MODE_n==1. The PMIC shall allow tSETTLE time for SW[A..D] adjustments to settle. The PWR_GOOD output status shall be locked during the tSETTLE time to prevent any glitches. SECURE_MODE_n==1 && PWR_GOOD==1 && valid command && tSETTLE satisfied
(7)	The FSM shall exit the Regulation state and enter the Start state when the VIN_Mgmt is invalid and VIN Bulk voltage is invalid. The Start state will reset the CSRs to default values (ie. this transition will deassert VR_EN). (VIN_Mgmt < VIN_Mgmt_Thresh) && (VIN_Bulk < VIN_Bulk_Thresh)
(8)	The FSM will exit the Regulation state and enter the Bulk Control Link Monitor state when the VIN Bulk is valid and the PWR_GOOD signal is high. (VIN_Bulk $\geq$ VIN_Bulk_Thresh) && PWR_GOOD==1
(9)	While in the Bulk Link Control Monitor state the PMIC can continue to be configured if SECURE_MODE_n==1. The PMIC shall allow tSETTLE time for SW[A..D] adjustments to settle. The PWR_GOOD output status shall be locked during the tSETTLE time to prevent any glitches. SECURE_MODE_n==1 && PWR_GOOD==1 && valid command && tSETTLE satisfied
(10)	The FSM shall exit the Bulk Link Monitor state and return to the Configuration state if SECURE_MODE_n==1 and the VR_EN bit is cleared. The act of setting VR_EN==0 will turn off the SW[A..D] rail outputs. SECURE_MODE_n==1 && VR_EN==0

Table 29 — High Level State Transitions

Transition	Description
(11)	The FSM shall exit the Bulk Control Link Monitor state and enter the Start state when the VIN_MGMT is invalid and VIN Bulk voltage is invalid. The Start state will reset the CSRs to default values (ie. this transition will deassert VR_EN). (VIN_MGMT < VIN_Mgmt_Thresh) && (VIN_BULK < VIN_Bulk_Thresh)
(12)	The FSM shall exit the Bulk Control Link Monitor state and enter the Configuration state when the VIN_Mgmt is valid and VIN_Bulk voltage is invalid or PWR_GOOD input pin is deasserted (i.e., any of the four conditions as listed below). VIN_Mgmt >= VIN_Mgmt_Thresh && VIN_Bulk < VIN_Bulk_Thresh && PWR_GOOD==1 VIN_Mgmt >= VIN_Mgmt_Thresh && VIN_Bulk >= VIN_Bulk_Thresh && PWR_GOOD==0 VIN_Mgmt >= VIN_Mgmt_Thresh && VIN_Bulk < VIN_Bulk_Thresh && PWR_GOOD==0 VIN_Mgmt < VIN_Mgmt_Thresh && VIN_Bulk >= VIN_Bulk_Thresh && PWR_GOOD==0
(13)	The FSM shall exit the Configuration state and enter the Start state if any of the LDOs are invalid. VOUT_1.8V or VOUT_1.0V LDOs is not valid

1. At first power on, VIN_Mgmt_Thresh = 2.8 V for PMIC to turn on. For normal operation, the VIN_Mgmt tolerance is as defined in Table 1.
2. At first power on, VIN_Bulk_Thresh = $R1A[7:5] + 1.0$ V.

8.13 Function Interrupt - PWR_GOOD and GSI_n Output Signals

This section defines the output functionality of GSI_n pin and PWR_GOOD (Table 145, “Register 0x32” [5] = ‘0’) pin.

When mask register bits are not set, the PMIC asserts its GSI_n output and PWR_GOOD output signals as shown in Table 30 when any event occurs. The table also highlights 11 events that causes PMIC to generate internally VR Disable command. For remaining events that does not trigger internal VR Disable command, the PMIC continues to operate as normal.

Table 30 — Events Interrupt Summary

Event	Status Bit	Clear Bit	Mask Bit	Threshold Bits	Trigger VR Disable?	PWR_GOOD Output	GSI_n Output
VIN Bulk Power Good	R08 [7]	R10 [7]	R15 [7]	R1A [7:5]	No	Low	Low
VIN Bulk Over Voltage	R08 [0]	R10 [0]	R15 [0]	R1B [7]	Yes	Low	Low
VIN Mgmt Over Voltage	R08 [1]	R10 [1]	R15 [1]	R1B [5]	No	High	Low
SWA Output Power Good	R08 [5]	R10 [5]	R15 [5]	R21 [1:0]; R22 [7:6]	No	Low	Low
SWB Output Power Good	R08 [4]	R10 [4]	R15 [4]	R23 [1:0]; R24 [7:6]	No	Low	Low
SWC Output Power Good	R08 [3]	R10 [3]	R15 [3]	R25 [1:0]; R26 [7:6]	No	Low	Low
SWD Output Power Good	R08 [2]	R10 [2]	R15 [2]	R27 [1:0]; R28 [7:6]	No	Low	Low
1.8 V LDO Power Good	R09 [5]	R11 [5]	R16 [5]	R1A [2]	No	Low	Low
1.0 V LDO Power Good	R33 [2]	R14 [2]	R19 [2]	R1A [0]	No	Low	Low
VBias LDO Power Good	R09 [6]	R11 [6]	R16 [6]	R1A [3]	No	Low	Low
SWA Output Over Voltage	R0A [7]	R12 [7]	R17 [7]	R22 [5:4]	Yes	Low	Low
SWB Output Over Voltage	R0A [6]	R12 [6]	R17 [6]	R24 [5:4]	Yes	Low	Low
SWC Output Over Voltage	R0A [5]	R12 [5]	R17 [5]	R26 [5:4]	Yes	Low	Low
SWD Output Over Voltage	R0A [4]	R12 [4]	R17 [4]	R28 [5:4]	Yes	Low	Low
SWA Output Under Voltage	R0B [3]	R13 [3]	R18 [3]	R22 [3:2]	Yes	Low	Low
SWB Output Under Voltage	R0B [2]	R13 [2]	R18 [2]	R24 [3:2]	Yes	Low	Low
SWC Output Under Voltage	R0B [1]	R13 [1]	R18 [1]	R26 [3:2]	Yes	Low	Low
SWD Output Under Voltage	R0B [0]	R13 [0]	R18 [0]	R28 [3:2]	Yes	Low	Low
VBias LDO Output or VIN_Bulk Input Under Voltage	R33 [3]	R14 [3]	R19 [3]	Vendor Specific	Yes	Low	Low
SWA Output Current Limit	R0B [7]	R13 [7]	R18 [7]	R20 [7:6]	No	High	Low
SWB Output Current Limit	R0B [6]	R13 [6]	R18 [6]	R20 [5:4]	No	High	Low
SWC Output Current Limit	R0B [5]	R13 [5]	R18 [5]	R20 [3:2]	No	High	Low
SWD Output Current Limit	R0B [4]	R13 [4]	R18 [4]	R20 [1:0]	No	High	Low
SWA Output High Current/Power	R09 [3]	R11 [3]	R16 [3]	R1C [7:2]	No	High	Low
SWB Output High Current/Power	R09 [2]	R11 [2]	R16 [2]	R1D [7:2]	No	High	Low
SWC Output High Current/Power	R09 [1]	R11 [1]	R16 [1]	R1E [7:2]	No	High	Low
SWD Output High Current/Power	R09 [0]	R11 [0]	R16 [0]	R1F [7:2]	No	High	Low
High Temperature Warning	R09 [7]	R11 [7]	R16 [7]	R1B [2:0]	No	High	Low
Critical Temperature	R08[6]	N/A	N/A	R2E [2:0]	Yes	Low	Low
VIN Mgmt to VIN Bulk Switchover	R09 [4]	R11 [4]	R16 [4]	R2F [7]	No	High	Low
Valid VIN Mgmt in Switchover State	R33 [4]	R14 [4]	R19 [4]	N/A	No	High	Low
PEC Error	R0A [3]	R12 [3]	R17 [3]	N/A	No	High	Low
Parity Error	R0A [2]	R12 [2]	R17 [2]	N/A	No	High	Low

The host is expected to read appropriate status registers to determine and isolate the cause of the GSI_n signal assertion or PWR_GOOD signal assertion. The host may attempt to clear or mask the appropriate corresponding interrupt event. The PMIC keeps the GSI_n signal asserted or PWR_GOOD signal asserted until the appropriate corresponding registers are explicitly cleared or masked by the host. Table 31 and Table 32 shows the PMIC's response of GSI_n signal and PWR_GOOD output signal for each event before and after host issues the Clear command. The Table 31 and Table 32 assumes that all mask bits are either '0' or '1' for simplicity.

Table 31 — PMIC Response for Clear Command by Host - 1

	Event Occurred; All Mask Bits = '0'		Clear Command; Event Not Present; All Mask Bits = '0'		Event Occurred; All Mask Bits = '1'		Clear Command; Event Not Present; All Mask Bits = '1'	
			R2F [1:0] = '00' or '01' or '10'		R2F [1:0] = '00'		R2F [1:0] = '00'	
Event	PWR_GO OD Output	GSI_n Output	PWR_GO OD Output	GSI_n Output	PWR_GO OD Output	GSI_n Output	PWR_GO OD Output	GSI_n Output
VIN_Bulk Power Good	Low	Low	High	High	Low	High	High	High
VIN_Bulk Over Voltage	Low	Low	Low	High	Low	High	Low	High
VIN_Mgmt Over Voltage	High	Low	High	High	High	High	High	High
SWA Output Power Good	Low	Low	High	High	Low	High	High	High
SWB Output Power Good	Low	Low	High	High	Low	High	High	High
SWC Output Power Good	Low	Low	High	High	Low	High	High	High
SWD Output Power Good	Low	Low	High	High	Low	High	High	High
1.8 V LDO Power Good	Low	Low	High	High	Low	High	High	High
1.0 V LDO Power Good	Low	Low	High	High	Low	High	High	High
VBias LDO Power Good	Low	Low	High	High	Low	High	High	High
SWA Output Over Voltage	Low	Low	Low	High	Low	High	Low	High
SWB Output Over Voltage	Low	Low	Low	High	Low	High	Low	High
SWC Output Over Voltage	Low	Low	Low	High	Low	High	Low	High
SWD Output Over Voltage	Low	Low	Low	High	Low	High	Low	High
SWA Output Under Voltage	Low	Low	Low	High	Low	High	Low	High
SWB Output Under Voltage	Low	Low	Low	High	Low	High	Low	High
SWC Output Under Voltage	Low	Low	Low	High	Low	High	Low	High
SWD Output Under Voltage	Low	Low	Low	High	Low	High	Low	High
VBias LDO Output or VIN_Bulk Input Under Voltage	Low	Low	Low	High	Low	High	Low	High
SWA Output Current Limit	High	Low	High	High	High	High	High	High
SWB Output Current Limit	High	Low	High	High	High	High	High	High
SWC Output Current Limit	High	Low	High	High	High	High	High	High
SWD Output Current Limit	High	Low	High	High	High	High	High	High
SWA Output High Current/Power	High	Low	High	High	High	High	High	High
SWB Output High Current/Power	High	Low	High	High	High	High	High	High
SWC Output High Current/Power	High	Low	High	High	High	High	High	High
SWD Output High Current/Power	High	Low	High	High	High	High	High	High
High Temperature Warning	High	Low	High	High	High	High	High	High
Critical Temperature	Low	Low	P/C	P/C	Low	Low	P/C	P/C
VIN_Mgmt to VIN_Bulk Switchover	High	Low	High	High	High	High	High	High
Valid VIN_Mgmt in Switchover	High	Low	High	High	High	High	High	High
PEC Error	High	Low	High	High	High	High	High	High
Parity Error	High	Low	High	High	High	High	High	High

Table 32 — PMIC Response for Clear Command by Host - 2

	Event Occurred; All Mask Bits = '1'		Clear Command; Event Not Present; All Mask Bits = '1'		Event Occurred; All Mask Bits = '1'		Clear Command; Event Not Present; All Mask Bits = '1'	
	R2F [1:0] = '01'		R2F [1:0] = '01'		R2F [1:0] = '10'		R2F [1:0] = '10'	
Event	PWR_GO OD Output	GSI_n Output	PWR_GO OD Output	GSI_n Output	PWR_GO OD Output	GSI_n Output	PWR_GO OD Output	GSI_n Output
VIN Bulk Power Good	High	Low	High	High	High	High	High	High
VIN Bulk Over Voltage	Low	Low	Low	High	Low	High	Low	High
VIN Mgmt Over Voltage	High	Low	High	High	High	High	High	High
SWA Output Power Good	High	Low	High	High	High	High	High	High
SWB Output Power Good	High	Low	High	High	High	High	High	High
SWC Output Power Good	High	Low	High	High	High	High	High	High
SWD Output Power Good	High	Low	High	High	High	High	High	High
1.8 V LDO Power Good	High	Low	High	High	High	High	High	High
1.0 V LDO Power Good	High	Low	High	High	High	High	High	High
VBias LDO Power Good	High	Low	High	High	High	High	High	High
SWA Output Over Voltage	Low	Low	Low	High	Low	High	Low	High
SWB Output Over Voltage	Low	Low	Low	High	Low	High	Low	High
SWC Output Over Voltage	Low	Low	Low	High	Low	High	Low	High
SWD Output Over Voltage	Low	Low	Low	High	Low	High	Low	High
SWA Output Under Voltage	Low	Low	Low	High	Low	High	Low	High
SWB Output Under Voltage	Low	Low	Low	High	Low	High	Low	High
SWC Output Under Voltage	Low	Low	Low	High	Low	High	Low	High
SWD Output Under Voltage	Low	Low	Low	High	Low	High	Low	High
VBias LDO Output or VIN Bulk Input Under Voltage	Low	Low	Low	High	Low	High	Low	High
SWA Output Current Limit	High	Low	High	High	High	High	High	High
SWB Output Current Limit	High	Low	High	High	High	High	High	High
SWC Output Current Limit	High	Low	High	High	High	High	High	High
SWD Output Current Limit	High	Low	High	High	High	High	High	High
SWA Output High Current/Power	High	Low	High	High	High	High	High	High
SWB Output High Current/Power	High	Low	High	High	High	High	High	High
SWC Output High Current/Power	High	Low	High	High	High	High	High	High
SWD Output High Current/Power	High	Low	High	High	High	High	High	High
High Temperature Warning	High	Low	High	High	High	High	High	High
Critical Temperature	Low	Low	P/C	P/C	Low	Low	P/C	P/C
VIN Mgmt to VIN Bulk Switchover	High	Low	High	High	High	High	High	High
Valid VIN Mgmt in Switchover	High	Low	High	High	High	High	High	High
PEC Error	High	Low	High	High	High	High	High	High
Parity Error	High	Low	High	High	High	High	High	High

Note that when host masks any of the event in appropriate register, it only masks the assertion of GSI_n output signal or assertion of PWR_GOOD output signal. The PMIC functional behavior remains the same as noted for each event other than assertion of GSI_n output signal and assertion of PWR_GOOD output signal.

8.14 Input Power Good Status

There is one possibility where PMIC recognizes the input supply fail.

1. VIN_Bulk goes below the threshold set in register Table 121, “Register 0x1A” [7:5].

When this event occurs for a period longer than $t_{\text{Input_PWR_GOOD_GSI_Assertion}}$ time then PMIC sets the register Table 103, “Register 0x08” [7] and drives GSI_n and PWR_GOOD output signal as shown in Table 30 at the same time. The PMIC allows access to all registers and PMIC continues to operate as normal as long as VIN_Bulk input remains above 4.25 V. See also Section 8.18. The host is responsible for taking any specific action. The host may clear the VIN_Bulk input power good status register by writing ‘1’ to register Table 103, “Register 0x08” [7] or by writing ‘1’ to global status clear register Table 115, “Register 0x14” [0]. If the input power not good condition is still present then PMIC will continue to drive GSI_n and PWR_GOOD output signal as in Table 30 and the status register Table 103, “Register 0x08” [7] will remain at ‘1’. If the input power not good condition persists, the host may set the appropriate mask register to remove GSI_n or PWR_GOOD output signal as shown in Table 31 and Table 32.

Note that after VR enable command, when VIN_Mgmt input goes below the threshold set in register Table 142, “Register 0x2F” [7], it is reported as switchover event as described in Section 8.22. Prior to VR Enable command, the VIN_Mgmt is always required to be above 2.8 V to guarantee PMIC’s functionality as described in Section 8.2.

8.15 Input Over Voltage Protection

An input over voltage protection mechanism is implemented to limit the voltages to the PMIC. The PMIC actively monitors the input voltage VIN_Bulk and VIN_Mgmt rail.

There are two possibilities where PMIC recognizes the input over voltage event.

1. VIN_Mgmt input goes above the threshold set in register Table 122, “Register 0x1B” [5].
2. VIN_Bulk input goes above the threshold set in register Table 122, “Register 0x1B” [7].

When either one or both event occurs for a period longer than $t_{\text{Input_OV_GSI_Assertion}}$ time then PMIC sets the register Table 103, “Register 0x08” [1:0] accordingly and drives GSI_n output signal as shown in Table 30 at the same time. Note that at this point, the PMIC does not assert PWR_GOOD output signal. The PMIC allows access to all registers and PMIC continues to operate as normal. The host is responsible for taking any specific action. The host may clear the VIN_Mgmt or VIN_Bulk input over voltage status register by writing ‘1’ to register Table 111, “Register 0x10” [1:0] appropriately or by writing ‘1’ to global status clear register Table 115, “Register 0x14” [0]. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 103, “Register 0x08” [1:0] will remain at ‘1’.

In programmable mode (i.e., Table 142, “Register 0x2F” [2] = ‘1’), if VIN_Bulk input supply over voltage condition persists greater than $t_{\text{Input_OV_VR_Disable}}$ time then PMIC internally generates VR Disable command and disables all of its switching output regulators and asserts PWR_GOOD signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the PWR_GOOD signal assertion and GSI_n signal assertion. Once host determines the cause, the host must first clear the VIN_Bulk input over voltage status register as well as any other relevant status registers individually or by writing ‘1’ to global status clear register Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted. If the input over voltage condition is still present then PMIC will continue to assert GSI_n output signal and the status register Table 103, “Register 0x08” [0] will remain at ‘1’. Once the status register is cleared and GSI_n output signal is de-asserted, the host may re-enable the PMIC’s output switching regulator by issuing VR Enable command. The PMIC enables output switching regulators and ensures PWR_GOOD signal is floated when all of its output regulators are normal and input over voltage condition is no longer present.

In secure mode (i.e., Table 142, “Register 0x2F” [2] = ‘0’), if VIN_Bulk input supply over voltage condition persists greater than tInput_OV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators and asserts PWR_GOOD signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking the action of power cycling the PMIC.

8.16 Output Power Good Status

The PMIC provides the voltage tolerance information to host that its output regulator may have crossed the desired voltage tolerance from its nominal programmed setting. The nominal programmed setting for output regulator SWA, SWB, SWC and SWD is programmed in register Table 128, “Register 0x21,” [7:1], Table 130, “Register 0x23,” [7:1], Table 132, “Register 0x25,” [7:1] and Table 134, “Register 0x27,” [7:1] respectively. The PMIC offers the PWR_GOOD condition to be set independently for low side and high side.

In addition, PMIC has three LDO regulators: VBias, VOUT_1.8V and VOUT_1.0V

There are five possibilities where PMIC recognizes the output power good event for any output regulator.

1. Output voltage goes below the threshold set in register Table 128, “Register 0x21,” [0] for SWA or Table 130, “Register 0x23,” [0] for SWB or Table 132, “Register 0x25,” [0] for SWC or Table 134, “Register 0x27,” [0] for SWD.
2. Output voltage goes above the threshold set in register Table 129, “Register 0x22,” [7:6] for SWA or Table 131, “Register 0x24,” [7:6] for SWB or Table 133, “Register 0x26,” [7:6] for SWC or Table 135, “Register 0x28,” [7:6] for SWD.
3. LDO output VBias goes below the threshold set in register Table 121, “Register 0x1A” [3].
4. LDO output VOUT_1.8V goes below the threshold set in register Table 121, “Register 0x1A” [2].
5. LDO output VOUT_1.0V goes below the threshold set in register Table 121, “Register 0x1A” [0].

When either event occurs for a period longer than Output_PWR_GOOD_GSI_Assertion time then PMIC sets the register Table 103, “Register 0x08” [5:2] or Table 104, “Register 0x09” [6:5] or Table 146, “Register 0x33” [2] appropriately and drives PWR_GOOD and GSI_n output signal as shown in Table 30 at the same time. The PMIC may continue to operate but DDR5 DIMM functionality may not be guaranteed. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine and identify the cause of the PWR_GOOD signal assertion and GSI_n signal assertion. Once host determines the cause, the host may clear the appropriate status register individually or by writing ‘1’ to global status clear register Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted and PWR_GOOD signal to be de-asserted. If the output power not good condition is still present then PMIC will continue to assert GSI_n output signal and assert PWR_GOOD signal and the appropriate status register Table 103, “Register 0x08” [5:2] or Table 104, “Register 0x09” [6:5] or Table 146, “Register 0x33” [2] will remain at ‘1’. If the output power not good condition persists, the host may set the appropriate mask register to remove GSI_n or PWR_GOOD output signal as shown in Table 31 and Table 32.

8.17 Output Over Voltage Protection

An output over voltage protection mechanism is implemented to limit the voltages on the PMIC output regulators. The PMIC actively monitors the output voltage on each enabled regulators.

There are four possibilities where PMIC recognizes the over voltage event.

1. SWA output regulator goes above the threshold set in register Table 129, “Register 0x22,” [5:4].
2. SWB output regulator goes above the threshold set in register Table 131, “Register 0x24,” [5:4].
3. SWC output regulator goes above the threshold set in register Table 133, “Register 0x26,” [5:4].
4. SWD output regulator goes above the threshold set in register Table 135, “Register 0x28,” [5:4].

8.17.1 DDR5 RDIMM/LRDIMM Environment - Table 171, “Register 0x4F” [7] = ‘0’:

In programmable mode (i.e., Table 142, “Register 0x2F” [2] = ‘1’), if any output over voltage condition persists greater than tOutput_OV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators, sets register Table 105, “Register 0x0A” [7:4] appropriately, asserts PWR_GOOD and asserts GSI_n output signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the PWR_GOOD signal assertion and GSI_n signal assertion. Once host determines the cause, the host must first clear the appropriate output over voltage status register as well as any other relevant status registers individually or by writing ‘1’ to global status clear register Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted. Once the status register is cleared and GSI_n output signal is de-asserted, the host may re-enable the PMIC’s output switching regulator by issuing VR Enable command. The PMIC enables output switching regulators and ensures PWR_GOOD signal is floated when all of its output regulators are normal.

In secure mode (i.e., Table 142, “Register 0x2F” [2] = ‘0’), if any output over voltage condition persists greater than tOutput_OV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators, sets register Table 105, “Register 0x0A” [7:4] appropriately, asserts PWR_GOOD and asserts GSI_n output signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking the action of power cycling the PMIC.

8.17.2 DDR5 NVDIMM or Other Custom Environment - Table 171, “Register 0x4F” [7] = ‘1’

In programmable mode (i.e., Table 142, “Register 0x2F” [2] = ‘1’), if any output over voltage condition persists greater than tOutput_OV_VR_Disable time then PMIC internally generates VR Disable command to disable only the affected switching output regulator, sets register Table 105, “Register 0x0A” [7:4] appropriately, asserts PWR_GOOD and asserts GSI_n output signal and continues to operate normal on other output regulators. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the PWR_GOOD signal assertion and GSI_n signal assertion. Once host determines the cause, the host must first clear the appropriate output over voltage status register as well as any other relevant status registers individually or by writing ‘1’ to global status clear register Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be deasserted. Once the status register is cleared and GSI_n output signal is de-asserted, the host may re-enable the PMIC’s output switching regulator by issuing VR Enable command. The PMIC enables output switching regulator and floats PWR_GOOD signal when all of its output regulators are normal. Note that in this case, though Host issues VR Enable command, it only turns on the only affected regulator that was disabled.

The secure mode ((i.e., Table 142, “Register 0x2F” [2] = ‘0’)) is not allowed when Table 171, “Register 0x4F” [7] = ‘1’.

8.18 Output Under Voltage & VIN_Bulk Under Voltage Lockout Protection

An output under voltage lockout protection mechanism is implemented to limit the voltages on the PMIC output regulators. The PMIC actively monitors the output voltage on each enabled regulators.

There are five possibilities where PMIC recognizes the under voltage lockout event.

1. SWA output regulator goes below the threshold set in register Table 129, “Register 0x22,” [3:2].
2. SWB output regulator goes below the threshold set in register Table 131, “Register 0x24,” [3:2].
3. SWC output regulator goes below the threshold set in register Table 133, “Register 0x26,” [3:2].
4. SWD output regulator goes below the threshold set in register Table 135, “Register 0x28,” [3:2].
5. VBIAS LDO output regulator goes below the vendor specific threshold or VIN_Bulk Input Voltage goes below vendor specific threshold.

8.18.1 DDR5 RDIMM/LRDIMM Environment - Table 171, “Register 0x4F” [7] = ‘0’:

In programmable mode (i.e., Table 142, “Register 0x2F” [2] = ‘1’), if any output under voltage condition (among five possibilities listed above) or VIN_Bulk input voltage condition as listed above persists greater than tOutput_UV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators, sets register Table 106, “Register 0x0B” [3:0], Table 146, “Register 0x33” [3] appropriately, asserts PWR_GOOD and asserts GSI_n output signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the PWR_GOOD signal assertion and GSI_n signal assertion. Once host determines the cause, the host must first clear the appropriate output under voltage status register as well as any other relevant status registers individually or by writing ‘1’ to global status clear register Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted. Once the status register is cleared and GSI_n output signal is de-asserted, the host may re-enable the PMIC’s output switching regulator by issuing VR Enable command assuming valid VIN_Bulk input voltage. The PMIC enables output switching regulators and floats PWR_GOOD signal when all of its output regulators are normal.

In secure mode (i.e., Table 142, “Register 0x2F” [2] = ‘0’), if any output under voltage condition (among five possibilities listed above) or VIN_Bulk input voltage condition as listed above persists greater than tOutput_UV_VR_Disable time then PMIC internally generates VR Disable command and disables all of its switching output regulators, sets register Table 106, “Register 0x0B” [3:0], Table 146, “Register 0x33” [3] appropriately, asserts PWR_GOOD and asserts GSI_n output signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking the action of power cycling the PMIC.

8.18.2 DDR5 NVDIMM or Other Custom Environment - Table 171, “Register 0x4F” [7] = ‘1’

In programmable mode (i.e., Table 142, “Register 0x2F” [2] = ‘1’), if any output under voltage condition (First four possibilities listed above) as listed above persists greater than tOutput_UV_VR_Disable time then PMIC internally generates VR Disable command to disable only the affected switching output regulator, sets register Table 106, “Register 0x0B” [3:0] appropriately, asserts PWR_GOOD and asserts GSI_n output signal and continues to operate normal on other output regulators. Note that if the fifth condition (VBIAS LDO output goes below vendor specific threshold or VIN_Bulk input goes below vendor specific threshold) listed above persists greater than tOutput_UV_VR_Disable time, then the PMIC internally generates VR Disable command and disables all of its switching output regulators, sets register Table 106, “Register 0x0B” [3:0] and Table 146, “Register 0x33” [3] appropriately, asserts PWR_GOOD and asserts GSI_n output signal. The PMIC keeps its VOUT_1.8V and VOUT_1.0V LDO output regulators active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the PWR_GOOD signal assertion and GSI_n signal assertion. Once host determines the cause, the host must first clear the appropriate output under voltage status register as well as any other relevant status registers individually or by writing ‘1’ to global status clear register Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted. Once the

status register is cleared and GSI_n output signal is de-asserted, the host may re-enable the PMIC's output switching regulator by issuing VR Enable command. The PMIC enables output switching regulator and ensures PWR_GOOD signal is floated when all of its output regulators are normal. Note that in this case, though Host issues VR Enable command, it only turns on the only affected regulator that was disabled.

The secure mode (i.e., Table 142, "Register 0x2F" [2] = '0') is not allowed when Table 171, "Register 0x4F" [7] = '1'.

8.19 Output Current Limiter Warning Event

The PMIC has output current limiter mechanism to limit the current on the PMIC output voltage regulators.

There are four possibilities where PMIC recognizes the current limiter event.

1. SWA output regulator current goes above the threshold set in register Table 127, "Register 0x20" [7:6].
2. SWB output regulator current goes above the threshold set in register Table 127, "Register 0x20" [5:4].
3. SWC output regulator current goes above the threshold set in register Table 127, "Register 0x20" [3:2].
4. SWD output regulator current goes above the threshold set in register Table 127, "Register 0x20" [1:0].

When either event occurs for a period longer than tOutput_Current_Limiter time then PMIC sets the register Table 106, "Register 0x0B" [7:4] appropriately, drives GSI_n output signal as shown in Table 30 at the same time. The PMIC continues to operate as normal. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the GSI_n signal assertion. Once host determine the cause, the host may clear the appropriate output current limiter status register as well as any other status registers individually or by writing '1' to global status clear register in Table 115, "Register 0x14" [0] which triggers the GSI_n signal to be de-asserted. If the output current limiter condition is still present then PMIC will continue to assert GSI_n output signal and the appropriate status register in Table 106, "Register 0x0B" [7:4] will remain at '1'. If the output current limiter condition persists, the host may set the appropriate mask register to remove the GSI_n output signal as shown in Table 31 and Table 32.

8.20 Output High Current Consumption Warning Event

The PMIC supports high output current consumption warning mechanism for each of its regulator output. If enabled, the PMIC actively monitors the average output current of the regulator.

There are four possibilities where PMIC recognizes the high output current consumption.

1. SWA output regulator average current goes above the threshold set in register Table 123, "Register 0x1C" [7:2].
2. SWB output regulator average current goes above the threshold set in register Table 124, "Register 0x1D" [7:2].
3. SWC output regulator average current goes above the threshold set in register Table 125, "Register 0x1E" [7:2].
4. SWD output regulator average current goes above the threshold set in register Table 126, "Register 0x1F" [7:2].

When either event occurs then PMIC sets the register Table 104, "Register 0x09" [3:0] appropriately, drives GSI_n output signal as shown in Table 30 at the same time. The PMIC continues to operate as normal. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the GSI_n signal assertion. Once host determines the cause, the host may clear the appropriate output current consumption warning status register as well as any other status registers individually or by writing '1' to global status clear register in Table 115, "Register 0x14" [0] which triggers the GSI_n signal to be de-asserted. If the output current consumption warning condition is still present then PMIC will continue to assert GSI_n output signal and the appropriate status register in Table 104, "Register 0x09" [3:0] will remain at '1'. If the output current consumption warning condition persists, the host may set the appropriate mask register to remove GSI_n output signal as shown in Table 31 and Table 32.

8.21 PMIC High Temperature Warning and Critical Temperature Protection

The PMIC provides a high temperature warning mechanism as well as critical temperature shutdown. There are two registers associated with PMIC temperature: The high temperature warning threshold register Table 122, “Register 0x1B” [2:0] and shutdown temperature threshold register Table 141, “Register 0x2E” [2:0]. The value programmed in the shutdown temperature register must be equal or greater than value programmed in a warning threshold register.

There is one possibility where PMIC recognizes the high temperature event.

1. The PMIC temperature goes above the threshold set in register Table 122, “Register 0x1B” [2:0].

When the above event occurs for a period longer than $t_{\text{High_Temp_Warning}}$ time, the PMIC sets the register Table 104, “Register 0x09” [7] and drives GSI_n output signal as shown in Table 30 at the same time. The PMIC continues to operate as normal. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the GSI_n signal assertion. Once host determines the cause, the host may clear the temperature warning status register as well as any other status registers individually or by writing ‘1’ to global status clear register in Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted. If the high temperature warning condition is still present then PMIC will continue to assert GSI_n output signal and the appropriate status register in Table 104, “Register 0x09” [7] will remain at ‘1’. If the high temperature warning condition persists, the host may set the appropriate mask register to remove GSI_n output signal as shown in Table 31 and Table 32.

If the PMIC temperature goes above the threshold set in register Table 141, “Register 0x2E” [2:0] for a period longer than $t_{\text{Shut_Down_Temp}}$ time, the PMIC internally generates VR Disable command and disables all of its switching output regulators as well as Vbias voltage regulator (optional), sets the code in register Table 100, “Register 0x05” [2:0], updates Table 103, “Register 0x08” [6], drives GSI_n and PWR_GOOD output signal as shown in Table 30 at the same time. The PMIC keeps its VOUT_1.8V LDO and VOUT_1.0V LDO output regulator active. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host is expected to monitor the temperature status registers. When the temperature drops below the threshold, the host must re-start the PMIC by going through the power cycle of the VIN_Mgmt input supply. If the PMIC is in VIN_Bulk input supply switchover state, the host must re-start the PMIC by going through the power cycle of the VIN_Bulk input supply.

8.22 VIN_Mgmt to VIN_Bulk Input Supply Switchover Event

After VR Enable command is registered, the PMIC automatically switches over from VIN_Mgmt to VIN_Bulk input supply under following condition.

1. VIN_Mgmt goes below the threshold set in register Table 142, “Register 0x2F” [7].

When the above event occurs for a period longer than $t_{\text{Input_PWR_GOOD_GSI_Assertion}}$ time then PMIC sets the register Table 104, “Register 0x09” [4] and drives GSI_n output signal as shown in Table 30 at the same time. The PMIC continues to operate as normal. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the GSI_n signal assertion. Once host determines the cause, the host may clear the status register individually or by writing ‘1’ to global status clear register in Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted. No further action is needed by the host or the PMIC at this point.

Note that this event is treated differently by the PMIC. When host clears this event, the PMIC must remove the GSI_n signal assertion even though PMIC does not see valid VIN_Mgmt. This is to simplify host because host knows that there is no VIN_Mgmt and yet host expects the PMIC (as well as system) to continue to run normal and host should not have to worry about masking this event in PMIC. It is assumed that at some point VIN_Mgmt supply will come back up again, PMIC will detect it and assert GSI_n output signal as described in Section 8.23. At this point, PMIC will be ready to assert GSI_n output signal again if VIN_Mgmt input supply goes below the threshold set in register Table 142, “Register 0x2F” [7].

8.23 Valid VIN_Mgmt Supply Detection in Switchover Mode

When PMIC is in switchover mode as described in Section 8.22, the VIN_Mgmt input supply may power back up at any time. When VIN_Mgmt input supply re-powers backup, the PMIC sets the register Table 146, “Register 0x33” [4] and drives GSI_n output signal as shown in Table 30 at the same time. The PMIC continues to operate as normal and automatically switches back to VIN_Mgmt input supply. The PMIC allows access to all registers. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the GSI_n signal assertion. Once host determines the cause, the host may clear the status register individually or by writing ‘1’ to global status clear register in Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted. No further action is needed by the host or the PMIC at this point.

Note that this event is treated differently by the PMIC. When host clears this event, the PMIC must remove the GSI_n signal assertion even though PMIC still sees valid VIN_Mgmt. This is to simplify host because host knows that VIN_Mgmt input supply is back and so host expects the PMIC (as well as system) to continue to run normal and host should not have to worry about masking this event in PMIC. It is assumed that if at some point VIN_Mgmt supply goes below the threshold again, PMIC will detect it and assert GSI_n output signal as described in Section 8.22. At this point, PMIC will be ready to assert GSI_n output signal again if VIN_Mgmt input supply re-powers back up again.

8.24 Packet Error Code (PEC) & Parity Error Event

In I3C Basic mode, on PMIC’s primary management interface, PEC function and Parity function can be enabled. If enabled, when PMIC detects either PEC error or Parity Error, the PMIC sets the register Table 105, “Register 0x0A” [3:2] appropriately, drives GSI_n output signal as shown in Table 30 and it continues to operate as normal and allows access to all registers. See Section 11.8 to 11.9 for additional details. The host is responsible for taking any specific action. The host may query the PMIC register space to determine the cause of the GSI_n signal assertion. Once host determines the cause, the host may clear the status register individually or by writing ‘1’ to global status clear register in Table 115, “Register 0x14” [0] which triggers the GSI_n signal to be de-asserted. No further action is needed by the host from this point on.

9 Analog to Digital Converter (ADC)

The PMIC supports analog to digital converter (ADC) to monitor input supply voltages (VIN_Bulk and VIN_Mgmt) as well as output voltage regulator voltage (SWA, SWB, SWC, SWD, VBias, VOUT_1.8V and VOUT_1.0V). The register Table 143, “Register 0x30” [7:3] allows to enable the ADC and select the desire input supply voltage or output supply voltage. The register Table 144, “Register 0x31” [7:0] provides the actual voltage measurement. The accuracy of the voltage measurement is as following:

- Switch Output Voltage Regulator SWA, SWB, SWC (Output Voltage Range: 1050 mV to 1160 mV): ± 1 LSB
- Switch Output Voltage Regulator SWA, SWB, SWC (Output Voltage Range outside of 1050 mV to 1160 mV): ± 3 LSB
- Switch Output Voltage Regulator SWD (Output Voltage Range: 1750 mV to 1850 mV): ± 1 LSB
- Switch Output Voltage Regulator SWD (Output Voltage Range outside of 1750 mV to 1850 mV): ± 3 LSB
- VOUT_1.8V, VOUT_1.0V: ± 3 LSB
- VIN_Bulk, VIN_Mgmt Input Voltage: ± 3 LSB
- VBias Output Voltage, VIN_Mgmt Input Voltage: ± 6 LSB

The PMIC also monitors output voltage regulator current or power (SWA, SWB, SWC and SWD) and updates registers Table 107, “Register 0x0C” [7:0] for SWA, Table 108, “Register 0x0D” [5:0] for SWB, Table 109, “Register 0x0E” [5:0] for SWC and Table 110, “Register 0x0F” [5:0] for SWD. The register Table 122, “Register 0x1B” [6] allows host to select whether PMIC should report current measurements or power measurements. The current or power measurement reported in this registers are an average measurement over time period defined in register Table 143, “Register 0x30” [1:0]. If Table 122, “Register 0x1B” [6] = ‘1’, the register Table 121, “Register 0x1A” [1] allows host to select whether PMIC should report individual rail power or total power in Table 107, “Register 0x0C” [7:0]. The register update frequency of this register is configured in Table 143, “Register 0x30” [1:0]. The internal sampling rate of the PMIC is vendor specific. The accuracy of the current (≥ 0.5 A) or corresponding power measurement is ± 3 LSB or ± 6 LSB respectively. The accuracy of the current measurement (< 0.5 A) is ± 4 LSB or corresponding power measurement is ± 7 LSB respectively.

If register Table 121, “Register 0x1A” [1] = ‘1’, the accuracy of total power reported in register Table 107, “Register 0x0C” = ± 12 LSB

9.1 PMIC Address ID (PID)

The PMIC has PID input pin which allows to assign up to three different unique ID for I²C and I3C Basic protocol. The PID input pin is shared with SWD_FB_N pin.

At first power on, when VIN_Mgmt input is applied, the PMIC automatically senses its ID. The PMIC also checks the configuration register Table 171, “Register 0x4F” [1].

If SWD output regulator is enabled and intended to operate in a single ended remote sensing mode, the PMIC offers three different ID as shown in Table 33. If SWD output regulator is not enabled, the PMIC still offers three different ID as shown in Table 33.

If SWD output regulator is enabled and intended to operate in a differential remote sensing mode, there is only one default ID for the PMIC as shown in Table 33. This means, there can be only one PMIC on the DIMM (or I²C and I3C Basic bus).

Table 33 — PMIC ID

R4F [1] =	PID Pin Connection on DIMM Board	PMIC ID	Comment
‘0’	short to GND	PID = 1001	PMIC can be configured
	Floating	PID = 1000	
	short to 1.8	PID = 1100	Connected to PMIC’s VOUT 1.8V Rail
‘1’	Connect to Differential Sensing GND	PID = 1001	SWD differential sensing

9.2 Error Injection

The PMIC offers error injection capability for the purpose of debug, test and validation at various stages. The error injection function is only applicable after PMIC has received VR Enable command and PMIC is in programmable mode of operation. If any of the output rails are not enabled through power on config 0 to power on config3 sequence registers, the error injection on that output rail does not apply.

When host enables the error injection function, the PMIC simply follows the behavior as described in Table 30 (Triggers VR Disable, drives PWR_GOOD and GSI_n signal). In error injection mode, the PMIC does update the error log registers (Table 99, “Register 0x04” to Table 102, “Register 0x07”) when it simulates VR Disable sequence. The host can erase the error log registers (Table 99, “Register 0x04” to Table 102, “Register 0x07”) by writing 0x74 to register R39.

To exit the mode error injection mode of operation, the PMIC requires power cycle of VIN_Bulk and VIN_Mgmt input supply.

10 I²C and I3C Basic Interface Operation

At power on, by default, the PMIC device comes up in legacy I²C mode of operation. Following applies in I²C mode:

1. The max operation speed is limited to 1 MHz
2. In-band interrupts are not supported
3. Bus reset is supported.
4. Parity check is not supported except for supported CCCs.
5. Packet Error check is not supported.

The PMIC device shall operate in the legacy I²C mode until put into I3C Basic mode via command.

The host may put the PMIC device in I3C Basic mode by issuing SETAASA CCC.

Following applies in I3C Basic mode.

1. The max operation speed is up to 12.5 MHz
2. In-band interrupts are supported
3. Bus reset is supported.
4. Parity check is always enabled by default.
5. Packet error check is supported and by default is disabled.

11 Device Interface - Protocol

The 7-bit serial address of the PMIC device applies to both I²C and I3C Basic mode of operation identically.

11.1 Serial Address of PMIC Device

The PMIC device 4-bit binary value (LID) is:

- If PID pin is connected to GND on PCB: ‘1001’
- If PID pin is tied to 1.8 V on PCB: ‘1100’
- If PID pin is floating on PCB: ‘1000’

The PMIC device samples the status of the PID pin on power up. The sampled status of the PID pin is used to select one of the three possible unique LID code for the device. The selected LID code either ‘1001’ or ‘1100’ or ‘1000’ is merged with a 3 bit HID code Table 147, “Register 0x34” [3:1] to establish the 7-bit address code the device. For example, with the default setting in Table 147, “Register 0x34” [3:1] = ‘111’; if the PID pin is connected to GND, the device address shall be ‘1001 111’.

Table 34 — 7-bit Address of PMIC Device

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
1	x	0	x	1	1	1	R/W
PMIC Device Type ID (LID)				Host ID (HID)			Read/ Write

11.2 Switch from I²C Mode to I3C Basic Mode

By default when PMIC first powers on, it operates in legacy I²C mode. The PMIC device shall operate in I²C mode until put into I3C Basic mode via command.

In I²C mode, the host is allowed to issued only 3 CCCs (DEVCTRL, SETHID, SETAASA). All other CCCs are not supported and the PMIC device may simply ignore it. The Host must issue DEVCTRL & SETHID CCC first (if required) followed by SETAASA CCC.

The Host puts the PMIC device in I3C Basic mode by issuing SETAASA CCC.

When SETAASA CCC is registered by the PMIC device, it updates the Table 145, “Register 0x32” [6] to ‘1’.

When SETHID CCC is registered by the PMIC device, it updates the Table 147, “Register 0x34” [3:1].

11.3 Switch from I3C Basic Mode to I²C Mode

The Host can put the PMIC device back in I²C mode from I3C Basic mode at any time by issuing RSTDAA CCC.

When RSTDAA CCC is registered by the PMIC device, it updates the Table 145, “Register 0x32” [6] to ‘0’.

11.4 I²C Slave Protocol

The PMIC device operate on a standard I²C serial interface. Transactions where the PMIC device is the targeted slave device begin with the Host issuing a START condition followed by a 7-bit PMIC device address then a read or write bit, RW. All data are transmitted with the most significant bit MSB first. During the address followed by R/W bit transmission, the PMIC device typically replies with an ACK unless there are exceptional conditions when it may passively assert a NACK.

The PMIC device host region registers that are write protected in secure mode of operation, the PMIC ACKs the host request but the PMIC does not execute the operation internally.

Similarly, regardless of secure mode or programmable mode of operation, without the correct password, all DIMM vendor and vendor specific region registers are write protected and PMIC ACKs the host request but the PMIC does not execute the operation internally.

The PMIC device accepts 1 byte of address which covers 256 bytes of registers. The PMIC device register space does not require page selection process as all registers are within first 256 bytes.

11.4.1 Write Operation - Data Packet

Table 35 — Write Command Data Packet

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S or Sr ¹	1	X	0	X	HID			W=0	A	
	Address [7:0]								A	
	Data								A	
	...								A	
	Data								A	

1. In I²C mode, Start or Repeat Start operation followed by 7'h7E with W=0 is only allowed for the purpose of issuing CCCs that are allowed in I²C mode. Any other operation including another Repeat Start is considered an illegal operation.

11.4.2 Read Operation - Data Packet

Table 36 — Read Command Data Packet

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S or Sr ¹	1	X	0	X	HID			W=0	A	
	Address [7:0]								A	
Sr	1	X	0	X	HID			R=1	A ²	
	Data								A	
	...								A	
	Data								N ³	P

1. In I²C mode, Start or Repeat Start operation followed by 7'h7E with W=0 is only allowed for the purpose of issuing CCCs that are allowed in I²C mode. Any other operation including another Repeat Start is considered an illegal operation.
2. If slave device NACKs during Repeat Start for any reason, the host may re-try Repeat Start again. The host can do the Repeat Start as many times it may desire. The PMIC may eventually ACK.
3. When PMIC device reaches last byte within the region (either Host region or DIMM vendor region), it will continue to return data but returned data will be 0x00 if there is no valid password for DIMM vendor region or Vendor specific region. Once the address counter reaches R255, it will reset to address R00 and it will continue to return the data. Only Host can perform STOP operation.

11.4.3 Default Read Address Pointer Mode

During normal operation of the DDR5 DIMM, the host periodically may poll critical information from the same location. An example may be the PMIC device's status registers or current or power measurement register readout. To help improve the efficiency of the I²C bus protocol, the PMIC offers a default read address pointer mode so that whenever the PMIC device sees the STOP operation on its SCL and SDA bus, its read address pointer is always resets to default address. The default read pointer address mode is enabled through register Table 152, "Register 0x3A" [6] and default starting address for read operation is selectable through register Table 152, "Register 0x3A" [5:4]. This allows host to read the read command data packet as shown in Table 37. The default read address pointer reduces the packet overhead by 2 bytes. The host typically enables this mode at last after VR Enable command when the normal operation of the DDR5 DIMM begins.

Table 37 — Read Command Data Packet w/ Default Address Pointer Mode

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N	Stop
S or Sr	1	X	0	X	HID			R=1	A	
	Data								A	
	...								A	
	Data								N ¹	P

1. When PMIC device reaches last byte within the region (either Host region or DIMM vendor region), it will continue to return data but returned data will be 0x00 if there is no valid password for DIMM vendor region or Vendor specific region. Once the address counter reaches R255, it will reset to address R00 and it will continue to return the data. Only Host can perform STOP operation.

11.5 I3C Basic Slave Protocol

The PMIC device operate on a standard I3C Basic serial interface. Transactions where the PMIC device is the targeted slave device begin with the Host issuing a START condition followed by a 7-bit PMIC device address then a read or write bit, RW. All data are transmitted with the most significant bit MSB first. During the address followed by R/W bit transmission, the PMIC device typically replies with an ACK unless there are exceptional conditions when it may passively assert a NACK. See Table 38. The “T” bit carries Parity information from the Host for each byte.

The PMIC device host region registers that are write protected in secure mode of operation, the PMIC does not execute the operation internally.

Similarly, regardless of secure mode or programmable mode of operation, without the correct password, all DIMM vendor and vendor specific region registers are write protected and PMIC does not execute the operation internally.

The Packet Error Code (PEC) function is disabled by default when the PMIC device is put in I3C Basic mode. The host may optionally enable this function through Table 147, “Register 0x34” [7] or DEVCTRL CCC. If enabled, the PEC is appended at the end of all transactions. If PEC is enabled, the host must complete the burst length as indicated in CMD field. In other words, the host must not interrupt the burst length pre-maturely for Write operation.

11.5.1 Write Operation - Data Packet

Table 38 — Write Command Data Packet; PEC Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			W=0	A ^{1,2,3}	
	Address [7:0]								T	
	Data								T	
	...								T	
	Data								T	

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC device NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. The PMIC device does not check for parity error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The PMIC device ignores the entire packet until STOP or next Repeat Start operation.
4. Repeat Start or Repeat Start with 7'h7E.

Table 39 — Write Command Data Packet; PEC Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			W=0	A ^{1,2,3}	
	Address [7:0]								T	
	CMD			W=0	0000			T		
	Data								T	
	...								T	
	Data								T	
	PEC								T	Sr ⁴ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC device NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. The PMIC device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The PMIC device ignores the entire packet until STOP or next Repeat Start operation.
4. Repeat Start or Repeat Start with 7'h7E.

The host may optionally allow PMIC device to request IBI. For this case, the transactions to the PMIC device begin with the I3C host issuing a START condition followed by 7'h7E and then write bit. If PMIC device has a pending IBI, it transmits its 7-bit device select code followed by R=1. If PMIC device has no pending IBI, there is no action taken by PMIC. The Table 40 & Table 41 shows the I3C Basic bus write command data packet with optional IBI header for PEC disabled and PEC enabled case respectively. Note that in Table 41, PEC calculation does not include IBI header byte (7'h7E followed by W=0).

Table 40 — Write Command Data Packet w/ IBI Header; No Pending IBI, PEC Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ^{1,2}	
Sr	1	X	0	X	HID			W=0	A ^{2,3,4}	
	Address [7:0]								T	
	Data								T	
	...								T	
	Data								T	Sr ⁵ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (Repeat Start)
2. The PMIC device NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK) and See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
4. The PMIC device does not check for parity error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The PMIC device ignores the entire packet until STOP or next Repeat Start operation.
5. Repeat Start or Repeat Start with 7'h7E.

Table 41 — Write Command Data Packet w/ IBI Header; No Pending IBI, PEC Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ^{1,2}	
Sr	1	X	0	X	HID			W=0	A ^{2,3,4}	
	Address [7:0]								T	
	CMD			W=0	0000			T		
	Data								T	
	...								T	
	Data								T	
	PEC								T	Sr ⁵ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (Repeat Start)
2. The PMIC device NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK) and See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
4. The PMIC device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The PMIC device ignores the entire packet until STOP or next Repeat Start operation.
5. Repeat Start or Repeat Start with 7'h7E.

11.5.2 Read Operation - Data Packet

The PMIC device operate on a standard I3C Basic serial interface. Transactions where the PMIC device is the targeted slave device begin with the Host issuing a START condition followed by a 7-bit PMIC device address then a read or write bit, RW. All data are transmitted with the most significant bit MSB first. During the address followed by R/W bit transmission, the PMIC device typically replies with an ACK unless there are exceptional conditions when it may passively assert a NACK. See Table 42. The “T” bit carries Parity information from the Host for each byte prior to Repeat START. After Repeat START, “T” bit carries information from PMIC device to Host indicating Continuous (‘1’) or Stop (‘0’) whether it is transmitting the last byte or not.

The Packet Error Code (PEC) function is disabled by default when the PMIC device is put in I3C Basic mode. The host may optionally enable this function through Table 147, “Register 0x34” [7] or DEVCTRL CCC. If enabled, the PEC is appended as shown in Table 42. If PEC is enabled, the host must complete the burst length as indicated in CMD field. In other words, the host must not interrupt the burst length pre-maturely for Read operation.

Table 42 — Read Command Data Packet; PEC Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			W=0	A ^{1,2,3}	
	Address [7:0]								T	
Sr	1	X	0	X	HID			R=1	A/N ^{4,5}	
	Data								T=1	
	...								T=1	
	Data								T=1 ^{6,7}	Sr ⁸ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC device NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. The PMIC device does not check for parity error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The PMIC device ignores the entire packet until STOP or next Repeat Start operation.
4. If slave device NACKs during Repeat Start for any reason, the host may re-try Repeat Start again. The host can do the Repeat Start as many times it may desire. If slave device NACKs due to parity error in previous bytes, it will always NACK regardless of how many times the Host tries Repeat Start. If there were no parity errors, the PMIC may eventually ACK.
5. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK).
6. See Figure 21 to see how Host ends slave device operation.
7. When PMIC device reaches last byte within the region (either Host region or DIMM vendor region), it will continue to return data but returned data will be 0x00 if there is no valid password for DIMM vendor region or Vendor specific region. Once the address counter reaches R255, it will reset to address R00 and it will continue to return the data. Only Host can perform STOP operation.
8. Repeat Start or Repeat Start with 7'h7E.

Table 43 — Read Command Data Packet; PEC Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			W=0	A ^{1,2,3}	
	Address [7:0]								T	
	CMD			R=1	0000			T		
	PEC								T	
Sr	1	X	0	X	HID			R=1	A/N ^{4,5}	
	Data								T=1	
	...								T=1	
	Data								T=1	
	PEC								T=0 ⁶	Sr ⁷ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC device NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. The PMIC device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The PMIC device ignores the entire packet until STOP or next Repeat Start operation.
4. If slave device NACKs during Repeat Start for any reason, the host may re-try Repeat Start again. The host can do the Repeat Start as many times it may desire. If slave device NACKs due to PEC error or parity error in previous bytes, it will always NACK regardless of how many times the Host tries Repeat Start. If there were no parity or PEC errors, the PMIC may eventually ACK. The PEC calculation by the slave device only includes device select code of the ACK response of the Repeat start operation. In other words, if there are more than one Repeat Start operation, the slave device includes device select of only the last Repeat Start from the Host when it ACKs in PEC calculation and all other NACK responses of the device select code of the Repeat Start are not included in PEC calculation.
5. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK).
6. See Figure 22 to see how slave device ends the operation followed by Host STOP operation.
7. Repeat Start or Repeat Start with 7'h7E.

The host may optionally allow PMIC device to request IBI. For this case, the transactions to the PMIC device begin with the I3C Basic host issuing a START condition followed by 7'h7E and then write bit. If PMIC device has a pending IBI, it transmits its 7-bit device select code followed by R=1. If PMIC device has no pending IBI, there is no action taken by PMIC. The Table 44 & Table 45 shows the I3C Basic bus read command data packet with optional IBI header for PEC disabled and PEC enabled case respectively. Note that in Table 45, PEC calculation does not include IBI header byte (7'h7E followed by W=0).

Table 44 — Read Command Data Packet w/ IBI Header; No Pending IBI, PEC Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ^{1,2}	
Sr	1	X	0	X	HID			W=0	A ^{2,3,4}	
	Address [7:0]								T	
Sr	1	X	0	X	HID			R=1	A/N ^{5,6}	
	Data								T=1	
	...								T=1	
	Data								T=1 ^{7,8}	Sr ⁹ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC device NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK) and See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
4. The PMIC device does not check for parity error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The PMIC device ignores the entire packet until STOP or next Repeat Start operation.
5. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK).
6. If slave device NACKs during Repeat Start for any reason, the host may re-try Repeat Start again. The host can do the Repeat Start as many times it may desire. If slave device NACKs due to parity error in previous bytes, it will always NACK regardless of how many times the Host tries Repeat Start.
7. See Figure 21 to see how Host ends slave device operation.
8. When PMIC device reaches last byte within the region (either Host region or DIMM vendor region), it will continue to return data but returned data will be 0x00 if there is no valid password for DIMM vendor region or Vendor specific region. Once the address counter reaches R255, it will reset to address R00 and it will continue to return the data. Only Host can perform STOP operation.
9. Repeat Start or Repeat Start with 7'h7E.

Table 45 — Read Command Data Packet w/IBI Header; No Pending IBI, PEC Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ^{1,2}	
Sr	1	X	0	X	HID			W=0	A ^{2,3,4}	
	Address [7:0]								T	
	CMD			R=1	0000			T		
	PEC								T	
Sr	1	X	0	X	HID			R=1	A/N ^{5,6}	
	Data								T=1	
	...								T=1	
	Data								T=1	
	PEC								T=0 ⁷	

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC device NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK) and See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
4. The PMIC device does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issued by the host does not match with its own device code. The PMIC device ignores the entire packet until STOP or next Repeat Start operation.
5. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK).
6. If slave device NACKs during Repeat Start for any reason, the host may re-try Repeat Start again. The host can do the Repeat Start as many times it may desire. If slave device NACKs due to PEC error or parity error in previous bytes, it will always NACK regardless of how many times the Host tries Repeat Start. If there were no parity or PEC errors, the PMIC may eventually ACK. The PEC calculation by the slave device only includes device select code of the ACK response of the Repeat start operation. In other words, if there are more than one Repeat Start operation, the slave device includes device select of only the last Repeat Start from the Host when it ACKs in PEC calculation and all other NACK responses of the device select code of the Repeat Start are not included in PEC calculation.
7. See Figure 22 to see how slave device ends the operation followed by Host STOP operation.
8. Repeat Start or Repeat Start with 7'h7E.

11.5.3 Default Read Address Pointer Mode

This mode works the same exact way as explained in Section 11.4.3. Table 46 and Table 47 shows the read command data packet for PEC function disabled and enabled respectively. When PEC function is enabled, Table 152, “Register 0x3A” [3:2] sets the number of bytes that PMIC device sends out followed by the PEC calculation. If PEC is enabled, the host must complete the burst length as indicated in Table 152, “Register 0x3A” [3:2] register. In other words, the host must not interrupt the burst length pre-maturely for default address pointer read operation.

Table 46 — Read Command Data Packet w/ Read Address Pointer Mode; PEC Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			R=1	A ¹	
	Data								T=1	
	...								T=1	
	Data								T=1 ^{2,3}	Sr ⁴ or P

1. The PMIC device NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. See Figure 21 to see how Host ends slave device operation.
3. When PMIC device reaches last byte within the region (either Host region or DIMM vendor region), it will continue to return data but returned data will be 0x00 if there is no valid password for DIMM vendor region or Vendor specific region. Once the address counter reaches R255, it will reset to address R00 and it will continue to return the data. Only Host can perform STOP operation.
4. Repeat Start or Repeat Start with 7'h7E.

Table 47 — Read Command Data Packet w/ Read Address Pointer Mode; PEC Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			R=1	A ¹	
	Data								T=1	
	...								T=1	
	Data								T=1	
	PEC								T=0 ²	Sr ³ or P

1. The PMIC device NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. See Figure 22 to see how slave device ends the operation followed by STOP operation
3. Repeat Start or Repeat Start with 7'h7E.

Table 48 — Read Command Data Packet w/ Read Address Pointer & IBI Header; No Pending IBI; PEC Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ^{1,2}	
Sr	1	X	0	X	HID			R=1	A/N ^{2,3}	
	Data								T=1	
	...								T=1	
	Data								T=1 ^{4,5}	Sr ⁶ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (Repeat Start).
2. The PMIC device NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK).
4. See Figure 21 to see how Host ends slave device operation.
5. When PMIC device reaches last byte within the region (either Host region or DIMM vendor region), it will continue to return data but returned data will be 0x00 if there is no valid password for DIMM vendor region or Vendor specific region. Once the address counter reaches R255, it will reset to address R00 and it will continue to return the data. Only Host can perform STOP operation.
6. Repeat Start or Repeat Start with 7'h7E.

Table 49 — Read Command Data Packet w/ Read Address Pointer & IBI Header; No Pending IBI; PEC Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ^{1,2}	
Sr	1	X	0	X	HID			R=1	A/N ^{2,3}	
	Data								T=1	
	...								T=1	
	Data								T=1	
	PEC								T=0 ⁴	

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (Repeat Start).
2. The PMIC device NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK).
4. See Figure 22 to see how slave device ends the operation followed by STOP operation
5. Repeat Start or Repeat Start with 7'h7E.

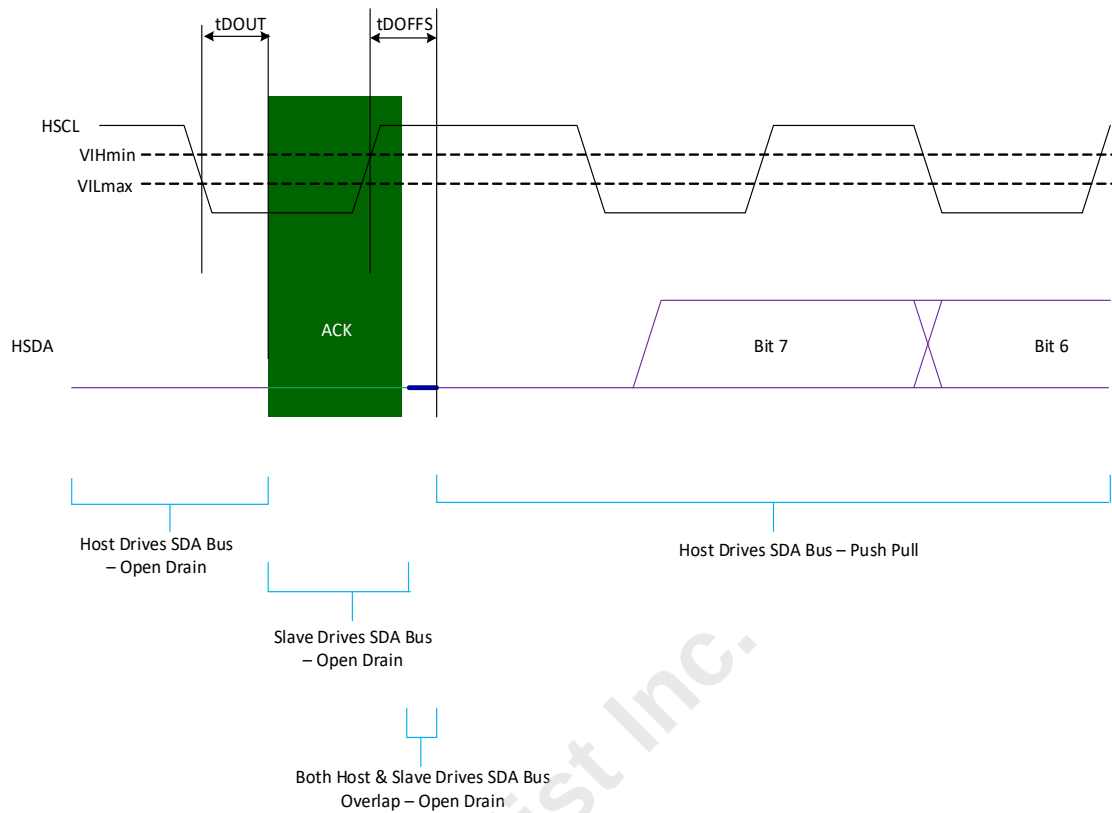


Figure 18 — Slave Open Drain to Host Push Pull Hand Off Operation

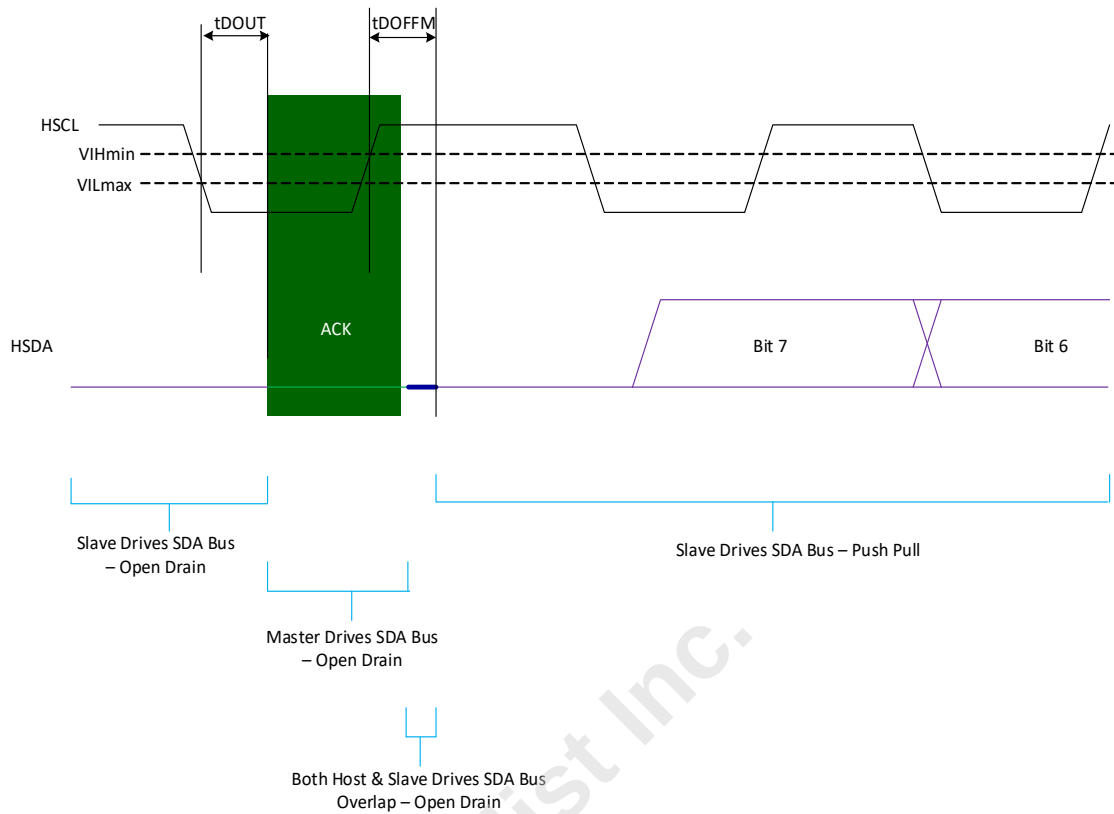


Figure 19 — Master Open Drain (ACK) to Slave Push Pull Hand Off Operation

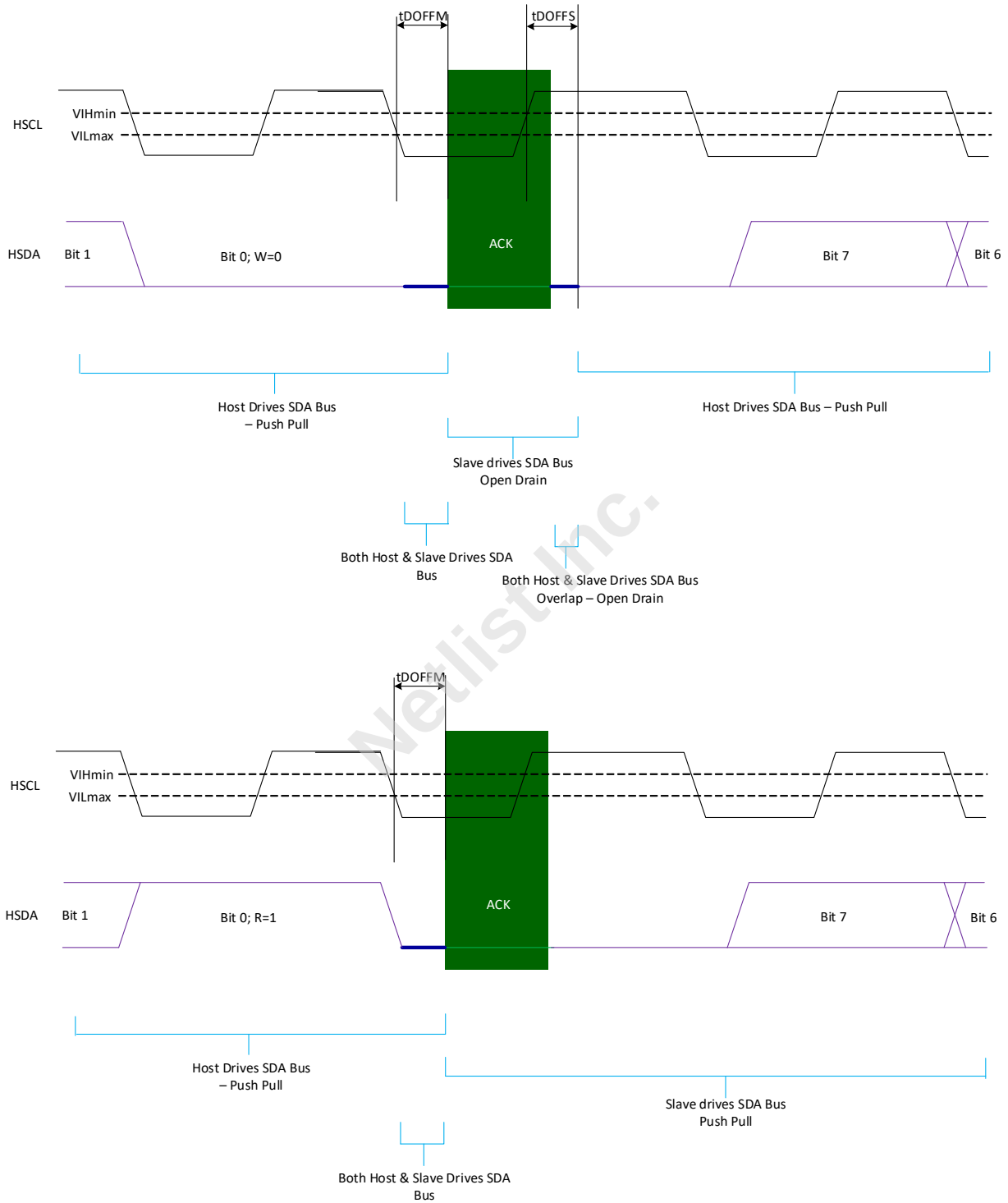


Figure 20 — Host Push Pull to Slave Open Drain Hand Off Operation

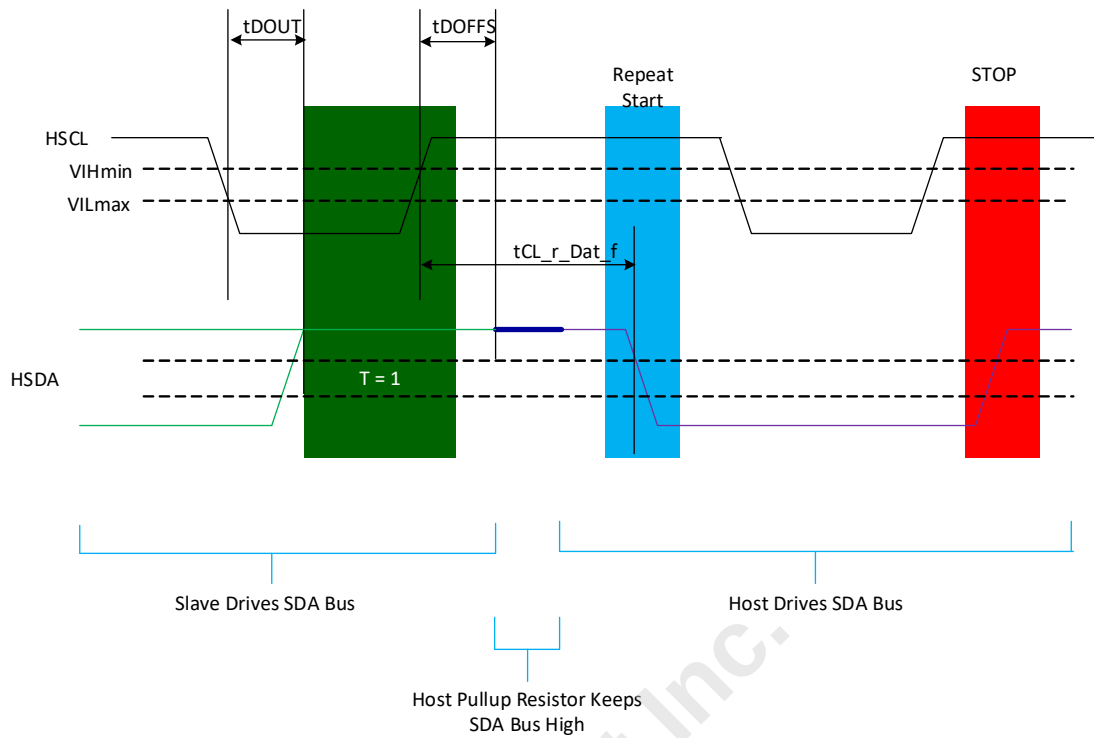


Figure 21 — T=1; Host Ends Read with Repeated START and STOP Waveform

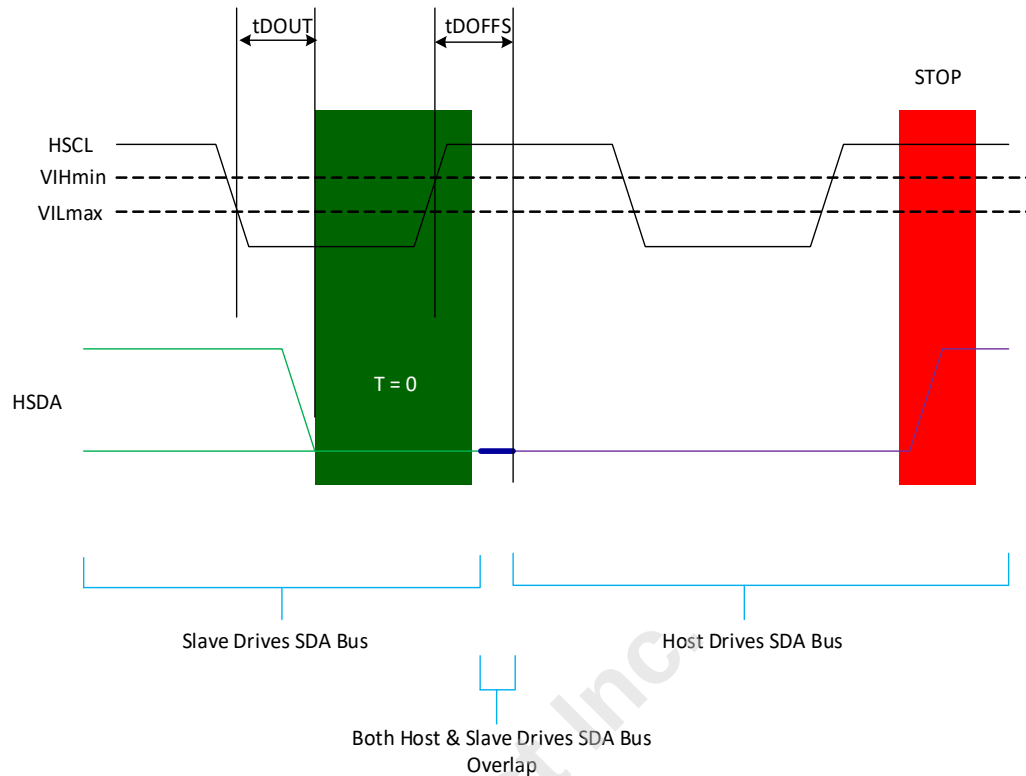


Figure 22 — T=0; Slave Ends Read; Host Generates STOP

11.6 In Band Interrupt (IBI)

In I²C mode, in band interrupt function is not supported. Only I3C Basic mode supports in band interrupt function.

11.6.1 Enabling & Disabling In Band Interrupt Function

By default, IBI function is disabled. The PMIC device enables the IBI when it registers ENEC CCC. Once enabled, the PMIC device sends an IBI when an event occurs.

- When Table 147, “Register 0x34” [6] = ‘1’, the device sends the IBI at next available opportunity when any of the register bits in Table 103, “Register 0x08” [7:0], Table 104, “Register 0x09” [7:0], Table 105, “Register 0x0A” [7:2], Table 106, “Register 0x0B” [7:0] and Table 146, “Register 0x33” [4:2] is set to ‘1’. The device also sets Table 105, “Register 0x0A” [1] to ‘1’ and updates Pending Interrupt Bits [3:0] = ‘0001’ for GETSTATUS CCC.
- When Table 147, “Register 0x34” [6] = ‘0’, the device does not send the IBI regardless of the register bits in Table 103, “Register 0x08” [7:0], Table 104, “Register 0x09” [7:0], Table 105, “Register 0x0A” [7:2], Table 106, “Register 0x0B” [7:0] and Table 146, “Register 0x33” [4:2]. However, the device set Table 105, “Register 0x0A” [1] to ‘1’ and updates Pending Interrupt Bits [3:0] = ‘0001’ for GETSTATUS CCC.

11.6.2 Mechanics of Interrupt Generation

Event interrupts may be generated by the local device if IBI is enabled. When there is a pending interrupt (i.e. Table 105, “Register 0x0A” [1] = ‘1’) and if IBI is enabled (i.e., Table 147, “Register 0x34” [6] = ‘1’) the PMIC device requests an interrupt after detecting START condition by transmitting its 7-bit binary address (LID bits followed by HID bits) followed by R/W = ‘1’ on the SDA bus serially (synchronized by SCL falling transitions).

If PMIC device detects no START condition but if the I3C bus (SDA and SCL) has been inactive (no edges seen) for t_{AVAL} period, then PMIC device may assert SDA low by $t_{\text{IBI_ISSUE}}$ time to request an interrupt. When the PMIC device requests an interrupt, the Host toggles the SCL. The PMIC device transmits its 7-bit binary address (LID bits followed by HID bits) followed by R/W bit = ‘1’ to the Host.

When the PMIC device requests an interrupt, the host may take one of the two actions below.

- The Host sends ACK on 9th bit to accept the interrupt request. At this point, if the PMIC device confirms that it has won the arbitration, the PMIC device transmits the IBI payload as shown in Table 50 and Table 51 for PEC disabled and PEC enabled configuration respectively. See Figure 23. Figure 23 just shows only first two data bits of the MDB byte to illustrate the timing. The interrupt payload contains MDB followed by Table 103, “Register 0x08”, Table 104, “Register 0x09”, Table 105, “Register 0x0A”, Table 106, “Register 0x0B” & Table 146, “Register 0x33” bytes. The host then issues the STOP command. Note the timing waveform in Figure 23. The host then accepts the IBI payload if it sends an ACK on 9th bit to accept the interrupt request. The host can interrupt the IBI payload at T bit. If host stops the IBI payload at T bit in the middle of payload, the PMIC retains the IBI status flag (Table 105, “Register 0x0A” [1]) and Pending Interrupt Bits [3:0] internally and waits for the next opportunity to request an interrupt. If the PMIC device successfully transmits the entire IBI payload, it then clears IBI status flag (Table 105, “Register 0x0A” [1] = ‘0’) and Pending Interrupt Bits [3:0] = ‘0000’ on its own and does not request for an IBI again unless there is another different event occurs; for another same event, the device does not request for an IBI.
- The Host sends NACK on the 9th bit as shown in Figure 24 followed by a STOP command. In this case, the PMIC device does not transmit the IBI payload and waits for the next opportunity to request an interrupt. At this point, though Host sent an NACK, it does have a knowledge of which PMIC device sent the IBI request. The PMIC device retains the IBI status flag (Table 105, “Register 0x0A” [1] = ‘1’) and Pending Interrupt Bits [3:0] = ‘0001’

Table 50 — Slave Device IBI Payload Packet; PEC is Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/T	Stop
S	1	X	0	X	HID			R=1	A ¹	
	MDB = 0x00								T=1	
	R08 [7:0]								T=1	
	R09 [7:0]								T=1	
	R0A [7:0]								T=1	
	R0B [7:0]								T=1	
	R33 [7:0]								T=0 ²	P

1. See Figure 19 to see how the transition occurs from Host Open Drain (ACK) to Slave Push Pull Operation (1st bit of MDB Byte bit [7]).
2. See Figure 22 to see how slave device ends the operation followed by Host STOP operation.

Table 51 — Slave Device IBI Payload Packet; PEC is Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/T	Stop
S	1	X	0	X	HID			R=1	A ¹	
	MDB 0x00								T=1	
	R08 [7:0]								T=1	
	R09 [7:0]								T=1	
	R0A [7:0]								T=1	
	R0B [7:0]								T=1	
	R33 [7:0]								T=1	
	PEC								T=0 ²	P

1. See Figure 19 to see how the transition occurs from Host Open Drain (ACK) to Slave Push Pull Operation (1st bit of MDB Byte bit [7]).
2. See Figure 22 to see how slave device ends the operation followed by Host STOP operation.

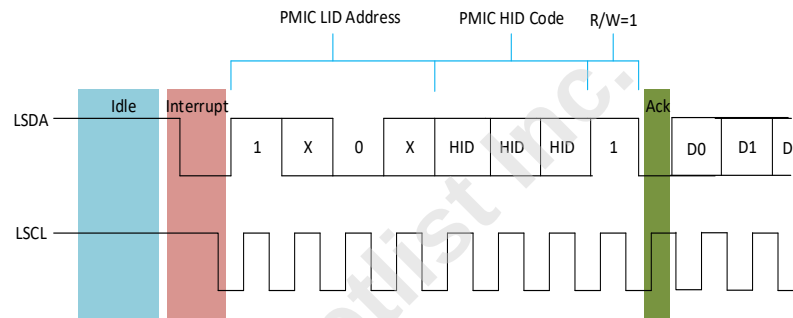


Figure 23 — PMIC Requests Interrupt, Host Ack followed by PMIC Device IBI Payload

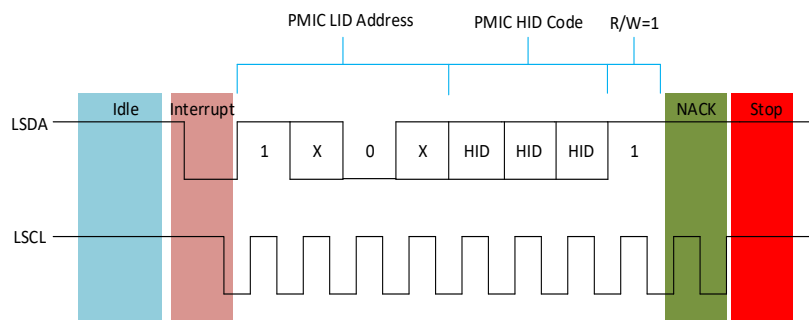


Figure 24 — PMIC Requests Interrupt; Host NACK followed by STOP

11.6.3 Interrupt Arbitration

As there are multiple devices I3C Basic bus, multiple device may request an interrupt when the Host I3C Basic bus is inactive for t_{AVAL} period. Arbitration process is required.

For DDR5 DIMM application environment, there could be up to total of 13 difference devices including the PMIC on I3C bus.

On a typical DDR5 DIMM application environment, all devices have the same 3-bit HID code. Hence the arbitration is always won by the lowest 4-bit LID code. For example, if one slave device has LID code of '0010' and other device (PMIC) has a LID code of '1001', through the arbitration process, the slave device LID code of '0010' wins. The PMIC device with a LID code of '1001' must release the bus and wait for next opportunity to request an interrupt. Table 52 shows the arbitration priority based on the LID code for all devices. The Green color cells in Table 52 are the likely devices that will be on a standard DDR5 RDIMM or DDR5 LRDIMM. The Olive color cells in Table 52 do not apply.

Table 52 — Interrupt Arbitration - Among All Devices

Device	LID Code	HID Code = '111'	Arbitration Priority
N/A	0000	N/A	N/A
RFU	0001	111	1
TS0	0010	111	2
RFU	0011	111	3
RFU	0100	111	4
RFU	0101	111	5
TS1	0110	111	6
RFU	0111	111	7
PMIC1	1000	111	8
PMIC0	1001	111	9
SPD Hub	1010	N/A	N/A
RCD	1011	111	10
PMIC2	1100	111	11
RFU	1101	111	12
RFU	1110	111	13
N/A	1111	N/A	N/A

In an uncommon but possible scenario would be that at the exact same time as when the Hub or local slave devices (i.e PMIC) are requesting an interrupt, the host is starting an operation to the Hub or local slave devices (i.e PMIC). When this happens, Host also gets involved in the arbitration process along with the Hub or the local slave devices (PMIC). During the arbitration phase, there will be always only one winning device and it could be either Hub or the local slave device (i.e PMIC) or the Host.

If the host wins during the arbitration phase, it continues with normal operation. The losing Hub or local slave device (i.e PMIC) waits for next opportunity to send an interrupt.

If the host loses during the arbitration phase, it must let go of the bus. When the host loses during the arbitration, the host must let the Hub or local slave device (i.e. PMIC) finish sending their 4-bit LID code followed by 3-bit HID code followed by R/W = '1'. At this point, during the 9th bit, the host has two options to take the action as noted:

- Host sends an ACK to accept the interrupt and hence accepts the IBI payload from the winning Hub or local slave device (i.e. PMIC). After the IBI payload, the host issues STOP operation.
- Host sends a NACK followed by STOP operation.

In a rare but still possible scenario would be that at the exact same time as when the PMIC is requesting an interrupt, the host is starting an operation to the same PMIC. When this happens, neither Host nor the PMIC knows it is a winner until the 8th bit and Host always wins. This is because, the PMIC sends R=1 (8th bit) during the interrupt. The host sets W=0 (8th bit) during the operation. As a result, the host wins and the PMIC must let go of the bus and wait for the next opportunity to send an interrupt.

In an extreme rare but still possible scenario would be that at the same exact time as when PMIC device is requesting an interrupt, the host is requesting a read operation with the default read address pointer mode to the PMIC device. When this happens, there is no winning device. This is the only time there is no winning device. This is because, the PMIC device sends R=1 (8th bit) during the interrupt and Host also sends R=1 for read request with default read address pointer mode. As a result, there is no winner because Host is waiting for PMIC to ACK and PMIC is waiting for Host to ACK. In this case, neither Host nor PMIC will ACK. Since there is no ACK (i.e. NACK) by either device, the Host must time out and repeat the read request with Repeat Start. When Host repeats the read request with Repeat Start, the PMIC does not send an interrupt because of Repeat Start.

11.6.4 Clearing Device Status and IBI Status Registers

The PMIC device provides the IBI status in Table 105, "Register 0x0A" [1] by setting it to '1'. The PMIC device clears the IBI status register Table 105, "Register 0x0A" [1] to '0' automatically when it sends a complete IBI (including payload and without interruption) and it also clears Pending Interrupt Bits [3:0] to '0000'. Once IBI status register is cleared, the PMIC does not request for an IBI again unless another event occurs.

The PMIC device provides the device status in Table 103, "Register 0x08" [7:0], Table 104, "Register 0x09" [7:0], Table 105, "Register 0x0A" [7:2], Table 106, "Register 0x0B" [7:0] and Table 146, "Register 0x33" [4:2] registers. The status information in Table 103, "Register 0x08" [7:0], Table 104, "Register 0x09" [7:0], Table 105, "Register 0x0A" [7:2], Table 106, "Register 0x0B" [7:0] and Table 146, "Register 0x33" [4:2] registers are latched and remains set even after the PMIC device sends IBI payload and clears the IBI status register Table 105, "Register 0x0A" [1] to '0'. The host must explicitly clear the status register through Clear command by writing '1' for appropriate status or by issuing a Global clear command.

After Host issues clear command, if the condition is no longer present, the PMIC device clears the appropriate status register, clears the IBI status register to '0' and Pending Interrupt Bits [3:0] to '0000' even if the PMIC device has not sent the IBI. After Host issues clear command, if the condition is still present, the device will again set the appropriate status register, sets the IBI status register to '1' and Pending Interrupt Bits [3:0] to '0001' even if the device has already sent the IBI and entire IBI payload.

11.7 Packet Error Check (PEC) Function

In I²C mode, packet error checking is not supported. Only I3C Basic mode supports packet error checking.

The PMIC device implement an 8-bit Packet Error Code (PEC) which is appended at the end of all transactions if PECs is enabled through DEVCTRL CCC or by directly writing '1' to Table 147, "Register 0x34" [7]. The PEC is a CRC-8 value calculated on all the messages bytes except for START, STOP, REPEATED START conditions or T-bits, ACK and NACK and IBI header (7'h7E followed W=0) bits.

The polynomial for CRC-8 calculations is:

- $C(X) = X^8 + X^2 + X^1 + 1$

The seed value for PEC function is all zero.

When Host calculates PEC for PMIC device, it includes LID and HID bits followed by R/W bit.

11.8 Parity Error Check Function

In I²C mode, parity error checking is not supported except for supported CCCs. Only I3C Basic mode supports parity error checking.

By default, when PMIC device is put in I3C Basic mode, parity function is automatically enabled. The host can disable the function after it is enabled. Host can also disable the parity function with DEVCTRL CCC or by directly writing '1' to Table 147, "Register 0x34" [5]. When parity function is disabled, the PMIC device simply ignores the "T" bit information from the Host. The host may actually choose to compute the parity and send that information during "T" bit or simply drive static low or high in "T" bit.

The PMIC device implements ODD parity. If an odd number of bits in the byte are '1', the parity bit value is '0'. If even number of bits in the byte are '1', the parity bit value is '1'. The host computes the parity and sends during "T" bit.

11.9 Packet Error Check & Parity Error Handling

There are two types of error checking done by the PMIC device. Parity error checking and Packet Error checking. By default, the parity error checking is always enabled and packet error checking is disabled. The host may enable the packet error checking at any time. The parity error is checked for each byte in a packet except for the device select code byte from the host. The host sends parity error information in “T” bit.

I3C Basic defines S0, S1, S2, S3, S4, S5, S6 error detection for slave devices. Only S1 and S2 error detection is supported by the PMIC for parity checking. All other errors are not supported and not applicable.

11.9.1 Write Command Data Packet Error Handling - PEC Disabled

The PMIC device checks for the parity error for each byte in a packet that it receives from the host except for the device select code byte that it receives from the host as shown in Table 53.

Table 53 — Write Command Data Packet; PEC Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			W=0	A ^{1,2,3}	
	Address [7:0]								T	
	Data								T	
	...								T	
	Data								T	Sr ⁴ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. The PMIC does not check for parity error in subsequent bytes when it determines the 7-bit device select code issues by the host does not match with its own device code. The PMIC ignores the entire packet until STOP or next Repeat Start operation.
4. Repeat Start or Repeat Start with 7'h7E.

Write command - if no parity error:

- The PMIC device executes the command.

Write command - if parity error:

- The PMIC device discards the byte in the packet that had a parity error.
- The PMIC device discards all subsequent bytes in that packet until the STOP operation. The PMIC device may or may not check parity for all sub-sequent bytes in that packet.
- Note that as the packet contains more than one byte, if first byte had no parity error but the second byte had a parity error, the PMIC device may or may not execute the first byte operation but second byte and all subsequent bytes operations are discarded.
- The PMIC device sets the Table 105, “Register 0x0A” [2:1] to ‘11’; P_Err in GETSTATUS CCC to ‘1’; updates Pending Interrupt Bits [3:0] in GETSTATUS CCC to ‘0001’; asserts GSI_n pin if enabled and waits for the next opportunity to send an in band interrupt if IBI is enabled.

11.9.2 Read Command Data Packet Error Handling - PEC Disabled

The PMIC device checks for parity error for each byte in a packet except for the device select code byte that it receives from the host prior to Repeat Start as shown in Table 54.

The PMIC device does not compute the parity when it sends the data to the Host. The does not check for parity error for the bytes shown in Table 54. The device sends Continuous ('1') or Stop ('0') information during "T" bit when PMIC device is sending the read data.

Table 54 — Read Command Data Packet; PEC Disabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			W=0	A ^{1,2,3}	
	Address [7:0]								T	
Sr	1	X	0	X	HID			R=1	A/N ^{4,5}	
	Data								T=1	
	...								T=1	
	Data								T=1 ^{6,7}	Sr ⁸ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. The PMIC does not check for parity error in subsequent bytes when it determines the 7-bit device select code issues by the host does not match with its own device code. The PMIC ignores the entire packet until STOP or next Repeat Start operation.
4. If slave device NACKs during Repeat Start for any reason, the host may re-try Repeat Start again. The host can do the Repeat Start as many times it may desire. If slave device NACKs due to parity error in previous bytes, it will always NACK regardless of how many times the Host tries Repeat Start. If there were no parity errors, the PMIC may eventually ACK.
5. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK).
6. See Figure 21 to see how Host ends slave device operation.
7. When PMIC device reaches last byte within the region (either Host region or DIMM vendor region), it will continue to return data but returned data will be 0x00 if there is no valid password for DIMM vendor region or Vendor specific region. Once the address counter reaches R255, it will reset to address R00 and it will continue to return the data. Only Host can perform STOP operation.
8. Repeat Start or Repeat Start with 7'h7E.

Read Command - If no parity error:

- The PMIC sends ACK back to the host when Host perform Start Repeat operation.
- The PMIC device executes the command and sends the data as shown in Table 54.

Read Command - If parity error:

- The PMIC device discards the byte in the packet that had a parity error.
- The PMIC device sends NACK back to the host when Host performs a Start Repeat operation. This is shown in the **RED color** cell in Table 54 above. The NACK represents either a parity error in one of the two bytes or that PMIC is not able to start the read operation. The host may re-try Repeat Start again. The host may do the Repeat Start as many times as it may desire. If the PMIC slave device NACKs due to parity error in a previous byte from the host, it will always NACK regardless of how many times Host tries Repeat Start.
- The PMIC does not send the data shown in Table 54 and instead expects Host to perform STOP operation.
- The PMIC device sets Table 105, "Register 0x0A" [2:1] to '11'; P_Err in GETSTATUS CCC to '1'; updates Pending Interrupt Bits [3:0] in GETSTATUS CCC to '0001'; asserts GSI_n pin if enabled and waits for the next opportunity to send an in band interrupt if IBI is enabled.

11.9.3 Write Command Data Packet Error Handling - PEC Is Enabled

The PMIC device checks for the parity error for each byte in a packet that it receives from the host except for the device select code byte that it receives from the host as shown in Table 55. Further, the PMIC device checks for the packet error for the entire packet (from Start condition until last byte of Data) that it receives from the host as shown in Table 55.

Table 55 — Write Command Data Packet; PEC Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			W=0	A ^{1,2,3}	
	Address [7:0]								T	
	CMD			W=0	0	0	0	0	T	
	Data								T	
	...								T	
	Data								T	
	PEC								T	Sr ⁴ or P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. The PMIC does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issues by the host does not match with its own device code. The PMIC ignores the entire packet until STOP or next Repeat Start operation.
4. Repeat Start or Repeat Start with 7'h7E.

Write command - if no parity error:

- The PMIC device waits for the entire packet. If no error in packet, the PMIC device executes the command. If there is an error in the packet, the PMIC device discards the entire packet and does not execute the packet and waits for STOP; sets the Table 105, "Register 0x0A" [3,1] to '11'; PEC_Err in GETSTATUS CCC to '1'; updates Pending Interrupt Bits [3:0] in GETSTATUS CCC to '0001'; asserts GSI_n pin if enabled and waits for the next opportunity to send an in band interrupt if IBI is enabled.

Write command - if parity error:

- The PMIC device discards that byte and the entire packet until STOP operation.
- The PMIC device sets the Table 105, "Register 0x0A" [2:1] to '11'; P_Err in GETSTATUS CCC to '1'; updates Pending Interrupt Bits [3:0] in GETSTATUS CCC to '0001'; asserts GSI_n pin if enabled and waits for the next opportunity to send an in band interrupt if IBI is enabled.
- The PMIC device may or may not check the error for the packet. If the PMIC device checks for the packet error, likely it will detect an error in the packet and the device may also set Table 105, "Register 0x0A" [3] & PEC_Err in GETSTATUS CCC as well.

11.9.4 Read Command Data Packet Error Handling - PEC Is Enabled

The PMIC device checks for parity error for each byte in a packet except for the device select code byte that it receives from the host prior to Repeat Start as shown in Table 56.

The PMIC device does not compute the parity when it sends the data to the Host. The does not check for parity error for the bytes shown in Table 56. The device sends Continuous ('1') or Stop ('0') information during "T" bit when PMIC device is sending the read data.

The PMIC device checks for the PEC error in a packet that it receives from Host from Start condition to Repeat Start (from first device select code followed by the address offset and CMD byte).

The PMIC device computes the packet error code for the entire packet starting with Repeat Start (device select code and the data PMIC device transmits back to Host).

Table 56 — Read Command Data Packet; PEC Enabled

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	X	0	X	HID			W=0	A ^{1,2,3}	
	Address [7:0]								T	
	CMD			R=1	0	0	0	0	T	
	PEC								T	
Sr	1	X	0	X	HID			R=1	A/N ^{4,5}	
	Data								T=1	Sr ⁷ or P
	...								T=1	
	Data								T=1	
	PEC								T=0 ⁶	

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation (1st bit of Addr, bit [7]).
2. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. The PMIC does not check for parity or PEC error in subsequent bytes when it determines the 7-bit device select code issues by the host does not match with its own device code. The PMIC ignores the entire packet until STOP or next Repeat Start operation.
4. If slave device NACKs during Repeat Start for any reason, the host may re-try Repeat Start again. The host can do the Repeat Start as many times it may desire. If slave device NACKs due to PEC error or parity error in previous bytes, it will always NACK regardless of how many times the Host tries Repeat Start. If there were no parity or PEC errors, the PMIC may eventually ACK. The PEC calculation by the slave device only includes device select code of the ACK response of the Repeat start operation. In other words, if there are more than one Repeat Start operation, the slave device includes device select of only the last Repeat Start from the Host when it ACKs in PEC calculation and all other NACK responses of the device select code of the Repeat Start are not included in PEC calculation.
5. See Figure 20 to see how the transition occurs from Host Push Pull Operation to Slave Open Drain (ACK).
6. See Figure 22 to see how slave device ends the operation followed by Host STOP operation.
7. Repeat Start or Repeat Start with 7'h7E.

Read command - If no parity error & no PEC error

- The PMIC device sends ACK back to the host when Host perform a Start Repeat operation.
- The PMIC device executes the command and sends the data as shown in Table 56.
- The PMIC computes PEC for the bytes (from Start condition to PEC byte prior to Repeat Start) shown in the cells in Table 56.

Read command - if parity error or PEC error

- The PMIC device discards the byte in the packet that had a parity error.
- The PMIC device discards second byte in that packet if a parity error occurred in first byte. The PMIC device may or may not check parity for the second byte in that packet.
- The PMIC device discards the packet if there is a PEC error.
- The PMIC sends NACK back to the host when Host perform Start Repeat operation. This is shown in the **RED color** cell in Table 56 above. The NACK represents either PEC error or a parity error in one of the three bytes or that PMIC is not able to start the read operation. The host may re-try Repeat Start again. The host may do the Repeat Start as many times it may desire. The PEC calculation by PMIC device only includes device select code of the ACK responses of the Repeat Start operation. In other words, if there are more than one Repeat Start operation, the PMIC device includes the device select of only the last Repeat Start from the Host when it ACKs in PEC calculation and other NACK responses of the device select codes of the Repeat Start are not included in PEC calculation. If the PMIC slave device NACKs due to PEC error or a parity error in a previous bytes from Host, it will always NACK regardless of how many times Host tries Repeat Start.
- The PMIC device does not send any data shown in Table 56 and instead expects Host to perform STOP operation.
- The PMIC device sets Table 105, “Register 0x0A” [3:2] accordingly and Table 105, “Register 0x0A” [1] to ‘1’; P_Err, PEC_Err in GETSTATUS CCC to ‘1’ accordingly; updates Pending Interrupt Bits [3:0] in GETSTATUS CCC to ‘0001’; asserts GSI_n pin if enabled and waits for the next opportunity to send an in band interrupt if IBI is enabled.

11.10 CCC Packet Error Handling

Parity error and PEC error detected in a CCC packet are handled the same way as described for normal Read/Write operations.

11.11 Error Reporting

All error conditions detected by the PMIC devices are captured in Table 103, “Register 0x08” [7:0], Table 104, “Register 0x09” [7:0], Table 105, “Register 0x0A” [7:1] Table 106, “Register 0x0B” [7:0] and Table 146, “Register 0x33” [4:2] registers.

There are four different possible ways error information can be communicated to the host.

1. The host makes the read request to Table 103, “Register 0x08”, Table 104, “Register 0x09”, Table 105, “Register 0x0A”, Table 106, “Register 0x0B” and Table 146, “Register 0x33” registers.
2. The host starts any transactions with Start condition followed by 7'h7E IBI header (Only applicable in I3C Basic mode).
3. The PMIC device sends in band interrupt if enabled, when its SCL and SDA input has been idle for t_{AVAL} time (Only applicable in I3C Basic mode).
4. The PMIC device asserts GSI_n pin if enabled.

11.12 I3C Basic Common Command Codes (CCC)

The I3C Basic spec lists large number of Common Command Codes (CCC). Not all CCC are required to be supported. The PMIC device NACKs for all unsupported CCC. The PMIC supports CCC as listed in Table 57 below.

The PMIC device requires STOP operation in between when switching from CCC operation to private device specific Write or Read or Default Read Address Pointer mode operation and vice versa. In other words, any CCC operation must be followed by STOP operation before continuing to any device specific Write or Read or Default Read Address Pointer mode operation. Similarly, any device specific Write or Read or Default Read Address Pointer mode operation must be followed by STOP operation before continuing to any CCC operation. The PMIC device also requires STOP operation between any direct CCC to broadcast CCC.

The PMIC device does allow Repeat Start operation between any direct CCC to any other direct CCC or between any broadcast CCC to any other broadcast CCC or between any private Write or Read or Default Read Address Pointer mode operation to any other private Write or Read or Default Read Address Pointer mode operation.

Table 57 — PMIC CCC Support Requirement

CCC	Mode	Code	Description	Note
ENEC	Broadcast	0x00	Enable Event Interrupts	
	Direct	0x80		
DISEC	Broadcast	0x01	Disable Event Interrupts	
	Direct	0x81		
RSTDAA	Broadcast	0x06	Put the device in I ² C Mode (aka: Reset Dynamic Address Assignment)	
	Direct	0x86		
SETAASA	Broadcast	0x29	Put the device in I3C Basic Mode (aka: Set All Addresses to Static Address)	
GETSTATUS	Direct	0x90	Get Device Status	
DEVCAP	Direct	0xE0	Get Device Capability	1
SETHID	Broadcast	0x61	PMIC updates 3-bit HID field	1
DEVCTRL	Broadcast	0x62	Configure SPD Hub and all devices behind Hub	1

1. JEDEC specific CCC.

11.12.1 ENEC CCC

The ENEC CCC is only supported after device is put in I3C Basic mode. In I²C mode, it is illegal for host to issue this CCC. When ENEC CCC is registered by the PMIC, it updates Table 147, “Register 0x34” [6] = ‘1’ and it takes in effect at the next Start operation (i.e., after STOP operation). Table 58 to Table 61 shows an example of a single ENEC CCC. Table 62 shows the encoding definition for ENEC CCC.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 58 — ENEC CCC - Broadcast

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x00 (Broadcast)								T	
	0x00							ENINT	T	Sr ² or P

1. The PMIC NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 59 — ENEC CCC - Broadcast w/ PEC

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x00 (Broadcast)								T	
	0x00							ENINT	T	
	PEC								T	Sr ² or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 60 — ENEC CCC - Direct

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x80 (Direct)								T	
Sr	DevID[6:0]							W=0	A ^{1,2}	
	0x00							ENINT	T	Sr ³ or P

1. The PMIC NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. The PMIC device does not check for parity error in subsequent bytes when it determines 7-bit device select code issued by the Host does not match with its own device code. The PMIC device ignores the entire packet until STOP operation or next Repeat Start operation.
3. Repeat Start or Repeat Start with 7'h7E.

Table 61 — ENEC CCC - Direct w/ PEC

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x80 (Direct)								T	
	PEC								T	
Sr	DevID[6:0]							W=0	A ^{1,2}	
	0x00							ENINT	T	
	PEC								T	Sr ³ or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. The PMIC device does not check for parity error in subsequent bytes when it determines 7-bit device select code issued by the Host does not match with its own device code. The PMIC device ignores the entire packet until STOP operation or next Repeat Start operation.
3. Repeat Start or Repeat Start with 7'h7E.

Table 62 — ENEC CCC Byte Encoding

Bit	Encoding	Notes
ENINT	0 = No Action 1 = Enable IBI Interrupt	It is illegal for Host to issue ENEC CCC with ENINT bit = '0'

11.12.2 DISEC CCC

The DISEC CCC is only supported after device is put in I3C Basic mode. In I²C mode, it is illegal for host to issue this CCC. When DISEC CCC is registered by the PMIC, it updates Table 147, "Register 0x34" [6] = '0' and it takes in effect at the next Start operation (i.e., after STOP operation). Table 63 to Table 66 shows an example of a single DISEC CCC. Table 67 shows the encoding definition for DISEC CCC.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 63 — DISEC CCC - Broadcast

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x01 (Broadcast)								T	
	7'h00							DISINT	T	

1. The PMIC NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 64 — DISEC CCC - Broadcast w/ PEC

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x01 (Broadcast)								T	
	7'h00							DISINT	T	
	PEC								T	
										Sr ² or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 65 — DISEC CCC - Direct

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x81 (Direct)								T	
Sr	DevID[6:0]							W=0	A ^{1,2}	
	0x00							DISINT	T	
										Sr ³ or P

1. The PMIC NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. The PMIC device does not check for parity error in subsequent bytes when it determines 7-bit device select code issued by the Host does not match with its own device code. The PMIC device ignores the entire packet until STOP operation or next Repeat Start operation.
3. Repeat Start or Repeat Start with 7'h7E.

Table 66 — DISEC CCC - Direct w/PEC

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x81 (Direct)								T	
	PEC								T	
Sr	DevID[6:0]							W=0	A ^{1,2}	
	0x00							DISINT	T	
	PEC								T	
										Sr ³ or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. The PMIC device does not check for parity error in subsequent bytes when it determines 7-bit device select code issued by the Host does not match with its own device code. The PMIC device ignores the entire packet until STOP operation or next Repeat Start operation.
3. Repeat Start or Repeat Start with 7'h7E.

Table 67 — DISEC CCC Byte Encoding

Bit	Encoding	Notes
DISINT	0 = No Action 1 = Disable IBI Interrupt	It is illegal for Host to issue DISEC CCC with DISINT bit = '0'

11.12.3 RSTDAA CCC

The RSTDAA CCC is only supported after device is put in I3C Basic mode. In I²C mode, this CCC is ignored. When RSTDAA CCC is registered by the PMIC, it updates Table 145, “Register 0x32” [6] = ‘0’ and it takes in effect at the next Start operation (i.e., after STOP operation). Further it disables IBI & PEC function (Table 147, “Register 0x34” [7:6] = ‘00’) and clears parity function Table 147, “Register 0x34” [5] = ‘0’).

Table 68 to Table 71 shows an example of a single RSTDAA CCC.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 68 — RSTDAA CCC - Broadcast

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x06 (Broadcast)								T	Sr ² or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 69 — RSTDAA CCC - Broadcast w/ PEC

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x06 (Broadcast)								T	
	PEC								T	Sr ² or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 70 — RSTDAA CCC - Direct

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x86 (Direct)								T	
Sr	DevID[6:0]							W=0	A ^{1,2}	Sr ³ or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. The PMIC device does not check for parity error in subsequent bytes when it determines 7-bit device select code issued by the Host does not match with its own device code. The PMIC device ignores the entire packet until STOP operation or next Repeat Start operation.
3. Repeat Start or Repeat Start with 7'h7E.

Table 71 — RSTDAA CCC - Direct w/ PEC

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x86 (Direct)								T	
	PEC								T	
Sr	DevID[6:0]							W=0	A ^{1,2}	
	PEC								T	Sr ³ or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. The PMIC device does not check for parity error in subsequent bytes when it determines 7-bit device select code issued by the Host does not match with its own device code. The PMIC device ignores the entire packet until STOP operation or next Repeat Start operation.
3. Repeat Start or Repeat Start with 7'h7E.

11.12.4 SETAASA CCC

The SETAASA CCC is only supported when device is in I²C mode. In I²C mode, when host issues this CCC, to guarantee that this CCC is registered by the device without any error, the host shall limit the maximum speed operation for this CCC to 1 MHz. In I³C Basic mode, this CCC is ignored. When SETAASA CCC is registered by the PMIC, it updates Table 145, "Register 0x32" [6] = '1' and it takes in effect at the next Start operation (i.e., after STOP operation). Table 72 shows an example of a single SETAASA CCC.

SETAASA CCC does not support PEC function as device is in I²C mode and there is no PEC function in I²C mode.

Table 72 — SETAASA CCC - Broadcast

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A	
	0x29 (Broadcast)								T	P

11.12.5 GETSTATUS CCC

The GETSTATUS CCC is supported in I3C Basic mode. In I²C mode, this CCC is ignored (i.e., it is not executed internally and the Repeat Start byte arriving after the 0x90 GETSTATUS CCC code is not acknowledged and host must do STOP operation. Table 73 to Table 74 shows an example of a single GETSTATUS CCC.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 73 — GETSTATUS CCC - Direct

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x90 (Direct)								T	
Sr	DevID[6:0]								R=1 A	
	PEC_Err	0	0	0	0	0	0	0	T	
	0	0	P_Err	0	Pending Interrupt				T	Sr ² or P

1. The PMIC NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 74 — GETSTATUS CCC - Direct w/ PEC

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ¹	
	0x90 (Direct)								T	
	PEC								T	
Sr	DevID[6:0]							R=1	A	
	PEC_Err	0	0	0	0	0	0	0	T	
	0	0	P_Err	0	Pending Interrupt				T	
	PEC								T	Sr ² or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 75 — GETSTATUS CCC Byte Encoding

Bit	Encoding	Notes
PEC_Err	0 = No Error 1 = PEC Error Occurred	This register is cleared when Host issues clear command to Table 113, “Register 0x12” [3] for PEC error
P_Err	0 = No Error 1 = Protocol Error; Parity Error occurred	This register is cleared when Host issues clear command to Table 113, “Register 0x12” [2] for Parity error.
Pending Interrupt	0000 = No Pending Interrupt 0001 = Pending Interrupt All other encodings are reserved	This register is cleared when Host issues clear command to any appropriate device status register that causes IBI status register to get cleared.

When the PMIC device responds to GETSTATUS CCC, after it completes the response, the PEC_Err, P_Err and Pending Interrupt Bits [3:0] do not automatically get cleared. The host must explicitly clear the appropriate status register through Clear command by writing ‘1’ to corresponding register or by issuing Global Clear command. Once the PMIC device clears the appropriate status register, only then PEC_Err, P_err and Pending Interrupt Bits [3:0] gets cleared.

After host issues clear command, if the condition is still present, the device will again set the appropriate status register, sets the IBI status register to ‘1’ and Pending Interrupt Bits [3:0] to ‘0001’.

11.12.6 DEVCAP CCC

The DEVCAP CCC is only supported after device is put in I3C Basic mode. In I²C mode, it is illegal for host to issue this CCC. Table 76 to Table 77 shows an example of a single DEVCAP CCC. Table 78 defines the encoding for DEVCAP CCC.

If PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7’h7E with W=0 byte in PEC calculation.

Table 76 — DEVCAP CCC - Direct

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0xE0 (Direct)								T	
Sr	DevID[6:0]							R=1	A ¹	
	MSB (Each bit defines capability)								T	
	LSB (Each bit defines capability)								T	
										Sr ² or P

1. The PMIC NACKs if there is a parity error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7’h7E.

Table 77 — DEVCAP CCC - Direct w/ PEC

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0xE0 (Direct)								T	
	PEC								T	
Sr	DevID[6:0]							R=1	A ¹	
	MSB (Each bit defines capability)								T	
	LSB (Each bit defines capability)								T	
	PEC								T	Sr ² or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. Repeat Start or Repeat Start with 7'h7E.

Table 78 — DEVCAP CCC Byte Encoding

Bit	Encoding	Notes
MSB [7]	RFU	Coded as '0'
MSB[6]	RFU	Coded as '0'
MSB[5]	RFU	Coded as '0'
MSB[4]	RFU	Coded as '0'
MSB[3]	RFU	Coded as '0'
MSB[2]	0 = No Support for Timer based Reset 1 = Supports Timer based Reset	Coded as '1'
MSB[1:0]	RFU	Coded as '00'
LSB[7:0]	RFU	Coded as '0x00'

11.12.7 SETHID CCC

The SETHID CCC is supported only when device is in I²C mode. In I²C mode, when host issues this CCC, to guarantee that this CCC is registered by the device without any error, the host shall limit the maximum speed operation for this CCC to 1 MHz. In I³C Basic mode, it is illegal for host to issue this CCC. When SETHID CCC is registered by the PMIC, it updates Table 147, "Register 0x34" [3:1] with the HID code received by the PMIC and it takes in effect at the next Start operation (i.e., after STOP operation). Table 79 shows an example of a single SETHID CCC. As the device is in I²C mode when SETHID CCC is issued, the PEC function is not supported.

Once PMIC receives SETHID CCC and updates its 3-bit HID code, after the Stop operation, PMIC device only responds to updated 7-bit address. The 4-bit LID code of the PMIC device remains as is.

The Host may issue SETHID CCC more than one time.

Table 79 — SETHID CCC - Broadcast

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A	
	0x61 (Broadcast)								T	
	0	0	0	0	HID[2:0]		0		T	P

11.12.8 DEVCTRL CCC

On a typical I3C Basic bus there can be up to 120 devices. For DDR5 DIMM application environment, there are up to 8 SPD5 Hub devices and behind each SPD5 Hub devices, there are 4 local slave devices totaling up to 40 or more devices on I3C Basic bus. For certain operation such as enable or disable functions that are common to all devices (i.e., Packet Error Check), the host must go through one device at a time which takes significant amount of time at initial power up. Further, it requires additional complexity on the host because it must speak different protocol depending on how it may access the device until all devices are configured identically.

To help expedite this configuration operation and to simplify the host complexity, the device supports the SPDCTRL CCC. The DEVCTRL CCC is supported either in I²C mode or I3C Basic mode of operation. In I²C mode, when host issues this CCC, to guarantee that this CCC is registered by the device without any error, the host shall limit the maximum speed operation for this CCC to 1 MHz. Table 80 to Table 81 shows an example of a single DEVCTRL CCC.

In I3C mode only, if PEC function is enabled, the PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

The host shall pay attention to DEVCTRL CCC. If DEVCTRL CCC is used to access device specific registers (e.g. RegMod = '1'), the host shall still follow any device specific register restriction. For example, if device specific register requires STOP operation for device to take in the effect of the setting, the host must also use STOP operation when using DEVCTRL CCC to access device specific register.

Table 80 — DEVCTRL CCC - Broadcast

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x62 (Broadcast)								T	
	AddrMask[2:0]			StartOffset[1:0]		PEC BL[1:0]		RegMod	T	
	DevID[6:0]							0	T ²	
	Byte 0 Data Payload								T	
	Byte 1 Data Payload								T	
	Byte 2 Data Payload								T	
	Byte 3 Data Payload								T	Sr ³ or P

1. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
2. An exception is made for DEVCTRL CCC. The PMIC does not report parity error when it determines 7-bit device select code issues by the host does not match with its own device code. If 7-bit device select code does not match but if parity is still valid, the device does not check for parity error in subsequent bytes; ignores the entire packet and waits until STOP or Repeat Start operation.
3. Repeat Start or Repeat Start with 7'h7E.

Table 81 — DEVCTRL CCC - Broadcast w/ PEC¹

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S or Sr	1	1	1	1	1	1	0	W=0	A ²	
	0x62 (Broadcast)								T	
	AddrMask[2:0]			StartOffset[1:0]		PEC BL[1:0]		RegMod	T	
	DevID[6:0]							0	T ³	
	Byte 0 Data Payload								T	
	Byte 1 Data Payload								T	
	Byte 2 Data Payload								T	
	Byte 3 Data Payload								T	
PEC								T	Sr ⁴ or P	

1. DEVCTRL CCC with PEC check is only supported in I3C mode.
2. The PMIC NACKs if there is a parity or PEC error in a previous transaction when host performs consecutive transactions with Repeat Start.
3. An exception is made for DEVCTRL CCC. The PMIC does not report parity error when it determines 7-bit device select code issues by the host does not match with its own device code. The device does not check for PEC as all subsequent bytes are discarded due to parity error. If 7-bit device select code does not match but if parity is still valid, the device does not check for parity error in subsequent bytes; ignores the entire packet and waits until STOP or Repeat Start operation.
4. Repeat Start or Repeat Start with 7'h7E.

Table 82 — DEVCTRL CCC Command Definition

Parameter	Definition
AddrMask[2:0]	Broadcast, Unicast or Multicast Command Selection 000 = Unicast Command; PMIC device responds if DevID[6:0] field matches with PMIC device's own 7-bit address (4-bit LID + 3-bit HID) 011 = Multicast Command; PMIC device and possible other device responds if DevID[6:3] field matches with PMIC device's own 4-bit LID address 111 = Broadcast Command; All devices responds to this command All other encodings are reserved
StartOffset[1:0]	Only applicable if RegMod = '0' Identifies the starting Byte (Byte 0 or Byte 1 or Byte 2 or Byte 3) for DEVCTRL CCC. Host can start at any Byte (from Byte 0 to Byte 3) and has continuous access to next byte until STOP operation. If Byte 3 is reached, the host is responsible for applying STOP operation. 00 = Byte 0 01 = Byte 1 10 = Byte 2 11 = Byte 3
PEC BL[1:0]	Only applicable if RegMod = '0' and PEC function is enabled. Identifies the burst length just for this DEVCTRL CCC. The device uses the setting in this field to know when the PEC byte is expected after the data bytes. 00 = 1 Byte 01 = 2 Byte 10 = 3 Byte 11 = 4 Byte
RegMod	Identifies if DEVCTRL is going to be used for General Registers as identified in Byte 0 to Byte 3 or device specific address offset register. 0 = Access to General Registers in Byte 0 to Byte 3 (i.e., StartOffset[1:0] = Valid) 1 = Device Specific Offset Address (i.e StartOffset[1:0] & PECBL[1:0] is a don't care and does not apply). The Host shall NOT use RegMod = '1' with Broadcast Command if there are different types of devices on the I3C Basic bus.
DevID[6:0]	Identifies 7-bit device address. Device responds to DEVCTRL CCC data packet depending on AddrMask[2:0]. If AddrMask[2:0] = '111', DevID[6:0] is a don't care and device always responds. If AddrMask[2:0] = '000', DevID[6:0] must match for device to respond If AddrMask[2:0] = '011', DevID[6:3] must match for device to respond. DevID[2:0] is don't care. For any other codes for AddrMask[2:0], the device always NACKs.

Table 83 — DEVCTRL CCC Data Payload Definition

Byte #	Bit #	Function	Definition	Comment
Byte 0	[7]	PEC Enable	0 = Disable 1 = Enable	Table 147, “Register 0x34” [7] is updated
	[6]	Parity Disable	0 = Enable 1 = Disable	Table 147, “Register 0x34” [5] is updated
	[5:2]	RFU	RFU	
	[1]	VR Enable	0 = VR Disable 1 = VR Enable	Table 145, “Register 0x32” [7] is updated.
	[0]	RFU	RFU	
Byte 1	[7:4]	RFU	RFU	
	[3]	Global & IBI Clear	0 = No Action 1 = Clear All Event and pending IBI ¹	Table 115, “Register 0x14” [0] is updated.
	[2:0]	RFU	RFU	
Byte 2	[7:0]	RFU	RFU	
Byte 3	[7:0]	RFU	RFU	

1. After slave device clears the event, the device can still have certain registers set to ‘1’ if the event is still present in which case, the device will generate an IBI again at the next opportunity.

11.12.8.1 DEVCTRL CCC Examples - RegMod = ‘0’

Table 84 shows an example of DEVCTRL CCC data packet. It assumes that all devices on the bus are already in I3C Basic mode with PEC function disabled and parity function enabled. In this example, the Host uses DEVCTRL CCC as Multicast command. Host sends Multicast command to all devices with 4-bit LID code of ‘1001’ on I3C Basic bus to do VR Enable followed by all devices with 4-bit LID code of ‘0110’ to disable parity function. The host sends AddrMask = ‘011’ to indicate Multicast command with DevID[6:3] match; StartOffset = ‘00’ to indicate starting Byte 0 and RegMod = ‘0’ to indicate general register. Upon receiving this command, all devices with DevID[6:3] that matches to ‘1001’ will do the VR Enable command and DevID[6:3] that matches to ‘0110’ will disable the parity function.

Table 84 — DEVCTRL CCC Example - Multicast Command to '1001' & '0110' Devices

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ¹	
	0x62 (Broadcast)								T	
	011			00		00		0	T	
	1001 000							0	T	
	0000 0010								T	
Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x62 (Broadcast)								T	
	011			00		00		0	T	
	0110 000							0	T	
	0100 0000								T	P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation.

Table 85 shows an example of DEVCTRL CCC data packet. It assumes that all devices on the bus are already in I3C Basic mode with PEC function disabled and parity function enabled. In this example, the Host uses DEVCTRL CCC as Broadcast command to enable PEC function. The host sends AddrMask = '111' to indicate Broadcast command; StartOffset = '00' to indicate starting Byte 0 and RegMod = '0' to indicates general register. Upon receiving this command, all devices will enable PEC function.

Table 85 — DEVCTRL CCC Example - Broadcast Command to all Devices

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ¹	
	0x62 (Broadcast)								T	
	111			00		00		0	T	
	0000 000							0	T	
	1000 0000								T	P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation.

Table 86 shows an example of DEVCTRL CCC data packet. It assumes that all devices on the bus are already in I3C Basic mode with PEC function disabled and parity function enabled. In this example, the Host uses DEVCTRL CCC as Unitcast command to enable VR on DIMM5. The host sends AddrMask = '000' to indicate Unicast command; StartOffset = '00' to indicate starting Byte 0 and RegMod = '0' to indicates general register. Upon receiving this command, PMIC on DIMM5 will enable its regulator.

Table 86 — DEVCTRL CCC Example - Unicast Command to PMIC on DIMM5

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ¹	
	0x62 (Broadcast)								T	
	000			00		00		0	T	
	1001 101							0	T	
	0000 0010								T	

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation.

11.12.8.2 DEVCTRL CCC Examples - RegMod = '1'

Table 87 shows an example of DEVCTRL CCC data packet for the purpose of configuring device specific address offset register. It assumes that all devices on the bus are already in I3C Basic mode with PEC function enabled and parity function enabled. In this example, the Host sends Multicast command to all devices with 4-bit LID code of '0010' on the I3C Basic bus to write to address offset of 0x1C and 0x1D with data 0xFF and 0x55 respectively followed by all devices with 4-bit LID of '1001' on the I3C Basic bus to write to address offset of 0x15 with data 0x78.

The PEC calculation starts with Start or Repeat Start operation but does not include 7'h7E with W=0 byte in PEC calculation.

Table 87 — DEVCTRL CCC Example - Multicast Command to '0010' & '1001' Devices

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ¹	
	0x62 (Broadcast)								T	
	011			00		00		1	T	
	0010 000							0	T	
	0001 1100 (address offset 0x1C)								T	
	0010 0000 (CMD field = 2 bytes of data)								T	
	1111 1111 (data)								T	
	0101 0101 (data)								T	
	PEC								T	
Sr	1	1	1	1	1	1	0	W=0	A ¹	
	0x62 (Broadcast)								T	
	011			00		00		1	T	
	1001 000							0	T	
	0001 0101 (address offset 0x15)								T	
	0000 0000 (CMD field = 1 byte of data)								T	
	0111 1000 (data)								T	
	PEC								T	
										P

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation.

Table 88 shows an example of DEVCTRL CCC data packet for the purpose of configuring device specific address offset register. It assumes that all devices on the bus are already in I3C Basic mode with PEC function disabled and parity function enabled. In this example, the Host sends Multicast command to all devices with 4-bit LID code of '1001' on the I3C Basic bus to write to address offset of 0x13 with data 0xFF and it continues to write data 0x01 to the next address.

Table 88 — DEVCTRL CCC Example - Multicast Command to '1001' Devices

Start	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	A/N/T	Stop
S	1	1	1	1	1	1	0	W=0	A ¹	
	0x62 (Broadcast)								T	
	011			00		00		1	T	
	1001 000							0	T	
	0001 0011 (address offset 0x13)								T	
	1111 1111 (data)								T	
	0000 0001 (data)								T	

1. See Figure 18 to see how the transition occurs from Slave Open Drain (ACK) to Host Push Pull Operation.

11.13 IO Operation

At power on, by default, the PMIC device comes up in legacy I²C mode of operation with Open Drain IO for its interface. The maximum speed is limited to 1 MHz and supported IO voltage levels are from 1.0 V to 3.3 V.

After power on, the host may put the PMIC device in I3C Basic mode of operation.

In I3C Basic mode, the host may drive the SCL clock input of the PMIC device using either Push-Pull output driver or using the open-drain output driver. It is expected that for all DDR5 DIMM family environment, the host may always drive the SCL clock input using a Push-Pull output driver.

To support in band interrupt, the PMIC device supports dynamic switching between Open Drain mode and Push Pull mode on its SCL and SDA bus for various event. The Table 89 below describes the different mode of operation by the PMIC device for each cycle.

Table 89 — PMIC Device Dynamic IO Operation Mode Switching

	Open Drain Mode	Push Pull Mode
START + Device Select Code	Yes	No
START + 7'h7E IBI Header Byte	Yes	No
REPEAT START + Device Select Code	No	Yes
REPEAT START + 7'h7E Header Byte	No	Yes
CCC Bytes (i.e after 7'h7E+W=0+ACK)	No	Yes
STOP	No	Yes
ACK/NACK Responses	Yes	No
Command, Block Address, Address Operation	No	Yes
Interrupt Request by Slave + Device Select Code	Yes	No
IBI Payload	No	Yes
Write Data, T-bit sequence	No	Yes
Read Data, T-bit sequence	No	Yes
PEC, T-bit sequence	No	Yes

11.14 Bus Clear

The PMIC device supports the following described Bus Clear feature in I²C mode only. Any attempt by host to perform I²C Bus clear on a slave device in I3C mode may result in an active drive bus contention on the SDA data line.

There may be abnormal circumstances when the host abruptly stops clocking SCL while the slave device is in the middle of outputting data for read operation. For these type of events, the SDA data line may appear as stuck low as the device is expecting to receive more clock pulses from the host. Eventually when the host has control of the SCL clock, the host may optionally clear the device that is stuck low on the SDA data line by sending continuous 18 clock pulses without driving the SDA data line followed by STOP operation. The device floats the SDA line within 18 clock pulses and returns to the Idle state. The device is ready for normal new transaction with Start condition.

11.15 Bus Reset

To prevent a malfunctioning device from locking up the I²C bus or I3C Basic bus, a bus reset mechanism is defined. It uses a timeout mechanism on SCL as shown in Figure 25 to force a device bus reset. All devices on a I²C or I3C Basic bus reset simultaneously. Bus reset operation works same way regardless of whether device is operating in I²C or I3C Basic mode.

To guarantee the device resets I²C bus or I3C Basic bus, the SCL clock input Low time has to be greater than or equal to $t_{\text{TIMEOUT(Max)}}$.

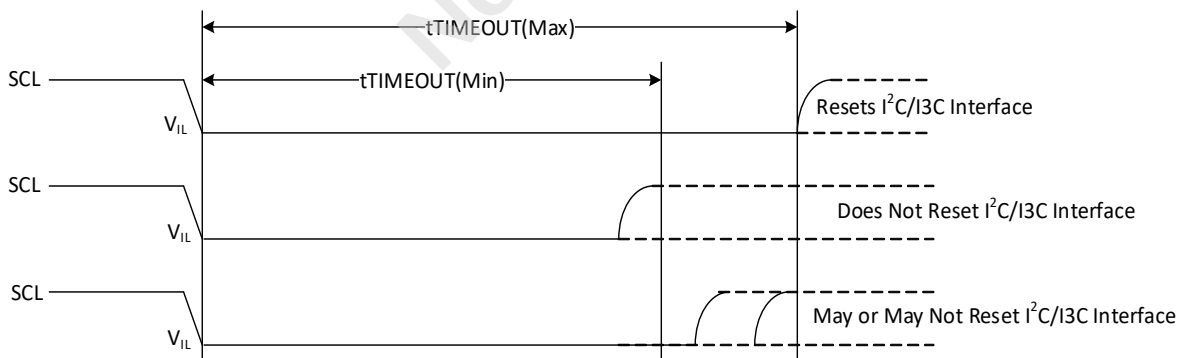
The PMIC device does not reset I²C bus or I3C Basic bus if the SCL clock input Low time is less than $t_{\text{TIMEOUT(Min)}}$.

If the SCL clock input Low time is between $t_{\text{TIMEOUT(Min)}}$ and $t_{\text{TIMEOUT(Max)}}$, the PMIC device does not guarantee and it may or may not reset the I²C bus or I3C Basic bus.

When RESET, the PMIC device takes following action.

1. Interface and any pending command or transactions are cleared
2. All internal register values are preserved unless noted otherwise in item # 3 below.
3. Device returns to I²C mode of operation; Table 147, “Register 0x34” [3:1] resets to ‘111’; Table 147, “Register 0x34” [7:5] resets to ‘000’; Table 145, “Register 0x32” [6] to ‘0’; Table 105, “Register 0x0A” [3:2] to ‘00’.
4. Device does not re-sample PID pin.
5. Device floats the SDA pin such that it gets pulled High by external/other device pullup.
6. Device treats bus reset as STOP operation.

Figure 25 — I²C or I3C Basic Bus Reset - PMIC Device



11.16 Command Truth Table

The command truth table as shown in Table 90 only applies in I3C Basic mode with PEC enabled. In I²C mode and I3C Basic mode with PEC disabled, the command truth table does not apply.

Table 90 — For I3C Mode Only w/ PEC Enabled - Command Truth Table

TS5 Command	Command Name	CMD Code	RW	Address
		2nd Byte Bits [7:5]	2nd Byte Bit [4]	1st Byte Bits [7:0]
Write 1 Byte to Register	W1R	000	0	V
Read 1 Byte from Register	R1R		1	V
Write 2 Byte to Register	W2R	001	0	V
Read 2 Byte from Register	R2R		1	V
Write 4 Byte to Register	W4R	010	0	V
Read 4 Byte from Register	R4R		1	V
Write 16 Byte to Register	W16R	011	0	V
Read 16 Byte from Register	R16R		1	V
Reserved	RSVD	100 to 111	RSVD	RSVD

12 Device Package and Pinout

The PMIC device is packaged in 5 mm x 5 mm FCQFN.

12.1 Package Pinout

The PMIC pinout is shown in Figure 26 & Figure 27 below for TOP view and BOTTOM view respectively.

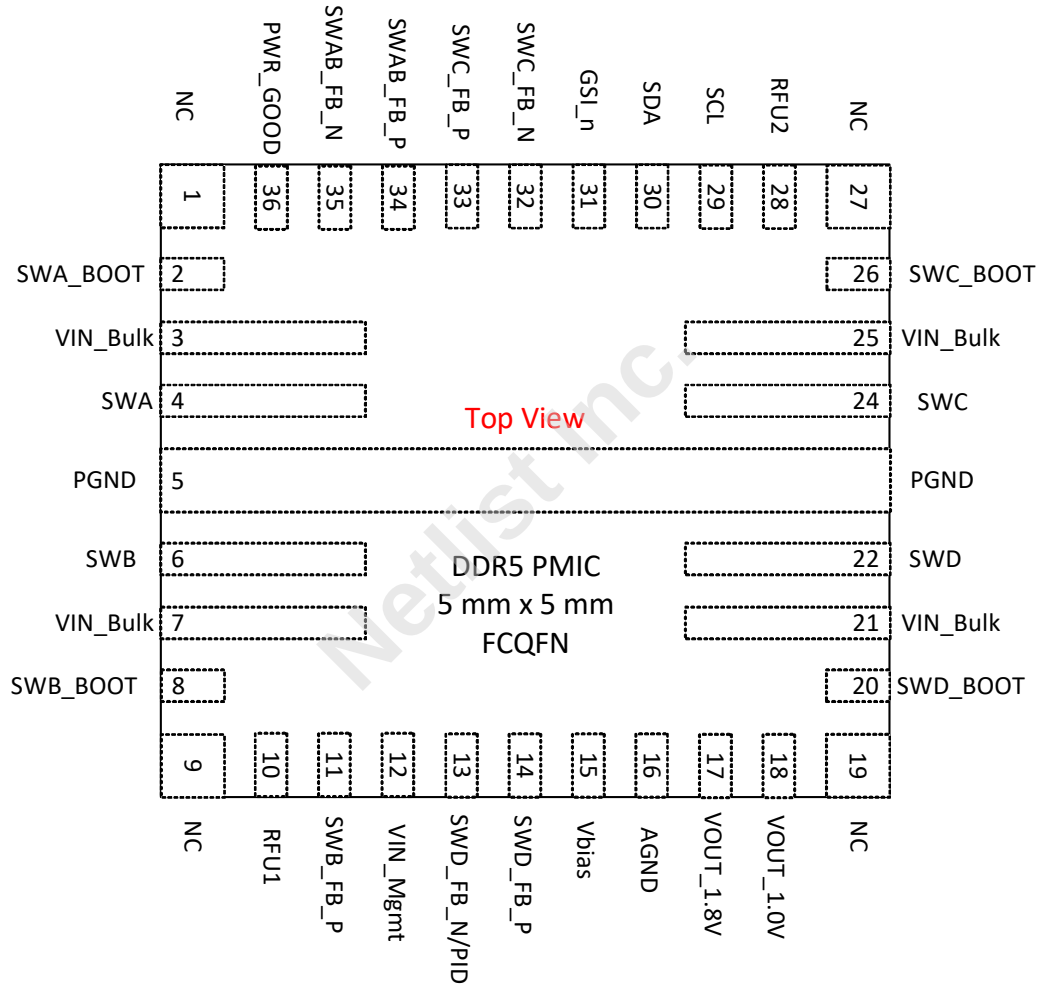


Figure 26 — PMIC Pinout - TOP View

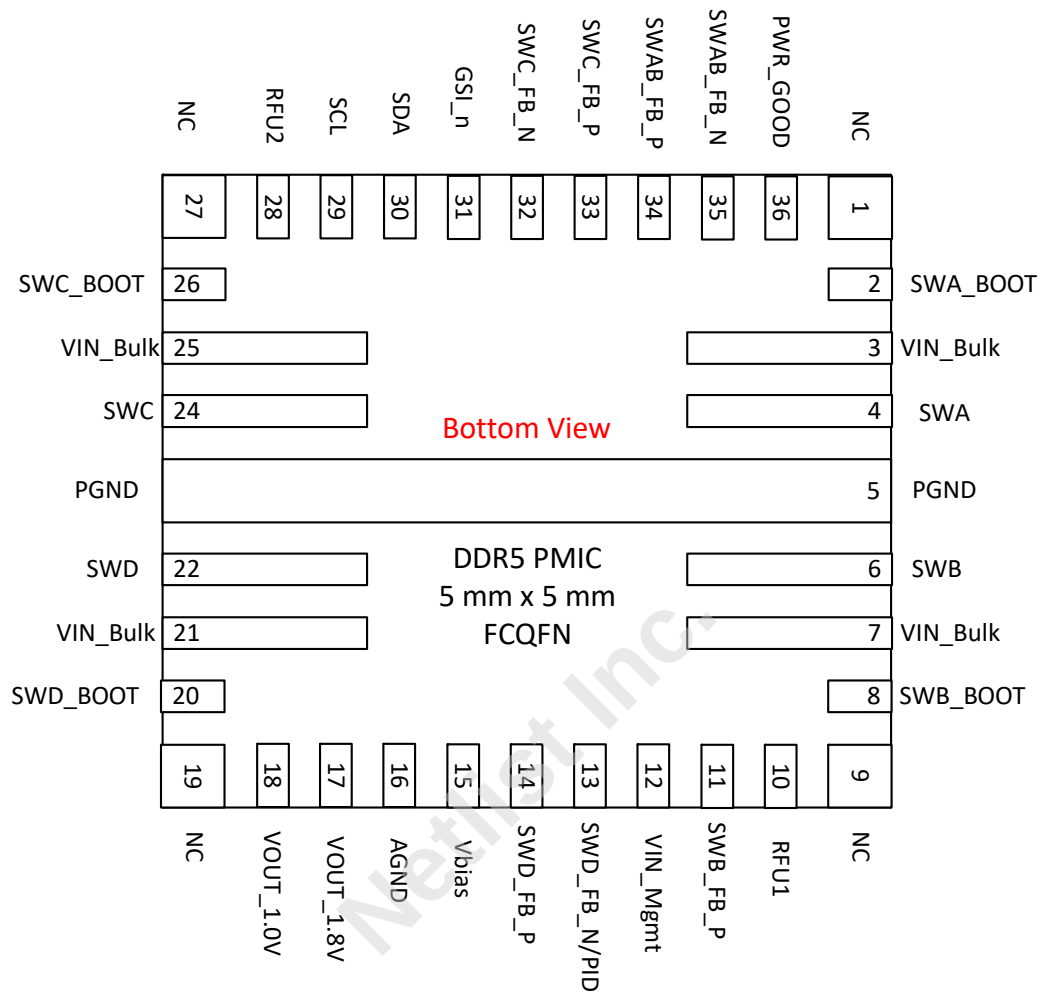


Figure 27 — PMIC Pinout - BOTTOM View

12.2 PMIC Package Pin List.

Table 91 — PMIC Pin Description

Pin Name	Type	Description
VIN_Bulk (4x)	I	12 V power input supply pin to the PMIC. All four VIN_Bulk input pins must be connected to the 12V input supply even if one or more output regulators are not intended to be used.
VIN_Mgmt	I	3.3 V power input supply pin to the PMIC for VOUT_1.8V & VOUT_1.0V LDO output, side band management access, internal memory read operation. Vendor usage of 3.3V input supply may vary.
PWR_GOOD	IO	Power good indicator. Open drain output. The PMIC ensures this pin high when VIN_Bulk input supply, as well as all enabled output buck regulators and all LDO regulators tolerance threshold is maintained as configured in the appropriate register. The PMIC drives this pin low when VIN_Bulk input goes below the threshold or when any of the enabled output buck regulator exceeds the thresholds configured in the appropriate register or when any LDO output regulator exceeds the threshold configured in the appropriate register. Input; The PMIC disables its output regulator when this pin is low. The LDO outputs shall remain on.
GSI_n	O	General Status Interrupt. Open drain output. The PMIC asserts this pin low to communicate any one or more critical event to host. This pin stays asserted until the appropriate registers are explicitly cleared.
VBias	PWR	Vbias Voltage generated by PMIC.
SCL	I	I ² C/I ³ C Basic Clock Input for management bus.
SDA	IO	I ² C/I ³ C Basic Data Input/Output for management bus.
SWA	O	Output switch node A buck regulator. This pin connects to L1 power inductor. In single phase regulator mode of operation, the SWA output must not be connected to either SWB or SWC output even if they are configured as same exact output voltage.
SWA_BOOT	PWR	Bootstrap node for SWA high side NMOS driver. This pin connects to SWA through a high quality capacitor.
SWAB_FB_P	I	In single phase regulator mode or dual phase regulator mode of operation, this pin connects to DIMM power plane load.
SWAB_FB_N	I	In single phase regulator mode or dual phase regulator mode of operation, this pin connects to DIMM ground plane.
SWB	O	Output switch node B buck regulator. This pin connects to L2 power inductor. In single phase regulator mode of operation, the SWB output must not be connected to either SWA or SWC output even if they are configured as same exact output voltage.
SWB_BOOT	PWR	Bootstrap node for SWB high side NMOS driver. This pin connects to SWB through a high quality capacitor.
SWB_FB_P	I	In single phase regulator mode of operation, this pin connects to DIMM power plane load. If not used, this pin must be connected to GND. In dual phase regulator mode of operation, this pin must be connected to GND.
SWC	O	Output switch node C buck regulator. This pin connects to L3 power inductor. The output of SWC must not be connected to the output of either SWA or SWB or SWA+SWB even if they are configured as same exact output voltage.
SWC_BOOT	PWR	Bootstrap node for SWC high NMOS driver. This pin connects to SWC through a high quality capacitor.
SWC_FB_P	I	This pin connects to DIMM power plane load.
SWC_FB_N	I	This pin connects to DIMM ground plane.
SWD	O	Output switch node D buck regulator. This pin connects to L4 power inductor.
SWD_BOOT	PWR	Bootstrap node for SWD high side NMOS driver. This pin connects to SWD through a high quality capacitor.

Table 91 — PMIC Pin Description

Pin Name	Type	Description
SWD_FB_P	I	This pin connects to DIMM power plane load.
SWD_FB_N/ PID	I	This pin connects to DIMM ground plane when PMIC is used in differential remote sensing mode of operation. This pin connects to DIMM ground plane or VOUT_1.8V rail or left floating when PMIC is used in a single ended remote sensing mode of operation.
VOUT_1.8V	O	1.8V LDO Output.
VOUT_1.0V	O	1.0 V LDO Output.
AGND	PWR	Analog Ground. Connects to DIMM ground plane.
PGND	PWR	Power Ground. Connects to DIMM ground plane.
RFU1, RFU2		This pins must be connected to GND.

12.3 Package Mechanical Drawing

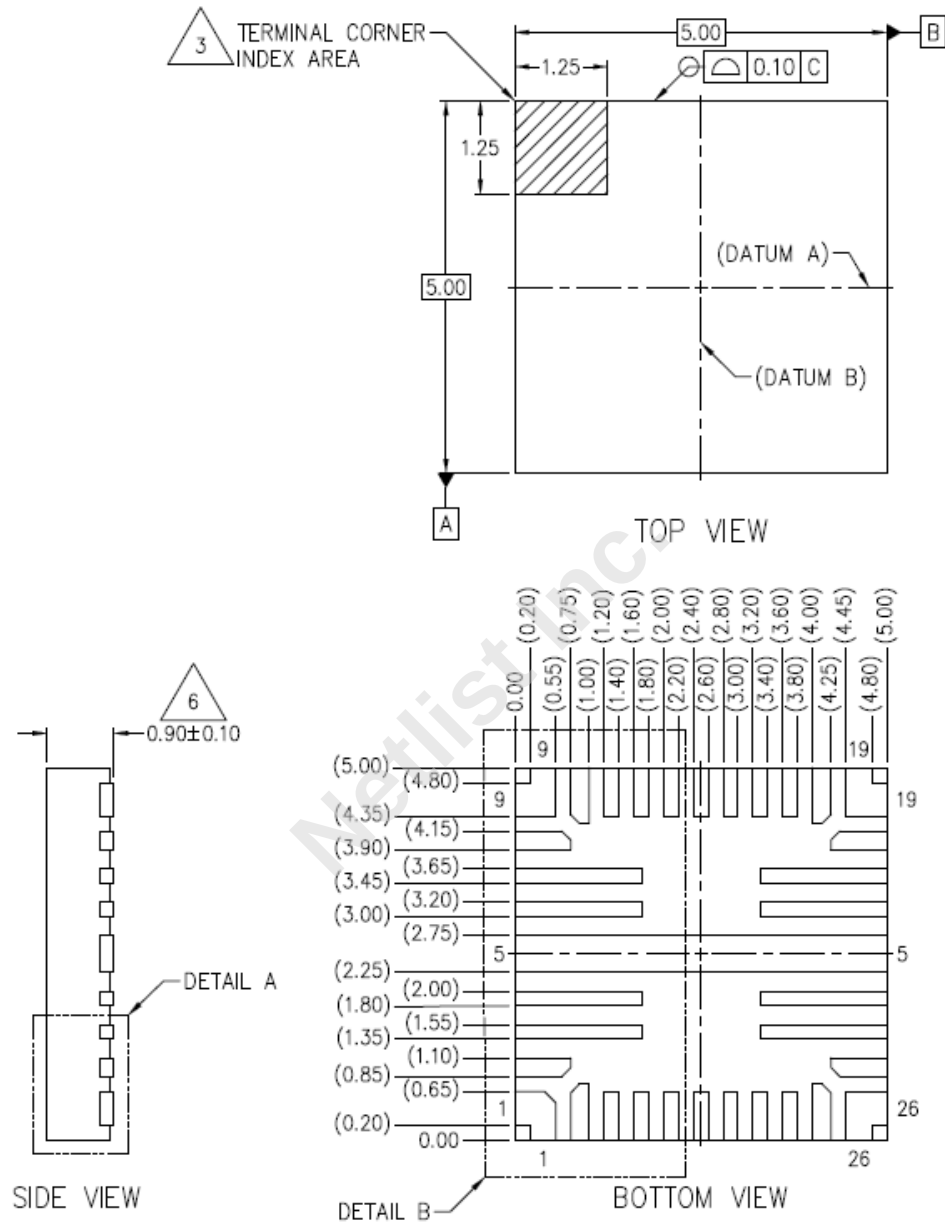


Figure 28 — Package Mechanical Outline

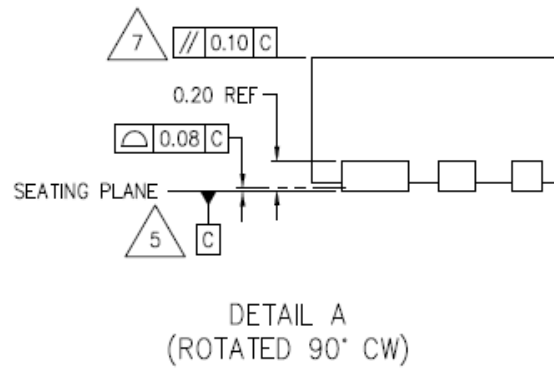


Figure 29 — Detail A Drawing

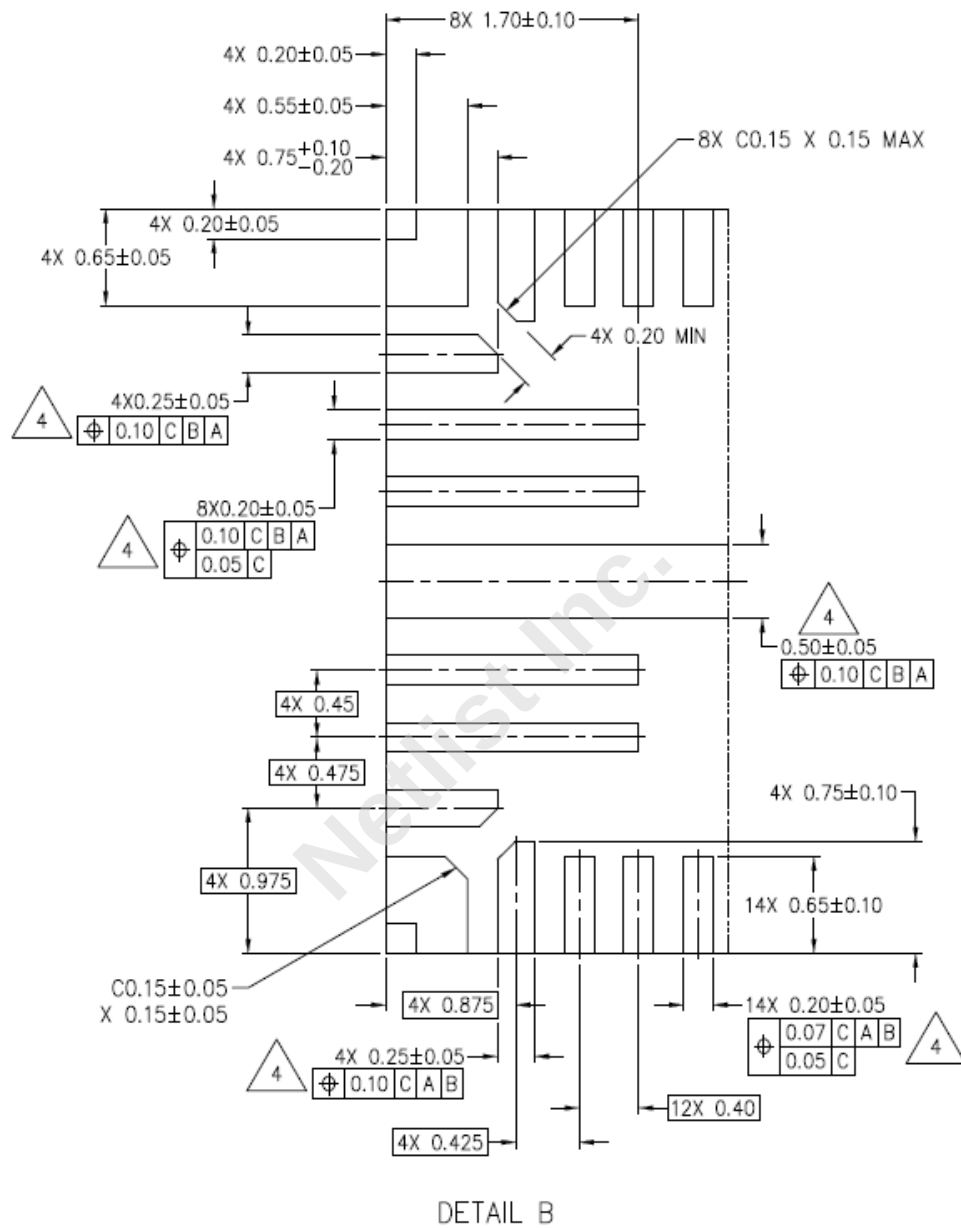


Figure 30 — Detail B Drawing

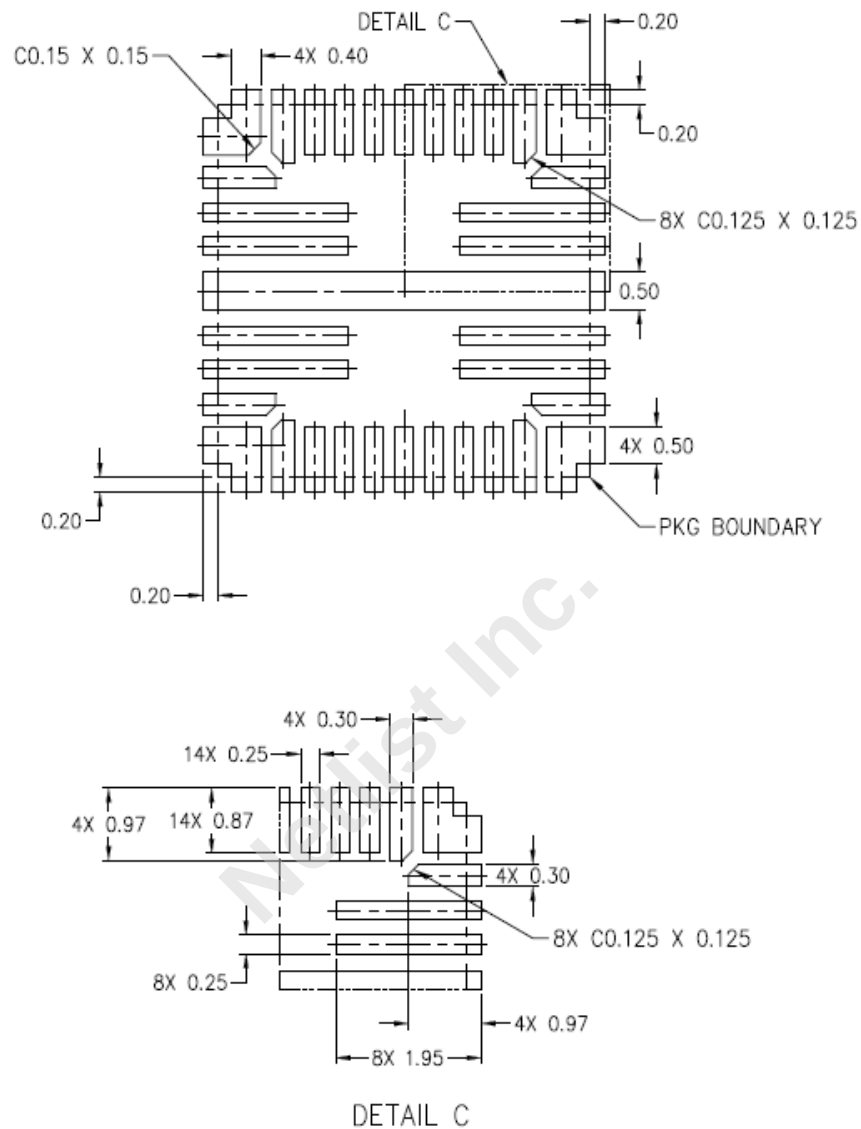


Figure 31 — Reference PCB Land Pattern

13 Inductor Specification

13.1 Mechanical Specification

The inductor package dimensions and its recommended land patterns are defined in Table 92 below.

Table 92 — Inductor Mechanical Specification

Package Size		Reference Drawing	Recommended Land Pattern
L (mm)	4.3 Max	Figure 32 (Left Picture)	Figure 32 (Right Picture)
W (mm)	4.3 Max		
H (mm)	2.0 Max		

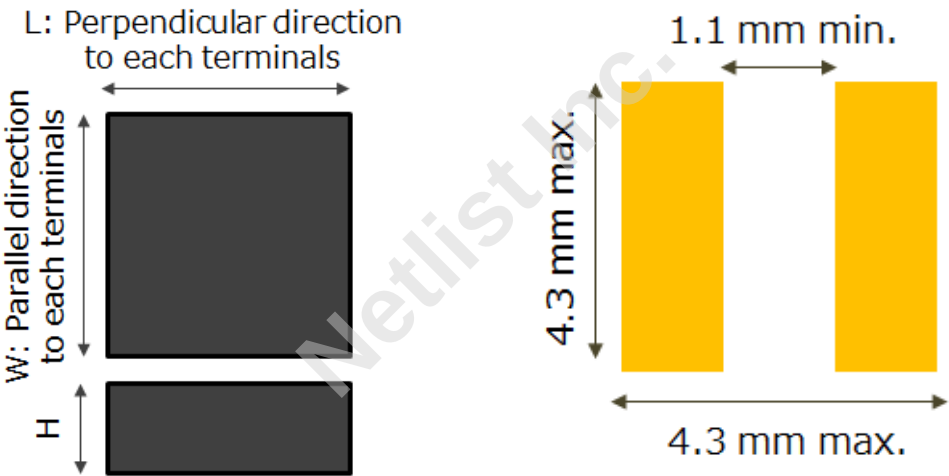


Figure 32 — Reference Drawing & Recommended Land Pattern

13.2 Electrical Specification

The inductor electrical specifications are defined in Table 93.

Table 93 — Inductor Electrical Specification¹

Package Height (mm)	L @ 0.5-1 MHz; 0 Bias + 20 % (μH)	Max DCR (mΩ)	Max ACR @ 1 MHz ^{2,3} (mΩ)	Min L @ 3.5 A (Ipeakmax of PMIC5010) ⁴ (μH)	Min L @ 6 A (Ipeakmax of PMIC5000) ⁵ (μH)
2.0 Max	0.47	6.9	90	0.30	0.30
	0.68	8.4	145	0.38	0.38
	1.0	15.5	250	0.56	0.56
	1.2	16.5	300	0.67	0.67

1. Test condition: Ambient Temperature = 20 ± 2 °C; Ambient Humidity = $65 \pm 5\%$ Rh
2. ACR definition: $ACR = R_s @ 1 \text{ MHz} - DCR$. Measured current (1 MHz/sinusoidal): 0.52A rms for 0.47 μH, 0.36A rms for 0.68 μH, 0.25 A rms for 1.0 μH and 0.2 A rms for 1.2 μH; with no DC Bias for all cases.
3. For Rs measurement, it is recommended to measure by Iwatsu SY-8218 (BH Analyzer, with NF IE-1125B), its upper compatible instruments or other instruments which is guaranteed on the measurement accuracy by inductor vendors.
4. Minimum inductance is defined at DC bias current given by definition in Ipeakmax of PMIC5010; Table 5, Table 8 and Table 10.
5. Minimum inductance is defined at DC bias current given by definition in Ipeakmax of PMIC5000; Table 3, Table 7 and Table 9.

14 Registers Space

14.1 Register Attribute Definition

All volatile registers have Base Attributes as defined in Table 94. Some register attributes are further modified with Attribute Modifiers, as defined in Table 95.

Table 94 — Register Base Attributes

Attribute	Abbreviation	Description
Read Only	RO	This bit can be read by software. Writes have no effect.
Read/Write	RW	This bit can be read or written by host.
Write Only	W	This bit can only be written by host. Read from this bit returns '0'.
Reserved	RV	This bit is reserved for future expansion and its value must not be modified by host. The bit will return '0' when read. Write has no effect.

Table 95 — Register Attribute Modifier

Attribute	Abbreviation	Description
Write '1' Only	1O	This bit can only be set (i.e., write '1') but not reset (i.e., write '0'). Write '0' has no effect.
Protected	P	This bit is protected by the password registers. This bit cannot be written to unless the password code has been written into the password registers.
Persistent	E	This bit is persistent during power cycle.

14.2 Register Map Breakdown

Table 96 — Register Map Breakdown

Register Range	Region	Comments
0x00 - 0x3F	Host Region	Host Accessible Registers
0x40 - 0x6F	DIMM Vendor Region	DIMM Vendor Registers - Non Volatile Memory Allows DIMM vendors to program the PMIC for a given DRAM/DIMM vendor designs. These are password protected registers and password is selected by DIMM vendor. Under normal operation, these registers are not used by any host. These registers require password for read access. Access to these registers without correct password will return all data as '0'. These registers require complete power cycle before it takes in effect. Changing these registers under normal operation is considered an illegal operation.
0x70 - 0xFF	Vendor Specific Region	Vendor Specific Registers - Non Volatile Memory These are vendor specific password protected registers. Under normal operation these registers are not used by any host. These registers require password for read access. Access to these registers without correct password will return all data as '0'.

14.3 Register Memory Protection

The PMIC DIMM vendors registers (0x40 - 0x6F) are password protected registers. Both Read and Write access to DIMM vendor registers are blocked unless it is unlocked by providing the correct password. The default password for DIMM vendor registers is 0x9473. The PMIC offers DIMM vendors to select their own password for DIMM vendor registers.

14.3.1 Steps to Access DIMM Vendor Region Registers

The steps to access the DIMM vendor registers are as following:

1. Write to register Table 149, "Register 0x37" = 8 bit password LSB code.
2. Write to register Table 150, "Register 0x38" = 8 bit password MSB code.
3. Write to register Table 151, "Register 0x39" = 0x40.
4. Perform Read operations to DIMM vendor registers as desired.
5. Write to register Table 151, "Register 0x39" = 0x00.

14.3.2 Steps to Change DIMM Vendor Region Password

By default, the DIMM vendor region register password is 0x9473. The steps to change the password from default password are as following:

1. Write to register Table 149, "Register 0x37" = 0x73.
2. Write to register Table 150, "Register 0x38" = 0x94.
3. Write to register Table 151, "Register 0x39" = 0x40.
4. Write to register Table 149, "Register 0x37" = New 8 bit password LSB code as desired by DIMM vendor.
5. Write to register Table 150, "Register 0x38" = New 8 bit password MSB code as desired by DIMM vendor.
6. Write to register Table 151, "Register 0x39" = 0x80.
7. Wait 200 ms.
8. Write to register Table 151, "Register 0x39" = 0x00.
9. Power cycle the PMIC. (Remove VIN_Bulk and VIN_Mgmt supply from the PMIC. The new password is in effect after the power cycle.

To change the password again from this point on, repeat steps 1 to 8 but note that in steps 1 and 2 current password is required.

14.3.3 Steps to Burn or Program DIMM Vendor Region Registers

The steps to burn or to program the DIMM vendor registers are as following:

1. Write to register Table 149, "Register 0x37" = 8 bit password LSB code.
2. Write to register Table 150, "Register 0x38" = 8 bit password MSB code.
3. Write to register Table 151, "Register 0x39" = 0x40.
4. Programming DIMM vendor registers are done at block level. Block 40 addresses: 0x40 - 0x4F; Block 50 addresses: 0x50 - 0x5F; Block 60 addresses: 0x60 - 0x6F. Perform write operation to each block as desired.
5. Burn each block one at a time: Block 40 addresses: Write register Table 151, "Register 0x39" = 0x81. Block 50 addresses: Write register Table 151, "Register 0x39" = 0x82. Block 60 addresses: Write register Table 151, "Register 0x39" = 0x85.
6. Wait time 200 ms.
7. To check if programming is complete: Perform read from register Table 151, "Register 0x39". The code 0x5A indicates it is complete. It takes 200 ms per page to program.
8. To verify if programming is done correctly: Perform read operation from appropriate block addresses.
9. Write to register Table 151, "Register 0x39" = 0x00.

14.3.4 Host Region Register Map

Table 97 — Register Color Coding Scheme

Region	Register Range	Restriction
Host Region + DIMM Vendor Region + Vendor specific Region	Table 116, "Register 0x15" to Table 142, "Register 0x2F" Table 145, "Register 0x32" [7,5:0] Table 157, "Register 0x40" to Register 0x6F Register 0x70 to Register 0xFF	Register Modification is NOT allowed in Secure Mode
Host Region	Table 127, "Register 0x20" to Table 140, "Register 0x2D,"	Registers are copied from DIMM Vendor Region Setting at power on

Table 98 — Host Region - Register Map

Register	Attribute	Description
0x00 to 0x03	RV	R00 [7:0] to R03 [7:0] - Reserved
Table 99, "Register 0x04"	ROE	R04 [7] Global Error Count R04 [6:4] Global Error History Log R04 [3:0] Reserved
Table 100, "Register 0x05"	ROE	R05 [7] Reserved R05 [6:3] Power On Reset - SWA, SWB, SWC & SWD Power Not Good R05 [2:0] Power On Reset - High Level Status Code
Table 101, "Register 0x06"	ROE	R06 [7:4] Power On Reset - SWA, SWB, SWC & SWD Under Voltage Lockout R06 [3:0] Power On Reset - SWA, SWB, SWC & SWD Over Voltage
Table 102, "Register 0x07"	ROE	R07 [7:0] Reserved

Table 98 — Host Region - Register Map

Register	Attribute	Description
Table 103, “Register 0x08”	RO	R08 [7] VIN_Bulk Input Power Good Status R08[6] Critical Temperature Shutdown Status R08 [5:2] SWA, SWB, SWC, SWD Output Power Good Status R08 [1] VIN_Mgmt Input Over Voltage Status R08 [0] VIN_Bulk Input Over Voltage Status
Table 104, “Register 0x09”	RO	R09 [7] PMIC High Temperature Warning Status R09 [6] VBias Power Good Status R09 [5] VOUT_1.8V Output Power Good Status R09 [4] VIN_Mgmt to VIN_Bulk Input Supply Switchover Status R09 [3:0] SWA, SWB, SWC & SWD High Output Current Consumption Warning Status
Table 105, “Register 0x0A”	RO	R0A [7:4] SWA, SWB, SWC, SWD Output Over Voltage Status R0A [3] PEC Error Status R0A [2] Parity Error Status R0A [1] IBI Status R0A [0] Reserved
Table 106, “Register 0x0B”	RO	R0B [7:4] SWA, SWB, SWC & SWD Output Current Limiter Warning Status R0B [3:0] SWA, SWB, SWC & SWD Output Under Voltage Lockout Status
Table 107, “Register 0x0C”	RO	R0C [7:0] SWA Output Current or Power or Total Output Power Measurement
Table 108, “Register 0x0D”	RO	R0D [7:6] Reserved R0D [5:0] SWB Output Current or Power Measurement
Table 109, “Register 0x0E”	RO	R0E [7:6] Reserved R0E [5:0] SWC Output Current or Power Measurement
Table 110, “Register 0x0F”	RO	R0F [7:6] Reserved R0F [5:0] SWD Output Current or Power Measurement
Table 111, “Register 0x10”	IO	R10 [7] Clear VIN_Bulk Input Power Good Status R10 [6] Reserved R10 [5:2] Clear SWA, SWB, SWC & SWD Output Power Good Status R10 [1] Clear VIN_Mgmt Input Over Voltage Status R10 [0] Clear VIN_Bulk Input Over Voltage Status
Table 112, “Register 0x11”	IO	R11 [7] Clear PMIC High Temperature Warning Status R11 [6] Clear VBias Power Good Status R11 [5] Clear VOUT_1.8V Output Power Good Status R11 [4] Clear VIN_Mgmt to VIN_Bulk Input Supply Switchover Status R11 [3:0] Clear SWA, SWB, SWC & SWD High Output Current Consumption Warning Status
Table 113, “Register 0x12”	IO	R12 [7:4] Clear SWA, SWB, SWC, SWD Output Over Voltage Status R12 [3] Clear PEC Error R12 [2] Clear Parity Error R12 [1:0] Reserved
Table 114, “Register 0x13”	IO	R13 [7:4] Clear SWA, SWB, SWC & SWD Output Current Limiter Warning Status R13 [3:0] Clear SWA, SWB, SWC & SWD Output Under Voltage Lockout Status

Table 98 — Host Region - Register Map

Register	Attribute	Description
Table 115, “Register 0x14”	IO	R14 [7:5] Reserved R14 [4] Clear VIN_Mgmt Power Good Status in Switchover Mode R14 [3] Clear VBias Output or VIN_Bulk Input Under Voltage Lockout Status R14 [2] Clear VOUT_1.0V Output Power Good Status R14 [1] Reserved R14 [0] Clear Global Status
Table 116, “Register 0x15”	RW	R15 [7] Mask VIN_Bulk Input Power Good Status R15 [6] Reserved R15 [5:2] Mask SWA, SWB, SWC & SWD Output Power Good Status R15 [1] Mask VIN_Mgmt Input Over Voltage Status R15 [0] Mask VIN_Bulk Input Over Voltage Status
Table 117, “Register 0x16”	RW	R16 [7] Mask PMIC High Temperature Warning Status R16 [6] Mask VBias Power Good Status R16 [5] Mask VOUT_1.8V Output Power Good Status R16 [4] Mask VIN_Mgmt to VIN_Bulk Input Supply Switchover Status R16 [3:0] Mask SWA, SWB, SWC & SWD High Output Current Consumption Warning Status
Table 118, “Register 0x17”	RW	R17 [7:4] Mask SWA, SWB, SWC, SWD Output Over Voltage R17 [3] Mask PEC Error Status R17 [2] Mask Parity Error Status R17 [1:0] Reserved
Table 119, “Register 0x18”	RW	R18 [7:4] Mask SWA, SWB, SWC & SWD Output Current Limiter Warning Status R18 [3:0] Mask SWA, SWB, SWC & SWD Output Under Voltage Lockout Status
Table 120, “Register 0x19”	RW	R19 [7:5] Reserved R19 [4] Mask VIN_Mgmt Power Good Status Switchover Mode R19 [3] Mask Vbias Output or VIN_Bulk Input Under Voltage Lockout Status R19 [2] Mask VOUT_1.0V Output Power Good Status R19 [1:0] Reserved
Table 121, “Register 0x1A”	RW	R1A [7:5] VIN_Bulk Input Power Good Threshold Voltage R1A [4] Reserved R1A [3] VBias Power Good Threshold Voltage R1A [2] VOUT_1.8 V Power Good Threshold Voltage R1A [1] Output Power Measurement Select R1A [0] VOUT_1.0 V Power Good Threshold Voltage
Table 122, “Register 0x1B”	RW	R1B [7] VIN_Bulk Input Over Voltage Threshold R1B [6] Current or Power Meter Select R1B [5] VIN_Mgmt Input Over Voltage Threshold R1B [4] Global Mask Control for PWR_GOOD Output Pin R1B [3] GSI_n Pin Enable R1B [2:0] PMIC High Temperature Warning Threshold
Table 123, “Register 0x1C”	RW	R1C [7:2] SWA Output High Current Threshold R1C [1:0] Reserved
Table 124, “Register 0x1D”	RW	R1D [7:2] SWB Output High Current Threshold R1D [1:0] Reserved
Table 125, “Register 0x1E”	RW	R1E [7:2] SWC Output High Current Threshold R1E [1:0] Reserved

Table 98 — Host Region - Register Map

Register	Attribute	Description
Table 126, “Register 0x1F,”	RW	R1F [7:2] SWD Output High Current Threshold R1F [1:0] Reserved
Table 127, “Register 0x20,”	RW	R20 [7:6] SWA Output Current Limiter Warning Threshold R20 [5:4] SWB Output Current Limiter Warning Threshold R20 [3:2] SWC Output Current Limiter Warning Threshold R20 [1:0] SWD Output Current Limiter Warning Threshold
Table 128, “Register 0x21,”	RW	R21 [7:1] SWA Voltage Setting R21 [0] SWA Power Good Low Side Threshold
Table 129, “Register 0x22,”	RW	R22 [7:6] SWA Power Good High Side Threshold R22 [5:4] SWA Over Voltage Threshold R22 [3:2] SWA Under Voltage Lockout Threshold R22 [1:0] SWA Soft Stop Time
Table 130, “Register 0x23,”	RW	R23 [7:1] SWB Voltage Setting R23 [0] SWB Power Good Low Side Threshold
Table 131, “Register 0x24,”	RW	R24 [7:6] SWB Power Good High Side Threshold R24 [5:4] SWB Over Voltage Threshold R24 [3:2] SWB Under Voltage Lockout Threshold R24 [1:0] SWB Soft Stop Time
Table 132, “Register 0x25,”	RW	R25 [7:1] SWC Voltage Setting R25 [0] SWC Power Good Low Side Threshold
Table 133, “Register 0x26,”	RW	R26 [7:6] SWC Power Good High Side Threshold R26 [5:4] SWC Over Voltage Threshold R26 [3:2] SWC Under Voltage Lockout Threshold R26 [1:0] SWC Soft Stop Time
Table 134, “Register 0x27,”	RW	R27 [7:1] SWD Voltage Setting R27 [0] SWD Power Good Low Side Threshold
Table 135, “Register 0x28,”	RW	R28 [7:6] SWD Power Good High Side Threshold R28 [5:4] SWD Over Voltage Threshold R28 [3:2] SWD Under Voltage Lockout Threshold R28 [1:0] SWD Soft Stop Time
Table 136, “Register 0x29,”	RW	R29 [7:6] SWA Mode Select R29 [5:4] SWA Switching Frequency R29 [3:2] SWB Mode Select R29 [1:0] SWB Switching Frequency
Table 137, “Register 0x2A,”	RW	R2A [7:6] SWC Mode Select R2A [5:4] SWC Switching Frequency R2A [3:2] SWD Mode Select R2A [1:0] SWD Switching Frequency
Table 138, “Register 0x2B,”	RW	R2B [7:6] VOUT_1.8V LDO Setting R2B [5:3] Voltage Range Selection for SWA, SWB and SWC R2B [2:1] VOUT_1.0V LDO Setting R2B [0] Voltage Range Selection for SWD
Table 139, “Register 0x2C,”	RW	R2C [7:5] SWA Soft Start Time R2C [4] Reserved R2C [3:1] SWB Soft Start Time R2C [0] Reserved
Table 140, “Register 0x2D,”	RW	R2D [7:5] SWC Soft Start Time R2D [4] Reserved R2D [3:1] SWD Soft Start Time R2D [0] Reserved

Table 98 — Host Region - Register Map

Register	Attribute	Description
Table 141, “Register 0x2E”	RW	R2E [7:3] Reserved R2E [2:0] PMIC Shutdown temperature threshold
Table 142, “Register 0x2F”	RW	R2F [7] VIN_Mgmt Input Supply Switchover Threshold Voltage R2F [6:3] SWA, SWB, SWC & SWD Enable R2F [2] Secure Mode Enable R2F [1:0] Mask Bits Register Control
Table 143, “Register 0x30”	RW	R30 [7] ADC Enable R30 [6:3] ADC Select R30 [2] Reserved R30 [1:0] ADC Register Update Frequency
Table 144, “Register 0x31”	RO	R31 [7:0] ADC Read Out
Table 145, “Register 0x32”	RW, RO	R32 [7] VR Enable R32 [6] Management Interface Selection R32 [5] PWR_GOOD Signal IO Type R32 [4:3] PMIC Power Good Output Signal Control R32 [2:0] Reserved
Table 146, “Register 0x33”	RO	R33 [7:5] Temperature Measurement R33 [4] VIN_Mgmt Power Good Status in Switchover Mode Only R33 [3] VBIAS Output or VIN_Bulk Input Under Voltage Lockout Status R33 [2] VOUT_1.0V Output Power Good Status R33 [1:0] Reserved
Table 147, “Register 0x34”	RW, RO	R34 [7] PEC Enable R34 [6] IBI Enable R34 [5] Parity Disable R34 [4] Reserved R34 [3:1] HID_CODE R34 [0] Reserved
Table 148, “Register 0x35”	RW	R35 [7] Error Injection Enable R35 [6:4] Output Rail Selection R35 [3] Over and Under Voltage Select R35 [2:0] Misc. Error Injection Type
0x36	RV	R36 [7:0] Reserved
Table 149, “Register 0x37”	WO	R37 [7:0] Password Lower Byte 0
Table 150, “Register 0x38”	WO	R38 [7:0] Password Upper Byte 1
Table 151, “Register 0x39”	RW	R39 [7:0] Command Codes
Table 152, “Register 0x3A”	RW	R3A [7] Reserved R3A [6] Default Read Address Pointer Enable R3A [5:4] Default Read Address Pointer Selection R3A [3:2] Burst Length for Default Read Address Pointer Mode in PEC Enabled Mode R3A [1:0] Reserved
Table 153, “Register 0x3B”	ROE	R3B [7:6] Reserved R3B [5:4] Major Revision ID R3B [3:1] Minor Revision ID R3B [0] PMIC Current Capability
Table 154, “Register 0x3C”	ROE	R3C [7:0] VENDOR_ID_BYTE0
Table 155, “Register 0x3D”	ROE	R3D [7:0] VENDOR_ID_BYTE1
0x3E to 0x3F	RV	R3E [7:0] to 0x3F Reserved

14.3.5 Host Region Registers

14.3.5.1 Status Registers

The PMIC offers status registers that are grouped into four different categories.

1. Global History of Error Log Register (Table 99, “Register 0x04” [7:4])
2. Error Log Registers (Table 100, “Register 0x05” [6:0], Table 101, “Register 0x06” [7:0], Table 102, “Register 0x07” [7:0]; Table 102, “Register 0x07” [7:0] is currently defined as Reserved)
3. Real time Status Registers (Table 103, “Register 0x08” [7:0], Table 104, “Register 0x09” [7:0], Table 105, “Register 0x0A” [7:1], Table 106, “Register 0x0B” [7:0], Table 146, “Register 0x33” [4:2])
4. Periodic Status Registers (Table 107, “Register 0x0C” [7:0], Table 108, “Register 0x0D” [5:0], Table 109, “Register 0x0E” [5:0], Table 110, “Register 0x0F” [5:0], Table 146, “Register 0x33” [7:5])

Global History of Error Log Registers (Table 99, “Register 0x04” [7:4]) - This register records the PMIC state at each abnormal power down cycle. This register reports the cumulative error of each abnormal power down sequence. The PMIC writes this register on its own when it internally generates VR Disable command on its own due to failure. The host can erase this register in MTP memory and clear the status register by writing the code 0x74 in Table 151, “Register 0x39”.

Error Log Registers (Table 100, “Register 0x05” [6:0], Table 101, “Register 0x06” [7:0] to Table 102, “Register 0x07” [7:0]) - These registers record the PMIC state at each power down sequence. The PMIC may report abnormal power down sequence or normal power down sequence. The PMIC writes this register on its own when it internally generates VR Disable command on its own due to failure. These registers are updated at power down cycle, if update is needed by the PMIC on its own. The host can clear the status register by writing the code 0x74 in Table 151, “Register 0x39”.

Real Time Status Registers (Table 103, “Register 0x08” [7:0], Table 104, “Register 0x09” [7:0], Table 105, “Register 0x0A” [7:1], to Table 106, “Register 0x0B” [7:0], Table 146, “Register 0x33” [4:2]): These registers are updated to ‘1’ any time based on any event that occurs. The status registers will remain at ‘1’ even if the failing condition is no longer present, until the Clear Register command is received by the PMIC. The GSI_n interrupt or PWR_GOOD interrupt may be generated by the PMIC at the same time depending on the type of event. The interrupts are only generated if they are not masked. The status registers Table 103, “Register 0x08” [7], Table 104, “Register 0x09” [5] and Table 146, “Register 0x33” [2] is only valid once valid VIN_Bulk and VIN_Mgmt input supply is valid at the PMIC input pin. The remaining status registers are valid after VR Enable command is registered.

Periodic Status Registers (Table 107, “Register 0x0C” [7:0], Table 108, “Register 0x0D” [5:0], Table 109, “Register 0x0E” [5:0], Table 110, “Register 0x0F” [5:0], Table 146, “Register 0x33” [7:5]) - These registers are updated periodically. These registers are only valid after VR Enable command is registers.

All Read Only (RO) registers except for registers Table 107, “Register 0x0C” [7:0], Table 108, “Register 0x0D” [5:0], Table 109, “Register 0x0E” [5:0] & Table 110, “Register 0x0F” [5:0] are one time latched registers. In other words, once PMIC sets those register flag, the host must explicitly clear those registers appropriately. The PMIC does not automatically update the registers on its own even if the event that triggered the status is no longer present. The registers Table 107, “Register 0x0C” [7:0], Table 108, “Register 0x0D” [5:0], Table 109, “Register 0x0E” [5:0] & Table 110, “Register 0x0F” [5:0] are dynamically updated by the PMIC at certain frequency and they represent the status at that point.

Table 99 — Register 0x04

R04			
Bits	Attribute	Default	Description^{1,2}
7	ROE	0	R04 [7]: GLOBAL_ERROR_COUNT Global Error Count Since Last Erase Operation ³ 0 = No Error or Only 1 Error since last Erase operation 1 = > 1 Error Count since last Erase operation
6	ROE	0	R04 [6]: GLOBAL_ERROR_LOG_BUCK_OV_OR_UV Global Error Log History for Buck Regulator Output Over or Under Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred
5	ROE	0	R04 [5]: GLOBAL_ERROR_LOG_VIN_BULK_OVER_VOLTAGE Global Error Log History for VIN_Bulk Over Voltage ⁴ 0 = No Error Occurred 1 = Error Occurred
4	ROE	0	R04 [4]: GLOBAL_ERROR_LOG_CRITICAL_TEMPERATURE Global Error Log History for Critical Temperature ⁴ 0 = No Error Occurred 1 = Error Occurred
3:0	RV	0	R04 [3:0]: Reserved

1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory.
2. Host must explicitly perform Erase operation to erase this entire register Table 99, “Register 0x04” [7:0] via Table 151, “Register 0x39”. The PMIC needs minimum of 100 ms for Erase operation.
3. PMIC counts the error since last erase operation and if more than one error occurs, it sets this bit to ‘1’. Host must explicitly perform Erase operation to erase this entire register Table 99, “Register 0x04” [7:0].
4. PMIC sets the bit when error occurs.

Table 100 — Register 0x05

R05			
Bits	Attribute	Default	Description^{1,2}
7	RV	0	R05 [7]: Reserved
6	ROE	0	R05 [6]: SWA_POWER_GOOD PMIC Power On - SWA Power Not Good ³ 0 = Normal Power On 1 = SWA Power Not Good
5	ROE	0	R05 [5]: SWB_POWER_GOOD PMIC Power On - SWB Power Not Good ^{3,4} 0 = Normal Power On 1 = SWB Power Not Good
4	ROE	0	R05 [4]: SWC_POWER_GOOD PMIC Power On - SWC Power Not Good ³ 0 = Normal Power On 1 = SWC Power Not Good
3	ROE	0	R05 [3]: SWD_POWER_GOOD PMIC Power On - SWD Power Not Good ³ 0 = Normal Power On 1 = SWD Power Not Good
2:0	ROE	0	R05 [2:0]: PMIC_ERROR_LOG PMIC Power On - High Level Status Bit to Previous Last Known Power Cycle 000 = Normal Power On 001 = Reserved 010 = Buck Regulator Output Over or Under Voltage ⁵ 011 = Critical Temperature 100 = VIN_Bulk Input Over Voltage 101 = Reserved 110 = Reserved 111 = Reserved

1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory.
2. This entire register status reflects previous power down cycle of the PMIC and is updated by the PMIC on its own at each power cycle, if update is needed. Because this register is updated only if there is an update needed, there is no NVM life time impact. This register is cleared when host issues the erase command via Table 151, "Register 0x39". The PMIC needs minimum of 100 ms for Erase operation.
3. This register is set only if PMIC generates internal VR Disable command due to fault condition.
4. Only applicable if Table 171, "Register 0x4F" [0] = '0'.
5. This code is a logical OR function of Table 101, "Register 0x06" [7:0] register bits.

Table 101 — Register 0x06

R06			
Bits	Attribute	Default	Description ^{1,2}
7	ROE	0	R06 [7]: SWA_UNDER_VOLTAGE_LOCKOUT PMIC Power On - SWA Under Voltage Lockout 0 = Normal Power On 1 = SWA Under Voltage Lockout
6	ROE	0	R06 [6]: SWB_UNDER_VOLTAGE_LOCKOUT PMIC Power On - SWB Under Voltage Lockout ³ 0 = Normal Power On 1 = SWB Under Voltage Lockout
5	ROE	0	R06 [5]: SWC_UNDER_VOLTAGE_LOCKOUT PMIC Power On - SWC Under Voltage Lockout 0 = Normal Power On 1 = SWC Under Voltage Lockout
4	ROE	0	R06 [4]: SWD_UNDER_VOLTAGE_LOCKOUT PMIC Power On - SWD Under Voltage Lockout 0 = Normal Power On 1 = SWD Under Voltage Lockout
3	ROE	0	R06 [3]: SWA_OVER_VOLTAGE PMIC Power On - SWA Over Voltage 0 = Normal Power On 1 = SWA Over Voltage
2	ROE	0	R06 [2]: SWB_OVER_VOLTAGE PMIC Power On - SWB Over Voltage ³ 0 = Normal Power On 1 = SWB Over Voltage
1	ROE	0	R06 [1]: SWC_OVER_VOLTAGE PMIC Power On - SWC Over Voltage 0 = Normal Power On 1 = SWC Over Voltage
0	ROE	0	R06 [0]: SWD_OVER_VOLTAGE PMIC Power On - SWD Over Voltage 0 = Normal Power On 1 = SWD Over Voltage

1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory.
2. This entire register status reflects previous power down cycle of the PMIC and is updated by the PMIC on its own at each power cycle, if update is needed. Because this register is updated only if there is an update needed, there is no NVM life time impact. This register is cleared when host issues the erase command via Table 151, "Register 0x39". The PMIC needs minimum of 100 ms for Erase operation.
3. Only applicable if Table 171, "Register 0x4F" [0] = '0'.

Table 102 — Register 0x07

R07			
Bits	Attribute	Default	Description ^{1,2}
7:0	ROE	0	R07 [7:0]: Reserved

1. The PMIC always attempts to write this register into its non-volatile memory. However, it may not be guaranteed depending on how fast the host may shut off the input power to the PMIC. The PMIC needs minimum of 5.0V for VIN_Bulk voltage and 100 ms duration from PWR_GOOD signal assertion to guarantee the write operation into non-volatile memory.
2. This entire register status reflects previous power down cycle of the PMIC and is updated by the PMIC on its own at each power cycle, if update is needed. Because this register is updated only if there is an update needed, there is no NVM life time impact. This register is cleared when host issues the erase command via Table 151, “Register 0x39”. The PMIC needs minimum of 50 ms for Erase operation.

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Table 103 — Register 0x08

R08			
Bits	Attribute	Default	Description
7	RO	0	R08 [7]: VIN_BULK_INPUT_POWER_GOOD_STATUS VIN_Bulk Input Power Good Status ¹ 0 = Power Good 1 = Power Not Good
6	RO	0	R08 [6]: CRITICAL_TEMP_SHUTDOWN_STATUS Critical Temperature Shutdown Status ² 0 = No Critical Temperature Shutdown 1 = Critical Temperature Shutdown
5	RO	0	R08 [5]: SWA_OUTPUT_POWER_GOOD_STATUS Switch Node A Output Power Good Status ³ 0 = Power Good 1 = Power Not Good
4	RO	0	R08 [4]: SWB_OUTPUT_POWER_GOOD_STATUS Switch Node B Output Power Good Status ⁴ 0 = Power Good 1 = Power Not Good
3	RO	0	R08 [3]: SWC_OUTPUT_POWER_GOOD_STATUS Switch Node C Output Power Good Status ⁵ 0 = Power Good 1 = Power Not Good
2	RO	0	R08 [2]: SWD_OUTPUT_POWER_GOOD_STATUS Switch Node D Output Power Good Status ⁶ 0 = Power Good 1 = Power Not Good
1	RO	0	R08 [1]: VIN_MGMT_INPUT_OVER_VOLTAGE_STATUS VIN_Mgmt Input Supply Over Voltage Status ⁷ 0 = No Over Voltage 1 = Over Voltage
0	RO	0	R08 [0]: VIN_BULK_INPUT_OVER_VOLTAGE_STATUS VIN_Bulk Input Supply Over Voltage Status ⁸ 0 = No Over Voltage 1 = Over Voltage

1. This register is set when VIN_Bulk input goes below the threshold setting in register Table 121, “Register 0x1A” [7:5].
2. This register is set when PMIC temperature goes above the threshold setting in register Table 141, “Register 0x2E” [2:0].
3. This register is set when SWA output voltage goes either below the threshold setting in register Table 128, “Register 0x21,” [1:0] or above the threshold setting in register Table 129, “Register 0x22,” [7:6].
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’. This register is set when SWB output goes either below the threshold setting in register Table 130, “Register 0x23,” [1:0] or above the threshold setting in register Table 131, “Register 0x24,” [7:6].
5. This register is set when SWC output goes either below the threshold setting in register Table 132, “Register 0x25,” [1:0] or above the threshold setting in register Table 133, “Register 0x26,” [7:6].
6. This register is set when SWD output goes either below the threshold setting in register Table 134, “Register 0x27,” [1:0] or above the threshold setting in register Table 135, “Register 0x28,” [7:6].
7. This register is set when VIN_Mgmt input voltage goes above the threshold setting in register Table 122, “Register 0x1B” [5].
8. This register is set when VIN_Bulk input voltage goes above the threshold setting in register Table 122, “Register 0x1B” [7].

Table 104 — Register 0x09

R09			
Bits	Attribute	Default	Description
7	RO	0	R09 [7]: PMIC_HIGH_TEMP_WARNING_STATUS PMIC High Temperature Warning Status ¹ 0 = Temperature Below the Warning Threshold 1 = Temperature Exceeded the Warning Threshold
6	RO	0	R09 [6]: VBIAS_POWER_GOOD_STATUS VBias Power Good Status ² 0 = Power Good 1 = Power Not Good
5	RO	0	R09 [5]: VOUT_1.8V_OUTPUT_POWER_GOOD_STATUS VOUT_1.8V LDO Output Power Good Status ³ 0 = Power Good 1 = Power Not Good
4	RO	0	R09 [4]: VIN_MGMT_TO_VIN_BULK_INPUT_SUPPLY_SWITCHOVER_STATUS VIN_Mgmt to VIN_Bulk Input Supply Automatic Switchover Status ⁴ 0 = VIN_Mgmt Input Supply is Present 1 = VIN_Mgmt Input Supply is Removed (i.e using VIN_Bulk Input Supply)
3	RO	0	R09 [3]: SWA_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Switch Node A High Output Current Consumption Warning Status ⁵ 0 = No High Current Consumption Warning 1 = High Current Consumption Warning
2	RO	0	R09 [2]: SWB_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Switch Node B High Output Current Consumption Warning Status ^{6,7} 0 = No High Current Consumption Warning 1 = High Current Consumption Warning
1	RO	0	R09 [1]: SWC_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Switch Node C High Output Current Consumption Warning Status ⁸ 0 = No High Current Consumption Warning 1 = High Current Consumption Warning
0	RO	0	R09 [0]: SWD_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Switch Node D High Output Current Consumption Warning Status ⁹ 0 = No High Current Consumption Warning 1 = High Current Consumption Warning

1. This register is set when PMIC temperature goes above the threshold setting in Table 122, “Register 0x1B” [2:0].
2. This register is set when VBias voltage goes below the threshold setting in register Table 121, “Register 0x1A” [3].
3. This register is set when VOUT_1.8V output exceeds the threshold setting in register Table 121, “Register 0x1A” [2].
4. This register is set when VIN_Mgmt input supply goes below the threshold setting in register Table 142, “Register 0x2F” [7].
5. This register is set when SWA output current consumption goes above the threshold setting in Table 123, “Register 0x1C” [7:2].
6. This register is set when SWB output current consumption goes above the threshold setting in Table 124, “Register 0x1D” [7:2]. If Table 171, “Register 0x4F” [0] = ‘1’, the setting in Table 124, “Register 0x1D” [7:2] must be identical as Table 123, “Register 0x1C” [7:2].
7. This register is applicable regardless of the setting in Table 171, “Register 0x4F” [0]
8. This register is set when SWC output current consumption goes above the threshold setting in Table 125, “Register 0x1E” [7:2].
9. This register is set when SWD output current consumption goes above the threshold setting in Table 126, “Register 0x1F” [7:2].

Table 105 — Register 0x0A

R0A			
Bits	Attribute	Default	Description
7	RO	0	R0A [7]: SWA_OUTPUT_OVER_VOLTAGE_STATUS Switch Node A Output Over Voltage Status ¹ 0 = No Over Voltage 1 = Over Voltage
6	RO	0	R0A [6]: SWB_OUTPUT_OVER_VOLTAGE_STATUS Switch Node B Output Over Voltage Status ² 0 = No Over Voltage 1 = Over Voltage
5	RO	0	R0A [5]: SWC_OUTPUT_OVER_VOLTAGE_STATUS Switch Node C Output Over Voltage Status ³ 0 = No Over Voltage 1 = Over Voltage
4	RO	0	R0A [4]: SWD_OUTPUT_OVER_VOLTAGE_STATUS Switch Node D Output Over Voltage Status ⁴ 0 = No Over Voltage 1 = Over Voltage
3	RO	0	R0A [3]: PEC_ERROR_STATUS Packet Error Code Status ^{5,6} 0 = No PEC Error 1 = PEC Error
2	RO	0	R0A [2]: PARITY_ERROR_STATUS T Bit Parity Error Status ^{6,7} 0 = No Parity Error 1 = Parity Error
1	RO	0	R0A [1]: IBI_STATUS In Band Interrupt Status 0 = No Pending IBI 1 = Pending IBI
0	RV	0	R0A [0]: Reserved

1. This register is set when SWA output voltage goes above the threshold setting in Table 129, "Register 0x22," [5:4].
2. Only applicable if Table 171, "Register 0x4F" [0] = '0'. This register is set when SWB output voltage goes above the threshold setting in Table 131, "Register 0x24," [5:4]. If Table 171, "Register 0x4F" [0] = '1', the setting in Table 131, "Register 0x24," [5:4] must be identical as Table 129, "Register 0x22," [5:4].
3. This register is set when SWC output voltage goes above the threshold setting in Table 133, "Register 0x26," [5:4].
4. This register is set when SWD output voltage goes above the threshold setting in Table 135, "Register 0x28," [5:4].
5. Applicable in I3C Basic Mode Only and if enabled in register Table 147, "Register 0x34" [7].
6. This register is updated when PMIC device goes through bus reset as described in Section 11.15.
7. Applicable in I3C Basic Mode and if enabled in register Table 147, "Register 0x34" [5]. Also applicable in I²C mode for supported CCC.

Table 106 — Register 0x0B

R0B			
Bits	Attribute	Default	Description
7	RO	0	R0B [7]: SWA_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Switch Node A Output Current Limiter Warning Status ¹ 0 = No Current Limiter Event 1 = Current Limiter Event
6	RO	0	R0B [6]: SWB_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Switch Node B Output Current Limiter Warning Status ² 0 = No Current Limiter Event 1 = Current Limiter Event
5	RO	0	R0B [5]: SWC_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Switch Node C Output Current Limiter Warning Status ³ 0 = No Current Limiter Event 1 = Current Limiter Event
4	RO	0	R0B [4]: SWD_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Switch Node D Output Current Limiter Warning Status ⁴ 0 = No Current Limiter Event 1 = Current Limiter Event
3	RO	0	R0B [3]: SWA_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Switch Node A Output Under Voltage Lockout Status ⁵ 0 = No Under Voltage Lockout 1 = Under Voltage Lockout
2	RO	0	R0B [2]: SWB_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Switch Node B Output Under Voltage Lockout Status ⁶ 0 = No Under Voltage Lockout 1 = Under Voltage Lockout
1	RO	0	R0B [1]: SWC_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Switch Node C Output Under Voltage Lockout Status ⁷ 0 = No Under Voltage Lockout 1 = Under Voltage Lockout
0	RO	0	R0B [0]: SWD_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Switch Node D Output Under Voltage Lockout Status ⁸ 0 = No Under Voltage Lockout 1 = Under Voltage Lockout

1. This register is set when SWA output current goes above the threshold setting in Table 127, "Register 0x20" [7:6].
2. This register is applicable regardless of the setting in Table 171, "Register 0x4F" [0]. This register is set when SWB output current goes above the threshold setting in Table 127, "Register 0x20" [5:4]. If Table 171, "Register 0x4F" [0] = '1', the setting in Table 127, "Register 0x20" [5:4] must be identical as Table 127, "Register 0x20" [7:6].
3. This register is set when SWC output current goes above the threshold setting in Table 127, "Register 0x20" [3:2].
4. This register is set when SWD output current goes above the threshold setting in Table 127, "Register 0x20" [1:0].
5. This register is set when SWA output voltage goes below the threshold setting in Table 129, "Register 0x22," [3:2].
6. Only applicable if Table 171, "Register 0x4F" [0] = '0'. This register is set when SWB output voltage goes below the threshold setting in Table 131, "Register 0x24,," [3:2]. If Table 171, "Register 0x4F" [0] = '1', the setting in Table 131, "Register 0x24,," [3:2] must be identical as Table 129, "Register 0x22,," [3:2].
7. This register is set when SWC output voltage goes below the threshold setting in Table 133, "Register 0x26," [3:2].
8. This register is set when SWD output voltage goes below the threshold setting in Table 135, "Register 0x28," [3:2].

Table 107 — Register 0x0C

R0C			
Bits	Attribute	Default	Description ¹
7:0	RO	0	R0C [7:0]: SWA_OUTPUT_CURRENT_POWER_MEASUREMENT If Table 121, “Register 0x1A”[1] = ‘0’: Switch Node A Output Current or Output Power² Measurement³ 0000 0000 = Un-defined 0000 0001 = 0.125 A or 125 mW 0000 0010 = 0.25 A or 250 mW 0000 0011 = 0.375 A or 375 mW 0000 0100 = 0.5 A or 500 mW 0000 0101 = 0.625 A or 625 mW 0000 0110 = 0.75 A or 750 mW 0000 0111 = 0.875 A or 875 mW 0000 1000 = 1.0 A or 1000 mW 0000 1001 = 1.125 A or 1125 mW .. 0011 0111 = 6.875 A or 6875 mW 0011 1000 = 7.0 A or 7000 mW 0011 1001 = 7.125 A or 7125 mW 0011 1010 = 7.25 A or 7250 mW 0011 1011 = 7.375 A or 7375 mW 0011 1100 = 7.5 A or 7500 mW 0011 1101 = 7.625 A or 7625 mW 0011 1110 = 7.75 A or 7750 mW 0011 1111 > = 7.875 A or 7875 mW All other encodings are reserved If Table 121, “Register 0x1A”[1] = ‘1’: Sum of SWA, SWB, SWC and SWD Output Power⁴ 0000 0000 = Undefined 0000 0001 = 125 mW 0000 0010 = 250 mW 0000 0011 = 375 mW 0000 0100 = 500 mW ... 1111 1100 = 31500 mW 1111 1101 = 31625 mW 1111 1110 = 31750 mW 1111 1111 > = 31875 mW

1. The PMIC reports current or power measurement as long as there is no output over voltage or output under voltage lockout event that triggers the VR Disable command. If PMIC triggers VR Disable command on its own, PMIC does not report current or power measurement. For all other events that causes PMIC's power good status as Not Good, the PMIC continues to provide current or power measurement.
2. If Table 122, “Register 0x1B” [6] = ‘0’, the PMIC reports current measurement. If Table 122, “Register 0x1B” [6] = ‘1’, the PMIC reports power measurement.
3. If Table 171, “Register 0x4F” [0] = ‘1’, host adds the current or power reported in Table 107, “Register 0x0C” [7:0] & Table 108, “Register 0x0D” [5:0] for total current or power consumption.
4. Register Table 122, “Register 0x1B” [6] must be configured as ‘1’.

Table 108 — Register 0x0D

R0D			
Bits	Attribute	Default	Description ¹
7:6	RV	0	R0D [7:6]: Reserved
5:0	RO	0	R0D [5:0]: SWB_OUTPUT_CURRENT_POWER_MEASUREMENT Switch Node B Output Current or Output Power ² Measurement ³ 000000 = Un-defined 000001 = 0.125 A or 125 mW 000010 = 0.25 A or 250 mW 000011 = 0.375 A or 375 mW 000100 = 0.5 A or 500 mW 000101 = 0.625 A or 625 mW 000110 = 0.75 A or 750 mW 000111 = 0.875 A or 875 mW 001000 = 1.0 A or 1000 mW 001001 = 1.125 A or 1125 mW ... 110111 = 6.875 A or 6875 mW 111000 = 7.0 A or 7000 mW 111001 = 7.125 A or 7125 mW 111010 = 7.25 A or 7250 mW 111011 = 7.375 A or 7375 mW 111100 = 7.5 A or 7500 mW 111101 = 7.625 A or 7625 mW 111110 = 7.75 A or 7750 mW 111111 >= 7.875 A or 7875 mW

1. The PMIC reports current or power measurement as long as there is no output over voltage or output under voltage lockout event that triggers the VR Disable command. If PMIC triggers VR Disable command on its own, PMIC does not report current or power measurement. For all other events that causes PMIC's power good status as Not Good, the PMIC continues to provide current or power measurement.
2. If Table 122, "Register 0x1B" [6] = '0', the PMIC reports current measurement. If Table 122, "Register 0x1B" [6] = '1', the PMIC reports power measurement.
3. If Table 171, "Register 0x4F" [0] = '1', host adds the current or power reported in Table 107, "Register 0x0C" [7:0] & Table 108, "Register 0x0D" [5:0] for total current or power consumption.

Table 109 — Register 0x0E

R0E			
Bits	Attribute	Default	Description ¹
7:6	RV	0	R0E [7:6]: Reserved
5:0	RO	0	R0E [5:0]: SWC_OUTPUT_CURRENT_POWER_MEASUREMENT Switch Node C Output Current or Output Power ² Measurement 000000 = Un-defined 000001 = 0.125 A or 125 mW 000010 = 0.25 A or 250 mW 000011 = 0.375 A or 375 mW 000100 = 0.5 A or 500 mW 000101 = 0.625 A or 625 mW 000110 = 0.75 A or 750 mW 000111 = 0.875 A or 875 mW 001000 = 1.0 A or 1000 mW 001001 = 1.125 A or 1125 mW 001010 = 1.25 A or 1250 mW ... 110111 = 6.875 A or 6875 mW 111000 = 7.0 A or 7000 mW 111001 = 7.125 A or 7125 mW 111010 = 7.25 A or 7250 mW 111011 = 7.375 A or 7375 mW 111100 = 7.5 A or 7500 mW 111101 = 7.625 A or 7625 mW 111110 = 7.75 A or 7750 mW 111111 >= 7.875 A or 7875 mW

1. The PMIC reports current or power measurement as long as there is no output over voltage or output under voltage lockout event that triggers the VR Disable command. If PMIC triggers VR Disable command on its own, PMIC does not report current or power measurement. For all other events that causes PMIC's power good status as Not Good, the PMIC continues to provide current or power measurement.
2. If Table 122, "Register 0x1B" [6] = '0', the PMIC reports current measurement. If Table 122, "Register 0x1B" [6] = '1', the PMIC reports power measurement.

Table 110 — Register 0x0F

R0F			
Bits	Attribute	Default	Description ¹
7:6	RV	0	R0F [7:6]: Reserved
5:0	RO	0	R0F [5:0]: SWD_OUTPUT_CURRENT_POWER_MEASUREMENT Switch Node D Output Current or Output Power ² Measurement 000000 = Un-defined 000001 = 0.125 A or 125 mW 000010 = 0.25 A or 250 mW 000011 = 0.375 A or 375 mW 000100 = 0.5 A or 500 mW 000101 = 0.625 A or 625 mW 000110 = 0.75 A or 750 mW 000111 = 0.875 A or 875 mW 001000 = 1.0 A or 1000 mW 001001 = 1.125 A or 1125 mW 110111 = 6.875 A or 6875 mW 111000 = 7.0 A or 7000 mW 111001 = 7.125 A or 7125 mW 111010 = 7.25 A or 7250 mW 111011 = 7.375 A or 7375 mW 111100 = 7.5 A or 7500 mW 111101 = 7.625 A or 7625 mW 111110 = 7.75 A or 7750 mW 111111 >= 7.875 A or 7875 mW

1. The PMIC reports current or power measurement as long as there is no output over voltage or output under voltage lockout event that triggers the VR Disable command. If PMIC triggers VR Disable command on its own, PMIC does not report current or power measurement. For all other events that causes PMIC's power good status as Not Good, the PMIC continues to provide current or power measurement.
2. If Table 122, "Register 0x1B" [6] = '0', the PMIC reports current measurement. If Table 122, "Register 0x1B" [6] = '1', the PMIC reports power measurement.

14.3.5.2 Clear Registers

For each Real Time Status Registers (Table 103, “Register 0x08” [7:0], Table 104, “Register 0x09” [7:0], Table 105, “Register 0x0A” [7:1], Table 106, “Register 0x0B” [7:0] and Table 146, “Register 0x33” [4:2]), the PMIC offers a way to clear the status of each event. The clear registers are Table 111, “Register 0x10” [7,5:0], Table 112, “Register 0x11” [7:0], Table 113, “Register 0x12” [7:2], Table 114, “Register 0x13” [7:0] and to Table 115, “Register 0x14” [4:2,0] respectively. All clear registers are Write ‘1’ only registers. When ‘1’ is written to any of the clear registers, the PMIC updates the status registers to default state and removes the interrupt condition on GSI_n and PWR_GOOD output signal assuming that event is no longer present. If the failing condition is still present, the status register will still remain at ‘1’. Note that GSI_n and PWR_GOOD interrupt is only applicable if that event is not masked. GSI_n output signal can be disabled.

When ‘1’ is written to any of the clear registers, there are three categories of response by the PMIC.

1. PMIC removes GSI_n interrupt (PWR_GOOD interrupt is not applicable. Related status registers are: Table 111, “Register 0x10” [1], Table 112, “Register 0x11” [7,4:0], Table 114, “Register 0x13” [7:4], Table 115, “Register 0x14” [4].
2. PMIC removes GSI_n and PWR_GOOD interrupt. Related status registers are: Table 111, “Register 0x10” [7,5:2], Table 112, “Register 0x11” [6:5], Table 115, “Register 0x14” [2].
3. PMIC only removes GSI_n interrupt and does not remove PWR_GOOD interrupt. Related status registers are: Table 111, “Register 0x10” [0], Table 113, “Register 0x12” [7:4], Table 114, “Register 0x13” [3:0], Table 115, “Register 0x14” [3]. The host is expected to either power cycle the PMIC or re-issue the VR Enable command if PMIC is in programmable mode.

The PMIC offers a Global Clear command by writing ‘1’ to registers Table 115, “Register 0x14” [0]. This command works same way as individual clear command. This command can alternatively be used by the host if more than one clear command is required to different registers.

Table 111 — Register 0x10

R10			
Bits	Attribute	Default	Description ¹
7	IO	0	R10 [7]: CLEAR_VIN_BULK_INPUT_POWER_GOOD_STATUS Clear VIN_Bulk Input Power Good Status. 1 = Clear Register Table 103, “Register 0x08” [7] ²
6	RV	0	R10 [6]: Reserved
5	IO	0	R10 [5]: CLEAR_SWA_OUTPUT_POWER_GOOD_STATUS Clear SWA Output Power Good Status. 1 = Clear Register Table 103, “Register 0x08” [5] ²

Table 111 — Register 0x10

R10			
Bits	Attribute	Default	Description ¹
4	1O	0	R10 [4]: CLEAR_SWB_OUTPUT_POWER_GOOD_STATUS Clear SWB Output Power Good Status. ³ 1 = Clear Register Table 103, “Register 0x08” [4] ²
3	1O	0	R10 [3]: CLEAR_SWC_OUTPUT_POWER_GOOD_STATUS Clear SWC Output Power Good Status. 1 = Clear Register Table 103, “Register 0x08” [3] ²
2	1O	0	R10 [2]: CLEAR_SWD_OUTPUT_POWER_GOOD_STATUS Clear SWD Output Power Good Status. 1 = Clear Register Table 103, “Register 0x08” [2] ²
1	1O	0	R10 [1]: CLEAR_VIN_MGMT_INPUT_OVER_VOLTAGE_STATUS Clear VIN_Mgmt Input Supply Over Voltage Status. 1 = Clear Register Table 103, “Register 0x08” [1] ²
0	1O	0	R10 [0]: CLEAR_VIN_BULK_INPUT_OVER_VOLTAGE_STATUS Clear VIN_Bulk Input Supply Over Voltage Status. 1 = Clear Register Table 103, “Register 0x08” [0] ²

1. Table 111, “Register 0x10” [7:0] are self clearing bits.
2. See Table 31 and Table 32 for GSI_n and POWER_GOOD output signal status change.
3. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 112 — Register 0x11

R11			
Bits	Attribute	Default	Description ¹
7	1O	0	R11 [7]: CLEAR_PMIC_HIGH_TEMP_WARNING_STATUS Clear PMIC High Temperature Warning Status. 1 = Clear Register Table 104, “Register 0x09” [7] ²
6	1O	0	R11 [6]: CLEAR_VBIAS_POWER_GOOD_STATUS Clear VBIAS Power Good Status. 1 = Clear Register Table 104, “Register 0x09” [6] ²
5	1O	0	R11 [5]: CLEAR_VOUT_1.8V_OUTPUT_POWER_GOOD_STATUS Clear VOUT_1.8V Output Power Good Status. 1 = Clear Register Table 104, “Register 0x09” [5] ²
4	1O	0	R11 [4]: CLEAR_VIN_MGMT_TO_VIN_BULK_INPUT_SWITCHOVER_STATUS Clear VIN_Mgmt to VIN_Bulk Input Supply Switchover Status. 1 = Clear Register Table 104, “Register 0x09” [4] ²
3	1O	0	R11 [3]: CLEAR_SWA_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Clear Switch Node A High Output Current Consumption Warning Status. 1 = Clear Register Table 104, “Register 0x09” [3] ²
2	1O	0	R11 [2]: CLEAR_SWB_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Clear Switch Node B High Output Current Consumption Warning Status. ³ Clear Register Table 104, “Register 0x09” [2] ²
1	1O	0	R11 [1]: CLEAR_SWC_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Clear Switch Node C High Output Current Consumption Warning Status. 1 = Clear Register Table 104, “Register 0x09” [1] ²
0	1O	0	R11 [0]: CLEAR_SWD_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Clear Switch Node D High Output Current Consumption Warning Status. 1 = Clear Register Table 104, “Register 0x09” [0] ²

1. Table 112, “Register 0x11” [7:0] are self clearing bits.

2. See Table 31 and Table 32 for GSI_n and POWER_GOOD output signal status change.

3. This register is applicable regardless of the setting in Table 171, “Register 0x4F” [0].

Table 113 — Register 0x12

R12			
Bits	Attribute	Default	Description ¹
7	1O	0	R12 [7]: CLEAR_SWA_OUTPUT_OVER_VOLTAGE_STATUS Clear Switch Node A Output Over Voltage Status. 1 = Clear Register Table 105, “Register 0x0A” [7] ²
6	1O	0	R12 [6]: CLEAR_SWB_OUTPUT_OVER_VOLTAGE_STATUS Clear Switch Node B Output Over Voltage Status. ³ 1 = Clear Register Table 105, “Register 0x0A” [6] ²
5	1O	0	R12 [5]: CLEAR_SWC_OUTPUT_OVER_VOLTAGE_STATUS Clear Switch Node C Output Over Voltage Status. 1 = Clear Register Table 105, “Register 0x0A” [5] ²
4	1O	0	R12 [4]: CLEAR_SWD_OUTPUT_OVER_VOLTAGE_STATUS Clear Switch Node D Output Over Voltage Status. 1 = Clear Register Table 105, “Register 0x0A” [4] ²
3	1O	0	R12 [3]: CLEAR_PER_ERROR_STATUS Clear PEC Error Status. 1 = Clear Register Table 105, “Register 0x0A” [3]
2	1O	0	R12 [2]: CLEAR_PARITY_ERROR_STATUS Clear Parity Error Status. 1 = Clear Register Table 105, “Register 0x0A” [2]
1:0	RV	0	R12 [1:0]: Reserved

1. Table 113, “Register 0x12” [7:0] are self clearing bits.

2. See Table 31 and Table 32 for GSI_n and POWER_GOOD output signal status change.

3. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 114 — Register 0x13

R13			
Bits	Attribute	Default	Description ¹
7	1O	0	R13 [7]: CLEAR_SWA_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Clear Switch Node A Output Current Limiter Warning Status. 1 = Clear Register Table 106, “Register 0x0B” [7] ²
6	1O	0	R13 [6]: CLEAR_SWB_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Clear Switch Node B Output Current Limiter Warning Status. ³ 1 = Clear Register Table 106, “Register 0x0B” [6] ²
5	1O	0	R13 [5]: CLEAR_SWC_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Clear Switch Node C Output Current Limiter Warning Status. 1 = Clear Register Table 106, “Register 0x0B” [5] ²
4	1O	0	R13 [4]: CLEAR_SWD_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Clear Switch Node D Output Current Limiter Warning Status. 1 = Clear Register Table 106, “Register 0x0B” [4] ²
3	1O	0	R13 [3]: CLEAR_SWA_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Clear Switch Node A Output Under Voltage Lockout Status. 1 = Clear Register Table 106, “Register 0x0B” [3] ²
2	1O	0	R13 [2]: CLEAR_SWB_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Clear Switch Node B Output Under Voltage Lockout Status. ⁴ 1 = Clear Register Table 106, “Register 0x0B” [2] ²
1	1O	0	R13 [1]: CLEAR_SWC_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Clear Switch Node C Output Under Voltage Lockout Status. 1 = Clear Register Table 106, “Register 0x0B” [1] ²
0	1O	0	R13 [0]: CLEAR_SWD_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Clear Switch Node D Output Under Voltage Lockout Status. 1 = Clear Register Table 106, “Register 0x0B” [0] ²

1. Table 114, “Register 0x13” [7:0] are self clearing bits.

2. See Table 31 and Table 32 for GSI_n and POWER_GOOD output signal status change.

3. This register is applicable regardless of the setting in Table 171, “Register 0x4F” [0]

4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 115 — Register 0x14

R14			
Bits	Attribute	Default	Description ¹
7:5	RV	0	R14 [7:5]: Reserved
4	IO	0	R14 [4]: CLEAR_VIN_MGMT_POWER_GOOD_STATUS_SWITCHOVER_MODE Clear Valid VIN_Mgmt Power Good Status in Switchover Mode. 1 = Clear Register Table 146, “Register 0x33” [4] ²
3	IO	0	R14 [3]: CLEAR_VBIAS_OUTPUT_OR_VIN_BULK_UNDER_VOLTAGE_LOCKOUT_STATUS Clear VBIAS Output or VIN_Bulk Input Under Voltage Lockout Status. 1 = Clear Register Table 146, “Register 0x33” [3] ²
2	IO	0	R14 [2]: CLEAR_VOUT_1.0V_OUTPUT_POWER_GOOD_STATUS Clear VOUT_1.0V Output Power Good Status. 1 = Clear Register Table 146, “Register 0x33” [2] ²
1	RV	0	R14 [1]: Reserved
0	IO	0	R14 [0]: GLOBAL_CLEAR_STATUS Clear all ³ status bits. 1 = Clear all status bits ⁴

1. Table 115, “Register 0x14” [4:2, 0] are self clearing bits.
2. See Table 31 and Table 32 for GSI_n and POWER_GOOD output signal status change.
3. All status bits in register Table 111, “Register 0x10” [7:5:0], Table 112, “Register 0x11” [7:0], Table 113, “Register 0x12” [7:2], Table 114, “Register 0x13” [7:0] & Table 115, “Register 0x14” [4:2].
4. See Table 31 and Table 32 for GSI_n and POWER_GOOD output signal status change.

14.3.5.3 Mask Registers

For each Real Time Status Registers (Table 103, “Register 0x08” [7:0], Table 104, “Register 0x09” [7:0], Table 105, “Register 0x0A” [7:1], Table 106, “Register 0x0B” [7:0], Table 146, “Register 0x33” [4:2]), the PMIC offers a way to mask the status of each event interrupt. The mask registers are Table 116, “Register 0x15” [7:5:0], Table 117, “Register 0x16” [7:0], Table 118, “Register 0x17” [7:2], Table 119, “Register 0x18” [7:0] and Table 120, “Register 0x19” [4:2] respectively. The mask registers only masks the event interrupt on GSI_n and PWR_GOOD signal.

There is also a global mask bits register control Table 142, “Register 0x2F” [1:0] to control the GSI_n and PWR_GOOD output signal. When all mask registers are Table 116, “Register 0x15” [7:5:0], Table 117, “Register 0x16” [7:0], Table 118, “Register 0x17” [7:2], Table 119, “Register 0x18” [7:0] configured as ‘0’, the setting in Table 142, “Register 0x2F” [1:0] does not matter. The setting in Table 142, “Register 0x2F” [1:0] only matters when one or more mask registers Table 116, “Register 0x15” [7:5:0], Table 117, “Register 0x16” [7:0], Table 118, “Register 0x17” [7:2], Table 119, “Register 0x18” [7:0] are configured to ‘1’.

For any failure events that causes the PMIC to generate VR Disable command on its own, the mask register bits (Table 116, “Register 0x15” [0], Table 118, “Register 0x17” [7:4], Table 119, “Register 0x18” [3:0], Table 120, “Register 0x19” [3], Table 142, “Register 0x2F” [1:0]) do not apply and PMIC will assert PWR_GOOD output signal regardless of the setting in mask registers. The PMIC still updates the status registers appropriately when any event occurs. When masked, the host is expected to read the status registers periodically to learn if any of the event has occurred or not. The host can mask or un-mask each event individually. The host can mask or un-mask at any time in programmable mode. In Secure mode of operation, the mask registers are locked.

Table 116 — Register 0x15

R15			
Bits	Attribute	Default	Description
7	RW	1	R15 [7]: MASK_VIN_BULK_INPUT_POWER_GOOD_STATUS Mask VIN_Bulk Input Power Good Status Event. 0 = Do Not Mask VIN_Bulk Input Power Good Status Event 1 = Mask VIN_Bulk Input Power Good Status Event ¹
6	RV	0	R15 [6]: Reserved
5	RW	1	R15 [5]: MASK_SWA_OUTPUT_POWER_GOOD_STATUS Mask SWA Output Power Good Status Event. 0 = Do Not Mask SWA Output Power Good Status Event 1 = Mask SWA Output Power Good Status Event ¹
4	RW	1	R15 [4]: MASK_SWB_OUTPUT_POWER_GOOD_STATUS Mask SWB Output Power Good Status Event. ² 0 = Do Not Mask SWB Output Power Good Status Event 1 = Mask SWB Output Power Good Status Event ¹
3	RW	1	R15 [3]: MASK_SWC_OUTPUT_POWER_GOOD_STATUS Mask SWC Output Power Good Status Event. 0 = Do Not Mask SWC Output Power Good Status Event 1 = Mask SWC Output Power Good Status Event ¹
2	RW	1	R15 [2]: MASK_SWD_OUTPUT_POWER_GOOD_STATUS Mask SWD Output Power Good Status Event. 0 = Do Not Mask SWD Output Power Good Status Event 1 = Mask SWD Output Power Good Status Event ¹
1	RW	0	R15 [1]: MASK_VIN_MGMT_INPUT_OVER_VOLTAGE_STATUS Mask VIN_Mgmt Input Supply Over Voltage Status Event. 0 = Do Not Mask VIN_Mgmt Input Supply Over Voltage Status Event 1 = Mask VIN_Mgmt Input Supply Over Voltage Status Event ³
0	RW	0	R15 [0]: MASK_VIN_BULK_INPUT_OVER_VOLTAGE_STATUS Mask VIN_Bulk Input Supply Over Voltage Status Event. 0 = Do Not Mask VIN_Bulk Input Supply Over Voltage Status Event 1 = Mask VIN_Bulk Input Supply Over Voltage Status Event ³

1. Not assert GSI_n or assert POWER_GOOD output signal.

2. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

3. Not assert GSI_n output signal.

Table 117 — Register 0x16

R16			
Bits	Attribute	Default	Description
7	RW	0	R16 [7]: MASK_PMIC_HIGH_TEMP_WARNING_STATUS Mask PMIC High Temperature Warning Status Event. 0 = Do Not Mask PMIC High Temperature Warning Status Event 1 = Mask PMIC High Temperature Warning Status Event ¹
6	RW	0	R16 [6]: MASK_VBIAS_POWER_GOOD_STATUS Mask VBias Power Good Status Event. 0 = Do Not Mask VBias Power Good Status Event 1 = Mask VBias Power Good Status Event ²
5	RW	1	R16 [5]: MASK_VOUT_1.8V_OUTPUT_POWER_GOOD_STATUS Mask VOUT_1.8V Output Power Good Status Event. 0 = Do Not Mask 1.8V Output Power Good Status Event 1 = Mask 1.8V Output Power Good Status Event ²
4	RW	0	R16 [4]: MASK_VIN_MGMT_TO_VIN_BULK_SWITCHOVER_STATUS Mask VIN_Mgmt to VIN_Bulk Input Supply Switchover Status Event. 0 = Do Not Mask VIN_Mgmt to VIN_Bulk Input Supply Switchover Status Event 1 = Mask VIN_Mgmt to VIN_Bulk Input Supply Switchover Status Event ¹
3	RW	0	R16 [3]: MASK_SWA_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Mask SWA High Output Current Consumption Warning Status Event. 0 = Do Not Mask SWA Output Current Consumption Warning Status Event 1 = Mask SWA Output Current Consumption Warning Status Event ¹
2	RW	0	R16 [2]: MASK_SWB_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Mask SWB High Output Current Consumption Warning Status Event. ³ 0 = Do Not Mask SWB Output Current Consumption Warning Status Event 1 = Mask SWB Output Current Consumption Warning Status Event ¹
1	RW	0	R16 [1]: MASK_SWC_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Mask SWC High Output Current Consumption Warning Status Event. 0 = Do Not Mask SWC Output Current Consumption Warning Status Event 1 = Mask SWC Output Current Consumption Warning Status Event ¹
0	RW	0	R16 [0]: MASK_SWD_HIGH_OUTPUT_CURRENT_CONSUMPTION_WARNING_STATUS Mask SWD High Output Current Consumption Warning Status Event. 0 = Do Not Mask SWD Output Current Consumption Warning Status Event 1 = Mask SWD Output Current Consumption Warning Status Event ¹

1. Not assert GSI_n output signal.

2. Not assert GSI_n or assert POWER_GOOD output signal

3. This register is applicable regardless of the setting in Table 171, “Register 0x4F” [0].

Table 118 — Register 0x17

R17			
Bits	Attribute	Default	Description
7	RW	0	R17 [7]: MASK_SWA_OUTPUT_OVER_VOLTAGE_STATUS Mask SWA Output Over Voltage Status Event. 0 = Do Not Mask SWA Output Over Voltage Status Event 1 = Mask SWA Output Over Voltage Status Event ¹
6	RW	0	R17 [6]: MASK_SWB_OUTPUT_OVER_VOLTAGE_STATUS Mask SWB Output Over Voltage Status Event. ² 0 = Do Not Mask SWB Output Over Voltage Status Event 1 = Mask SWB Output Over Voltage Status Event ¹
5	RW	0	R17 [5]: MASK_SWC_OUTPUT_OVER_VOLTAGE_STATUS Mask SWC Output Over Voltage Status Event. 0 = Do Not Mask SWC Output Over Voltage Status Event 1 = Mask SWC Output Over Voltage Status Event ¹
4	RW	0	R17 [4]: MASK_SWD_OUTPUT_OVER_VOLTAGE_STATUS Mask SWD Output Over Voltage Status Event. 0 = Do Not Mask SWD Output Over Voltage Status Event 1 = Mask SWD Output Over Voltage Status Event ¹
3	RW	0	R17 [3]: MASK_PEC_ERROR_STATUS Mask PEC Error Event for GSI_n output Only ³ 0 = Do Not Mask PEC Error Status Event 1 = Mask PEC Error Status
2	RW	0	R17 [2]: MASK_PARITY_ERROR_STATUS Mask Parity Error Event for GSI_n output Only ³ 0 = Do Not Mask Parity Error Status Event 1 = Mask Parity Error Status
1:0	RV	0	R17 [1:0]: Reserved

1. Not assert GSI_n output signal.

2. Only applicable if Table 171, "Register 0x4F" [0] = '0'.

3. Only applicable when PMIC is in I3C Basic Mode. This Mask register only masks the GSI_n output. Does not apply to IBI.

Table 119 — Register 0x18

R18			
Bits	Attribute	Default	Description
7	RW	0	R18 [7]: MASK_SWA_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Mask SWA Output Current Limiter Warning Status Event. 0 = Do Not Mask SWA Output Current Limiter Warning Status Event 1 = Mask SWA Output Current Limiter Warning Status Event ¹
6	RW	0	R18 [6]: MASK_SWB_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Mask SWB Output Current Limiter Warning Status Event. ² 0 = Do Not Mask SWB Output Current Limiter Warning Status Event 1 = Mask SWB Output Current Limiter Warning Status Event ¹
5	RW	0	R18 [5]: MASK_SWC_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Mask SWC Output Current Limiter Warning Status Event. 0 = Do Not Mask SWC Output Current Limiter Warning Status Event 1 = Mask SWC Output Current Limiter Warning Status Event ¹
4	RW	0	R18 [4]: MASK_SWD_OUTPUT_CURRENT_LIMITER_WARNING_STATUS Mask SWD Output Current Limiter Warning Status Event. 0 = Do Not Mask SWD Output Current Limiter Warning Status Event 1 = Mask SWD Output Current Limiter Warning Status Event ¹
3	RW	0	R18 [3]: MASK_SWA_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Mask SWA Output Under Voltage Lockout Status Event. 0 = Do Not Mask SWA Output Under Voltage Lockout Status Event 1 = Mask SWA Output Under Voltage Lockout Status Event ¹
2	RW	0	R18 [2]: MASK_SWB_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Mask SWB Output Under Voltage Lockout Status Event. ³ 0 = Do Not Mask SWB Output Under Voltage Lockout Status Event 1 = Mask SWB Output Under Voltage Lockout Status Event ¹
1	RW	0	R18 [1]: MASK_SWC_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Mask SWC Output Under Voltage Lockout Status Event. 0 = Do Not Mask SWC Output Under Voltage Lockout Status Event 1 = Mask SWC Output Under Voltage Lockout Status Event ¹
0	RW	0	R18 [0]: MASK_SWD_OUTPUT_UNDER_VOLTAGE_LOCKOUT_STATUS Mask SWD Output Under Voltage Lockout Status Event. 0 = Do Not Mask SWD Output Under Voltage Lockout Status Event 1 = Mask SWD Output Under Voltage Lockout Status Event ¹

1. Not assert GSI_n output signal.

2. This register is applicable regardless of the setting in Table 171, “Register 0x4F” [0].

3. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 120 — Register 0x19

R19			
Bits	Attribute	Default	Description
7:5	RV	0	R19 [7:5]: Reserved
4	RW	0	R19 [4]: MASK_VIN_MGMT_POWER_GOOD_STATUS_SWITCHOVER_MODE Mask VIN_Mgmt Input Supply Power Good Status in Switchover Mode Only 0 = Do Not Mask VIN_Mgmt Input Power Supply Power Good Status Event in Switchover Mode 1 = Mask VIN_Mgmt Input Power Supply Power Good Status Event in Switchover Mode ¹
3	RW	0	R19 [3]: MASK_VBIAS_OUTPUT_OR_VIN_BULK_UNDER_VOLTAGE_LOCKOUT_STATUS Mask VBIAS Output or VIN_Bulk Input Under Voltage Lockout Event 0 = Do Not Mask Vbias Output or VIN_Bulk Input Under Voltage Lockout Event 1 = Mask Vbias Output or VIN_Bulk Input Under Voltage Lockout Event ¹
2	RW	1	R19 [2]: MASK_VOUT_1.0V_OUTPUT_POWER_GOOD_STATUS Mask VOUT_1.0V Output Power Good Status Event. 0 = Do Not Mask VOUT_1.0V Output Power Good Status Event 1 = Mask VOUT_1.0V Output Power Good Status Event ²
1:0	RV	0	R19 [1:0]: Reserved

1. Not assert GSI_n output signal.

2. Not assert GSI_n or POWER_GOOD output signal.

14.3.5.4 Threshold & Configuration Registers

Table 121 — Register 0x1A

R1A			
Bits	Attribute	Default	Description
7:5	RW	110	R1A [7:5]: VIN_BULK_POWER_GOOD_THRESHOLD_VOLTAGE VIN Bulk Input Threshold Voltage for Input Power Good Status for input supply 000 = Reserved 001 = 9.5 V 010 = 8.5 V 011 = 7.5 V 100 = 6.5 V 101 = 5.5 V 110 = 4.25 V ^{1,2,3} 111 = Reserved
4	RV	0	R1A [4]: Reserved
3	RW	0	R1A [3]: VBIAS_POWER_GOOD_THRESHOLD_VOLTAGE VBias LDO Output Threshold Voltage for Power Good Status 0 = Vendor Specific 1 = Reserved
2	RW	0	R1A [2]: VOUT_1.8V_POWER_GOOD_THRESHOLD_VOLTAGE VOUT 1.8V LDO Output Threshold Voltage for Power Good Status 0 = 1.6 V 1 = Reserved
1	RW	0	R1A [1]: OUTPUT_POWER_SELECT Switch Regulator Output Power Select ⁴ 0 = Report individual power for each rail in R0C, R0D, R0E & R0F 1 = Report total power of each rail in R0C ⁵
0	RW	0	R1A [0]: VOUT_1.0V_POWER_GOOD_THRESHOLD_VOLTAGE VOUT 1.0V LDO Output Threshold Voltage for Power Good Status 0 = -10% from the setting in Table 173, “Register 0x51” [2:1] 1 = -15% from the setting in Table 173, “Register 0x51” [2:1]

1. If VIN_Bulk input voltage falls below this threshold, the PMIC may not guarantee the operation.
2. If VIN_Bulk voltage continues to fall below vendor specific UVLO, the PMIC triggers an VR Disable command and executes power down sequence.
3. If VIN_Bulk voltage threshold is higher than default setting of this register, the PMIC continues to operate but updates its PWR_GOOD status register and GSI_n/PWR_GOOD signal is asserted when VIN_Bulk falls below the threshold setting.
4. This register is only applicable if Table 122, “Register 0x1B” [6] = ‘1’.
5. Host should only read Table 107, “Register 0x0C” [7:0] for total power. The register contents of Table 108, “Register 0x0D”, Table 109, “Register 0x0E” & Table 110, “Register 0x0F” may not be valid.

Table 122 — Register 0x1B

R1B			
Bits	Attribute	Default	Description
7	RW	0	R1B [7]: VIN_BULK_OVER_VOLTAGE_THRESHOLD VIN_Bulk Input Over Voltage Threshold Setting For GSI_n Assertion 0 = 14.5 V 1 = 16.0 V
6	RW	0	R1B [6]: CURRENT_OR_POWER_METER_SELECT PMIC Output Regulator Measurement - Current or Power Meter 0 = Report Current Measurements in registers ¹ 1 = Report Power Measurements in registers ¹
5	RW	0	R1B [5]: VIN_MGMT_OVER_VOLTAGE_THRESHOLD VIN_Mgmt Input Over Voltage Threshold 0 = 3.8 V 1 = 3.7 V
4	RW	0	R1B [4]: GLOBAL_PWR_GOOD_PIN_STATUS_MASK Global Mask PWR_GOOD Output Pin ² 0 = Not Masked 1 = Masked
3	RW	0	R1B [3]: GSI_N_PIN_ENABLE Enable GSI_n Pin ³ 0 = Disable GSI_n Pin 1 = Enable GSI_n Pin
2:0	RW	101	R1B [2:0]: PMIC_HIGH_TEMPERATURE_WARNING_THRESHOLD PMIC High Temperature Warning Threshold ⁴ 000 = Reserved 001 = PMIC temperature $\geq 85^{\circ}\text{C}$ 010 = PMIC temperature $\geq 95^{\circ}\text{C}$ 011 = PMIC temperature $\geq 105^{\circ}\text{C}$ 100 = PMIC temperature $\geq 115^{\circ}\text{C}$ 101 = PMIC temperature $\geq 125^{\circ}\text{C}$ 110 = PMIC temperature $\geq 135^{\circ}\text{C}$ 111 = Reserved

1. Table 107, "Register 0x0C" [7:0], Table 108, "Register 0x0D" [5:0], Table 109, "Register 0x0E" [5:0], Table 110, "Register 0x0F" [5:0].
2. Mask POWER_GOOD output signal for all appropriate register bits in Table 116, "Register 0x15" [7,5:0], Table 117, "Register 0x16" [7:0], Table 118, "Register 0x17" [7:4], Table 119, "Register 0x18" [7:0] and Table 120, "Register 0x19" [4:2]. Mask Register Control Table 142, "Register 0x2F" [1:0] still applies when Global PWR_GOOD output Mask register is set to '1'.
3. This register can be used as Global Mask Function for GSI_n pin. If disabled, this masks GSI_n output signal for all register bits in Table 116, "Register 0x15" [7, 5:0], Table 117, "Register 0x16" [7:0], Table 118, "Register 0x17" [7:4], Table 119, "Register 0x18" [7:0] and Table 120, "Register 0x19" [4:2].
4. The tolerance of the temperature warning threshold is $\pm 5^{\circ}\text{C}$ from the selected setting.

Table 123 — Register 0x1C

R1C			
Bits	Attribute	Default	Description
7:2	RW	011000	R1C [7:2]: SWA_OUTPUT_HIGH_CURRENT_CONSUMPTION_WARNING_THRE SHOLD SWA Output High Current Consumption Warning Threshold 000000 = Un-defined 000001 => 0.125 A 000010 => 0.25 A 000011 => 0.375 A 000100 => 0.5 A or 000101 => 0.625 A 000110 => 0.75 A 000111 => 0.875 A 001000 => 1.0 A 001001 => 1.125 A ... 010111 => 2.875 A 011000 => 3.0 A 011001 => 3.125 A ... 110111 => 6.875 A 111000 => 7.0 A 111001 => 7.125 A 111010 => 7.25 A 111011 => 7.375 A 111100 => 7.5 A 111101 => 7.625 A 111110 => 7.75 A 111111 => 7.875 A
1:0	RV	0	R1C [1:0]: Reserved

Table 124 — Register 0x1D

R1D			
Bits	Attribute	Default	Description
7:2	RW	011000	R1D [7:2]: SWB_OUTPUT_HIGH_CURRENT_CONSUMPTION_WARNING_THRESHOLD SWB Output High Current Consumption Warning Threshold ¹ 000000 = Un-defined 000001 => 0.125 A 000010 => 0.25 A 000011 => 0.375 A 000100 => 0.5 A or 000101 => 0.625 A 000110 => 0.75 A 000111 => 0.875 A 001000 => 1.0 A 001001 => 1.125 A ... 010111 => 2.875 A 011000 => 3.0 A ... 110111 => 6.875 A 111000 => 7.0 A 111001 => 7.125 A 111010 => 7.25 A 111011 => 7.375 A 111100 => 7.5 A 111101 => 7.625 A 111110 => 7.75 A 111111 => 7.875 A
1:0	RV	0	R1D [1:0]: Reserved

1. This register is applicable regardless of the setting in Table 171, "Register 0x4F" [0]. For dual phase operation, this register should be configured identically as Table 123, "Register 0x1C" [7:2].

Table 125 — Register 0x1E

R1E			
Bits	Attribute	Default	Description
7:2	RW	011000	R1E [7:2]: SWC_OUTPUT_HIGH_CURRENT_CONSUMPTION_WARNING_THRE SHOLD SWC Output High Current Consumption Warning Threshold 000000 = Un-defined 000001 => 0.125 A 000010 => 0.25 A 000011 => 0.375 A 000100 => 0.5 A or 000101 => 0.625 A 000110 => 0.75 A 000111 => 0.875 A 001000 => 1.0 A 001001 => 1.125 A ... 010111 => 2.875 A 011000 => 3.0 A 011001 => 3.125 A ... 110111 => 6.875 A 111000 => 7.0 A 111001 => 7.125 A 111010 => 7.25 A 111011 => 7.375 A 111100 => 7.5 A 111101 => 7.625 A 111110 => 7.75 A 111111 => 7.875 A
1:0	RV	0	R1E [1:0]: Reserved

Table 126 — Register 0x1F

R1F			
Bits	Attribute	Default	Description
7:2	RW	011000	R1F [7:2]: SWD_OUTPUT_HIGH_CURRENT_CONSUMPTION_WARNING_THRESHOLD SWD Output High Current Consumption Warning Threshold 000000 = Un-defined 000001 => 0.125 A 000010 => 0.25 A 000011 => 0.375 A 000100 => 0.5 A or 000101 => 0.625 A 000110 => 0.75 A 000111 => 0.875 A 001000 => 1.0 A 001001 => 1.125 A ... 010111 => 2.875 A 011000 => 3.0 A 011001 => 3.125 A ... 110111 => 6.875 A 111000 => 7.0 A 111001 => 7.125 A 111010 => 7.25 A 111011 => 7.375 A 111100 => 7.5 A 111101 => 7.625 A 111110 => 7.75 A 111111 => 7.875 A
1:0	RV	0	R1F [1:0]: Reserved

Table 127 — Register 0x20¹

R20			
Bits	Attribute	Default	Description
7:6	RW	0	R20 [7:6]: SWA_OUTPUT_CURRENT_LIMITER_WARNING_THRESHOLD_SETTING PMIC5000, COT Ivalley_limit 00 = 4.0 A 01 = 4.5 A 10 = 5.0 A 11 = 5.5 A PMIC5010, COT Ivalley_limit 00 = 2.0 A 01 = 2.5 A 10 = 3.0 A 11 = 3.5 A
5:4	RW	0	R20 [5:4]: SWB_OUTPUT_CURRENT_LIMITER_WARNING_THRESHOLD_SETTING PMIC5000, COT Ivalley_limit:² 00 = 4.0 A 01 = 4.5 A 10 = 5.0 A 11 = 5.5 A PMIC5010, COT Ivalley_limit 00 = 2.0 A 01 = 2.5 A 10 = 3.0 A 11 = 3.5 A
3:2	RW	0	R20 [3:2]: SWC_OUTPUT_CURRENT_LIMITER_WARNING_THRESHOLD_SETTING PMIC5000, COT Ivalley_limit: 00 = 4.0 A 01 = 4.5 A 10 = 5.0 A 11 = 5.5 A PMIC5010, COT Ivalley_limit 00 = 2.0 A 01 = 2.5 A 10 = 3.0 A 11 = 3.5 A

Table 127 — Register 0x20¹

R20			
Bits	Attribute	Default	Description
1:0	RW	0	R20 [1:0]: SWD_OUTPUT_CURRENT_LIMITER_WARNING_THRESHOLD_SETTING PMIC5000, COT Ivalley_limit: 00 = 4.0 A 01 = 4.5 A 10 = 5.0 A 11 = 5.5 A PMIC5010, COT Ivalley_limit 00 = 2.0 A 01 = 2.5 A 10 = 3.0 A 11 = 3.5 A

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 172, “Register 0x50”.
2. This register is applicable regardless of the setting in Table 171, “Register 0x4F” [0]. If Table 171, “Register 0x4F” [0] = ‘1’, this register must be configured identically as bits [7:6].

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Table 128 — Register 0x21^{1,2}

R21			
Bits	Attribute	Default	Description
7:1	RW	011 1100	R21 [7:1]: SWA_VOLTAGE_SETTING SWA Output Regulator Voltage Setting ^{3,4} 000 0000 = 800 mV ⁵ or 600 mV ⁶ 000 0001 = 805 mV or 605 mV 000 0010 = 810 mV or 610 mV ... 011 1100 = 1100 mV or 900 mV ... 111 1101 = 1425 mV or 1225 mV 111 1110 = 1430 mV or 1230 mV 111 1111 = 1435 mV or 1235 mV
0	RW	0	R21 [0]: SWA_POWER_GOOD_THRESHOLD_LOW_SIDE_VOLTAGE_SETTING SWA Output Threshold Low Side Voltage For Power Good Status 0 = -5% from the setting in Table 128, “Register 0x21,” [7:1] 1 = -7.5% from the setting in Table 128, “Register 0x21,” [7:1]

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 161, “Register 0x45”.
2. If required, the host must update the settings in register Table 128, “Register 0x21,” [0], Table 129, “Register 0x22,” [7:2] and Table 127, “Register 0x20” [7:6] first prior to updating the settings in the register Table 128, “Register 0x21,” [7:1].
3. PMIC guarantees efficiency spec and all electrical characteristics spec within a range of 1050 mV to 1160 mV.
4. After VR is enabled, the host may update this register to any new setting that it may desire. However to prevent false error trigger, the PMIC internally will increment the SWA output voltage setting by 5 mV at a time. The time it takes for PMIC to adjust the output voltage is 5 μ s for each 5 mV increment. The host must wait sufficient time for PMIC to adjust to final value. As an example, if host adjusts the output voltage by 50 mV from the original value, the host must wait minimum of 50 μ s before PMIC can guarantee the new output voltage. During this time, PMIC masks the PWR_GOOD output signal assertion to prevent any artificial error due to the crossing of the output voltage threshold setting. However, PMIC does assert PWR_GOOD output signal if there is any abnormal issues that triggers VR Disable command as described in Table 30. Further, PMIC does monitor PWR_GOOD input signal and executes power off config sequence registers if it is registered low when Table 145, “Register 0x32” [5] = ‘1’.
5. Table 138, “Register 0x2B,” [5] = ‘0’; 5 mV step size.
6. Table 138, “Register 0x2B,” [5] = ‘1’; 5 mV step size

Table 129 — Register 0x22^{1,2}

R22			
Bits	Attribute	Default	Description
7:6	RW	01	R22 [7:6]: SWA_POWER_GOOD_THRESHOLD_HIGH_SIDE_VOLTAGE_SETTING SWA Output Threshold High Side Voltage For Power Good Status 00 = +5% from the setting in Table 128, “Register 0x21,” [7:1] 01 = +7.5% from the setting in Table 128, “Register 0x21,” [7:1] 10 = +10% from the setting in Table 128, “Register 0x21,” [7:1] 11 = Reserved
5:4	RW	10	R22 [5:4]: SWA_OVER_VOLTAGE_THRESHOLD_SETTING SWA Output Regulator Threshold For Over Voltage Status ³ 00 = +7.5% from the setting in Table 128, “Register 0x21,” [7:1] 01 = +10% from the setting in Table 128, “Register 0x21,” [7:1] 10 = +12.5% from the setting in Table 128, “Register 0x21,” [7:1] 11 = Reserved
3:2	RW	00	R22 [3:2]: SWA_UNDER_VOLTAGE_LOCKOUT_THRESHOLD_SETTING SWA Output Regulator Threshold For Under Voltage Lockout Status 00 = -10% from the setting in Table 128, “Register 0x21,” [7:1] 01 = -12.5% from the setting in Table 128, “Register 0x21,” [7:1] 10 = Reserved 11 = Reserved
1:0	RW	00	R22 [1:0]: SWA_OUTPUT_SOFT_STOP_TIME SWA Output Regulator Soft Stop Time After VR Disable ⁴ 00 = 0.5 ms 01 = 1 ms 10 = 2 ms 11 = 4 ms

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 162, “Register 0x46”.
2. If required, the host must update the setting in register Table 128, “Register 0x21,” [0], Table 129, “Register 0x22,” [7:2] and Table 127, “Register 0x20” [7:6] first prior to updating the settings in the register Table 128, “Register 0x21,” [7:1].
3. The setting for the Over Voltage must be higher than Power Good High Side Voltage threshold in Table 129, “Register 0x22,” [7:6].
4. This is the time it takes for buck regulator to go from steady state voltage to 0 V.

Table 130 — Register 0x23^{1,2,3}

R23			
Bits	Attribute	Default	Description
7:1	RW	011 1100	R23 [7:1]: SWB_VOLTAGE_SETTING SWB Output Regulator Voltage Setting ^{4,5} 000 0000 = 800 mV ⁶ or 600 mV ⁷ 000 0001 = 805 mV or 605 mV 000 0010 = 810 mV or 610 mV ... 011 1100 = 1100 mV or 900 mV ... 111 1101 = 1425 mV or 1225 mV 111 1110 = 1430 mV or 1230 mV 111 1111 = 1435 mV or 1235 mV
0	RW	0	R23 [0]: SWB_POWER_GOOD_THRESHOLD_LOW_SIDE_VOLTAGE_SETTING SWB Output Threshold Low Side Voltage For Power Good Status 0 = -5% from the setting in Table 130, “Register 0x23,,” [7:1] 1 = -7.5% from the setting in Table 130, “Register 0x23,,” [7:1]

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 163, “Register 0x47”.
2. If required, the host must update the settings in register Table 130, “Register 0x23,,” [0], Table 131, “Register 0x24,,” [7:2] and Table 127, “Register 0x20” [5:4] first prior to updating the settings in the register Table 130, “Register 0x23,,” [7:1].
3. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.
4. PMIC guarantees efficiency spec and all electrical characteristics spec within a range of 1050 mV to 1160 mV.
5. After VR is enabled, the host may update this register to any new setting that it may desire. However to prevent false error trigger, the PMIC internally will increment the SWB output voltage setting by 5 mV at a time. The time it takes for PMIC to adjust the output voltage is 5 μ s for each 5 mV increment. The host must wait sufficient time for PMIC to adjust to final value. As an example, if host adjusts the output voltage by 50 mV from the original value, the host must wait minimum of 50 μ s before PMIC can guarantee the new output voltage. During this time, PMIC masks the PWR_GOOD output signal assertion to prevent any artificial error due to the crossing of the output voltage threshold setting. However, PMIC does assert PWR_GOOD output signal if there is any abnormal issues that triggers VR Disable command as described in Table 30. Further, PMIC does monitor PWR_GOOD input signal and executes power off config sequence registers if it is registered low when Table 145, “Register 0x32” [5] = ‘1’.
6. Table 138, “Register 0x2B,,” [4] = ‘0’; 5 mV step size.
7. Table 138, “Register 0x2B,,” [4] = ‘1’; 5 mV step size

Table 131 — Register 0x24^{1,2,3}

R24			
Bits	Attribute	Default	Description
7:6	RW	01	R24 [7:6]: SWB_POWER_GOOD_THRESHOLD_HIGH_SIDE_VOLTAGE_SETTING SWB Output Threshold High Side Voltage For Power Good Status 00 = +5% from the setting in Table 130, “Register 0x23,” [7:1] 01 = +7.5% from the setting in Table 130, “Register 0x23,” [7:1] 10 = +10% from the setting in Table 130, “Register 0x23,” [7:1] 11 = Reserved
5:4	RW	10	R24 [5:4]: SWB_OVER_VOLTAGE_THRESHOLD_SETTING SWB Output Regulator Threshold For Over Voltage Status ⁴ 00 = +7.5% from the setting in Table 130, “Register 0x23,” [7:1] 01 = +10% from the setting in Table 130, “Register 0x23,” [7:1] 10 = +12.5% from the setting in Table 130, “Register 0x23,” [7:1] 11 = Reserved
3:2	RW	00	R24 [3:2]: SWB_UNDER_VOLTAGE_LOCKOUT_THRESHOLD_SETTING SWB Output Regulator Threshold For Under Voltage Lockout Status 00 = -10% from the setting in Table 130, “Register 0x23,” [7:1] 01 = -12.5% from the setting in Table 130, “Register 0x23,” [7:1] 10 = Reserved 11 = Reserved
1:0	RW	00	R24 [1:0]: SWB_OUTPUT_SOFT_STOP_TIME SWB Output Regulator Soft Stop Time After VR Disable ⁵ 00 = 0.5 ms 01 = 1 ms 10 = 2 ms 11 = 4 ms

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 164, “Register 0x48”.
2. If required, the host must update the settings in register Table 130, “Register 0x23,” [0], Table 131, “Register 0x24,” [7:2] and Table 127, “Register 0x20” [5:4] first prior to updating the settings in the register Table 130, “Register 0x23,” [7:1].
3. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.
4. The setting for the Over Voltage must be higher than Power Good High Side Voltage threshold in Table 131, “Register 0x24,” [7:6].
5. This is the time it takes for buck regulator to go from steady state voltage to 0 V.

Table 132 — Register 0x25^{1,2}

R25			
Bits	Attribute	Default	Description
7:1	RW	011 1100	R25 [7:1]: SWC_VOLTAGE_SETTING SWC Output Regulator Voltage Setting ^{3,4} 000 0000 = 800 mV ⁵ or 600 mV ⁶ 000 0001 = 805 mV or 605 mV 000 0010 = 810 mV or 610 mV ... 011 1100 = 1100 mV or 900 mV ... 111 1101 = 1425 mV or 1225 mV 111 1110 = 1430 mV or 1230 mV 111 1111 = 1435 mV or 1235 mV
0	RW	0	R25 [0]: SWC_POWER_GOOD_THRESHOLD_LOW_SIDE_VOLTAGE_SETTING SWC Output Threshold Low Side Voltage For Power Good Status 0 = -5% from the setting in Table 132, “Register 0x25,” [7:1] 1 = -7.5% from the setting in Table 132, “Register 0x25,” [7:1]

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 165, “Register 0x49”.
2. If required, the host must update the settings in register Table 132, “Register 0x25,” [0], Table 133, “Register 0x26,” [7:2] and Table 127, “Register 0x20” [3:2] first prior to updating the settings in the register Table 132, “Register 0x25,” [7:1].
3. PMIC guarantees efficiency spec and all electrical characteristics spec within a range of 1050 mV to 1160 mV.
4. After VR is enabled, the host may update this register to any new setting that it may desire. However to prevent false error trigger, the PMIC internally will increment the SWC output voltage setting by 5 mV at a time. The time it takes for PMIC to adjust the output voltage is 5 μ s for each 5 mV increment. The host must wait sufficient time for PMIC to adjust to final value. As an example, if host adjusts the output voltage by 50 mV from the original value, the host must wait minimum of 50 μ s before PMIC can guarantee the new output voltage. During this time, PMIC masks the PWR_GOOD output signal assertion to prevent any artificial error due to the crossing of the output voltage threshold setting. However, PMIC does assert PWR_GOOD output signal if there is any abnormal issues that triggers VR Disable command as described in Table 30. Further, PMIC does monitor PWR_GOOD input signal and executes power off config sequence registers if it is registered low when Table 145, “Register 0x32” [5] = ‘1’.
5. Table 138, “Register 0x2B,” [3] = ‘0’; 5 mV step size.
6. Table 138, “Register 0x2B,” [3] = ‘1’; 5 mV step size

Table 133 — Register 0x26^{1,2}

R26			
Bits	Attribute	Default	Description
7:6	RW	01	R26 [7:6]: SWC_POWER_GOOD_THRESHOLD_HIGH_SIDE_VOLTAGE_SETTING SWC Output Threshold High Side Voltage For Power Good Status 00 = +5% from the setting in Table 132, “Register 0x25,” [7:1] 01 = +7.5% from the setting in Table 132, “Register 0x25,” [7:1] 10 = +10% from the setting in Table 132, “Register 0x25,” [7:1] 11 = Reserved
5:4	RW	10	R26 [5:4]: SWC_OVER_VOLTAGE_THRESHOLD_SETTING SWC Output Regulator Threshold For Over Voltage Status ³ 00 = +7.5% from the setting in Table 132, “Register 0x25,” [7:1] 01 = +10% from the setting in Table 132, “Register 0x25,” [7:1] 10 = +12.5% from the setting in Table 132, “Register 0x25,” [7:1] 11 = Reserved
3:2	RW	00	R26 [3:2]: SWC_UNDER_VOLTAGE_LOCKOUT_THRESHOLD_SETTING SWC Output Regulator Threshold For Under Voltage Lockout Status 00 = -10% from the setting in Table 132, “Register 0x25,” [7:1] 01 = -12.5% from the setting in Table 132, “Register 0x25,” [7:1] 10 = Reserved 11 = Reserved
1:0	RW	00	R26 [1:0]: SWC_OUTPUT_SOFT_STOP_TIME SWC Output Regulator Soft Stop Time After VR Disable ⁴ 00 = 0.5 ms 01 = 1 ms 10 = 2 ms 11 = 4 ms

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 166, “Register 0x4A”.
2. If required, the host must update the settings in register Table 132, “Register 0x25,” [0], Table 133, “Register 0x26,” [7:2] and Table 127, “Register 0x20” [3:2] first prior to updating the settings in the register Table 132, “Register 0x25,” [7:1].
3. The setting for the Over Voltage must be higher than Power Good High Side Voltage threshold in Table 133, “Register 0x26,” [7:6].
4. This is the time it takes for buck regulator to go from steady state voltage to 0 V.

Table 134 — Register 0x27^{1,2}

R27			
Bits	Attribute	Default	Description
7:1	RW	011 1100	R27 [7:1]: SWD_VOLTAGE_SETTING SWD Output Regulator Voltage Setting ^{3,4} 000 0000 = 1500 mV ⁵ or 2200 mV ⁶ 000 0001 = 1505 mV or 2205 mV 000 0010 = 1510 mV or 2210 mV ... 011 1100 = 1800 mV or 2500 mV ... 111 1101 = 2125 mV or 2825 mV 111 1110 = 2130 mV or 2830 mV 111 1111 = 2135 mV or 2835 mV
0	RW	0	R27 [0]: SWD_POWER_GOOD_THRESHOLD_LOW_SIDE_VOLTAGE_SETTING SWD Output Threshold Low Side Voltage For Power Good Status 0 = -5% from the setting in Table 134, “Register 0x27,” [7:1] 1 = -7.5% from the setting in Table 134, “Register 0x27,” [7:1]

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 167, “Register 0x4B”.
2. If required, the host must update the settings in register Table 134, “Register 0x27,” [0], Table 135, “Register 0x28,” [7:2] and Table 127, “Register 0x20” [1:0] first prior to updating the settings in the register Table 134, “Register 0x27,” [7:1].
3. PMIC guarantees efficiency spec and all electrical characteristics spec within a range of 1750 mV to 1850 mV.
4. After VR is enabled, the host may update this register to any new setting that it may desire. However to prevent false error trigger, the PMIC internally will increment the SWD output voltage setting by 5 mV at a time. The time it takes for PMIC to adjust the output voltage is 5 μ s for each 5 mV increment. The host must wait sufficient time for PMIC to adjust to final value. As an example, if host adjusts the output voltage by 50 mV from the original value, the host must wait minimum of 50 μ s before PMIC can guarantee the new output voltage. During this time, PMIC masks the PWR_GOOD output signal assertion to prevent any artificial error due to the crossing of the output voltage threshold setting. However, PMIC does assert PWR_GOOD output signal if there is any abnormal issues that triggers VR Disable command as described in Table 30. Further, PMIC does monitor PWR_GOOD input signal and executes power off config sequence registers if it is registered low when Table 145, “Register 0x32” [5] = ‘1’.
5. Table 138, “Register 0x2B,” [0] = ‘0’; 5 mV step size.
6. Table 138, “Register 0x2B,” [0] = ‘1’; 5 mV step size.

Table 135 — Register 0x28^{1,2}

R28			
Bits	Attribute	Default	Description
7:6	RW	01	R28 [7:6]: SWD_POWER_GOOD_THRESHOLD_HIGH_SIDE_VOLTAGE_SETTING SWD Output Threshold High Side Voltage For Power Good Status 00 = +5% from the setting in Table 134, “Register 0x27,” [7:1] 01 = +7.5% from the setting in Table 134, “Register 0x27,” [7:1] 10 = +10% from the setting in Table 134, “Register 0x27,” [7:1] 11 = Reserved
5:4	RW	10	R28 [5:4]: SWD_OVER_VOLTAGE_THRESHOLD_SETTING SWD Output Regulator Threshold For Over Voltage Status ³ 00 = +7.5% from the setting in Table 134, “Register 0x27,” [7:1] 01 = +10% from the setting in Table 134, “Register 0x27,” [7:1] 10 = +12.5% from the setting in Table 134, “Register 0x27,” [7:1] 11 = Reserved
3:2	RW	00	R28 [3:2]: SWD_UNDER_VOLTAGE_LOCKOUT_THRESHOLD_SETTING SWD Output Regulator Threshold For Under Voltage Lockout Status 00 = -10% from the setting in Table 134, “Register 0x27,” [7:1] 01 = -12.5% from the setting in Table 134, “Register 0x27,” [7:1] 10 = Reserved 11 = Reserved
1:0	RW	00	R28 [1:0]: SWD_OUTPUT_SOFT_STOP_TIME SWD Output Regulator Soft Stop Time After VR Disable ⁴ 00 = 1 ms 01 = 2 ms 10 = 4 ms 11 = 8 ms

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 168, “Register 0x4C”.
2. If required, the host must update the settings in register Table 134, “Register 0x27,” [0], Table 135, “Register 0x28,” [7:2] and Table 127, “Register 0x20” [1:0] first prior to updating the settings in the register Table 134, “Register 0x27,” [7:1].
3. The setting for the Over Voltage must be higher than Power Good High Side Voltage threshold in Table 135, “Register 0x28,” [7:6].
4. This is the time it takes for buck regulator to go from steady state voltage to 0 V.

Table 136 — Register 0x29^{1,2}

R29			
Bits	Attribute	Default	Description
7:6	RW	10	R29 [7:6]: SWA_MODE_SELECT SWA Output Regulator Mode Selection 00 = Reserved 01 = Reserved 10 = COT; DCM (Constant on Time; Discontinuous Current Mode) 11 = COT; Forced CCM (Constant on Time; Continuous Current Mode)
5:4	RW	01	R29 [5:4]: SWA_SWITCHING_FREQ SWA Output Regulator Switching Frequency 00 = 500 KHz 01 = 750 KHz 10 = 1000 KHz 11 = 1250 KHz
3:2	RW	10	R29 [1:0]: SWB_MODE_SELECT SWB Output Regulator Mode Selection ³ 00 = Reserved 01 = Reserved 10 = COT; DCM (Constant on Time; Discontinuous Current Mode) 11 = COT; Forced CCM (Constant on Time; Continuous Current Mode)
1:0	RW	01	R29 [1:0]: SWB_SWITCHING_FREQ SWB Output Regulator Switching Frequency ³ 00 = 500 KHz 01 = 750 KHz 10 = 1000 KHz 11 = 1250 KHz

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 169, “Register 0x4D”.
2. If required, the host must adjust this register first before issuing VR Enable command in Table 145, “Register 0x32” [7].
3. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 137 — Register 0x2A^{1,2}

R2A			
Bits	Attribute	Default	Description
7:6	RW	10	R2A [7:6]: SWC_MODE_SELECT Switch Node C Output Regulator Mode Selection 00 = Reserved 01 = Reserved 10 = COT; DCM (Constant on Time; Discontinuous Current Mode) 11 = COT; Forced CCM (Constant on Time; Continuous Current Mode)
5:4	RW	01	R2A [5:4]: SWC_SWITCHING_FREQ Switch Node C Output Regulator Switching Frequency 00 = 500 KHz 01 = 750 KHz 10 = 1000 KHz 11 = 1250 KHz
3:2	RW	10	R2A [3:2]: SWD_MODE_SELECT Switch Node D Output Regulator Mode Selection 00 = Reserved 01 = Reserved 10 = COT; DCM (Constant on Time; Discontinuous Current Mode) 11 = COT; Forced CCM (Constant on Time; Continuous Current Mode)
1:0	RW	01	R2A [1:0]: SWD_SWITCHING_FREQ Switch Node D Output Regulator Switching Frequency 00 = 500 KHz 01 = 750 KHz 10 = 1000 KHz 11 = 1250 KHz

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 170, “Register 0x4E”.
2. If required, the host must adjust this register first before issuing VR Enable command in Table 145, “Register 0x32” [7].

Table 138 — Register 0x2B^{1,2}

R2B			
Bits	Attribute	Default	Description
7:6	RW	01	R2B [7:6]: VOUT_1.8V_VOLTAGE_SETTING VOUT 1.8 V LDO Output Voltage Setting ³ 00 = 1.7 V 01 = 1.8 V 10 = 1.9 V 11 = Reserved
5	RW	0	R2B [5]: SWA_VOLTAGE_RANGE SWA Output Voltage Range Selection ⁴ 0 = Range: 800 mV to 1435 mV for SWA; 5 mV step size 1 = Range: 600 mV to 1235 mV for SWA; 5 mV step size
4	RW	0	R2B [4]: SWB_VOLTAGE_RANGE SWB Output Voltage Range Selection ^{5,6} 0 = Range: 800 mV to 1435 mV for SWB; 5 mV step size 1 = Range: 600 mV to 1235 mV for SWB; 5 mV step size
3	RW	0	R2B [3]: SWC_VOLTAGE_RANGE SWC Output Voltage Range Selection ⁷ 0 = Range: 800 mV to 1435 mV for SWC; 5 mV step size 1 = Range: 600 mV to 1235 mV for SWC; 5 mV step size
2:1	RW	01	R2B [2:1]: VOUT_1.0V_VOLTAGE_SETTING VOUT 1.0 V LDO Voltage Setting ⁸ 00 = 0.9 V 01 = 1.0 V 10 = 1.1 V 11 = 1.2 V
0	RW	0	R2B [0]: SWD_VOLTAGE_RANGE SWD Output Voltage Range Selection ⁹ 0 = Range: 1500 mV to 2135 mV for SWD; 5 mV step size 1 = Range: 2200 mV to 2835 mV for SWD; 5 mV step size

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 173, “Register 0x51”.
2. If required, the host must adjust this register first before issuing VR Enable command in Table 145, “Register 0x32” [7]. The host must also wait minimum of 5 μ s after the adjustment before issuing VR Enable command.
3. The VOUT_1.8V Power Good threshold in register Table 121, “Register 0x1A” [2] is always fixed regardless of the setting in this register.
4. Range & resolution selection applies to register Table 128, “Register 0x21,” [7:1].
5. Range & resolution selection applies to register Table 130, “Register 0x23,” [7:1].
6. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.
7. Range & resolution selection applies to register Table 132, “Register 0x25,” [7:1].
8. If required, the host must adjust this register one step at a time (0.1 V increment or decrement) to prevent false trigger of power good status and PWR_GOOD pin assertion. In other words, host should not increment or decrement 0.2 V or 0.3 V from its current setting.
9. Range & resolution selection applies to register Table 134, “Register 0x27,” [7:1].

Table 139 — Register 0x2C^{1,2}

R2C			
Bits	Attribute	Default	Description
7:5	RW	001	R2C [7:5]: SWA_OUTPUT_SOFT_START_TIME SWA Output Regulator Soft Start Time After VR Enable ³ 000 = 1 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms ... 111 = 14 ms
4	RV	0	R2C [4]: Reserved
3:1	RW	001	R2C [3:1]: SWB_OUTPUT_SOFT_START_TIME SWB Output Regulator Soft Start Time After VR Enable ^{3,4} 000 = 1 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms ... 111 = 14 ms
0	RV	0	R2C [0]: Reserved

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 178, “Register 0x5D”.
2. If required, the host must adjust this register first before issuing VR Enable command in Table 145, “Register 0x32” [7].
3. This is the time it takes for buck regulator to go from 0 V to steady state voltage (the actual voltage varies based on pre-programmed voltage).
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 140 — Register 0x2D^{1,2}

R2D			
Bits	Attribute	Default	Description
7:5	RW	001	R2D [7:5]: SWC_OUTPUT_SOFT_START_TIME SWC Output Regulator Soft Start Time After VR Enable ³ 000 = 1 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms ... 111 = 14 ms
4	RV	0	R2D [4]: Reserved
3:1	RW	001	R2D [3:1]: SWD_OUTPUT_SOFT_START_TIME SWD Output Regulator Soft Start Time After VR Enable ³ 000 = 1 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms ... 111 = 14 ms
0	RV	0	R2D [0]: Reserved

1. At first power on, this register is automatically configured identically by PMIC on its own as Table 179, “Register 0x5E”.
2. If required, the host must adjust this register first before issuing VR Enable command in Table 145, “Register 0x32” [7].
3. This is the time it takes for buck regulator to go from 0 V to steady state voltage (the actual voltage varies based on pre-programmed voltage)

Table 141 — Register 0x2E

R2E			
Bits	Attribute	Default	Description
7:3	RV	0	R2E [7:3]: Reserved
2:0	RW	100	R2E [2:0]: PMIC_SHUTDOWN_TEMPERATURE_THRESHOLD PMIC Shutdown Temperature Threshold 000 = PMIC Temperature $\geq 105^{\circ}\text{C}$ 001 = PMIC Temperature $\geq 115^{\circ}\text{C}$ 010 = PMIC Temperature $\geq 125^{\circ}\text{C}$ 011 = PMIC Temperature $\geq 135^{\circ}\text{C}$ 100 = PMIC Temperature $\geq 145^{\circ}\text{C}$ 101 = Reserved 110 = Reserved 111 = Reserved

Table 142 — Register 0x2F

R2F			
Bits	Attribute	Default	Description
7	RW	0	R2F [7]: VIN_MGMT_INPUT_SUPPLY_SWITCHOVER_THRESHOLD VIN_Mgmt Input Supply Switchover Voltage Threshold to VIN_Bulk Input Supply 0 = Vendor Specific ¹ 1 = Reserved
6	RW	0	R2F [6]: SWA_REGULATOR_CONTROL Disable SWA Regulator Output ^{2,3} 0 = Disable Switch Node A Output Regulator 1 = Enable Switch Node A Output Regulator
5	RW	0	R2F [5]: SWB_REGULATOR_CONTROL Disable SWB Regulator Output ^{2,3,4} 0 = Disable Switch Node B Output Regulator 1 = Enable Switch Node B Output Regulator
4	RW	0	R2F [4]: SWC_REGULATOR_CONTROL Disable SWC Regulator Output ^{2,3} 0 = Disable Switch Node C Output Regulator 1 = Enable Switch Node C Output Regulator
3	RW	0	R2F [3]: SWD_REGULATOR_CONTROL Disable SWD Regulator Output ^{2,3} 0 = Disable Switch Node D Output Regulator 1 = Enable Switch Node D Output Regulator
2	RW	0	R2F [2]: SECURE_MODE PMIC Mode Operation 0 = Secure Mode Operation ⁵ 1 = Programmable Mode Operation
1:0	RW	10	R2F [1:0]: MASK_BITS_REGISTER_CONTROL Mask Bits Register Control ⁶ 00 = Mask GSI_n Signal Only (PWR_GOOD Signal will assert) 01 = Mask PWR_GOOD Signal Only (GSI_n Signal will assert) 10 = Mask GSI_n and PWR_GOOD Signals (neither PWR_GOOD assert or GSI_n signal will assert) 11 = Reserved

1. The VIN_Mgmt input switchover voltage threshold range is vendor specific and can vary between 2.6 V to 2.9 V max.
2. This bit must be used only after power up sequence (after VR Enable command). At first power up, PMIC automatically updates the status of this register to '1' after VR Enable command. When VR Enable command is registered, the PMIC updates this register based on Power On Sequence Configuration (0 to 3) setting. If enabled in Power On Sequence Configuration 0 to 3 registers, only then, under programmable mode of operation, the PMIC's output regulator can be disabled by clearing this bit and they can be re-enabled again by setting this bit. The PMIC does not alter its Power Good output signal and keeps it asserted High. If any regulator is not enabled in Power on Sequence Configuration 0 to 3, it cannot be enabled using this register. For example, if only SWA is enabled and SWB, SWC & SWD is not enabled in Table 157, "Register 0x40" [7:0] to Table 160, "Register 0x43," [7:0] then only SWA can be disabled and then re-enabled again but SWB, SWC and SWD cannot be enabled using Table 142, "Register 0x2F" [6:3].
3. In programmable mode, after VR enable command, if any output regulators are disabled by clearing Table 142, "Register 0x2F" [6:3] and then if host issues VR Disable command or PMIC internally triggers VR Disable command, the PMIC keeps the disabled output regulator in Table 142, "Register 0x2F" [6:3] off and remaining output regulators are disabled by following the Power Off Sequence Configuration 0 to 3 settings.
4. Only applicable if Table 171, "Register 0x4F" [0] = '0'.
5. This bit must be configured before issuing VR Enable command. If this bit is configured to '0', when PMIC registers VR Enable command, all registers bits starting Table 116, "Register 0x15" to Table 142, "Register 0x2F", Table 145, "Register 0x32" in host region as well as Table 157, "Register 0x40" to Register 0x6F in the DIMM vendor region are secured and host cannot change unless the PMIC goes through power cycle.
6. Applies to Mask Registers Table 116, "Register 0x15" [7,5:0], Table 117, "Register 0x16" [7:0], Table 118, "Register 0x17" [7:2], Table 119, "Register 0x18" [7:0], Table 120, "Register 0x19" [4:2] when any one or more Mask registers are set to '1'. If all Mask registers are configured as '0', the setting in this register (Table 142, "Register 0x2F" [1:0]) does not matter.

Table 143 — Register 0x30

R30			
Bits	Attribute	Default	Description
7	RW	0	R30 [7]: ADC_ENABLE Enable ADC (Analog to Digital Conversion) 0 = Disable ¹ 1 = Enable
6:3	RW	0	R30 [6:3]: ADC_SELECT Input Selection for ADC Readout 0000 = SWA Output Voltage 0001 = SWB Output Voltage ² 0010 = SWC Output Voltage 0011 = SWD Output Voltage 0100 = Reserved 0101 = VIN_Bulk Input Voltage 0110 = VIN_Mgmt Input Voltage 0111 = VBIAS Output Voltage 1000 = VOUT_1.8V Output Voltage 1001 = VOUT_1.0V Output Voltage All other encodings are reserved.
2	RV	0	R30 [2]: Reserved
1:0	RW	0	R30 [1:0]: ADC_REGISTER_UPDATE_FREQUENCY ADC Current or Power Measurement Update Frequency ^{3,4} 00 = 1 ms 01 = 2 ms 10 = 4 ms 11 = 8 ms

1. Disables the ADC function completely. Applies to voltage readout in Table 144, “Register 0x31” [7:0] as well as current or power readout in Table 107, “Register 0x0C” [7:0], Table 108, “Register 0x0D” [5:0], Table 109, “Register 0x0E” [5:0] and Table 110, “Register 0x0F” [5:0]. Does not apply to thermal sensor temperature readout in Table 146, “Register 0x33” [7:5] as well as high temperature warning and critical temperature shutdown.
2. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’
3. For average output current or power measurement in registers Table 107, “Register 0x0C” [7:0], Table 108, “Register 0x0D” [5:0], Table 109, “Register 0x0E” [5:0] and Table 110, “Register 0x0F” [5:0].
4. This register represents how often the registers are updated. The internal sampling rate is vendor specific.

Table 144 — Register 0x31

R31			
Bits	Attribute	Default	Description
7:0	RO	0	R31 [7:0]: ADC_READ ADC Output Voltage Reading¹ (Applies to SW[A:D], VOUT_1.8V, VOUT_1.0V, VIN_Mgmt) 0000 0000 = Undefined 0000 0001 = 15 mV 0000 0010 = 30 mV .. 1111 1111 > = 3825 mV ADC Output Voltage Reading² (Applies to VIN_Bulk Input Voltage) 0000 0000 = Undefined 0000 0001 = 70 mV 0000 0010 = 140 mV .. 1111 1111 > = 17850 mV ADC Output Voltage Reading³ (Applies to VBias Output Voltage) 0000 0000 = Undefined 0000 0001 = 25 mV 0000 0010 = 50 mV .. 1111 1111 > = 6375 mV

1. Only valid when Table 143, “Register 0x30” [6:3] = ‘0000’ or ‘0001’ or ‘0010’ or ‘0011’ or ‘0110’ or ‘1000’ or ‘1001’.

2. Only valid when Table 143, “Register 0x30” [6:3] = ‘0101’.

3. Only valid when Table 143, “Register 0x30” [6:3] = ‘0111’.

Table 145 — Register 0x32

R32			
Bits	Attribute	Default	Description
7	RW	0	R32 [7]: VR_ENABLE PMIC Enable ^{1,2,3,4} 0 = PMIC Disable 1 = PMIC Enable
6	RO	0	R32 [6]: MANAGEMENT_INTERFACE_SELECTION PMIC Management Bus Interface Protocol Selection ⁵ 0 = I ² C Protocol (Max speed 1 MHz) 1 = I3C Basic Protocol
5	RW	1	R32 [5]: PWR_GOOD_IO_TYPE PMIC PWR_GOOD Output Signal Type 0 = Output Only 1 = Input & Output
4:3	RW	0	R32 [4:3]: PWR_GOOD_OUTPUT_SIGNAL_CONTROL PMIC PWR_GOOD Output Signal Control 0x = PMIC controls PWR_GOOD on its own based on internal status 10 = PWR_GOOD Output Low 11 = PWR_GOOD Output Float ⁶
2:0	RV	00	R32 [2:0]: Reserved

1. Host sets this bit at first power on. After this bit is set, the PMIC executes Power On Sequence configuration 0 (Table 157, “Register 0x40”) to Power On Sequence configuration 3 (Table 160, “Register 0x43,”) registers. At least one bit in Table 157, “Register 0x40” [6:3] must be set to ‘1’ to issue VR Enable command.
2. The host shall ensure that prior to issuing VR Enable command, there is no pending IBI interrupt (i.e Table 105, “Register 0x0A” [1] = ‘1’) status. After host issues VR Enable command, the PMIC may NACK any I²C or I3C Basic bus transaction by host until tPMIC_PWR_GOOD_OUT timing parameter is satisfied. The host shall not access any device specific registers or issue any CCCs until tPMIC_PWR_GOOD_OUT parameter is satisfied. The PMIC device may request for an IBI during power up sequence (i.e., during tPMIC_PWR_GOOD_OUT time) if there is any event.
3. After host issues VR Enable command, the PMIC may NACK any I2C or I3C Basic bus transactions by host until tPMIC_PWR_GOOD_OUT timing parameter is satisfied.
4. An exception to this register is applied. When PMIC is operating in secure mode of operation and PWR_GOOD input is low (If Table 145, “Register 0x32” [5] = ‘1’), the PMIC unlocks this register. The PMIC allows host to issue VR Enable command. After host issues VR Enable command, the PMIC re-locks this register.
5. This register is automatically updated when SETAASA CCC or RSTDAA CCC is registered by the PMIC device or when PMIC device goes through bus reset as described in Section 11.15 regardless of whether PMIC is in secure mode or programmable mode of operation. This register can be read by the Host through normal Read operation but it cannot be written with normal write operation either in I²C mode or I3C Basic mode of operation. When this register is updated, it takes in effect where there is a next START operation (i.e., after STOP operation).
6. When this encoding is set, the PMIC always floats the PWR_GOOD output signal even when there is an internal VR Disable command due to fault condition.

Table 146 — Register 0x33

R33			
Bits	Attribute	Default	Description
7:5	RO	0	R33 [7:5]: TEMPERATURE_MEASUREMENT PMIC Temperature ¹ 000 = $\leq 85^{\circ}\text{C}$ 001 = 85°C 010 = 95°C 011 = 105°C 100 = 115°C 101 = 125°C 110 = 135°C 111 = $> 140^{\circ}\text{C}$
4	RO	0	R33 [4]: VIN_MGMT_POWER_GOOD_STATUS_SWITCHOVER_MODE VIN_Mgmt Input Supply Power Good Status in Switchover Mode Only ² 0 = Power Not Good 1 = Power Good
3	RO	0	R33 [3]: VBias_OR_VIN_BULK_UNDER_VOLTAGE_LOCKOUT_STATUS VBias or VIN_Bulk Under Voltage Lockout Status ³ 0 = No Under Voltage Lockout 1 = Under Voltage Lockout
2	RO	0	R33 [2]: VOUT_1.0V_OUTPUT_POWER_GOOD_STATUS VOUT_1.0V LDO Output Power Good Status ⁴ 0 = Power Good 1 = Power Not Good
1:0	RV	0	R33 [1:0]: Reserved

1. The accuracy of the temperature readout code is $\pm 5^{\circ}\text{C}$.

2. This register has no meaning when PMIC is NOT in switchover mode. In switchover mode only, when PMIC detects VIN_Mgmt input supply from the platform, this bit is set to indicate that PMIC now recognizes valid VIN_Mgmt input power supply.

3. This register is set when VBias LDO output goes below vendor specific threshold or VIN_Bulk input goes below vendor specific threshold.

4. This register is set when VOUT_1.0V output drops below the threshold setting in register Table 121, "Register 0x1A" [0].

Table 147 — Register 0x34

R34			
Bits	Attribute	Default	Description¹
7	RW	0	R34 [7]: PEC_ENABLE Packet Error Code Enable ² (Applicable Only if R32 [6] = '1') 0 = Disable 1 = Enable
6	RO	0	R34 [6]: IBI_ENABLE In Band Interrupt Enable ³ (Applicable Only if R32 [6] = '1') 0 = Disable 1 = Enable
5	RW	0	R34 [5]: PARITY_DISABLE T Bit Parity Code Disable ² (Applicable Only if R32 [6] = '1'.) 0 = Enable 1 = Disable ⁴
4	RV	0	R34 [4]: Reserved
3:1	RW	111	R34 [3:1]: HID_CODE PMIC's 3-bit HID Code ⁵ 000 001 010 011 100 101 110 111
0	RV	0	R34 [0]: Reserved

1. The write (or update) transaction to this register must be followed by STOP operation to allow the PMIC device to update the setting.
2. This register is automatically updated when RSTDAA CCC is registered by the PMIC device or when PMIC device goes through bus reset as described in Section 11.15. This register can also be written or read by the Host through normal write/read operation either in I²C mode or I3C mode of operation.
3. This register is automatically updated when ENEC CCC or DISEC CCC or RSTDAA CCC is registered by the PMIC device or when PMIC device goes through bus reset as described in Section 11.15. This register can be read by the Host through normal Read operation but it cannot be written with normal write operation either in I²C mode or I3C mode of operation.
4. When Parity function is disabled, the PMIC simply ignores the "T" bit information from the Host. The host may actually choose to compute the parity and send that information in "T" bit or simply drive static low or high in "T" bit.
5. This register is updated when PMIC device receives SETHID CCC or when PMIC device goes through bus reset as described in Section 11.15.

Table 148 — Register 0x35

R35			
Bits	Attribute	Default	Description ^{1,2,3}
7	RW	0	R35 [7]: ERROR_INJECTION_ENABLE Error Injection Enable ⁴ 0 = Disable 1 = Enable
6:4	RW	0	R35 [6:4]: ERROR_INJECTION_RAIL_SELECTION Error Injection - Input Rail and Output Rail Selection ^{5,6} 000 = Undefined 001 = SWA Output Only 010 = SWB Output Only 011 = SWC Output Only 100 = SWD Output Only 101 = VIN_Bulk Input Only 110 = VIN_Mgmt Input Only 111 = Do Not Use
3	RW	0	R35 [3]: OVER_VOLTAGE_UNDER_VOLTAGE_SELECT Over Voltage or Under Voltage Selection for Bits [6:4] ⁷ 0 = Over Voltage 1 = Under Voltage ⁸
2:0	RW	0	R35[2:0]: MISC_ERROR_INJECTION_TYPE Miscellaneous Error Injection Type ⁹ 000 = Undefined 001 = VIN_Mgmt to VIN_Bulk Switchover 010 = Critical Temperature Shutdown 011 = High Temperature Warning Threshold 100 = VOUT_1.8V LDO Power Good 101 = High Current Consumption Warning ¹⁰ 110 = Reserved 111 = Current Limiter Warning ¹⁰

1. This register Table 148, “Register 0x35” [7:0] is only applicable in programmable mode after VR Enable command is executed by the PMIC.
2. When Error Injection Function is enabled, error log registers (Table 99, “Register 0x04” to Table 102, “Register 0x07”) are also updated (logged) when PMIC simulates VR Disable command. The host can erase the error log registers (Table 99, “Register 0x04” to Table 99, “Register 0x04”) by writing 0x74 to Table 151, “Register 0x39”.
3. To exit from Error Injection Mode, the PMIC must go through power cycle of both VIN_Bulk and VIN_Mgmt input supply.
4. The PMIC must be in programmable mode.
5. This register Table 148, “Register 0x35” [6:4] is only applicable if Table 148, “Register 0x35” [2:0] is ‘000’. Any value other than ‘000’ in both Table 148, “Register 0x35” [6:4] and Table 148, “Register 0x35” [2:0] is considered an illegal setting and PMIC operation is not guaranteed.
6. Only applicable if the selected output rails are enabled via VR Enable command. If dual phase regulator is selected, use SWA encoding to inject the error. Register bit [3] selects either over voltage or under voltage condition for the setting selected in this register.
7. This register Table 148, “Register 0x35” [3] is only applicable if bits [6:4] is anything other than ‘000’.
8. The under voltage selection only applies to SWx output rails and VIN_Bulk input. Does not apply to VIN_Mgmt input.
9. This register Table 148, “Register 0x35” [2:0] is only applicable if Table 148, “Register 0x35” [6:3] is ‘0000’. Any value other than ‘000’ in both Table 148, “Register 0x35” [6:4] and Table 148, “Register 0x35” [2:0] is considered an illegal setting and PMIC operation is not guaranteed.
10. Applies to all enabled SWx output regulators at the same time.

Table 149 — Register 0x37

R37			
Bits	Attribute	Default	Description
7:0	WO	-	R37 [7:0]: DIMM_VENDOR_MEMORY_REGION_PASSWORD_LOWER_BYTE DIMM Vendor Memory Region (0x40 - 0x6F) Password - Lower Byte [7:0] = Code

Table 150 — Register 0x38

R38			
Bits	Attribute	Default	Description
7:0	WO	-	R38 [7:0]: DIMM_VENDOR_MEMORY_REGION_PASSWORD_UPPER_BYTE DIMM Vendor Memory Region (0x40 - 0x6F) Password - Upper Byte [7:0] = Code

Table 151 — Register 0x39

R39			
Bits	Attribute	Default	Description
7:0	RW	0x00	R39 Codes: Host Region Codes: 0x74: Clear Registers R04 to R07, Erase MTP memory for R04 Register. DIMM Vendor Region (0x40 to 0x6F) Write Codes: 0x40: Unlock DIMM Vendor Region. Password needs to be present in R37 & R38 registers. 0x00: Lock DIMM Vendor Region. 0x80: Burn DIMM Vendor Region Password. New password needs to be present in R37 & R38. 0x81: Burn DIMM Vendor Region - 0x40 to 0x4F 0x82: Burn DIMM Vendor Region - 0x50 to 0x5F 0x85: Burn DIMM Vendor Region - 0x60 to 0x6F DIMM Vendor Region (0x40 to 0x6F) Read Codes: 0x5A: Burning is complete in DIMM Vendor region.

Table 152 — Register 0x3A

R3A			
Bits	Attribute	Default	Description¹
7	RV	0	R3A [7]: Reserved
6	RW	0	R3A [6]: DEFAULT_READ_ADDRESS_POINTER_ENABLE Enable Default Address Read Pointer when PMIC sees STOP operation 0 = Disable Default Address Pointer (address pointer is set by Host) ² 1 = Enable Default Address Pointer; Address selected by register bits [5:4] ³
5:4	RW	0	R3A [5:4]: DEFAULT_READ_STARTING_ADDRESS Default Read Address Pointer Selection when PMIC sees STOP operation ⁴ 00 = R08 01 = R0C 10 = Reserved 11 = Reserved
3:2	RW	0	R3A [3:2]: BURST_LENGTH_FOR_READ_DEFAULT_ADDR_POINTER Burst Length (# of Bytes) to be transferred for Read Default Address Pointer Mode ⁵ 00 = 2 Bytes 01 = 4 Bytes 10 = Reserved 11 = 16 Bytes
1:0	RV	0	R3A [1:0]: Reserved

1. The write (or update) transaction to this register must be followed by STOP operation to allow the PMIC device to update the setting.
2. The register setting in R3A [5:4] is a don't care.
3. This mode is only allowed when PEC function is disabled (i.e register R34 [7] = '0').
4. This register is only applicable if R3A [6] = '1'.
5. This register is only applicable if R3A [6] = '1' and R34 [7] = '1'.

Table 153 — Register 0x3B

R3B			
Bits	Attribute	Default	Description
7:6	ROE	0	R3B [7:6]: PMIC_PART_CAPABILITY_EXT PMIC Current Capability Extension. This register bits [7:6] and bit [0] provides 3 bit encoding as following: 000 = Small PMIC (Low Current) 001 = Big PMIC (High Current) 01x = Extreme PMIC (Highest Current) All other encodings are reserved.
5:4	ROE	-	R3B [5:4]: REVISION_ID_MAJOR_STEPPING Major Revision Stepping 00 = Revision 1 01 = Revision 2 10 = Revision 3 11 = Revision 4
3:1	ROE	-	R3B [3:1]: REVISION_ID_MINOR_STEPPING Minor Revision Stepping 000 = Revision 0 001 = Revision 1 010 = Revision 2 011 = Revision 3 All other encodings are reserved.
0	ROE	-	R3B [0]: PMIC_PART_CAPABILITY PMIC Current Capability. See also bits [7:6] definition. 0 = Small PMIC (Low Current) 1 = Big PMIC (High Current)

Table 154 — Register 0x3C

R3C			
Bits	Attribute	Default	Description
7:0	ROE	-	R3C [7:0]: VENDOR_ID_BYTE0 Vendor Identification Register Byte 0.

Table 155 — Register 0x3D

R3D			
Bits	Attribute	Default	Description
7:0	ROE	-	R3D [7:0]: VENDOR_ID_BYTE1 Vendor Identification Register Byte 1.

14.3.6 DIMM Vendor Region Registers

Table 156 — DIMM Vendor Region - Register Map

Register	Attribute	Description
Table 157, "Register 0x40"	RWPE	R40 [7:0] Power On Sequence Config 0
Table 158, "Register 0x41,"	RWPE	R41 [7:0] Power On Sequence Config 1
Table 159, "Register 0x42,"	RWPE	R42 [7:0] Power On Sequence Config 2
Table 160, "Register 0x43,"	RWPE	R43 [7:0] Power On Sequence Config 3
0x44	RV	R44 [7:0] Reserved
Table 161, "Register 0x45"	RWPE	R45 [7:1] SWA Voltage Setting R45 [0] SWA Power Good Low Side Threshold
Table 162, "Register 0x46"	RWPE	R46 [7:6] SWA Power Good High Side Threshold R46 [5:4] SWA Over Voltage Threshold R46 [3:2] SWA Under Voltage Lockout Threshold R46 [1:0] SWA Soft Stop Time
Table 163, "Register 0x47"	RWPE	R47 [7:1] SWB Voltage Setting R47 [0] SWB Power Good Low Side Threshold
Table 164, "Register 0x48"	RWPE	R48 [7:6] SWB Power Good High Side Threshold R48 [5:4] SWB Over Voltage Threshold R48 [3:2] SWB Under Voltage Lockout Threshold R48 [1:0] SWB Soft Stop Time
Table 165, "Register 0x49"	RWPE	R49 [7:1] SWC Voltage Setting R49 [0] SWC Power Good Low Side Threshold
Table 166, "Register 0x4A"	RWPE	R4A [7:6] SWC Power Good High Side Threshold R4A [5:4] SWC Over Voltage Threshold R4A [3:2] SWC Under Voltage Lockout Threshold R4A [1:0] SWC Soft Stop Time
Table 167, "Register 0x4B"	RWPE	R4B [7:1] SWD Voltage Setting R4B [0] SWD Power Good Low Side Threshold
Table 168, "Register 0x4C"	RWPE	R4C [7:6] SWD Power Good High Side Threshold R4C [5:4] SWD Over Voltage Threshold R4C [3:2] SWD Under Voltage Lockout Threshold R4C [1:0] SWD Soft Stop Time
Table 169, "Register 0x4D"	RWPE	R4D [7:6] SWA Mode Select R4D [5:4] SWA Switching Frequency R4D [3:2] SWB Mode Select R4D [1:0] SWB Switching Frequency
Table 170, "Register 0x4E"	RWPE	R4E [7:6] SWC Mode Select R4E [5:4] SWC Switching Frequency R4E [3:2] SWD Mode Select R4E [1:0] SWD Switching Frequency
Table 171, "Register 0x4F"	RWPE	R4F [7] Output Regulator Disable Contro R4F [6:5] Reserved R4F [4] SWA Remote Sensing Scheme R4F [3] Reserved R4F [2] SWC Remote Sensing Scheme R4F [1] SWD Remote Sensing Scheme R4F [0] SWA, SWB Single or Dual Phase Regulator Select
Table 172, "Register 0x50"	RWPE	R50 [7:6] SWA Current Limit Warning Threshold R50 [5:4] SWB Current Limit Warning Threshold R50 [3:2] SWC Current Limit Warning Threshold R50 [1:0] SWD Current Limit Warning Threshold

Table 156 — DIMM Vendor Region - Register Map

Register	Attribute	Description
Table 173, “Register 0x51”	RWPE	R51 [7:6] VOUT_1.8V LDO Setting R51 [5:3] Voltage Range Selection for SWA, SWB and SWC R51 [2:1] VOUT_1.0V LDO Setting R51 [0] Voltage Range Selection for SWD
0x52 to 0x57	RV	R52 [7:0] to R57 [7:0] Reserved
Table 174, “Register 0x58”	RWPE	R58 [7:0] Power Off Sequence Config 0
Table 175, “Register 0x59,”	RWPE	R59 [7:0] Power Off Sequence Config 1
Table 176, “Register 0x5A,”	RWPE	R5A [7:0] Power Off Sequence Config 2
Table 177, “Register 0x5B,”	RWPE	R5B [7:0] Power Off Sequence Config 3
0x5C	RV	R5C [7:0] Reserved
Table 178, “Register 0x5D”	RWPE	R5D [7:5] SWA Soft Start Time R5D [4] Reserved R5D [3:1] SWB Soft Start Time R5D [0] Reserved
Table 179, “Register 0x5E”	RWPE	R5E [7:5] SWC Soft Start Time R5E [4] Reserved R5E [3:1] SWD Soft Start Time R5E [0] Reserved
0x5F to 0x6F	RV	R5F [7:0] to R6F [7:0] Reserved

Table 157 — Register 0x40¹

R40			
Bits	Attribute	Default	Description
7	RWPE	0	R40 [7]: POWER_ON_SEQUENCE_CONFIG0 PMIC Power On Sequence Config 0 ² 0 = Do Not Execute Config 0 1 = Execute Config 0
6	RWPE	0	R40 [6]: POWER_ON_SEQUENCE_CONFIG0_SWA_ENABLE Enable SWA Output Regulator. 0 = Disable SWA Output Regulator 1 = Enable SWA Output Regulator
5	RWPE	0	R40 [5]: POWER_ON_SEQUENCE_CONFIG0_SWB_ENABLE Enable SWB Output Regulator. ³ 0 = Disable SWB Output Regulator 1 = Enable SWB Output Regulator
4	RWPE	0	R40 [4]: POWER_ON_SEQUENCE_CONFIG0_SWC_ENABLE Enable SWC Output Regulator. 0 = Disable SWC Output Regulator 1 = Enable SWC Output Regulator
3	RWPE	0	R40 [3]: POWER_ON_SEQUENCE_CONFIG0_SWD_ENABLE Enable SWD Output Regulator. 0 = Disable SWD Output Regulator 1 = Enable SWD Output Regulator
2:0	RWPE	001	R40 [2:0]: POWER_ON_SEQUENCE_CONFIG0_IDLE Idle time after Power On Sequence Config 0 ⁴ 000 = 0 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms 100 = 8 ms 101 = 10 ms 110 = 12 ms 111 = 24 ms

1. If more than one configuration register is used for power on sequence, first register must start at Table 157, “Register 0x40” and it must go in sequential order to Table 160, “Register 0x43,” to turn on all desired regulators. In other words, there must not be any gap of the register that is used for power on sequence.
2. If bit [7] = ‘0’, bits [6:3] must be programmed as ‘0000’. If bit [7] = ‘1’, at least one of the bits [6:3] must be programmed as ‘1’.
3. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’. This bit is a don’t care when Table 171, “Register 0x4F” [0] = ‘1’.
4. Idle time is the additional time after soft-start time expires. The PMIC waits sum of Soft Start time and Idle time before it executes the next power on sequence configuration register. If more than one regulators are enabled, the PMIC uses the largest value of the soft start time among the regulators that are enabled in this configuration register.

Table 158 — Register 0x41^{1,2}

R41			
Bits	Attribute	Default	Description
7	RWPE	0	R41 [7]: POWER_ON_SEQUENCE_CONFIG1 PMIC Power On Sequence Config 1 0 = Do Not Execute Config ³ 1 = Execute Config 1
6	RWPE	0	R41 [6]: POWER_ON_SEQUENCE_CONFIG1_SWA_ENABLE Enable SWA Output Regulator. 0 = Disable SWA Output Regulator 1 = Enable SWA Output Regulator
5	RWPE	0	R41 [5]: POWER_ON_SEQUENCE_CONFIG1_SWB_ENABLE Enable SWB Output Regulator. ⁴ 0 = Disable SWB Output Regulator 1 = Enable SWB Output Regulator
4	RWPE	0	R41 [4]: POWER_ON_SEQUENCE_CONFIG1_SWC_ENABLE Enable Switch Node C Output Regulator. 0 = Disable SWC Output Regulator 1 = Enable SWC Output Regulator
3	RWPE	0	R41 [3]: POWER_ON_SEQUENCE_CONFIG1_SWD_ENABLE Enable Switch Node D Output Regulator. 0 = Disable SWD Output Regulator 1 = Enable SWD Output Regulator
2:0	RWPE	001	R41 [2:0]: POWER_ON_SEQUENCE_CONFIG1_IDLE Idle time after Power On Sequence Config 1 ⁵ 000 = 0 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms 100 = 8 ms 101 = 10 ms 110 = 12 ms 111 = 24 ms

1. If any regulators are enabled in Table 157, “Register 0x40” [6:3], those regulators must be configured as ‘1’ in this sequence.
2. If more than one configuration register is used for power on sequence, first register must start at Table 157, “Register 0x40” and it must go in sequential order to Table 160, “Register 0x43,” to turn on all desired regulators. In other words, there must not be any gap of the register that is used for power on sequence.
3. If bit [7] = ‘0’, bits [6:3] must be programmed as ‘0000’. If bit [7] = ‘1’, at least one of the bits [6:3] must be programmed as ‘1’.
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’. This bit is a don’t care when Table 171, “Register 0x4F” [0] = ‘1’.
5. Idle time is the additional time after soft-start time expires. The PMIC waits sum of Soft Start time and Idle time before it executes the next power on sequence configuration register. If more than one regulators are enabled, the PMIC uses the largest value of the soft start time among the regulators that are enabled in this configuration register.

Table 159 — Register 0x42^{1,2}

R42			
Bits	Attribute	Default	Description
7	RWPE	0	R42 [7]: POWER_ON_SEQUENCE_CONFIG2 PMIC Power On Sequence Config 2 ³ 0 = Do Not Execute Config 2 1 = Execute Config 2
6	RWPE	0	R42 [6]: POWER_ON_SEQUENCE_CONFIG2_SWA_ENABLE Enable Switch Node A Output Regulator. 0 = Disable Switch Node A Output Regulator 1 = Enable Switch Node A Output Regulator
5	RWPE	0	R42 [5]: POWER_ON_SEQUENCE_CONFIG2_SWB_ENABLE Enable Switch Node B Output Regulator. ⁴ 0 = Disable Switch Node B Output Regulator 1 = Enable Switch Node B Output Regulator
4	RWPE	0	R42 [4]: POWER_ON_SEQUENCE_CONFIG2_SWC_ENABLE Enable Switch Node C Output Regulator. 0 = Disable Switch Node C Output Regulator 1 = Enable Switch Node C Output Regulator
3	RWPE	0	R42 [3]: POWER_ON_SEQUENCE_CONFIG2_SWD_ENABLE Enable Switch Node D Output Regulator. 0 = Disable Switch Node D Output Regulator 1 = Enable Switch Node D Output Regulator
2:0	RWPE	001	R42 [2:0]: POWER_ON_SEQUENCE_CONFIG2_IDLE Idle time after Power On Sequence Config 2 ⁵ 000 = 0 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms 100 = 8 ms 101 = 10 ms 110 = 12 ms 111 = 24 ms

1. If any regulators are enabled in Table 157, “Register 0x40” [6:3] and Table 158, “Register 0x41,” [6:3], those regulators must be configured as ‘1’ in this sequence.
2. If more than one configuration register is used for power on sequence, first register must start at Table 157, “Register 0x40” and it must go in sequential order to Table 160, “Register 0x43,” to turn on all desired regulators. In other words, there must not be any gap of the register that is used for power on sequence.
3. If bit [7] = ‘0’, bits [6:3] must be programmed as ‘0000’. If bit [7] = ‘1’, at least one of the bits [6:3] must be programmed as ‘1’.
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’. This bit is a don’t care when Table 171, “Register 0x4F” [0] = ‘1’.
5. Idle time is the additional time after soft-start time expires. The PMIC waits sum of Soft Start time and Idle time before it executes the next power on sequence configuration register. If more than one regulators are enabled, the PMIC uses the largest value of the soft start time among the regulators that are enabled in this configuration register.

Table 160 — Register 0x43^{1,2}

R43			
Bits	Attribute	Default	Description
7	RWPE	0	R43 [7]: POWER_ON_SEQUENCE_CONFIG3 PMIC Power On Sequence Config 3 0 = Do Not Execute Config 3 ³ 1 = Execute Config 3
6	RWPE	0	R43 [6]: POWER_ON_SEQUENCE_CONFIG3_SWA_ENABLE Enable Switch Node A Output Regulator. 0 = Disable Switch Node A Output Regulator 1 = Enable Switch Node A Output Regulator
5	RWPE	0	R43 [5]: POWER_ON_SEQUENCE_CONFIG3_SWB_ENABLE Enable Switch Node B Output Regulator. ⁴ 0 = Disable Switch Node B Output Regulator 1 = Enable Switch Node B Output Regulator
4	RWPE	0	R43 [4]: POWER_ON_SEQUENCE_CONFIG3_SWC_ENABLE Enable Switch Node C Output Regulator. 0 = Disable Switch Node C Output Regulator 1 = Enable Switch Node C Output Regulator
3	RWPE	0	R43 [3]: POWER_ON_SEQUENCE_CONFIG3_SWD_ENABLE Enable Switch Node D Output Regulator. 0 = Disable Switch Node D Output Regulator 1 = Enable Switch Node D Output Regulator
2:0	RWPE	001	R43 [2:0]: POWER_ON_SEQUENCE_CONFIG3_IDLE Idle time after Power On Sequence Config 3 ⁵ 000 = 0 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms 100 = 8 ms 101 = 10 ms 110 = 12 ms 111 = 24 ms

1. If any regulators are enabled in Table 157, “Register 0x40” [6:3], Table 158, “Register 0x41,” [6:3] and Table 159, “Register 0x42,” [6:3], those regulators must be configured as ‘1’ in this sequence.
2. If more than one configuration register is used for power on sequence, first register must start at Table 157, “Register 0x40” and it must go in sequential order to Table 160, “Register 0x43,” to turn on all desired regulators. In other words, there must not be any gap of the register that is used for power on sequence.
3. If bit [7] = ‘0’, bits [6:3] must be programmed as ‘0000’. If bit [7] = ‘1’, at least one of the bits [6:3] must be programmed as ‘1’.
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’. This bit is a don’t care when Table 171, “Register 0x4F” [0] = ‘1’.
5. Idle time is the additional time after soft-start time expires. The PMIC waits sum of Soft Start time and Idle time before it executes the next event. If more than one regulators are enabled, the PMIC uses the largest value of the soft start time among the regulators that are enabled in this configuration register..

Table 161 — Register 0x45

R45			
Bits	Attribute	Default	Description
7:1	RWPE	011 1100	R45 [7:1]: SWA_VOLTAGE_SETTING SWA Output Regulator Voltage Setting ¹ 000 0000 = 800 mV ² or 600 mV ³ 000 0001 = 805 mV or 605 mV 000 0010 = 810 mV or 610 mV ... 011 1100 = 1100 mV or 900 mV ... 111 1101 = 1425 mV or 1225 mV 111 1110 = 1430 mV or 1230 mV 111 1111 = 1435 mV or 1235 mV
0	RWPE	0	R45 [1:0]: SWA_POWER_GOOD_THRESHOLD_LOW_SIDE_VOLTAGE_SETTING SWA Output Threshold Low Side Voltage For Power Good Status 0 = - 5% from the setting in Table 161, “Register 0x45” [7:1] 1 = - 7.5% from the setting in Table 161, “Register 0x45” [7:1]

1. PMIC guarantees efficiency spec and all electrical characteristics spec within a range of 1050 mV to 1160 mV.

2. Table 173, “Register 0x51” [5] = ‘0’; 5 mV step size.

3. Table 173, “Register 0x51” [5] = ‘1’; 5 mV step size.

Table 162 — Register 0x46

R46			
Bits	Attribute	Default	Description
7:6	RWPE	01	R46 [7:6]: SWA_POWER_GOOD_THRESHOLD_HIGH_SIDE_VOLTAGE_SETTING Switch Node A Output Threshold High Side Voltage For Power Good Status 00 = +5% from the setting in Table 161, “Register 0x45” [7:1] 01 = +7.5% from the setting in Table 161, “Register 0x45” [7:1] 10 = +10% from the setting in Table 161, “Register 0x45” [7:1] 11 = Reserved
5:4	RWPE	10	R46 [5:4]: SWA_OVER_VOLTAGE_THRESHOLD_SETTING Switch Node A Output Regulator Threshold For Over Voltage Status ¹ 00 = +7.5% from the setting in Table 161, “Register 0x45” [7:1] 01 = +10% from the setting in Table 161, “Register 0x45” [7:1] 10 = +12.5% from the setting in Table 161, “Register 0x45” [7:1] 11 = Reserved
3:2	RWPE	00	R46 [3:2]: SWA_UNDER_VOLTAGE_LOCKOUT_THRESHOLD_SETTING Switch Node A Output Regulator Threshold For Under Voltage Lockout Status 00 = -10% from the setting in Table 161, “Register 0x45” [7:1] 01 = -12.5% from the setting in Table 161, “Register 0x45” [7:1] 10 = Reserved 11 = Reserved
1:0	RWPE	00	R46 [1:0]: SWA_OUTPUT_SOFT_STOP_TIME SWA Output Regulator Soft Stop Time After VR Disable ² 00 = 0.5 ms 01 = 1 ms 10 = 2 ms 11 = 4 ms

1. The setting for the Over Voltage must be higher than Power Good High Side Voltage threshold in Table 162, “Register 0x46”[7:6].
2. This is the time it takes for buck regulator to go from steady state voltage to 0 V.

Table 163 — Register 0x47

R47			
Bits	Attribute	Default	Description ¹
7:1	RWPE	011 1100	R47 [7:1]: SWB_VOLTAGE_SETTING SWB Output Regulator Voltage Setting ² 000 0000 = 800 mV ³ or 600 mV ⁴ 000 0001 = 805 mV or 605 mV 000 0010 = 810 mV or 610 mV ... 011 1100 = 1100 mV or 900 mV ... 111 1101 = 1425 mV or 1225 mV 111 1110 = 1430 mV or 1230 mV 111 1111 = 1435 mV or 1235 mV
0	RWPE	0	R47 [1]: SWB_POWER_GOOD_THRESHOLD_LOW_SIDE_VOLTAGE_SETTING SWB Output Threshold Low Side Voltage For Power Good Status 0 = -5% from the setting in Table 163, “Register 0x47” [7:1] 1 = -7.5% from the setting in Table 163, “Register 0x47” [7:1]

1. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

2. PMIC guarantees efficiency spec and all electrical characteristics spec within a range of 1050 mV to 1160 mV.

3. Table 173, “Register 0x51” [4] = ‘0’; 5mV step size.

4. Table 173, “Register 0x51” [4] = ‘1’; 5 mV step size.

Table 164 — Register 0x48

R48			
Bits	Attribute	Default	Description ¹
7:6	RWPE	01	R48 [7:6]: SWB_POWER_GOOD_THRESHOLD_HIGH_SIDE_VOLTAGE_SETTING SWB Output Threshold High Side Voltage For Power Good Status 00 = +5% from the setting in Table 163, “Register 0x47” [7:1] 01 = +7.5% from the setting in Table 163, “Register 0x47” [7:1] 10 = +10% from the setting in Table 163, “Register 0x47” [7:1] 11 = Reserved
5:4	RWPE	10	R48 [5:4]: SWB_OVER_VOLTAGE_THRESHOLD_SETTING SWB Output Regulator Threshold For Over Voltage Status ² 00 = +7.5% from the setting in Table 163, “Register 0x47” [7:1] 01 = +10% from the setting in Table 163, “Register 0x47” [7:1] 10 = +12.5% from the setting in Table 163, “Register 0x47” [7:1] 11 = Reserved
3:2	RWPE	00	R48 [3:2]: SWB_UNDER_VOLTAGE_LOCKOUT_THRESHOLD_SETTING SWB Output Regulator Threshold For Under Voltage Lockout Status 00 = -10% from the setting in Table 163, “Register 0x47” [7:1] 01 = -12.5% from the setting in Table 163, “Register 0x47” [7:1] 10 = Reserved 11 = Reserved
1:0	RWPE	00	R48 [1:0]: SWB_OUTPUT_SOFT_STOP_TIME SWB Output Regulator Soft Stop Time After VR Disable ³ 00 = 0.5 ms 01 = 1 ms 10 = 2 ms 11 = 4 ms

1. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

2. The setting for the Over Voltage must be higher than Power Good High Side Voltage threshold in Table 164, “Register 0x48”[7:6].

3. This is the time it takes for buck regulator to go from steady state voltage to 0 V.

Table 165 — Register 0x49

R49			
Bits	Attribute	Default	Description
7:1	RWPE	011 1100	R49 [7:1]: SWC_VOLTAGE_SETTING SWC Output Regulator Voltage Setting ¹ 000 0000 = 800 mV ² or 600 mV ³ 000 0001 = 805 mV or 605 mV 000 0010 = 810 mV or 610 mV ... 011 1100 = 1100 mV or 900 mV ... 111 1101 = 1425 mV or 1225 mV 111 1110 = 1430 mV or 1230 mV 111 1111 = 1435 mV or 1235 mV
0	RWPE	0	R49 [0]: SWC_POWER_GOOD_THRESHOLD_LOW_SIDE_VOLTAGE_SETTING Switch Node C Output Threshold Low Side Voltage For Power Good Status 0 = -5% from the setting in Table 165, “Register 0x49” [7:1] 1 = -7.5 from the setting in Table 165, “Register 0x49” [7:1]

1. PMIC guarantees efficiency spec and all electrical characteristics spec within a range of 1050 mV to 1160 mV.

2. Table 173, “Register 0x51” [3] = ‘0’; 5 mV step size.

3. Table 173, “Register 0x51” [3] = ‘1’; 5 mV step size.

Table 166 — Register 0x4A

R4A			
Bits	Attribute	Default	Description
7:6	RWPE	01	R4A [7:6]: SWC_POWER_GOOD_THRESHOLD_HIGH_SIDE_VOLTAGE_SETTING SWC Output Threshold High Side Voltage For Power Good Status 00 = +5% from the setting in Table 165, “Register 0x49” [7:1] 01 = +7.5% from the setting in Table 165, “Register 0x49” [7:1] 10 = +10% from the setting in Table 165, “Register 0x49” [7:1] 11 = Reserved
5:4	RWPE	10	R4A [5:4]: SWC_OVER_VOLTAGE_THRESHOLD_SETTING SWC Output Regulator Threshold For Over Voltage Status ¹ 00 = +7.5% from the setting in Table 165, “Register 0x49” [7:1] 01 = +10% from the setting in Table 165, “Register 0x49” [7:1] 10 = +12.5% from the setting in Table 165, “Register 0x49” [7:1] 11 = Reserved
3:2	RWPE	00	R4A [3:2]: SWC_UNDER_VOLTAGE_LOCKOUT_THRESHOLD_SETTING SWC Output Regulator Threshold For Under Voltage Lockout Status 00 = -10% from the setting in Table 165, “Register 0x49” [7:1] 01 = -12.5% from the setting in Table 165, “Register 0x49” [7:1] 10 = Reserved 11 = Reserved
1:0	RWPE	00	R4A [1:0]: SWC_OUTPUT_SOFT_STOP_TIME SWC Output Regulator Soft Stop Time After VR Disable ² 00 = 0.5 ms 01 = 1 ms 10 = 2 ms 11 = 4 ms

1. The setting for the Over Voltage must be higher than Power Good High Side Voltage threshold in Table 166, “Register 0x4A”[7:6].
2. This is the time it takes for buck regulator to go from steady state voltage to 0 V.

Table 167 — Register 0x4B

R4B			
Bits	Attribute	Default	Description
7:1	RWPE	011 1100	R4B [7:1]: SWD_VOLTAGE_SETTING SWD Output Regulator Voltage Setting ¹ 000 0000 = 1500 mV ² or 2200 mV ³ 000 0001 = 1505 mV or 2205 mV 000 0010 = 1510 mV or 2210 mV ... 011 1100 = 1800 mV or 2500 mV ... 111 1101 = 2125 mV or 2825 mV 111 1110 = 2130 mV or 2830 mV 111 1111 = 2135 mV or 2835 mV
0	RWPE	0	R4B [0]: SWD_POWER_GOOD_THRESHOLD_LOW_SIDE_VOLTAGE_SETTING SWD Output Threshold Low Side Voltage For Power Good Status 0 = -5% from the setting in Table 167, “Register 0x4B” [7:1] 1 = -7.5% from the setting in Table 167, “Register 0x4B” [7:1]

1. PMIC guarantees efficiency spec and all electrical characteristics spec within a range of 1750 mV to 1850 mV.

2. Table 173, “Register 0x51” [0] = ‘0’; 5 mV step size.

3. Table 173, “Register 0x51” [0] = ‘1’; 5 mV step size.

Table 168 — Register 0x4C

R4C			
Bits	Attribute	Default	Description
7:6	RWPE	01	R4C [7:6]: SWD_POWER_GOOD_THRESHOLD_HIGH_SIDE_VOLTAGE_SETTING SWD Output Threshold High Side Voltage For Power Good Status 00 = +5% from the setting in Table 167, “Register 0x4B” [7:1] 01 = +7.5% from the setting in Table 167, “Register 0x4B” [7:1] 10 = Reserved 11 = Reserved
5:4	RWPE	10	R4C [5:4]: SWD_OVER_VOLTAGE_THRESHOLD_SETTING SWD Output Regulator Threshold For Over Voltage Status ¹ 00 = +7.5% from the setting in Table 167, “Register 0x4B” [7:1] 01 = +10% from the setting in Table 167, “Register 0x4B” [7:1] 10 = +12.5% from the setting in Table 167, “Register 0x4B” [7:1] 11 = Reserved
3:2	RWPE	00	R4C [3:2]: SWD_UNDER_VOLTAGE_LOCKOUT_THRESHOLD_SETTING SWD Output Regulator Threshold For Under Voltage Lockout Status 00 = -10% from the setting in Table 167, “Register 0x4B” [7:1] 01 = -12.5% from the setting in Table 167, “Register 0x4B” [7:1] 10 = Reserved 11 = Reserved
1:0	RWPE	00	R4C [1:0]: SWD_OUTPUT_SOFT_STOP_TIME SWD Output Regulator Soft Stop Time After VR Disable ² 00 = 1 ms 01 = 2 ms 10 = 4 ms 11 = 8 ms

1. The setting for the Over Voltage must be higher than Power Good High Side Voltage threshold in Table 168, “Register 0x4C” [7:6].
2. This is the time it takes for buck regulator to go from steady state voltage to 0 V.

Table 169 — Register 0x4D

R4D			
Bits	Attribute	Default	Description
7:6	RWPE	10	R4D [7:6]: SWA_MODE_SELECT Switch Node A Output Regulator Mode Selection 00 = Reserved 01 = Reserved 10 = COT; DCM (Constant on Time; Discontinuous Current Mode) 11 = COT; CCM (Constant on Time; Continuous Current Mode)
5:4	RWPE	01	R4D [5:4]: SWA_SWITCHING_FREQ Switch Node A Output Regulator Switching Frequency 00 = 500 KHz 01 = 750 KHz 10 = 1000 KHz 11 = 1250 KHz
3:2	RWPE	10	R4D [1:0]: SWB_MODE_SELECT Switch Node B Output Regulator Mode Selection ¹ 00 = Reserved 01 = Reserved 10 = COT; DCM (Constant on Time; Discontinuous Current Mode) 11 = COT; CCM (Constant on Time; Continuous Current Mode)
1:0	RWPE	01	R4D [1:0]: SWB_SWITCHING_FREQ Switch Node B Output Regulator Switching Frequency ¹ 00 = 500 KHz 01 = 750 KHz 10 = 1000 KHz 11 = 1250 KHz

1. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 170 — Register 0x4E

R4E			
Bits	Attribute	Default	Description
7:6	RWPE	10	R4E [7:6]: SWC_MODE_SELECT Switch Node C Output Regulator Mode Selection 00 = Reserved 01 = Reserved 10 = COT; DCM (Constant on Time; Discontinuous Current Mode) 11 = COT; CCM (Constant on Time; Continuous Current Mode)
5:4	RWPE	01	R4E [5:4]: SWC_SWITCHING_FREQ Switch Node C Output Regulator Switching Frequency 00 = 500 KHz 01 = 750 KHz 10 = 1000 KHz 11 = 1250 KHz
3:2	RWPE	10	R4E [3:2]: SWD_MODE_SELECT Switch Node D Output Regulator Mode Selection 00 = Reserved 01 = Reserved 10 = COT; DCM (Constant on Time; Discontinuous Current Mode) 11 = COT; CCM (Constant on Time; Continuous Current Mode)
1:0	RWPE	01	R4E [1:0]: SWD_SWITCHING_FREQ Switch Node D Output Regulator Switching Frequency 00 = 500 KHz 01 = 750 KHz 10 = 1000 KHz 11 = 1250 KHz

Table 171 — Register 0x4F

R4F			
Bits	Attribute	Default	Description
7	RWPE	0	R4F [7]: OUTPUT_REGULAOTRS_DISABLE_CONTROL ¹ Output Regulator SWA, SWB, SWC and SWD Disable Control for OV and UV. 0 = Disable all switching regulators of PMIC ² 1 = Disable only the effected switching output regulator; Rest of the PMIC's switching regulators remains operations ³
6:5	RWPE	0	R4F [6:5]: Reserved
4	RWPE	0	R4F [4]: SWA_OUTPUT_REGULATOR_REMOTE_SENSING SWA Output Regulator Remote Sensing Scheme on DIMM 0 = Single Ended Remote Sensing on DIMM. 1 = Differential Remote Sensing on DIMM
3	RV	0	R4F [3]: Reserved
2	RWPE	0	R4F [2]: SWC_OUTPUT_REGULATOR_REMOTE_SENSING SWC Output Regulator Remote Sensing Scheme on DIMM 0 = Single Ended Remote Sensing on DIMM. 1 = Differential Remote Sensing on DIMM
1	RWPE	0	R4F [1]: SWD_OUTPUT_REGULATOR_REMOTE_SENSING SWD Output Regulator Remote Sensing Scheme on DIMM 0 = Single Ended Remote Sensing on DIMM; Use SWD_FB_N pin as PID input pin to determine the PMIC's ID ⁴ 1 = Differential Remote Sensing on DIMM ⁵ .
0	RWPE	0	R4F [0]: SWA_SWB_PHASE_MODE_SELECT Switch Node A and Switch Node B Phase Regulator Mode Selection. 0 = Single Phase Regulator Mode 1 = Dual Phase Regulator Mode

1. Applicable to Output Over Voltage and Output Under Voltage. For VIN_Bulk input over voltage or VBIAS under voltage or VIN_Bulk Input under voltage, this bit does not apply and the PMIC always disables all switching regulators.
2. For standard DDR5 RDIMM and DDR5 LRDIMM, this bit must be always configured to '0'.
3. For DDR5 NVDIMM or any other custom DIMM, this bit can be configured to either '0' or '1' as desired. This is applicable only after VR_ENABLE bit is set to '1'. Prior to VR_ENABLE bit is set to '1', PMIC keeps all buck regulator off,
4. Up to three different PMIC can be placed on DIMM module.
5. Only one PMIC can be placed on DIMM module.

Table 172 — Register 0x50

R50			
Bits	Attribute	Default	Description
7:6	RWPE	0	R50 [7:6]: SWA_OUTPUT_CURRENT_LIMITER_WARNING_THRESHOLD_SETTING PMIC5000 COT Mode, Ivalley_limit: 00 = 4.0 A 01 = 4.5 A 10 = 5.0 A 11 = 5.5 A PMIC5010 COT Mode, Ivalley_limit: 00 = 2.0 A 01 = 2.5 A 10 = 3.0 A 11 = 3.5 A
5:4	RWPE	0	R50 [5:4]: SWB_OUTPUT_CURRENT_LIMITER_WARNING_THRESHOLD_SETTING PMIC5000 COT Mode, Ivalley_limit¹: 00 = 4.0 A 01 = 4.5 A 10 = 5.0 A 11 = 5.5 A PMIC5010 COT Mode, Ivalley_limit: 00 = 2.0 A 01 = 2.5 A 10 = 3.0 A 11 = 3.5 A
3:2	RWPE	0	R50 [3:2]: SWC_OUTPUT_CURRENT_LIMITER_WARNING_THRESHOLD_SETTING PMIC5000 COT Mode, Ivalley_limit: 00 = 4.0 A 01 = 4.5 A 10 = 5.0 A 11 = 5.5 A PMIC5010 COT Mode, Ivalley_limit: 00 = 2.0 A 01 = 2.5 A 10 = 3.0 A 11 = 3.5 A

Table 172 — Register 0x50

R50			
Bits	Attribute	Default	Description
1:0	RWPE	0	R50 [1:0]: SWD_OUTPUT_CURRENT_LIMITER_WARNING_THRESHOLD_SETTING PMIC5000 COT Mode, Ivalley_limit: 00 = 4.0 A 01 = 4.5 A 10 = 5.0 A 11 = 5.5 A PMIC5010 COT Mode, Ivalley_limit: 00 = 2.0 A 01 = 2.5 A 10 = 3.0 A 11 = 3.5 A

1. This register is applicable regardless of the setting in Table 171, “Register 0x4F” [0].

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Table 173 — Register 0x51

R51			
Bits	Attribute	Default	Description
7:6	RWPE	01	R51 [7:6]: VOUT_1.8V_VOLTAGE_SETTING VOUT 1.8 V LDO Output Voltage Setting ¹ 00 = 1.7 V 01 = 1.8 V 10 = 1.9 V 11 = Reserved
5	RWPE	0	R51 [5]: SWA_VOLTAGE_RANGE SWA Output Voltage Range Selection ² 0 = Range: 800 mV to 1435 mV for SWA; 5 mV step size. 1 = Range: 600 mV to 1235 mV for SWA; 5 mV step size
4	RWPE	0	R51 [4]: SWB_VOLTAGE_RANGE SWB Output Voltage Range Selection ^{3,4} 0 = Range: 800 mV to 1435 mV for SWB; 5 mV step size 1 = Range: 600 mV to 1235 mV for SWB; 5 mV step size
3	RWPE	0	R51 [3]: SWC_VOLTAGE_RANGE SWC Output Voltage Range Selection ⁵ 0 = Range: 800 mV to 1435 mV for SWC; 5 mV step size 1 = Range: 600 mV to 1235 mV for SWC; 5 mV step size
2:1	RWPE	01	R51 [2:1]: VOUT_1.0V_VOLTAGE_SETTING VOUT 1.0 V LDO Voltage Setting 00 = 0.9 V 01 = 1.0 V 10 = 1.1 V 11 = 1.2 V
0	RWPE	0	R51 [0]: SWD_VOLTAGE_RANGE SWD Output Voltage Range Selection ⁶ 0 = Range: 1500 mV to 2135 mV for SWD; 5 mV step size 1 = Range: 2200 mV to 2835 mV for SWD; 5 mV step size

1. The VOUT_1.8V Power Good threshold in register Table 121, “Register 0x1A” [2] is always fixed regardless of the setting in this register.
2. Range & resolution selection applies to registers Table 161, “Register 0x45” [7:1].
3. Range & resolution selection applies to registers Table 163, “Register 0x47” [7:1].
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.
5. Range & resolution selection applies to registers Table 165, “Register 0x49” [7:1].
6. Range & resolution selection applies to registers Table 167, “Register 0x4B” [7:1].

Table 174 — Register 0x58¹

R58			
Bits	Attribute	Default	Description
7	RWPE	0	R58 [7]: POWER_OFF_SEQUENCE_CONFIG0 PMIC Power Off Sequence Config 0 0 = Do Not Execute Config 0 ² 1 = Execute Config 0
6	RWPE	0	R58 [6]: POWER_OFF_SEQUENCE_CONFIG0_SWA_DISABLE Disable SWA Output Regulator. 0 = Do Not Disable SWA Output Regulator 1 = Disable SWA Output Regulator
5	RWPE	0	R58 [5]: POWER_OFF_SEQUENCE_CONFIG0_SWB_DISABLE Disable Switch Node B Output Regulator. ³ 0 = Do Not Disable SWB Output Regulator 1 = Disable SWB Output Regulator
4	RWPE	0	R58 [4]: POWER_OFF_SEQUENCE_CONFIG0_SWC_DISABLE Disable Switch Node C Output Regulator. 0 = Do Not Disable SWC Output Regulator 1 = Disable SWC Output Regulator
3	RWPE	0	R58 [3]: POWER_OFF_SEQUENCE_CONFIG0_SWD_DISABLE Disable SWD Output Regulator. 0 = Do Not Disable SWD Output Regulator 1 = Disable SWD Output Regulator
2:0	RWPE	0	R58 [2:0]: POWER_OFF_SEQUENCE_CONFIG0_IDLE Idle time after Power Off Sequence Config 0 ⁴ 000 = 0 ms 001 = 1 ms 010 = 2 ms 011 = 3 ms 100 = 4 ms 101 = 5 ms 110 = 6 ms 111 = 7 ms

1. If more than one configuration register is used for power off sequence, first register must start at Table 174, “Register 0x58” and it must go in sequential order to Table 177, “Register 0x5B,” to turn off all regulators. In other words, there must not be any gap of the register that is used for power off sequence.
2. If bit [7] = ‘0’, bits [6:3] must be programmed as ‘0000’. If bit [7] = ‘1’, at least one of the bits [6:3] must be programmed as ‘1’.
3. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.
4. Idle time is the additional time after soft-stop time expires. The PMIC waits sum of soft stop time and idle time before it executes the next power off sequence configuration register. If more than one regulators are disabled, the PMIC uses the largest value of the soft stop time among the regulators that are disabled in this configuration register.

Table 175 — Register 0x59^{1,2}

R59			
Bits	Attribute	Default	Description
7	RWPE	0	R59 [7]: POWER_OFF_SEQUENCE_CONFIG1 PMIC Power Off Sequence Config1 ³ 0 = Do Not Execute Config 1 1 = Execute Config 1
6	RWPE	0	R59 [6]: POWER_OFF_SEQUENCE_CONFIG1_SWA_DISABLE Disable SWA Output Regulator. 0 = Do Not Disable SWA Output Regulator 1 = Disable SWA Output Regulator
5	RWPE	0	R59 [5]: POWER_OFF_SEQUENCE_CONFIG1_SWB_DISABLE Disable Switch Node B Output Regulator. ⁴ 0 = Do Not Disable SWB Output Regulator 1 = Disable SWB Output Regulator
4	RWPE	0	R59 [4]: POWER_OFF_SEQUENCE_CONFIG1_SWC_DISABLE Disable Switch Node C Output Regulator. 0 = Do Not Disable SWC Output Regulator 1 = Disable SWC Output Regulator
3	RWPE	0	R59 [3]: POWER_OFF_SEQUENCE_CONFIG1_SWD_DISABLE Disable Switch Node D Output Regulator. 0 = Do Not Disable SWD Output Regulator 1 = Disable SWD Output Regulator
2:0	RWPE	0	R59 [2:0]: POWER_OFF_SEQUENCE_CONFIG1_IDLE Idle time after Power Off Sequence Config 1 ⁵ 000 = 0 ms 001 = 1 ms 010 = 2 ms 011 = 3 ms 100 = 4 ms 101 = 5 ms 110 = 6 ms 111 = 7 ms

1. If any regulators are disabled in Table 174, “Register 0x58” [6:3], those regulators must be configured as ‘1’ in this sequence.
2. If more than one configuration register is used for power off sequence, first register must start at Table 174, “Register 0x58” and it must go in sequential order to Table 177, “Register 0x5B,” to turn off all regulators. In other words, there must not be any gap of the register that is used for power off sequence.
3. If bit [7] = ‘0’, bits [6:3] must be programmed as ‘0000’. If bit [7] = ‘1’, at least one of the bits [6:3] must be programmed as ‘1’.
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.
5. Idle time is the additional time after soft-stop time expires. The PMIC waits sum of soft stop time and idle time before it executes the next power off sequence configuration register. If more than one regulators are disabled, the PMIC uses the largest value of the soft stop time among the regulators that are disabled in this configuration register.

Table 176 — Register 0x5A^{1,2}

R5A			
Bits	Attribute	Default	Description
7	RWPE	0	R5A [7]: POWER_OFF_SEQUENCE_CONFIG2 PMIC Power Off Sequence Config 2 ³ 0 = Do Not Execute Config 2 1 = Execute Config 2
6	RWPE	0	R5A [6]: POWER_OFF_SEQUENCE_CONFIG2_SWA_DISABLE Disable SWA Output Regulator. 0 = Do Not Disable SWA Output Regulator 1 = Disable SWA Output Regulator
5	RWPE	0	R5A [5]: POWER_OFF_SEQUENCE_CONFIG2_SWB_DISABLE Disable Switch Node B Output Regulator. ⁴ 0 = Do Not Disable SWB Output Regulator 1 = Disable SWB Output Regulator
4	RWPE	0	R5A [4]: POWER_OFF_SEQUENCE_CONFIG2_SWC_DISABLE Disable Switch Node C Output Regulator. 0 = Do Not Disable SWC Output Regulator 1 = Disable SWC Output Regulator
3	RWPE	0	R5A [3]: POWER_OFF_SEQUENCE_CONFIG2_SWD_DISABLE Disable Switch Node D Output Regulator. 0 = Do Not Disable SWD Output Regulator 1 = Disable SWD Output Regulator
2:0	RWPE	0	R5A [2:0]: POWER_OFF_SEQUENCE_CONFIG2_IDLE Idle time after Power Off Sequence Config 2 ⁵ 000 = 0 ms 001 = 1 ms 010 = 2 ms 011 = 3 ms 100 = 4 ms 101 = 5 ms 110 = 6 ms 111 = 7 ms

1. If any regulators are disabled in Table 174, “Register 0x58” [6:3], and Table 175, “Register 0x59,” [6:3] those regulators must be configured as ‘1’ in this sequence.
2. If more than one configuration register is used for power off sequence, first register must start at Table 174, “Register 0x58” and it must go in sequential order to Table 177, “Register 0x5B,” to turn off all regulators. In other words, there must not be any gap of the register that is used for power off sequence.
3. If bit [7] = ‘0’, bits [6:3] must be programmed as ‘0000’. If bit [7] = ‘1’, at least one of the bits [6:3] must be programmed as ‘1’.
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.
5. Idle time is the additional time after soft-stop time expires. The PMIC waits sum of soft stop time and idle time before it executes the next power off sequence configuration register. If more than one regulators are disabled, the PMIC uses the largest value of the soft stop time among the regulators that are disabled in this configuration register.

Table 177 — Register 0x5B^{1,2}

R5B			
Bits	Attribute	Default	Description
7	RWPE	0	R5B [7]: POWER_OFF_SEQUENCE_CONFIG3 PMIC Power Off Sequence Config 3 ³ 0 = Do Not Execute Config 3 1 = Execute Config 3
6	RWPE	0	R5B [6]: POWER_OFF_SEQUENCE_CONFIG3_SWA_DISABLE Disable SWA Output Regulator. 0 = Do Not Disable SWA Output Regulator 1 = Disable SWA Output Regulator
5	RWPE	0	R5B [5]: POWER_OFF_SEQUENCE_CONFIG3_SWB_DISABLE Disable SWB Output Regulator. ⁴ 0 = Do Not Disable SWB Output Regulator 1 = Disable SWB Output Regulator
4	RWPE	0	R5B [4]: POWER_OFF_SEQUENCE_CONFIG3_SWC_DISABLE Disable Switch Node C Output Regulator. 0 = Do Not Disable SWC Output Regulator 1 = Disable SWC Output Regulator
3	RWPE	0	R5B [3]: POWER_OFF_SEQUENCE_CONFIG3_SWD_DISABLE Disable Switch Node D Output Regulator. 0 = Do Not Disable SWD Output Regulator 1 = Disable SWD Output Regulator
2:0	RV	0	R5B [2:0]: Reserved

1. If any regulators are disabled in Table 174, “Register 0x58” [6:3], Table 175, “Register 0x59,” [6:3] and Table 176, “Register 0x5A,” [6:3], those regulators must be configured as ‘1’ in this sequence.
2. If more than one configuration register is used for power off sequence, first register must start at Table 174, “Register 0x58” and it must go in sequential order to Table 177, “Register 0x5B,” to turn off all regulators. In other words, there must not be any gap of the register that is used for power off sequence.
3. If bit [7] = ‘0’, bits [6:3] must be programmed as ‘0000’. If bit [7] = ‘1’, at least one of the bits [6:3] must be programmed as ‘1’.
4. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 178 — Register 0x5D

R5D			
Bits	Attribute	Default	Description
7:5	RWPE	001	R5D [7:5]: SWA_OUTPUT_SOFT_START_TIME SWA Output Regulator Soft Start Time After VR Enable ¹ 000 = 1 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms ... 111 = 14 ms
4	RV	0	R5D [4]: Reserved
3:1	RWPE	001	R5D [3:1]: SWB_OUTPUT_SOFT_START_TIME SWB Output Regulator Soft Start Time After VR Enable ^{1,2} 000 = 1 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms ... 111 = 14 ms
0	RV	0	R5D [0]: Reserved

1. This is the time it takes for buck regulator to go from 0 V to steady state voltage (the actual voltage varies based on pre-programmed voltage).
2. Only applicable if Table 171, “Register 0x4F” [0] = ‘0’.

Table 179 — Register 0x5E

R5E			
Bits	Attribute	Default	Description
7:5	RWPE	001	R5E [7:5]: SWC_OUTPUT_SOFT_START_TIME SWC Output Regulator Soft Start Time After VR Enable ¹ 000 = 1 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms ... 111 = 14 ms
4	RV	0	R5E [4]: Reserved
3:1	RWPE	001	R5E [3:1]: SWD_OUTPUT_SOFT_START_TIME SWD Output Regulator Soft Start Time After VR Enable ¹ 000 = 1 ms 001 = 2 ms 010 = 4 ms 011 = 6 ms ... 111 = 14 ms
0	RV	0	R5E [0]: Reserved

1. This is the time it takes for buck regulator to go from 0 V to steady state voltage (the actual voltage varies based on pre-programmed voltage)



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